



Aum Sri Sairam

Sri Sathya Sai Institute of Educare

Chromepet, Chennai - 600 044

www.sssieducare.org



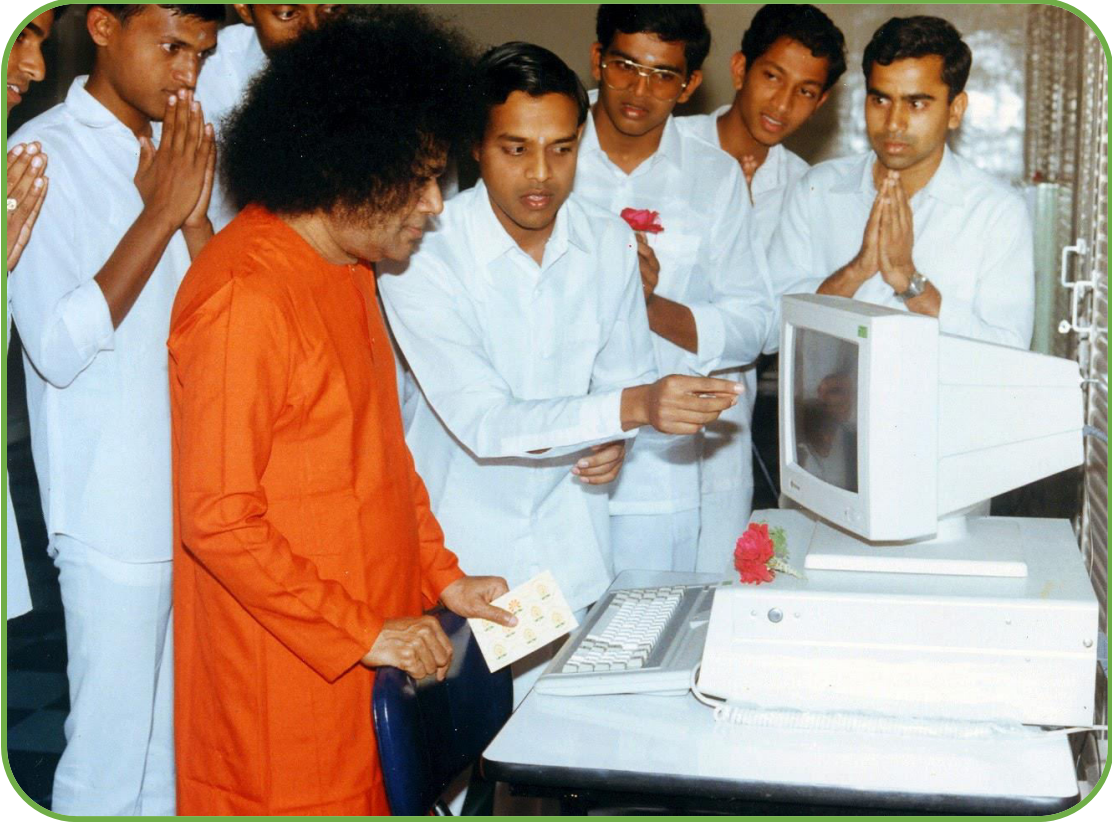
CS Centum for Sure

XII Computer Science Board Preparation Material 2022-23



**Published by
Department of Computer Science, SSSIE**

Aum Sri Sairam

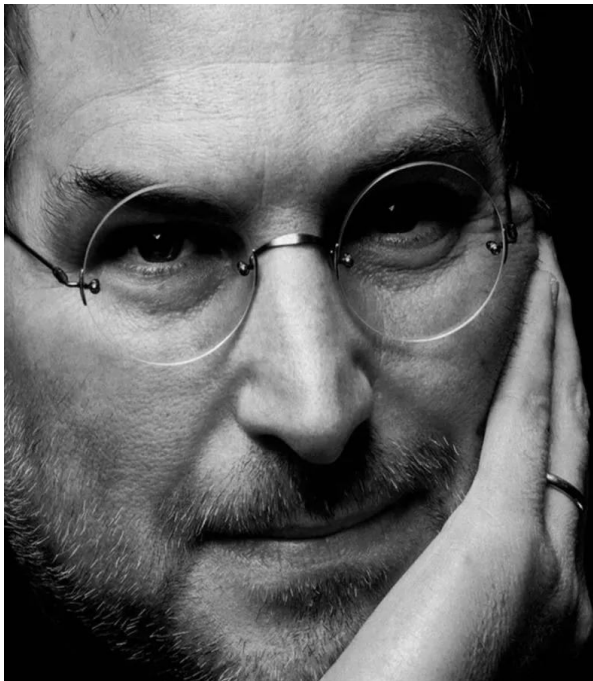


**FOLLOW THE 'INNERNET,'
NOT THE INTERNET.**

Bhagawan Sri Sathya Sai Baba

CS - Centum for Sure 2022-23

S.No.	Title	Page
Coffee with Coding		
1	PYTHON FUNDAMENTALS HANDOUT	1
2	DATA HANDLING HANDOUT	10
3	CONDITIONALS & LOOPS HANDOUT	18
4	SLTD CODE COMPARISON HANDOUT	31
5	STRINGS HANDOUT	45
6	DICTIONARY HANDOUT	52
7	FUNCTIONS HANDOUT	54
8	TEXT FILE CODE COMPARISON HANDOUT	57
9	CSV FILE HANDLING HANDOUT	63
10	BINARY FILE HANDLING HANDOUT	65
11	LIST MANIPULATION (DIGITS) HANDOUT	69
12	LIST MANIPULATION (STRINGS) HANDOUT	77
13	STACK HANDOUT	80
14	SQL MADE EASY	88
15	CONNECTIVITY HANDOUT	106
16	NETWORKING CASE STUDY HANDOUT	110
Tasty Theory		
17	VIVA HANDOUT - PYTHON REVIEW	114
18	VIVA HANDOUT - FUNCTIONS	129
19	VIVA HANDOUT - FILE HANDLING	137
20	VIVA HANDOUT - DATA STRUCTURE & STACK	141
21	VIVA HANDOUT - SQL	144
22	VIVA HANDOUT - CONNECTIVITY	150
23	VIVA HANDOUT - NETWORKING	152
24	NCERT FOCUS - PYTHON REVIEW	162
25	NCERT FOCUS - FLOW OF CONTROL	169
26	NCERT FOCUS - FUNCTIONS	171
27	NCERT FOCUS - STRINGS	173
28	NCERT FOCUS - LISTS	175
29	NCERT FOCUS - TUPLES & DICTIONARIES	177
30	NCERT FOCUS - FILE HANDLING	179
31	NCERT FOCUS - STACK	185
32	NCERT FOCUS - DATABASE CONCEPTS	187
33	NCERT FOCUS - NETWORKING	194
34	OUTSTANDING OUTPUTS	205
35	REJOICE & REASON	223
36	MAGICAL MCQs	226
37	THE BIBLE 2022-23	260



Steve Jobs

1955-2011

**"The ones who are crazy enough to think
that they can change the world,
are the ones who do."**

"If you are working on something that you really care about, you don't have to be pushed. The vision pulls you." – Steve Jobs

XII COMPUTER SCIENCE

Towards Getting Centum in CS – See Yes!

1. PYTHON FUNDAMENTALS HANDOUT

QUESTION-BASED MODELS AND NOTES

1) IDENTIFY THE TOKENS:

Identify the tokens:

a) for b) : c) % d) 34

TOKENS

- What are tokens?
- The smallest individual unit in a program is token or lexical unit

TOKENS OF PYTHON

- Keywords : while, for, break etc.
- Identifiers : _sai, all67 etc.
- Literals : 34, 6.7, True, None, [6,7,8] etc.
- Operators : &, %, <=, or etc.
- Expressions: a+b*c, a or b and not c etc.
- Punctuators: ' " # \ () [] @ , : . =

2) IDENTIFY THE OPERATORS

Which of the following is a valid arithmetic operators in Python

(i) // (ii) ? (iii) < (iv) not

Arithmetic Operators

Operator	Meaning	Example
+	Addition	4 + 7 → 11
-	Subtraction	12 - 5 → 7
*	Multiplication	6 * 6 → 36
/	Division	30 / 5 → 6
%	Modulus	10 % 4 → 2
//	Quotient	18 // 5 → 3
**	Exponent	3 ** 5 → 243

LOGICAL OPERATORS

LOGICAL OPERATORS	
and	Logical AND
or	Logical OR
not	Logical NOT

ASSIGNMENT OPERATORS

ASSIGNMENT OPERATORS	
=	assignment
/=	Assign quotient
*=	Assign product
+=	Assign sum
-=	Assign difference
%=	Assign remainder
**=	Assign exponent
//=	Assign floor division

RELATIONAL OPERATORS

RELATIONAL OPERATORS	
<	Less than
>	Greater than
<=	Less than or equal to
>=	Greater than or equal to
==	Equal to
!=	Not equal to

BITWISE OPERATORS

BITWISE OPERATORS	
&	Bitwise AND
^	Bitwise exclusive OR(XOR)
	Bitwise OR
<<	Bitwise Left Shift
>>	Bitwise Right Shift

SHIFT OPERATORS

SHIFT OPERATORS	
<<	Shift left
>>	Shift right

MEMBERSHIP OPERATORS

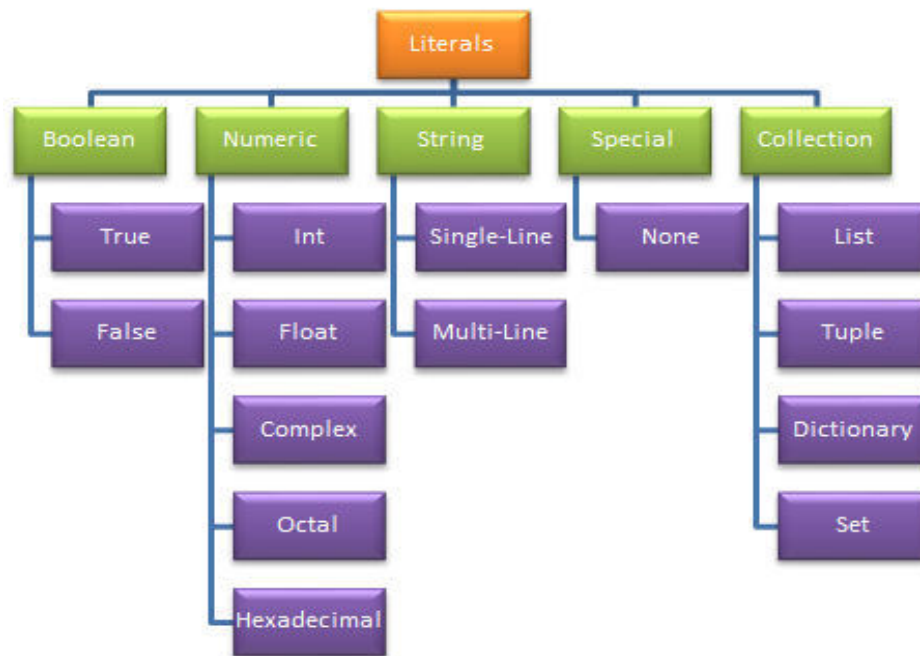
MEMBERSHIP OPERATORS	
in	Whether variable present in the sequence
not in	Whether variable not present in the sequence

IDENTITY OPERATORS

IDENTITY OPERATORS	
is	Is the identity same?
is not	Is the identity not same?

3) IDENTIFY THE LITERALS

Identify the literals: (i) 12 (ii) "sai" (iii) True (iv) 0o134 (v) 445E05



4) IDENTIFY THE KEYWORDS

Identify the valid keyword(s) from the following:

Marks1, \$Roll, Avg Marks, IF, _sales2008, while, name, break, Continue

```
help> keywords

Here is a list of the Python keywords. Enter any keyword to get more help.

False      def        if         raise
None       del        import    return
True       elif       in        try
and        else       is        while
as         except    lambda    with
assert     finally   nonlocal  yield
break      for       not
class     from     or
continue  global  pass

help> 
```

5) IDENTIFY THE IDENTIFIERS

Identify the valid identifier(s) from the following:

Marks1, \$Roll, Avg Marks, IF, _sales2008, while, name, break, Continue

RULES FOR FORMING IDENTIFIERS

- ✓first character must be a letter or underscore _
- ✓Upper and lower are different, Python is case-sensitive
- ✓Digits 0 to 9 can be used except as first character
- ✓Unlimited in length
- ✓cannot contain special symbols except _
- ✓Keywords cannot be used as identifiers

- Valid: Myfile, DATEg_7_99, _HJIO, FILE13
- Invalid: DATA-REC, 23LTE, break, my.file

6) IDENTIFY THE DATA TYPES:

Identify the correct data type: a) 'a' b) 5+7j c) (5,) d) {a:"ram"}

Name	Type	Description
Integers	int	Whole numbers, such as: 3 300 200
Floating point	float	Numbers with a decimal point: 2.3 4.6 100.0
Strings	str	Ordered sequence of characters: "hello" 'Sammy' "2000" "楽しい"
Lists	list	Ordered sequence of objects: [10,"hello",200.3]
Dictionaries	dict	Unordered Key:Value pairs: {"mykey": "value", "name": "Frankie"}
Tuples	tup	Ordered immutable sequence of objects: (10,"hello",200.3)
Sets	set	Unordered collection of unique objects: {"a","b"}
Booleans	bool	Logical value indicating True or False

7) IDENTIFY THE DATA TYPE FOR THE METHODS/FUNCTIONS BELOW:

a) extend(), b) setdefault(), c) print(), d) replace(), e) remove()

STRING METHODS:

len(), capitalize(), title(), lower(), upper(), count(), find(), index(),endswith(), startswith(), isalnum(),isalpha(),

isdigit(), islower(), isupper(), isspace(), lstrip(), rstrip(), strip(), replace(), join(), partition(), split()

LIST METHODS:

len(), list(), append(), extend(), insert(), count(), index(), remove(), pop(), reverse(), sort(), sorted(), min(), max(), sum()

TUPLE METHODS:

len(), tuple(), count(), index(), sorted(), min(), max(), sum()

DICTIONARY METHODS:

len(), dict(), keys(), values(), items(), get(), update(), del, clear(), fromkeys(), copy(), pop(), popitem(), setdefault(), max(), min(), count(), sorted(), copy()

BUILT-IN METHODS:

print(), input(), sum(), len(), id(), type(), list(), tuple(), int(), str(), sorted() etc.

8) IDENTIFY THE MODULES FOR THE METHODS/FUNCTIONS BELOW:

Name the Python Library module which need to be imported to invoke the following function:

a) pow(), b) randint(), c) dump(), d) writer()

math module

pi, e, sqrt(), ceil(), floor(), pow(), fabs(), sin(), cos(), tan()

random module

random(), randint(), randrange()

statistics module

mean(), median(), mode()

9) IDENTIFY THE TYPE OF NUMBERS

a)56 b)5+6j c)0x342F d)23.2

- 456 -Integer (Decimal)
- 0o456 – Integer(octal)
- 0x456 – Integer(Hexadecimal)
- True, False – Boolean integers
- 89.3 – Float
- 567E09, 454E-9 - Float
- 4+3j – Complex number

10) COMPLEX NUMBER BASED QUESTION

`a = 3+7.4j`

Write the real and imaginary part of the complex number above. Also, write the statement to print them.

```
>>>a.real #3.0
```

```
>>>a.imag #7.4
```

11) WRITING EQUIVALENT DECIMAL/OCTAL/HEXADECIMAL VALUES:

DECIMAL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
OCTAL	0	1	2	3	4	5	6	7	10	11	12	13	14	15	16	17
HEX	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

Write the equivalent decimal value for the hexadecimal number E:

Ans: 16

Python automatically converts octal and hex numbers to their equivalent decimal on the interpreter.

```
>>>a = 0o10
```

```
>>>print(a)      #prints 8 decimal equivalent
```

```
>>>a = 0xB
```

```
>>>print(a)      #prints 11 decimal equivalent
```

9) print() STATEMENT QUESTION MODELS

a) print() with end OR sep:

```
>>>print("sai" "ram")
```

sairam

```
>>>print("sai", "ram", end = '%')
```

sai ram%

```
>>>print("sai", "ram")
```

sai ram

```
>>>print("sai", "ram", sep = '%')
```

sai%ram

```
>>>print("sai", "ram", end = '%', sep = '%')
```

sai%ram%

b) print() WITH SINGLE AND MULTILINE STRINGS

```
>>>print("Sairam")      #Sairam
```

```
>>>print("Sai\          #Sairam
ram")
```

```
>>>print("'''Sai          #Sai
```

```
ram")
```

```
ram
```

(prints one below the other, due to '\n' created)

c) print() with escape sequence

```
>>>print("Selfless \n service")
```

```
Selfless
```

```
service
```

```
>>>print("Selfless \t service")
```

```
Selfless
```

```
service
```

```
>>>print("Selfless \n service \t is supreme")
```

```
Selfless
```

```
service
```

```
is supreme
```

```
>>>print("India\'s Capital is Chennai")
```

```
India's Capital is Chennai
```

```
>>> print("India\'s Capital is \"Chennai\"")
```

```
India's Capital is "Chennai"
```

10) input() BASED QUESTIONS:

string input

```
str = input("Enter your value:")
```

integer input

```
no = int(input("Enter your value:"))
```

float input

```
no = float(input("Enter your value:"))
```

list input

```
a = list(input("Enter your list values:"))
```

```
>>>sairam
```

```
>>>a
```

```
['s','a','i','r','a','m']
```

tuple input

```
a = tuple(input("Enter your tuple values:"))
```

```
>>>12345
```

```
>>>a
```

```
['1','2','3','4','5']
```

eval(input())

```
a = eval(input("Enter :"))
```

```
>>>Enter:[1,2,3,4,5]
```

```
a stores list [1,2,3,4,5]
```

```
a = eval(input("Enter:"))
```

```
>>>Enter: {1:'sai', 2:'baba'}
```

```
a stores dictionary {1:'sai', 2:'baba'}
```

(Directly we can store list/tuple etc. (not dict.))

Keeping serial no. while getting input:

```
for i in range(1,5):
```

```
    n = int(input("Enter subject " +str(i)+" Score:"))
```

We get output as below:

```
Enter subject 1 Score: 45
```

```
Enter subject 2 Score: 89
```

```
Enter subject 3 Score: 90
```

```
Enter subject 4 Score: 34
```

11) type() BASED QUESTION

Predict the output: type('CBSE')

```
>>>type(0)      #<class 'int'>
>>>type("0")    #<class 'string'>
>>>type("True")  #<class 'string'>
>>>type(True)    #<class 'bool'>
>>>type([1,2])   #<class 'list'>
>>>type((1,2))   #<class 'tuple'>
>>>type({1:6})   #<class 'dict'>
>>>type(type())  #TypeError
```

```
>>> type(print())
# <class 'NoneType'> #prints in next line
```

```
>>>type(9.4)      #<class 'float'>
>>>type(None)     #<class 'NoneType'>
>>> type(type(12)) #<class 'type'>
>>> type(type(type('sai')) # <class 'type'>
```

```
>>> type(print('sairam'))
sairam
<class 'NoneType'>
```

12) len() BASED QUESTION

a)Getting length of single objects

```
>>> len(1)      #TypeError
>>> len('1')     #1
>>> len('sairam') #6
>>> len([4,5])   #2
>>> len((5,2,'sai')) #3
>>> len({3:'ram', 2:'krish'}) # 2
>>> len([])      #0
```

```
>>> len({})      #0
>>> len({})      #0
>>> len([' '])   #1
>>> len([''])    #1
>>> len([""])    #0
>>> len("")      #0
>>> len(type('sairam')) # Type Error
```

b)Getting length of nested objects

```
>>> len([1,2,[3,6,7],[4,5,6,[3,1]]]) #4
>>> len((3,4,(3,2),(2,3,(4,5,(2,1))))) #4
```

c) Getting length after explicit conversion

```
>>> len(str(float(2)))
#3
>>> len(str((float(int(bool(len('sairam'))))))))
#3
```

```
>>> len(str(bool('True')))  
#4
```

d) Getting length of multiline strings

```
str = "Sairam"  
>>>len(str)  #6  
str = "Sai\'s ram"  
>>>len(str)  #9
```

```
str ="Sai\  
ram is\  
great"  
>>> len(str)  #14
```

```
str = "Sai ram"  
>>>len(str)  #7
```

```
str =""Sai  
ram is  
great"  
>>> len(str)  #16
```

13) None BASED QUESTION

```
a = None  
>>>a          #no output  
>>>print(a)    #prints None
```

```
>>>print(print("sairam"))  
sairam  
None
```

```
>>>print(print("sairam", end = " "))  
Sairam None
```

```
>>> str(print("sairam")) + "Baba"  
sairam  
'NoneBaba'
```

14) bool() BASED QUESTION

```
>>> bool(1)      # True  
>>> bool('1')    # True  
>>> bool('0')    # True  
>>> bool(0)      # False  
>>> bool(True)   # True  
>>> bool(False)  # False  
>>> bool('False') # False  
>>> bool('sai')  # True
```

```
>>> bool("")      # False  
>>> bool(' ')     # True  
>>> bool([])      # False  
>>> bool(())      # False  
>>> bool({1:'sai'}) # True  
>>> bool({})      # False  
>>> bool(float(int(0.456))) # False  
>>> bool(str(float(int(0.456)))) # True
```



```
>>> bool(print())    #prints newline and False
```

```
>>>bool(0j)          #False, complex no.
```

15) VARIABLE INITIALIZATION BASED QUESTIONS

Correct: a = 12

Incorrect: 12 = a

In a = 12, Which are Lvalue and Rvalue?

Lvalue: a

Rvalue: 12

Multiple assignment:

Correct: a=b=c=7

Invalid: a, b, c = 7

Multiple assignment with multiple values:

Correct: a,b,c = 1,2,3

Invalid: a=b=c=1,2,3

16) VARIABLE INTERNALS-BASED QUESTION

```
>>>a = 45
```

```
>>>type(a)  # type of the variable
```

```
>>>a        # value of the variable
```

```
>>>id(a)     # memory address of the variable
```

~~~~~

*"Success is not the key to happiness. Happiness is the key to success. If you love what you are doing, you will be successful." – Albert Schweitzer*

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS - See Yes!*

### 2. DATA HANDLING HANDOUT

#### QUESTION-BASED MODELS AND NOTES

#### 1) OPERATOR-BASED QUESTIONS

##### A) ARITHMETIC OPERATORS

```
>>>4/2      #2.0
>>>4//2     #2
>>>4.2//2   #2.0 (floor division)
>>>4//2.0   #2.0

>>>4*2      #8
>>>4**2     #16 (gives power)
>>>3**2     #9
>>>5**2     #25
>>>10%2     # 0      (gives the remainder)
>>>10.0%2   # 0.0
>>>10%2.0   # 0.0
>>> 17%3    # 2
>>>103%10   #3      (%10 used to get unit digit)
```

##### B) RELATIONAL OPERATORS

###### Numeric comparison

```
>>>4 < 8    # True
>>>4 == 3   # False
>>>5 != 4    # True
>>>8 >= 8   # True
```

#### Strings comparison

```
>>>"sai" > "Sai"    # True
```

```
>>>"Sun" > "moon"  # False
```

(Capital S is smaller than small s, comparison operator will be checking the ASCII value.)

```
>>>"sai" < "saibaba" # True
```

```
>>>"sai" < "Saibaba" #False
```

| Character | ASCII Value | Character | ASCII Value |
|-----------|-------------|-----------|-------------|
| A         | 65          | a         | 97          |
| B         | 66          | b         | 98          |
| C         | 67          | c         | 99          |
| D         | 68          | d         | 100         |
| E         | 69          | e         | 101         |
| F         | 70          | f         | 102         |
| G         | 71          | g         | 103         |
| H         | 72          | h         | 104         |
| I         | 73          | i         | 105         |
| J         | 74          | j         | 106         |
| K         | 75          | k         | 107         |
| L         | 76          | l         | 108         |
| M         | 77          | m         | 109         |
| N         | 78          | n         | 110         |
| O         | 79          | o         | 111         |
| P         | 80          | p         | 112         |
| Q         | 81          | q         | 113         |
| R         | 82          | r         | 114         |
| S         | 83          | s         | 115         |
| T         | 84          | t         | 116         |

|   |    |   |     |
|---|----|---|-----|
| U | 85 | u | 117 |
| V | 86 | v | 118 |
| W | 87 | w | 119 |
| X | 88 | x | 120 |
| Y | 89 | y | 121 |
| Z | 90 | z | 122 |

The ASCII value of space ' ' is 32, which is the difference between capital and small alphabets.

For example:

ASCII of A = 65, add 32 with 65 and you will get 97, which is the ASCII value of a,

### **ORD()**

To get ASCII value of any character, **ord()** method is used.

```
>>>ord('B')
```

```
66
```

```
>>>ord('b')
```

```
98
```

### **CHAR()**

To get the character of an ASCII code, **char()** method is used.

```
>>>char(65)
```

```
'A'
```

```
>>>char(99)
```

```
'c'
```

### **List/Tuple comparison**

```
>>>[1,2]==[1,2]
```

```
True
```

```
>>> [1,2]==[1,1]
```

```
False
```

```
>>> [1,2]>[3,2]
```

```
False
```

```
>>> (1,2)<(4,2)
```

```
True
```

```
>>> [1,2,3,4]>[5]
```

```
False
```

```
>>> ('a','b','c')>('H','J')
```

```
True
```

### **Dictionary comparison**

With dictionary, it produces TypeError

```
>>>{'a':1, 'b':2}<{'a':1, 'b':2} #produces error
```

### **Chained comparison**

(‘and’ logical operation is involved in between)

```
>>>4<6>2
```

```
True
```

```
>>>4>6>2
```

```
False
```

```
>>>4>6<2
```

```
False
```

```
>>>4<6<2
```

```
False
```

```
>>>[1,2]>[3,4]<[9,1]
```

```
False
```

```
>>> [1,2]>[3,4]<[9,1]
```

```
False
```

```
>>> [1,2]<[3,4]<[9,1]
```

```
True
```

```
>>>'a'>'A'<'c'
```

True

>>>'h'<'a'<'A'

False

>>>'h'>'a'>'A'

True

### C) LOGICAL OPERATORS

First, learn TRUTH table: (1 means True, 0 means False)

AND Truth Table

| Inputs |   | Output  |
|--------|---|---------|
| A      | B | Y = A.B |
| 0      | 0 | 0       |
| 0      | 1 | 0       |
| 1      | 0 | 0       |
| 1      | 1 | 1       |

OR Truth Table

| Inputs |   | Output  |
|--------|---|---------|
| A      | B | Y = A+B |
| 0      | 0 | 0       |
| 0      | 1 | 1       |
| 1      | 0 | 1       |
| 1      | 1 | 1       |

NOT gate Truth Table

| Input A | Output X=A |
|---------|------------|
| 0       | 1          |
| 1       | 0          |

**AND is adamant** (wants both operands to be True to result True)

**OR is full of Pyar** (If 1 operand is True, results True)

**Example for AND operation: Going to Parthi**

| PARENTS PERMISSION | TICKET | GOING TO PARTHI |
|--------------------|--------|-----------------|
| NO                 | NO     | NO              |
| NO                 | YES    | NO              |
| YES                | NO     | NO              |
| YES                | YES    | YES             |

**Example for OR operation: Attending online class**

| MOBILE | LAPTOP | ATTENDING ONLINE CLASS |
|--------|--------|------------------------|
| NO     | NO     | NO                     |
| NO     | YES    | YES                    |
| YES    | NO     | YES                    |
| YES    | YES    | YES                    |

### QUESTIONS & ANSWERS

>>> 4 or 2 #4

>>>4 and 2 #2

>>>4 or 0 #4

>>>4 and 0 #0

>>>0 or 4 #4

>>>0 and 4 #0

>>>"sai" or "baba" #"sai"

>>>"sai" and "baba" #"baba"

>>>"sai" and "" #" "

>>>not False # True

>>>not not True # True

>>>not not False # False

### D) MEMBERSHIP OPERATOR-BASED QUESTIONS

>>>"ram" in "Sairam"

True

>>>"baba" not in "Saibaba"

False

>>>"Sathya" in "Sathyasai"

False

>>>1 in [1,2,3]

True

>>> [1] in [1,2,3]

False

```
>>> [1] in [[1],2,3]
```

True

```
>>>'a' in 'baba'
```

True

### **E) IDENTITY OPERATOR-BASED QUESTIONS**

(checks the memory location)

```
>>>"ram" is "Sairam"
```

False

```
>>>"baba" is "baba"
```

True

```
>>>"Sathya" is not "Sathyasai"
```

True

```
>>>8 is not 12
```

True

### **Special feature of Identity operators**

```
a = 5
```

```
b = 5
```

```
>>>a == b
```

True

```
>>>a is b
```

True

But,

```
a = 257
```

```
b = 257
```

```
>>>a==b
```

True (values are same)

```
>>>a is b
```

False (location is different)

### **Reason:**

Initially, a and b point to the same interned object in memory, but when their values are outside the range of common integers (ranging from -5 to 256), they're stored at separate memory addresses.

### **Similarly,**

```
>>>a = 'sai'
```

```
>>>b = 'sai'
```

```
>>>a is b
```

True

```
>>>c = input("Enter your name:")
```

Enter your name:sai

```
>>>a is c
```

False

(Due to c stored in different location)

~~~~~

2) ORDER OF PRECEDENCE

Follow, **PECUM PAXOR NAO** and **OPERATOR ASSOCIATIVITY** while solving mixed expressions.

OPERATOR PRECEDENCE		
	OPERATOR	DESCRIPTION
P E C U M P A X O R	()	Parentheses
	**	Exponentiation
	~x	Bitwise NOT
	+x, -x	Unary + Unary -
	*, /, //, %	Multiplication, Division, FD, Modulus
	+, -	Addition, Subtraction
	&	Bitwise AND
	^	Bitwise XOR
		Bitwise OR
	<, <=, >, >=, <>, !=, ==, is, is not	Relational and identity operators
N A O	not x	Boolean NOT
	and	Boolean AND
	or	Boolean OR

HIGHEST

LOWEST

Activate Windows
Go to PC settings to activate Windows

- Associativity is the order in which an expression is evaluated.
- all operators have left-to-right associativity
- Only ** has right-to-left associativity

```
>>> 7 > 6 and 4 is 3
```

```
False
```

```
>>> 8/2+5//2-2
```

```
4.0
```

```
>>> 5**2+3**2 or 4**2
```

```
34
```

```
>>> 10 - 4 * 2
```

```
2
```

```
>>> (10 - 4) * 2
```

```
>>> 5 * (2 // 3)
```

```
0
```

```
>>> 2 ** 3 ** 2
```

```
512
```

```
>>> (2 ** 3) ** 2
```

```
64
```

```
>>> 24 ** 2 // 4 % 25 / 19 * 8
```

```
8.0
```

```
>>> 2**4 + (5 + 5)**(1 + 1)
```

```
116
```

```
>>> "sai" in "sairam" or "baba" not in "sairam"
```

```
True
```

```
>>> "Come" or 'go' and 'Baba' > 'Sairam'
```

```
'Come'
```


3) TYPE CONVERSION BASED QUESTION

```
>>> a = 2
>>> str(a)
'2'
>>> float(a)
2.0
>>> str(float(a))
'2.0'
>>> str(float(bool(a)))
'1.0'
>>> bool(str(float(a)))
True
>>> int(5/2)
2
>>> float(5//2)
2.0
>>> int(float(5//2))
2
```

4) CONVERT ELEMENTS INTO SEQUENCE DATA TYPE

a = 5, b = 6, c = 7, d = 8

Make list using these values.

```
lst = [a,b,c,d]
```

Make tuple using these values.

```
tup = (a,b,c,d)
```

Make dictionary using a,b,c,d as keys and 5,6,7 and 8 as values.

```
dic = {'a':5, 'b':6, 'c':7, 'd':8}
```

5) MUTABILITY AND IMMUTABILITY BASED QUESTION

```
>>>a = "Sairam"
```

>>>a[0] = 'J' produces type error as strings are immutable. (a = "Jairam" altogether we can change)

```
>>>a = [1,2,3,5]
```

```
>>>a[0] = 11
```

```
>>>a
```

```
[11,2,35]      # list gets changed as it is mutable.
```

6) math MODULE BASED QUESTION

```
>>> import math
```

```
>>> math.floor(5.2)
```

```
5
```

```
>>> math.ceil(5.2)
```

```
6
```

```
>>> round(5.2)
```

```
5  (does not need math module)
```

```
>>> round(5.7)
```

```
6
```

```
>>> math.pow(5,2)
```

```
25.0
```

```
>>> math.fabs(4)
```

```
4
```

(calculates how much distance the number is from zero)

```
>>> math.fabs(-6)
```

```
6.0
```

Other functions: sin(), cos(), tan(), log10(), exp() etc.

7) random MODULE BASED QUESTION

random.random() - generates random float value between 0.0 and 1.0 (less than 1.0)

```
>>> random.random()
```

```
0.5862529697448331
```

```
>>> random.random()
```

```
0.6214951713307775
```

>>> 5+random.random()*10 - generates float value from 5 to number less than 15

```
12.032685619720363
```

```
>>> 5+random.random()*10
```

```
13.618864604679896
```

>>> 21+random.random()*4 - generates float value from 21 to number less than 25

```
24.233493200778895
```

```
>>> 21+random.random()*4
```

```
23.684025732564205
```

random.randint(a,b)

>>> random.randint(4,10) -generates random whole numbers from 4 to 10, including 10 7

```
>>> random.randint(4,10)
```

```
8
```

```
>>> random.randint(4,10)
```

```
4
```

random.randrange(a,b)

>>> random.randrange(4) - generates random whole numbers from 0 to 3

```
2
```

>>> random.randrange(1,10) -generates random whole numbers from 1 to 9 4

```
>>> random.randrange(1,10)
```

```
7
```

8) statistics MODULE BASED QUESTION

```
import statistics
```

statistics.mean(<seq>)

returns the average value of the sequence passed

```
>>> a = [1,2,3,4,5]
```

```
>>> statistics.mean(a) # results 3
```

statistics.median(<seq>)

returns the middle value of the sequence passed

```
>>> a = [1,2,3,4,5]
```

```
>>> statistics.mean(a)
```

```
3
```

```
>>> a = [1,2,3,4,5,6]
```

```
>>> statistics.median(a)
```

```
3.5
```

statistics.mode(<seq>)

returns the most often repeated value of the sequence passed, other wise the first.

```
>>> a = [1,2,3,4,5,6,1]
```

```
>>> statistics.mode(a)
```

```
1
```

```
>>> a = [9,1,3,4,5,7]
```

```
>>> statistics.mode(a)
```

```
9
```

~~~~~

*"Practice does not make perfect. Only perfect practice makes perfect." - Vince Lombardi.*

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS - See Yes!*

### 3. CONDITIONALS & LOOPS HANDOUT

#### QUESTION-BASED MODELS AND NOTES

##### 1) 4 forms of if CONDITION

```
a = int(input("Enter a number:"))
```

###### i) only if

```
if a%2==0:  
    print("Even")
```

###### ii) if with else

```
if a%2==0:  
    print("Even")  
else:  
    print("Odd")
```

###### iii) if with elif

```
if a%2==0:  
    print("Even")  
elif a%2!=0:  
    print("Odd")
```

###### iii) if with elif and else:

```
if a%2==0:  
    print("Even")
```

```
elif a%2!=0:
```

```
    print("Odd")
```

```
else:
```

```
    print("Enter Correct value")
```

##### 2) Double if CONDITIONS

###### Form 1

```
a = 10  
if a%2==0:  
    a+=2  
if a%5==0:  
    a+=5  
print(a)          #prints 17
```

###### Form 2

```
a = 5  
if a%2==0:  
    a+=2  
else:  
    a+=3  
if a%5==0:  
    a+=5  
print(a)          #prints 13
```

##### 3) if with True:

###### example 1:

```
if True:  
    print("sai")  
else:  
    print("baba")  
#prints sai as if is True by default
```

### example 2:

```
a = 1
if a:
    print("sai")
else:
    print("baba")
#prints sai as a is a True value
```

### example 3:

```
a = 0
if a:
    print("sai")
else:
    print("baba")
#prints baba as a is a False value now
```

## 4) SEQUENCE CREATION USING range()

```
>>>a = range(5)
>>>print(a)
range(0, 5)

>>>a = range(1,5)
>>>print(a)
range(1, 5)

>>>a = range(1,5)
>>>print(type(a))
<class 'range'>
```

## 5) PRINTING NUMBERS THROUGH range()

### example 1

```
a = range(1,6)
for i in a:
    print(i, end=' ')
```

Output:

1 2 3 4 5

### example 2

```
a = range(1,10,2)
for i in a:
    print(i, end=' ')
```

Output:

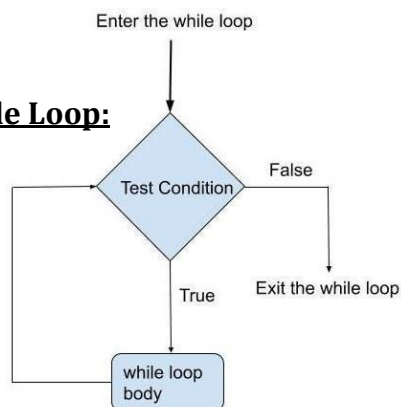
1 3 5 7 9

## 5) 2 FORMS OF LOOP:

### for Loop:



### While Loop:



## 6) for & while COMPARISON - ASCENDING

**a)Print numbers from 0 to 10 using for & while loop:**

```
for i in range(11):  
    print(i, end=' ')
```

```
i = 0  
while i<11:  
    print(i, end=' ')  
  
    i = i +1
```

Output: 0 1 2 3 4 5 6 7 8 9 10

**b)Print numbers from 1 to 10 using for & while loop:**

```
for i in range(1,11):  
    print(i, end=' ')
```

```
i = 1  
while i<11:  
    print(i, end=' ')  
  
    i = i +1
```

Output: 1 2 3 4 5 6 7 8 9 10

**c)Print odd numbers from 1 to 10 using for & while loop:**

```
for i in range(1,11,2):  
    print(i, end=' ')
```

```
i = 1  
while i<11:  
    print(i, end=' ')  
  
    i = i+2
```

Output: 1 3 5 7 9

**d)Print even numbers from 1 to 10 using for & while loop:**

```
for i in range(2,11,2):  
    print(i, end=' ')
```

```
i = 2  
while i<11:  
    print(i, end=' ')  
  
    i = i+2
```

Output: 2 4 6 8 10

**e) Print numbers ending with 3 from 1 to 50 using for & while loop::**

```
for i in range(1,51):  
    if i%10==3:  
        print(i, end=' ')
```

```
for i in range(3,51,10):  
    print(i, end=' ')
```

```
i = 1  
while i<51:  
    if i % 10==3:  
        print(i, end=' ')  
  
    i = i +1
```

```
i = 3  
while i<51:  
    print(i)  
  
    i = i +10
```

Output: 3 13 23 33 43

**f) Print multiples of 7 from 1 to 100 using for & while loop:**

```
for i in range(1,101):  
    if i%7==0:  
        print(i, end=' ')
```

```
for i in range(7,101,7):  
    print(i, end=' ')
```

```
i = 1  
while i<101:  
    if i % 7==0:  
        print(i, end=' ')  
    i = i +1
```

```
i = 7  
while i<101:  
    print(i, end=' ')  
    i = i +7
```

Output: 7 14 21 28 35 42 49 56 63 70 77 84 91 98

**7) for & while COMPARISON - DESCENDING****a)Print numbers from 10 to 1 using for & while loop:**

```
for i in range(10,0,-1):  
    print(i, end=' ')
```

```
i = 10  
while i>0:  
    print(i, end=' ')  
    i = i -1
```

Output: 10 9 8 7 6 5 4 3 2 1

**b) Print odd numbers between 1 and 10 in descending order using for & while loop:**

```
for i in range(9,0,-2):  
    print(i, end=' ')
```

```
i = 9  
while i>0:  
    print(i, end=' ')  
    i = i -2
```

Output: 9 7 5 3 1

**c) Print numbers ending with 3 from 1 to 50 in descending order using for & while loop:**

```
for i in range(50,0,-1):  
    if i%10==3:  
        print(i, end=' ')
```

```
for i in range(43,0,-10):  
    print(i, end=' ')
```

```
i = 50  
while i>0:  
    if i % 10==3:  
        print(i, end=' ')  
    i = i -1
```

```
i = 43  
while i>0:  
    print(i, end=' ')  
    i = i -10
```

Output: 9 7 5 3 1



## **7) TRAVERSING STRINGS - DIRECT TRAVERSAL**

```
str = "sairam"
```

### **Example 1**

```
for i in str:  
    print(i, end=' ')
```

Output: s a i r a m

### **Example 2**

```
for i in str:  
    print(i, '*', end = "")
```

Output: s \*a \*i \*r \*a \*m \*

### **Example 3**

```
for i in str:  
    print(str, '*', end = "")
```

Output: sairam \*sairam \*sairam \*sairam \*sairam  
\*sairam \*

## **8) TRAVERSING STRINGS - RANGE TRAVERSAL USING FOR LOOP**

```
str = 'sairam'
```

### **a) Print all letters**

```
for i in range(len(str)):  
    print(str[i], end=' ')
```

Output: s a i r a m

### **b) Print letters on even index**

```
for i in range(len(str)):  
    if i%2==0:  
        print(str[i], end=' ')
```

Output: s i a

### **c) Print letters on odd index**

```
for i in range(len(str)):  
    if i%2!=0:  
        print(str[i], end=' ')
```

Output: a r m

### **d) Print letters based on step value**

```
str = "sathyasaibaba"  
for i in range(0,len(str),3):  
    print(str[i], end=' ')
```

Output: s h s b a

### **e) Reverse the String**

```
for i in range(len(str)-1,-1,-1):  
    print(str[i], end=' ')
```

Output: a b a b i a s a y h t a s

### **f) Reverse the String with step value**

```
for i in range(len(str)-1,-1,-2):  
    print(str[i], end=' ')
```

Output: a a i s y t s

## **9) TRAVERSING STRINGS - TRAVERSING USING WHILE LOOP & LEN()**

```
str = 'sairam'
```

### **a) Print all letters**

```
i = 0
while i < len(str):
    print(str[i], end=' ')
    i = i + 1
```

Output: s a i r a m

### **b) Print letters on even index**

```
i = 0
while i < len(str):
    if i%2==0:
        print(str[i], end=' ')
    i = i + 1
```

Output: s i a

### **c) Print letters on odd index**

```
i = 0
while i < len(str):
    if i%2!=0:
        print(str[i], end=' ')
    i = i + 1
```

Output: a r m

### **d) Print letters based on step value**

```
str = "sathyasaibaba"
i = 0
while i < len(str):
    print(str[i], end=' ')
    i = i + 3
```

Output: s h s b a

### **e) Reverse the String**

```
i = len(str)-1
while i >= 0:
    print(str[i], end=' ')
    i = i - 1
```

Output: a b a b i a s a y h t a s

### **f) Reverse the String with step value**

```
i = len(str)-1
while i >= 0:
    print(str[i], end=' ')
    i = i - 2
```

Output: a a i s y t s

## **10) TRAVERSING LIST/TUPLE - DIRECT TRAVERSL**

### **example 1 - list**

```
lst = [1,2,3,4,5,6,7]
```

```
for i in lst:
```

```
    print(i, end=' ')
```

Output: 1 2 3 4 5 6 7

### **example 2 - tuple**

```
tup = (1,2,3,4,5,6,7)
```

```
for i in tup:
```

```
    print(i, end=' ')
```

Output: 1 2 3 4 5 6 7

### **example 3 - list**

```
lst = [1,2,3,4,5,6,7]
```

```
for i in lst:
```

```
    print(i, sep='#', end="")
```

Output: 1#2#3#4#5#6#7

### **example 4 - tuple**

```
tup = (1,2,3,4,5,6,7)
```

```
for i in tup:
```

```
    print(i, sep= '%', end="")
```

Output: 1%2%3%4%5%6%7

## **11) TRAVERSING LIST/TUPLE - RANGE TRAVERSL**

### **a) Print whole list**

```
lst = [1,2,3,4,5,6,7]
```

```
for i in range(len(lst)):
```

```
    print(lst[i], end=' ')
```

Output: 1 2 3 4 5 6 7

### **b) Print elements on odd index**

```
lst = [1,2,3,4,5,6,7]
```

```
for i in range(len(lst)):
```

```
    if i%2!=0:
```

```
        print(lst[i], end=' ')
```

Output: 2 4 6

### **c) Print list in reversed form**

```
lst = [1,2,3,4,5,6,7]
```

```
for i in range(len(lst)-1,-1,-1):
```

```
    print(lst[i], end=' ')
```

Output: 7 6 5 4 3 2 1

### **d) Print list in reversed form with step value**

```
lst = [1,2,3,4,5,6,7]
```

```
for i in range(len(lst)-1,-1,-3):
```

```
    print(lst[i], end=' ')
```

Output: 7 4 1

### **e) Print list in reversed form using while loop**

```
lst = [1,2,3,4,5,6,7]
```

```
i = len(lst)-1
```

```
while i>=0:
```

```
    print(lst[i], end=' ')
```

```
    i = i - 1
```

Output: 7 6 5 4 3 2 1

## 12) SUM OUTPUT USING LOOP WITH STEP-BY-STEP SOLUTION

### a) Print sum of numbers from 1 to 10

```
sum=0
```

```
for i in range(1,11):
```

```
    sum+=i
```

```
print(sum)
```

Output: 55

| i  | sum+=i  | sum |
|----|---------|-----|
| 1  | 0 + 1   | 1   |
| 2  | 1 + 2   | 3   |
| 3  | 3 + 3   | 6   |
| 4  | 6 + 4   | 10  |
| 5  | 10 + 5  | 15  |
| 6  | 15 + 6  | 21  |
| 7  | 21 + 7  | 28  |
| 8  | 28 + 8  | 36  |
| 9  | 36 + 9  | 45  |
| 10 | 45 + 10 | 55  |

### b) Print sum of even numbers from 10 to 1

```
sum = 0
```

```
i = 10
```

```
while i>0:
```

```
    if i%2==0:
```

```
        sum+=i
```

```
    i-=1
```

```
print(sum)
```

Output: 30

| i  | sum+=i | sum |
|----|--------|-----|
| 10 | 0 + 10 | 10  |
| 9  | -      |     |
| 8  | 10+8   | 18  |
| 7  | -      |     |
| 6  | 18+6   | 24  |
| 5  | -      |     |
| 4  | 24+4   | 28  |
| 3  | -      |     |
| 2  | 28+2   | 30  |
| 1  | -      |     |
|    |        |     |

### c) Print sum of odd numbers from 21 to 30

```
sum = 0
```

```
for i in range(21,31,2):
```

```
    sum+=i
```

```
print(sum)
```

Output: 125

| i  | sum+=i | sum |
|----|--------|-----|
| 21 | 0+21   | 21  |
| 23 | 21+23  | 44  |
| 25 | 44+25  | 69  |
| 27 | 69+27  | 96  |
| 29 | 96+29  | 125 |

#### d) Predict the output

a = 5

b = 10

sum = 0

i = 1

while i<11:

    if i%2==0:

        sum+=(a+b)

        print(i, sum)

    a+=1

    b-=2

    i+=1

print(sum)

| i  | a  | b  | sum+=(a+b) | sum |
|----|----|----|------------|-----|
| 1  | 5  | 10 | -          | 0   |
| 2  | 6  | 8  | 0+=(6+8)   | 14  |
| 3  | 7  | 6  | -          | 14  |
| 4  | 8  | 4  | 14+=(8+4)  | 26  |
| 5  | 9  | 2  | -          | 26  |
| 6  | 10 | 0  | 26+(10+0)  | 36  |
| 7  | 11 | -2 | -          | 36  |
| 8  | 12 | -4 | 36+(12-4)  | 44  |
| 9  | 13 | -6 | -          | 44  |
| 10 | 14 | -8 | 44+(14-8)  | 50  |

Output: 50

#### e) Predict the output

x = 1

y = 10

for i in range(9):

    if i%2!=0:

        print(i,x+1,y+2)

    x+=2

    y-=2

| i | x | y  | print(i, x+1,y+2) | Output | x+=2 | y-=2 |
|---|---|----|-------------------|--------|------|------|
| 0 | 1 | 10 | -                 |        | -    | -    |
| 1 | 1 | 10 | 1, 1+1, 10+2      | 1 2 12 | 3    | 8    |
| 2 | - | -  | -                 |        | -    | -    |
| 3 | 3 | 8  | 3, 3+1, 8+2       | 3 4 10 | 5    | 6    |
| 4 | - | -  | -                 |        | -    | -    |
| 5 | 5 | 6  | 5, 5+1, 6+2       | 5 6 8  | 7    | 4    |
| 6 | - | -  | -                 |        | -    | -    |
| 7 | 7 | 4  | 7, 7+1, 4+2       | 7 8 6  | 9    | 2    |
| 8 | - | -  | -                 |        | -    | -    |

Output:

1 2 12

3 4 10

5 6 8

7 8 6

### 13) SERIES ADDITION CODE

(assuming user enters n as 10)

Write a program to find sum of the series:

#### 1) $1 + 3 + 5 + 7 + \dots + n$

```
n = int(input("Enter your limit:"))
```

```
sum = 0
```

```
for i in range(n+1):
```

```
    if i%2!=0:
```

```
        sum+=i
```

```
print(sum)
```

| i  | sum+=i | sum |
|----|--------|-----|
| 0  | 0      | 0   |
| 1  | 0 + 1  | 1   |
| 2  | -      | -   |
| 3  | 1+3    | 4   |
| 4  | -      | -   |
| 5  | 4+5    | 9   |
| 6  | -      | -   |
| 7  | 9+7    | 16  |
| 8  | -      | -   |
| 9  | 16+9   | 25  |
| 10 | -      | -   |

#### 2) $2^5 + 4^5 + 6^5 + 8^5 + \dots + n^5$

(assuming user enters n as 10)

```
n = int(input("Enter your limit:"))
```

```
sum = 0
```

```
for i in range(2,n+1):
```

```
    if i%2==0:
```

```
        sum+=(i*i*i*i*i)
```

```
print(sum)
```

| i  | sum+=i               | sum    |
|----|----------------------|--------|
| 2  | $0+(2*2*2*2*2)$      | 32     |
| 3  | -                    | -      |
| 4  | $2+(4*4*4*4*4)$      | 1056   |
| 5  | -                    | -      |
| 6  | $6+(6*6*6*6*6)$      | 8832   |
| 7  | -                    | -      |
| 8  | $6+(8*8*8*8*8)$      | 41600  |
| 9  | -                    | -      |
| 10 | $6+(10*10*10*10*10)$ | 141600 |

#### Same calculation done using math:

```
import math
```

```
n = int(input("Enter your limit:"))
```

```
sum=0
```

```
for i in range(2,n+1):
```

```
    if i%2==0:
```

```
        sum+=(math.pow(i,5))
```

```
print(sum)
```

### 13) random OUTPUT

```
import random
```

```
for i in range(3):
```

```
    n = random.randint(1,4)
```

```
    print(n, end=' ')
```

a) 1 3 2   b) 4 2 1 c) 7 0 1 d) 1 2 3

| i | n = random.randint(1,4) |
|---|-------------------------|
| 0 | 1, 2, 3, 4              |
| 1 | 1, 2, 3, 4              |
| 2 | 1, 2, 3, 4              |

```
import random
```

```
for i in range(3):
```

```
    n = random.randrange(1,4)
```

```
    print(n, end='$ ')
```

| i | n = random.randrange(1,4)<br>with '\$' |
|---|----------------------------------------|
| 0 | 1\$ 2\$ 3\$                            |
| 1 | 1\$ 2\$ 3\$                            |
| 2 | 1\$ 2\$ 3\$                            |

a) 1 \$3\$ 2\$ b) 4\$ 2\$1\$ c) 1\$ 2\$ 3\$ d) 3\$ 0\$ 1\$

#### 14) NESTED LOOP OUTPUT

##### example 1

```
for i in range(3):
```

```
    for j in range(2):
```

```
        print(i+j, end=' ')
```

Output: 0 1 1 2 2 3

| i | j | i+j     |
|---|---|---------|
| 0 | 0 | 0+0 = 0 |
|   | 1 | 0+1 = 1 |
| 1 | 0 | 1+0 = 1 |
|   | 1 | 1+1 = 2 |
| 2 | 0 | 2+0=2   |
|   | 1 | 2+1=3   |

##### example 2

```
sum = 0
```

```
for i in range(10):
```

```
    if i%3==0:
```

```
        k = 0
```

```
        while k<=2:
```

```
            sum+=(i-k)
```

```
            k+=1
```

```
print(sum)
```

Output: 42

| i | k | sum+=(i-k) | sum |
|---|---|------------|-----|
| 0 | 0 | 0+=(0-0)   | 0   |
|   | 1 | 0+=(0-1)   | -1  |
|   | 2 | -1+=(0-2)  | -3  |
| 1 |   |            |     |
| 2 |   |            |     |
| 3 | 0 | -3+=(3-0)  | 0   |
|   | 1 | 0+=(3-1)   | 2   |
|   | 2 | 2+=(3-2)   | 3   |
| 4 |   |            |     |
| 5 |   |            |     |
| 6 | 0 | 3+=(6-0)   | 9   |
|   | 1 | 9+=(6-1)   | 14  |
|   | 2 | 14+=(6-2)  | 18  |
| 7 |   |            |     |
| 8 |   |            |     |
| 9 | 0 | 18+=(9-0)  | 27  |
|   | 1 | 27+=(9-1)  | 35  |
|   | 2 | 35+=(9-2)  | 42  |

### 15) NESTED LOOP - SITUATION BASED

Kumar wants to get admission at SAI INTERNATIONAL SCHOOL, CHENNAI. He needs to submit the average and total score of 5 subjects he obtained in class 8,9 and 10. Write code in Python to get correct input and produce the output.

```
for std in range(8,11):
    print("Enter marks scored in class:",std)
    tot=0
    avg=0
    for sub in range(1,6):
        score = int(input("Score obtained in Subject " +str(sub) + ":"))
        tot+=score
    print("Total score in class ",std,":",tot)
    avg = tot/5
    print("Average score in class",std,":",avg)
    print()
```

### 16) NESTED LOOP - PATTERN PRINTING

#### a) Pattern 1 (square/rectangle):

\* \* \* \*

\* \* \* \*

\* \* \* \*

\* \* \* \*

```
for i in range(4):
    for j in range(4):
        print("*",end=" ")
    print()
```

| i | j       | print("*", end= ' ') |
|---|---------|----------------------|
| 0 | 0 1 2 3 | * * * *              |
| 1 | 0 1 2 3 | * * * *              |
| 2 | 0 1 2 3 | * * * *              |
| 3 | 0 1 2 3 | * * * *              |



**b) Pattern 2(right half straight pyramid):**

```
*
* *
* * *
* * * *
* * * * *
```

```
for i in range(1,6):
    for j in range(i):
        print("*",end=" ")
    print()
```

| i | j in range(i) | print("*", end= ' ') |
|---|---------------|----------------------|
| 1 | 0,1           | *                    |
| 2 | 0,1,2         | * *                  |
| 3 | 0,1,2,3       | * * *                |
| 4 | 0,1,2,3,4     | * * * *              |
| 5 | 0,1,2,3,4,5   | * * * * *            |

**c) Pattern 3 (right half reverse pyramid):**

```
* * * * *
* * * *
* * *
* *
*
```

```
for i in range(1,6):
    for j in range(6-i):
        print("*",end=" ")
    print()
```

| i | j in range(6-i)       | print     |
|---|-----------------------|-----------|
| 1 | $6-1 = 5 = 0,1,2,3,4$ | * * * * * |
| 2 | $6-2 = 4 = 0,1,2,3$   | * * * *   |
| 3 | $6-3 = 3 = 0,1,2$     | * * *     |
| 4 | $6-4 = 2 = 0,1$       | * *       |
| 5 | $6-5 = 1 = 0$         | *         |

~~~~~

XII COMPUTER SCIENCE*Towards Getting Centum in CS - See Yes!***4. SLTD – STRING LIST TUPLE DICTIONARY – CODE COMPARISONS HANDOUT**

Operation	String	List	Tuple	Dictionary
Creating empty type	a = "	a = []	a = ()	a = {}
	a = ""			
	a = str()	a = list()	a = tuple()	a = dict()
With values	a = "sairam"	a = [1,2,3] b = ['s','a','i']	a = (1,2,3) b = ('s','a','i')	a = {1:'sai', 2:'baba', 3:'ram'} b = {'sai':1, 'baba':2, 'ram':3}
Finding length	>>>len(a) 6	>>>len(a) 3	>>>len(a) 3	>>>len(a) 3
Name of each data	elements/character	elements	elements	Key : value pairs
Accessing each element	>>>a[1]	>>>a[1]	>>>a[1]	>>>a[1]
	a	2	2	'sai'
	>>>a[-3]	>>>b[-3]	>>>b[-3]	>>>b['ram']
	r	's'	's'	3
Nested Feature	Nil	c = [1, [9,5], [1,2,3],[4]]	c = (1,(9,5), (1,2,3),(4,2))	c = {'name':'sai', 'score': { 'eng':40, 'sci':45, 'mat':60} }
Nested length	Nil	>>>len(c) 4	>>>len(c) 4	>>>len(c) 2

Operation	String	List	Tuple	Dictionary
Nested length	Nil	>>>len(c[2]) 3	>>>len(c[3]) 2	>>>len(c['score']) 3
Nesting with other data types	Nil	d = [1, [3,4], (5,6), {1:'sai'}]	d = (1, [3,4], (5,6), {1:'sai'})	d = {{1,2}:'sai', 2:[5,6], 'baba': {'a':'ram', 'b':'krish'}} (Keys must be immutable)
Mutability	Immutable >>>a = 'sai' >>>a[0] ='b' #TypeError	Mutable >>>a = [1,2,3] >>>a[0] = 55 >>>a [55,2,3]	Immutable >>>a = (1,2,3) >>>a[0] = 55 #TypeError	Mutable (partially, values alone are mutable, Keys are immutable) a = {1:'sai', 2:'baba', 3:'ram'} a[1] = 'krish' >>>a {1:'krish', 2:'baba', 3:'ram'}
Adding new elements	<u>Methods:</u> - Manual initialization - Concatenation (+) a = '' a+='sai' a+='r' a+='a' a+='m' >>>a 'sairam'	<u>Methods:</u> - append() - extend() - insert() - Manual initialization - Concatenation (+) append() a = [] a.append(1) >>>a [1]	<u>Methods:</u> - Manual initialization - Concatenation (+) a = () a.append(1) #AttributeError a = (1,2) b = (3,4) c = a+b c becomes (1, 2, 3, 4)	<u>Methods:</u> - Manual initialization - update() - setdefault() Manual a = {} a[1] = 'sai' a[2] = 'baba' a[3] = 'ram' >>>a {1:'sai', 2:'baba', 3:'ram'}

	a.append([2,3]) <pre>>>>a [1, [2,3]]</pre> extend() <pre>a.extend([4,5]) >>>a [1, [2,3], 4, 5] a.extend(6) #TypeError</pre> insert() <pre>a = [1,2,3] a.insert(2,5) >>>a [1,2,5,3] (in the place of index 2, insert 5, the element present in that place is pushed aside) pq: Insert means index first</pre>	update() <pre>a = {'1':'sai'} b = {'2':'baba'} a.update(b) >>>a {'1': 'sai', 2: 'baba'}</pre> setdefault() <u>method 1 (with new key)</u> <pre>a = {'1':'sai'} >>>a.setdefault(2,'baba') 'baba' #value returned >>>a {'1': 'sai', 2: 'baba'}</pre> <u>method 1 (with old key)</u> <pre>>>>a.setdefault(1,'rama') 'sai' (old value returned , no change is made to the dictionary)</pre> #Concatenation (+) cannot be done with dictionary
--	--	--

Operation	String	List	Tuple	Dictionary
Traversal	a) Direct Traversal 1 for i in 'sairam': print(i, end='') output:sairam	a) Direct Traversal 1 for i in [1, 2, 3, 4]: print(i, end=' ') output:1 2 3 4	a) Direct Traversal 1 for i in (1, 2, 3, 4): print(i, end=' ') output:1 2 3 4	a = {'sai': 2, 'baba': 3, 'ram': 1} a) Key Traversal <u>Method 1:</u> for i in a: print(i, end=' ') Output: 1 2 3 <u>Method 2:</u> for i in a.keys(): print(i, end=' ') Output: 1 2 3
	b) Direct Traversal 2 a = 'sairam' for i in a: print(i, end='') output:sairam	b) Direct Traversal 2 a = [1, 2, 3, 4] for i in a: print(i, end=' ') output:1 2 3 4	b) Direct Traversal 2 a = (1, 2, 3, 4) for i in a: print(i, end=' ') output:1 2 3 4	b) Value Traversal <u>Method 1 using:</u> for i in a: print(a[i], end=' ') Output: sai baba ram
	b) Range Traversal a = 'sairam' for i in range(len(a)): print(a[i], end='') output:sairam	c) Range Traversal a = [1, 2, 3, 4] for i in range(len(a)): print(a[i], end=' ') output:1 2 3 4	c) Range Traversal a = (1, 2, 3, 4) for i in range(len(a)): print(a[i], end=' ') output:1 2 3 4	<u>Method 2 using values():</u> for i in a.values(): print(i, end=' ') Output: sai baba ram

				c) Items Traversal a = {'sai': 2, 'baba': 3, 'ram': 1} for i in a.items(): print(i) Output: (1, 'sai') (2, 'baba') (3, 'ram')
				d) Keys & Values Traversal for x, y in a.items(): print(x,y) Output: 1 sai 2 baba 3 ram
Getting directly from user	a = input("Enter :") or a = str(input("Enter:"))	a = list(input("Enter:")) Enter:123 >>>a ['1', '2', '3']	a = tuple(input("Enter:")) Enter:123 >>>a ('1', '2', '3')	Nil
Printing Format	a = 'sai' print(a) #sai >>>a # 'sai'	a = [1,2, 'a', 'sai'] >>>a[0] 1	a = [1,2, 'a', 'sai'] >>>a[0] 1	a = {'sai': 2, 'baba': 3, 'ram': 1} >>>print(a) {'sai': 2, 'baba': 3, 'ram': 1}

	<pre>>>>print(a[0]) #s >>>a[0] #'s'</pre>	<pre>>>>a[3] 'sai' >>>print(a[3]) sai</pre>	<pre>>>>a[3] 'sai' >>>print(a[3]) sai</pre>	<pre>>>>print(a.keys()) dict_keys([1, 2, 3]) >>>print(a.values()) dict_values(['sai', 'baba', 'ram']) (keys/values printed as a list inside dict function) >>>a.items() dict_items([(1, 'sai'), (2, 'baba'), (3, 'ram')]) (The items are printed as a list of tuples inside dict)</pre>
Concatenation (Done only with same data type)	<pre>a = 'sai' b = 'ram' >>>a+b 'sairam' c = 2 >>>a+c #TypeError as c is different data type</pre>	<pre>a = [1,2] b = [3,4] c = [5,6] c =a+b+c c becomes [1, 2, 3, 4, 5, 6] d = 4 >>>a+d #TypeError as d is different data type</pre>	<pre>a = (1,2) b = (3,4) c = (5,6) c =a+b+c c becomes (1, 2, 3, 4, 5, 6) d = 4 >>>a+d #TypeError as d is different data type</pre>	<pre>a = {1:'sai'} b = {2:'baba'} a+b #not permitted #TypeError occurs</pre>
Replication (Done only with integer)	<pre>a = 'sai' >>>a*3 'saisaisai'</pre>	<pre>a = [1,2] >>>a*3 [1, 2, 1, 2, 1, 2]</pre>	<pre>a = (1,2) >>>a*3 (1, 2, 1, 2, 1, 2)</pre>	<pre>a = {1:'sai'} >>>a*3 #TypeError</pre>

Operation	String	List	Tuple	Dictionary
Shallow copy	a = 'sai' b = a >>>b sai	a = [1,2] b = a >>>b [1,2]	a = (1,2) b = a >>>b [1,2]	a = {1:'sai'} b = a >>>b {1:'sai'}
	b='baba' >>>a 'sai' >>>b 'baba' #new location # a and b points to same memory location #when b is changed , it does not change a, as they point to 2 different locations now. (Check memory address with the help of id() method.)	b = [4,5] >>>a [1,2] >>>b [4,5] #new location But, If the elements are changed individually, a = b #a gets [4,5] now a[0] = 9 >>>a [9, 5] >>>b [9, 5] Change is reflected on b also as they point to same location, i.e. shallow copy	b = (4,5) >>>a (1,2) >>>b (4,5) #new location Individual change is not allowed as it is an immutable type same like string. But, If the elements are changed individually, a = b #a gets {2:'ram'} now a[2] = 'krish' >>>a {2:'krish'} >>>b {2:'krish'} Change is reflected on b also as they point to same location, i.e. shallow copy	

Operation	String	List	Tuple	Dictionary
Deep copy	<pre> a = 'sai' b = a.copy() #AttributeError </pre>	<pre> a = [1,2] b = a.copy() >>>id(a) #2836075102336 >>>id(b) #2836075212352 #now both are in different location Now, a[0] = 9 >>>a # prints [9,2] >>>b # prints [1,2] #change is not reflected on b due to deep copy. </pre> We can also make deep copy through list() <pre> a = [1,2] b = list(a) >>>id(a) 1990356091008 >>>id(b) 1990328127104 </pre>	<pre> a = (1,2) b = a.copy() #AttributeError </pre>	<pre> a = {'1':'sai'} b = a.copy() id(a) #2109540637120 id(b) #2109569901504 #now both are in different location Now b[1] = 'baba' >>>b #prints {'1':'baba'} >>>a #prints {'1':'sai'} #change is not reflected on b due to deep copy. </pre> We can also make deep copy through dict() <pre> a = {'1':'sai'} b = dict(a) >>>id(a) 2570704219136 >>>id(b) 2570703908992 </pre>

Operation	String	List	Tuple	Dictionary
Replacing /Changing elements	<pre>a = 'sai' a[0] = 'b' # TypeError But, we can replace as a whole a = 'baba' >>>a # prints 'baba' Using replace() a = 'sai baba' a.replace('baba', 'rama') 'sai rama' #It returns the output immediately. If you check, >>>a 'sai baba' (change is seen only on the output, original string is not changed as it is immutable)</pre>	<pre>a = [1,2] a[0] = 7 >>>a [7, 2] #change is done through accessing the index</pre>	<pre>a = (1,2) a[0] = 7 # TypeError But, we can replace as a whole a = [4,5] >>>a [4,5]</pre>	<pre>a = {1:'sai'} a[1] = 'baba' >>>a {1:'baba'} Using update() b = {1:'krish', 2:'rama'} a.update(b) >>>a {1:'krish', 2:'rama'} #new key is added, old key is replaced</pre>

Operation	String	List	Tuple	Dictionary
Sorting elements	a = 'sairam' a.sort() #AttributeError Can be done through built-in function, sorted() <pre>>>>sorted(a) ['a', 'a', 'i', 'm', 'r', 's'] #the output is returned as a list #only output is sorted</pre> Descending sort <pre>>>>sorted(a, reverse=True) ['s', 'r', 'm', 'i', 'a', 'a']</pre> But, original value does not change, only output is sorted.	Ascending sort <pre>a = [9,3,5,1] >>>a.sort() #no output is returned.</pre> When you check, <pre>>>>a #prints [1, 3, 5, 9] #ascending by default #same as, a.sort(reverse=False)</pre> Descending sort <pre>>>>a.sort(reverse=True) >>>a #prints [9, 5, 3, 1] sorted() also can be used >>>sorted(a) #[1, 3, 5, 9] But, original value does not change, only output is sorted. Sort cannot be stored <pre>>>>b = a.sort(a) >>>b #No output comes.</pre> </pre>	a = (9,3,5,1) a.sort() #AttributeError Can be done through built-in function, sorted() <pre>>>>sorted(a) [1, 3, 5, 9] >>>sorted(a, reverse=True) [9, 5, 3, 1]</pre> But, original value does not change, only output is sorted.	a = {8:'sai', 1:'ram', 3:'baba'} <pre>>>>a.sort() #AttributeError</pre> Can be done through built-in function, sorted() But, only the keys are sorted are come as a list. <pre>sorted(a) [1, 3, 8] sorted(a, reverse = True) [8, 3, 1]</pre>

Operation	String	List	Tuple	Dictionary
Deleting elements	a = 'sairam' a.pop() #AttributeError But, can be done through, a = "	Methods used: • pop() • remove() • clear() • del pop() means popi (remove by index) remove() means remove your ego (remove the element itself) pop() a = [1,5,7,2] >>>a.pop() 2 #simple pop removes the last element >>>a.pop(1) #prints 5 >>>a [1, 7] #the deleted element is returned, can be stored ex: b = a.pop()	a = (1,4,5,6) a.pop() #AttributeError But, can be done through, a = () Indirectly done as follows: a = (1,4,5,6) a = list(a) >>>a [1, 4, 5, 6] >>>a.pop(2) 5 >>>a [1, 4, 6] a = tuple(a) >>>a (1, 4, 6) (Convert tuple to list, remove or change and then again convert to tuple.)	Methods used: • pop() • popitem() • clear() dictionary pop() is different than list pop() dictionary popitem() is same as list pop() popitem() <u>removes the last key:value</u> a = {'1':'sai', 2:'baba', 3: 'krish'} >>>a.popitem() (3, 'krish') #returned >>>a {1: 'sai', 2: 'baba'}

Operation	String	List	Tuple	Dictionary
		<u>remove()</u> a = [1,5,7,2] >>>a.remove(2) >>>a #prints [1, 5, 7] #output is not returned #cannot be stored		pop() a.pop() #TypeError, should be done passing the key >>>a.pop(1) 'sai' >>>a {2: 'baba'} clear() a = {'1':'sai', 2:'baba', 3: 'krish'} >>>a.clear() {} #all key:value pairs are removed

Operation	String	List	Tuple	Dictionary
Other notes	Conversions a = 'sairam'	Sorting a = [1,2,'s','A'] a.sort()	Conversions a = (1,2)	Conversions a = {1:'sai' , 2:'baba'}
	>>>list(a) ['s', 'a', 'i', 'r', 'a', 'm']	#TypeError due to change in data type	>>>list(a) [1, 2]	>>>str(a) "{1: 'sai', 2: 'baba'}"
	>>>tuple(a) ('s', 'a', 'i', 'r', 'a', 'm')	With only strings a = ['a', 'R','r'] >>>a.sort() >>>a ['R', 'a', 'r']	>>>str(a) '(1, 2)' >>>dict(a) #TypeError	>>>list(a) [1, 2] #only keys are taken >>>tuple(a) (1, 2)
	>>>dict(a) #ValueError			
	a = '1'	With complex No a = [1,6,True,6j,4.5] >>>a.sort() #TypeError	For List Conversions a = [1,2,3] >>>str(a) "[1,2,3]"	>>>int(a) #TypeError
	>>>int(a) 1			
	>>>list(a) ['1']	But if, a = [1, 6, True, 4.5] >>>a.sort() >>>a [1, True, 4.5, 6]	>>>tuple(a) (1,2,3) >>>dict(a) #TypeError	

Operation	String	List	Tuple	Dictionary
counting	a = 'sathyasaibaba'	a = [1,2,3,1,4,5,1]	a = (1,5,6,1,7,8,1)	a = {1:'sai', 2:'baba', 1:'ram'}
	a.count('a')	>>>a.count(1)	a.count(1)	>>>a.count(1)
	5	3	3	#AttributeError
	a.count('a', 5, 12) 3 #between 5 and 11 index (String count accepts 3 arguments, what to count, start and end range)	(list count accepts only one argument, range specification not allowed)	(tuple count accepts only one argument, range specification not allowed)	#illegal
Built-in Functions	>>>a.count('p')	>>>a.count(9)	>>>a.count(9)	
	0	0	0	
	a = 'sai'	a = [4,3,1,2]	a = (4,3,1,2)	a = {1:'sai', 2:'baba',4:'ram'}
	sum(a) #TypeError	>>>sum(a)	>>>sum(a)	>>>sum(a)
		10	10	7
	>>>min(a)	>>>max(a)	>>>max(a)	>>>max(a)
	'a'	4	4	4
	>>>max(a)	>>>min(a)	>>>min(a)	>>>min(a)
	's'	1	1	1

~~~~~

*"However difficult life may seem, there is always something you can do and succeed at."*  
- Stephen Hawking.

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS - See Yes!*

### 5.STRINGS HANDOUT

#### 1) STRING INDEXING

##### TWO WAY INDEXING

```
>>>name= "SAIRAM"
```

| FORWARD INDEXING → → → → →  |    |    |    |    |    |
|-----------------------------|----|----|----|----|----|
| 0                           | 1  | 2  | 3  | 4  | 5  |
| S                           | A  | I  | R  | A  | M  |
| -6                          | -5 | -4 | -3 | -2 | -1 |
| ← ← ← ← ← BACKWARD INDEXING |    |    |    |    |    |

```
>>> name[0] #prints S
>>> name[-6] #prints S
>>> name[5] #prints M
>>> name[-1] #prints M
>>> name[2] #prints I
>>> name[-16-3+16] #prints R
```

(Backward indexing is calculated as

$\text{len(string)} + \text{value} = \text{correct index}$ )

ex: If `name[-4]` is given,

$6 + (-4) = 2 = \text{name}[2] = \text{I}$

#### 2) STRING SLICING

```
>>>str= "SATHYASAI BABA"
```

Then,

```
0  1  2  3  4  5  6  7  8  9 10 11 12
S  A  T  H  Y  A  S  A  I  B  A  B  A
-13 -12 -11 -10 -9 -8 -7 -6 -5 -4 -3 -2 -1
```

##### Thumb Rule:

In slicing, whether forward or backward,

- If the direction is from **LEFT TO RIGHT** and the step value is **positive**, then we will get output. →
- If the direction is from **RIGHT TO LEFT** and the step value is **negative**, then we will get output. ←
- If the slicing fails in both ways, then only empty strings/list/tuple is printed

#### FORWARD SLICING

```
>>> str[0:15]
'SATHYA SAI BABA'

>>> str[0:6]
'SATHYA'

>>> str[0:15:2]
'STY A AA'

>>> str[0:len(str):2]
'STY A AA'

>>> str[0:15:3]
'SH IA'
```



### **FORWARD SLICING- REVERSE**

```
>>> str[15:0]
''
>>> str[14:0]
''
>>> str[14:0:-1]
'ABAB IAS AYHTA'
>>>str[14:-1:-1]
''
>>>str[14::-1]
'ABAB IAS AYHTAS'
>>> str[14:]
'A'
>>> str[14:0:-2]
'AA A YT'
```

### **BACWARD SLICING- STRAIGHT**

```
>>>str[-8:-5]
'SAI'
>>> str[-15:-1:2]
'STY A A'
```

### **BACWARD SLICING- REVERSE**

```
>>> str[-1::-1]
'ABAB IAS AYHTAS'
>>> str[-1:-16:-2]
'AA A YTS'
```

### **COLON : MODELS**

```
>>> str[:]
'SATHYA SAI BABA'
>>> str[:,]
'SATHYA SAI BABA'
>>> str[0:]
'SATHYA SAI BABA'
>>> str[-15:]
'SATHYA SAI BABA'
>>> str[:10]+str[10:]
'SATHYA SAI BABA'
>>> str[::-1]
'ABAB IAS AYHTAS'
>>> str[::-2]
'AA A YTS'
```

- Index out of range produces IndexError  
>>>str[22]  
IndexError: string index out of range
- Slice out of range produces empty string  
>>>str[21:25]  
''

### **3) STRING OPERATORS**

#### **+ CONCATENATION**

```
a = "sai"
b = "ram"
>>>print(a+b)
```

```
sairam
```

```
>>> "sai" + "123" + "baba"
```

```
'sai123baba'
```

```
>>> "sai" + 123
```

**TypeError produced**

## **+ REPLICATION**

```
a = "sai"
```

```
b = 2
```

```
>>> print(a*b)
```

```
saisai
```

```
>>> a = "sai"
```

```
>>> b = "ram"
```

```
>>> print(a+b*3)
```

```
sairamramram
```

(replication is done first, when both operators occur)

```
>>> "sai"*"baba"
```

**TypeError produced**

## **4) STRING FUNCTIONS/METHODS**

### **a) isalpha()**

```
>>> str="sairam"
```

```
>>> str.isalpha()
```

```
True
```

```
>>> str="sai ram"
```

```
>>> str.isalpha()
```

```
False
```

```
>>> str="sairam123"
```

```
>>> str.isalpha()
```

```
False
```

### **b) isalnum()**

```
>>> str = "sairam123"
```

```
>>> str.isalnum()
```

```
True
```

```
>>> str="sairam 123"
```

```
>>> str.isalnum()
```

```
False
```

```
>>> str="sairam"
```

```
>>> str.isalnum()
```

```
True
```

### **c) islower()**

```
>>> str="sairam"
```

```
>>> str.islower()
```

```
True
```

```
>>> str="Sairam"
```

```
>>> str.islower()
```

```
False
```

```
>>> str="sai ram"
```

```
>>> str.islower()
```

```
True
```

### **d) isupper()**

```
>>> str="sairam"
```

```
>>> str.isupper()
```

```
False
```

```
>>> str="Saiam"
```

```
>>> str.isupper()
```

```
False
```

```
>>> str = "SAIRAM"
```

```
>>> str.isupper()
```

```
True
```

```
>>> str = "SAI RAM"
```

```
>>> str.isupper()
```

```
True
```

```
>>> str = "SAI 2 RAM"
```

```
>>> str.isupper()
```

```
True
```

#### **e) isspace()**

```
>>> str = "sai ram"
```

```
>>> str.isspace()
```

```
False
```

```
>>> str[3].isspace()
```

```
True
```

#### **f) isdigit()**

```
>>> str = "sairam123"
```

```
>>> str.isdigit()
```

```
False
```

```
>>> str="123"
```

```
>>> str.isdigit()
```

```
True
```

#### **g) lower()**

```
>>> str="sairam"
```

```
>>> str.upper()
```

```
'SAIRAM'
```

```
>>> str="SAIRAM"
```

```
>>> str.upper()
```

```
'SAIRAM'
```

```
>>> str="SAI RAM123"
```

```
>>> str.upper()
```

```
'SAI RAM123'
```

#### **h) upper()**

```
>>> str = "sairam"
```

```
>>> str.lower()
```

```
'sairam'
```

```
>>> str = "SAIRAM"
```

```
>>> str.lower()
```

```
'sairam'
```

```
>>> str = "sairam 123"
```

```
>>> str.lower()
```

```
'sairam 123'
```

#### **i) lower()/upper() MAKING PERMANENT CHANGE**

```
>>> str="SAIRAM"
```

```
>>> str.lower()
```

```
'sairam'
```

```
>>> str
'SAIRAM'
```

```
>>> str=str.lower()
>>> str
'sairam'
```

#### **j) capitalize()**

```
>>> str = "sairam"
>>> str.capitalize()
'Sairam'
```

```
>>> str = "sathya sai baba"
>>> str.capitalize()
'Sathya sai baba'
>>> str = "SAIRAM"
>>> str.capitalize()
'Sairam'
```

#### **k) title()**

```
>>> str = "sathya sai baba"
>>> str.title()
'Sathya Sai Baba'
```

```
>>> str = "SAIRAM"
>>> str.title()
'Sairam'
```

#### **l) lstrip()**

##### **removing leading spaces**

```
>>> str = " sairam"
>>> str.lstrip()
'sairam'
```

##### **removing leading characters matching the string characters passed**

```
>>> str = "sathyasai"
>>> str.lstrip("tease")
'hyasai'
```

#### **m) rstrip()**

##### **removing trailing spaces**

```
>>> str = "sai ram "
>>> str.rstrip()
'sai ram'
```

##### **removing trailing characters matching with the string characters passed**

```
>>> str = "sathyasai"
>>> str.rstrip("saint")
'sathy'
```

#### **n) strip()**

##### **removing leading and trailing spaces**

```
>>> str = " sathya sai "
>>> str.strip()
'sathya sai'
```

##### **removing leading and trailing characters matching with the string characters passed**

```
>>> str = "sathya sai"
>>> str.strip("saint")
'hya '
```

```
>>> str="deepavali"
>>> str.strip("delhi")
'pava'
```

### **o) find()**

```
>>> str = "sathya sai baba"
>>> str.find("sai")
7
>>> str = "sathya sai baba"
>>> str.find("b")
11
>>> str.find("sai",5,10)
7
>>> str.find("sai",11,17)
-1
```

### **p) index()**

```
>>> str = "sathya sai baba"
>>> str.index("sai")
7
>>> str.index("b")
11
>>> str.index("sai", 5,10)
7
>>> str.index("sai", 11,17)
```

ValueError produced

### **q) replace()**

```
>>> str = "sai ram"
>>> str.replace("ram", "baba")
'sai baba'
```

```
>>> str.replace("sathya","krishna")
'sai ram'
```

```
>>> str
'sai ram'
```

```
>>> str = str.replace("ram", "krishna")
>>> str
'sai krishna'
```

```
>>> str = str.replace("sai ram", "sai hanuman")
>>> str
'sai hanuman'
```

### **r) join()**

```
>>> str = "sairam"
>>> "".join(str)
's*a*i*r*a*m'
```

### **s) startswith()**

```
a ='sai baba'
>>>a.startswith('s')
True
>>>a.startswith('p')
False
>>>a.startswith('sai')
True
>>>a.startswith('Sai')
False
```

### **t) endswith()**

```
a ='sai baba'
```

```
>>>a.endswith('a')
```

```
True
```

```
>>>a.endswith('s')
```

```
False
```

```
>>>a.endswith('aba')
```

```
True
```

```
>>>a.endswith('baba')
```

```
True
```

### u) split()

*Makes the string into a list taking the passed argument as the separator. What is passed will not appear in the list.*

```
a = 'sathya sai baba'
```

```
>>>a.split()
```

```
['sathya', 'sai', 'baba']
```

```
>>>a.split('t')
```

```
['sa', 'hya sai baba']
```

```
>>>a.split('sai')
```

```
['sathya ', ' baba']
```

```
>>>a.split(" ")
```

```
['sathya', 'sai', 'baba']
```

### v) partition()

*makes the given string into 3 parts of a tuple using the separated passed.*

```
a = 'sathya sai baba'
```

```
>>>a.partition('sai')
```

```
('sathya ', 'sai', ' baba')
```

```
>>>a.partition('t')
```

```
('sa', 't', 'hya sai baba')
```

```
>>>a.partition('baba')
```

```
('sathya sai ', 'baba', '')
```

```
>>>a.partition('sathya')
```

```
('', 'sathya', ' sai baba')
```

## 5) FUNCTIONS WITH ITERATION

### Counting uppercase letters

```
str = "Sai Ram"
```

```
count = 0
```

```
for i in str:
```

```
    if i.isupper():
```

```
        count+=1
```

```
print("No. of uppercase letters:", count)
```

**output:** No. of uppercase letters: 2

### Counting lowercase letters

```
str = "Sai Ram"
```

```
count = 0
```

```
for i in str:
```

```
    if i.islower():
```

```
        count+=1
```

```
print("No. of lowercase letters:", count)
```

**output:** No. of lowercase letters: 4

### Checking spaces

```
str = "Sai Ram"
```

```
for i in str:
```

```
    if i.isspace():
```

```
        print(str*2)
```

**output:** Sai RamSai Ram

## 6. DICTIONARY METHODS/FUNCTIONS

### 1) BASIC METHODS

```
dict1 = {1: "Python", 2: "Java", 3: "Ruby", 4:  
"Scala"}
```

#### a) copy()

```
dict2 = dict1.copy()  
print(dict2)  
{1: 'Python', 2: 'Java', 3: 'Ruby', 4: 'Scala'}
```

#### b) clear()

```
dict1.clear()  
print(dict1)  
{}
```

#### c) get()

```
print(dict2.get(1))  
Python
```

#### d) items()

```
print(dict2.items())  
dict_items([(1, 'Python'), (2, 'Java'), (3, 'Ruby'),  
(4, 'Scala')])
```

#### e) keys()

```
print(dict2.keys())  
dict_keys([1, 2, 3, 4])
```

#### f) pop()

```
dict2.pop(4)  
print(dict2)  
{1: 'Python', 2: 'Java', 3: 'Ruby'}
```

#### g) popitem()

```
dict2.popitem()  
print(dict2)  
{1: 'Python', 2: 'Java'}
```

#### h) update()

```
dict2.update({3: "Scala"})  
print(dict2)  
{1: 'Python', 2: 'Java', 3: 'Scala'}
```

#### i) values()

```
print(dict2.values())  
dict_values(['Python', 'Java', 'Scala'])
```

## 2) fromkeys()

returns the dictionary with key mapped and specific value. It creates a new dictionary from the given sequence with the specific value. The default value is None.

### With tuple

```
a = (1,2,3)
```

```
>>>dict.fromkeys(a)
```

```
{1: None, 2: None, 3: None}
```

```
>>>dict.fromkeys(a,'sai')
```

```
{1: 'sai', 2: 'sai', 3: 'sai'}
```

### With list

```
a = [1,2,3]
```

```
>>>dict.fromkeys(a)
```

```
{1: None, 2: None, 3: None}
```

### With string

```
a = 'sai'
```

```
>>>dict.fromkeys(a)
```

```
{'s': None, 'a': None, 'i': None}
```

## 3) del statement

```
a = {1:'sai',2 :'baba'}
```

```
del a[2]
```

```
a
```

```
{1: 'sai'}
```

```
del a
```

## 4) Other ways of creating dictionary

### from a list of tuples

```
Dict = dict([(1, 'Sai'), (2, 'Baba')])
```

```
print(Dict)
```

```
{1: 'Sai', 2: 'Baba'}
```

### from 2 lists using zip()

```
a = [1,2,3]
```

```
b = ['sai', 'baba', 'ram']
```

```
c = dict(zip(a,b))
```

```
>>>c
```

```
{1: 'sai', 2: 'baba', 3: 'ram'}
```

~~~~~


7. FUNCTIONS HANDOUT

1) 2 TYPES OF FUNCTIONS

void, non-return, non-fruitful function	non-void, return, fruitful function
<pre>def sai(): a = 2+3 print(a) sai()</pre>	<pre>def sai(): a = 2+3 return a ans = sai() print(ans)</pre>

2) Passing parameters

```
def printsai(n):  
  
    for i in range(n):  
  
        print("Sairam")
```

```
n = int(input("Enter number:"))  
  
printsai(n)
```

3) 3 TYPES OF PARAMETERS

- positional
- keyword
- default

a) Positional example

```
def sai(a,b):  
  
    print(a+b)  
  
x=1  
  
y=2  
  
sai(x,y)  
  
#a receives x, b receives y, the order is followed
```

b) Keyword example

```
def sai(a,b):  
  
    print(a+b)  
  
x=1  
  
y=2  
  
sai(b=x, a=y)  
  
#The order can be changed and sent, but the  
same argument in the function header must be  
used.
```

c) Default example

```
def sai(a,b=4,c=5):  
  
    print(a+b+c)  
  
x=1  
  
y=2  
  
z=3  
  
sai(x)  
  
#a gets x, b is 4, c is 5  
  
output:10
```

```
sai(x,y)
```

```
#a gets x, b gets y, c is 5
```

```
output:8
```

```
sai(x,y,z)
```

```
#a gets x, b gets y, c gets z
```

```
output:6
```

(The values can be missed for the arguments defaulted while calling the function)

4) USE OF global KEYWORD

without using global keyword

```
def calc():
```

```
    a = 3
```

```
    print("a during the function:",a)
```

```
a = 5
```

```
print("a before calling the function:",a)
```

```
calc()
```

```
print("a after calling the function:",a)
```

Output:

a before calling the function: 5

a during the function: 3

a after calling the function: 5

Using global keyword

```
def calc():
```

```
    global a
```

```
    a = 3
```

```
    print("a during the function:",a)
```

```
a = 5
```

```
print("a before calling the function:",a)
```

```
calc()
```

```
print("a after calling the function:",a)
```

Output:

a before calling the function: 5

a during the function: 3

a after calling the function: 3

5) PACKING AND UNPACKING

```
def demo():
```

```
    a = 1
```

```
    b = 2
```

```
    c = 3
```

```
    return a+b,b+a,c+a    #packing
```

```
ans = demo
```

```
print(ans)    # (3, 3, 4) is printed, a tuple
```

```
x,y,z = demo()    #unpacking
```

```
print(x)    #3
```

```
print(y)    #3
```

```
print(z)    #4
```

5) SCOPE OF A VARIABLE

```
def calc():  
    a = 3    #LOCAL a  
    print(a)  
  
a = 5    #GLOBAL a  
print(a)
```

6) PARAMETERS AND ARGUMENTS

```
def calc(a,b): #parameters  
    c= a+b  
    print(c)  
  
calc(7,2)    #arguments
```

7) ACTUAL AND FORMAL PARAMETERS /ARGUMENTS

```
def calc(a,b): #formal parameters  
    c= a+b  
    print(c)  
  
calc(7,2)    #actual parrameters
```

8) FLOW OF EXECUTION

a) Trace the flow of execution of the following code:

```
1  #function 0  
2  def fun0():  
3      print('Sairam')  
4      return 6  
5  
6  #function 1  
7  def fun1():  
8      #calling 0 function  
9      x = fun0()  
10     return x  
11  
12 # main program  
13 y = fun1()  
14 print(y)  
15
```

Answer:

2->7->13->7->9->2->3->4->9->10->13->14

b) Trace the flow of execution of the following program:

```
1  #this is python function  
2  
3  def fun2():  
4      #function2  
5      return 2  
6  
7  def fun1(x,y):  
8      z = fun2()  
9      return x+y+z  
10  
11 #main program  
12 a = int(input("Enter first number:"))  
13 b = int(input("Enter second number:"))  
14 ans = fun1(a,b)  
15 print(ans)
```

Answer:

3->7->12->13->14->7->8->3->5->8->9->14->15

~~~~~

**XII COMPUTER SCIENCE***Towards Getting Centum in CS - See Yes!***8. TEXT FILE HANDLING : 1) CHARACTER, WORD, LINE- CODE COMPARISON HANDOUT**

| Operation                  | Character - Based                                                                                                                                                                       | Word - Based                                                                                                                                                                                               | Line - Based                                                                                                                                                                            |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Counting all</b>        | <b>Count all characters</b><br>def fun():<br>count = 0<br>f= open("myfile.txt", "r")<br>data = f.read()<br>for i in data:<br>count+=1<br>print("No of Characters:", count)<br>f.close() | <b>Count all words</b><br>def fun():<br>count = 0<br>f = open("myfile.txt", "r")<br>data = f.read()<br>words = data.split()<br>for word in words:<br>count+=1<br>print("No of words:", count)<br>f.close() | <b>Count all lines</b><br>def fun():<br>count = 0<br>f = open("nowfile.txt", "r")<br>data = f.readlines()<br>for line in data:<br>count+=1<br>print("No of lines:", count)<br>f.close() |
| <b>Counting Particular</b> | <b>Counting letter 'a'</b><br>def fun():<br>count = 0<br>f = open("myfile.txt", "r")<br>data = f.read()                                                                                 | <b>Counting word 'is'</b><br>def fun():<br>count = 0<br>f = open("myfile.txt", "r")<br>data = f.read()<br>words = data.split()                                                                             | <b>Counting lines starting with 'b'</b><br>def fun():<br>count = 0<br>f = open("myfile.txt", "r")<br>data = f.readlines()                                                               |

|                        |                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        | <p>for letter in data:</p> <pre> if letter=='a':     count+=1 print("No of 'a':",count) </pre>                                                                                                                                                                                            | <p>for word in words:</p> <pre> if word=='is':     count+=1 print("No of 'is' :", count) </pre>                                                                                                                                                                                                                                                                            | <p>for line in data:</p> <pre> if line[0]=='b':     count+=1 print("No of lines:",count) </pre>                                                                                                                                                                                                |
| <b>Counting Vowels</b> | <p><b>Counting No. of vowels</b></p> <p>#counting all the individual vowels</p> <pre> def fun():     count = 0     f = open("myfile.txt", "r")     data = f.read()      for letter in data:         if letter in 'aeiouAEIOU':             count+=1     print("No of 'a':", count) </pre> | <p><b>Counting words with vowels</b></p> <p>#ex: tea, sai, baba, rangoli</p> <pre> def fun():     count = 0     f = open("myfile.txt", "r")     data = f.read()     words = data.split()      for word in words:         for letter in word:             if letter in 'aeiouAEIOU':                 count+=1                 break     print("No of 'is' :", count) </pre> | <p><b>Counting lines starting with vowels</b></p> <p>ex: Almighty is great</p> <pre> def fun():     count = 0     f = open("myfile.txt", "r")     data = f.readlines()      for line in data:         if line[0] in 'aeiouAEIOU':             count+=1     print("No of lines:", count) </pre> |

~~~~~

2) TEXT FILE HANDLING (READ/ COUNT/TRANSFER)

USING "r" MODE

A) CHARACTER BASED

a. COUNT/PRINT CHARACTERS

```
def fun():
    count = 0
    with open("myfile.txt","r") as f:
        data = f.read()
        for i in data:
            count+=1
        print("No of Characters:", count)
```

b. COUNT PARTICULAR CHARACTER

```
def fun():
    count = 0
    with open("myfile.txt","r") as f:
        data = f.read()
        for i in data:
            if i=='a':
                count+=1
        print("No of 'a' present:",count)
```

B) WORD BASED

c. COUNT/PRINT WORDS

```
def fun():
    count = 0
    with open("nowfile.txt","r") as f:
        data = f.read()
        words = data.split()
        for word in words:
            count+=1
        print("No of words present:",count)
```

d. COUNT PARTICULAR WORD

```
def fun():
    count = 0
    with open("nowfile.txt","r") as f:
        data = f.read()
        words = data.split()
        for word in words:
            if word=='is':
                count+=1
        print("No of 'is' present:", count)
```

e. COUNT WORDS STARTING WITH PARTICULAR LETTER

```
def fun():
    count = 0
    with open("nowfile.txt","r") as f:
        data = f.read()
        words = data.split()
        for word in words:
            if word[0]=='j':
                count+=1
        print("No of words starting with 'j':",
count)
```

f. COUNT WORDS ENDING WITH PARTICULAR LETTER

```
def fun():
    count = 0
    with open("nowfile.txt","r") as f:
        data = f.read()
        words = data.split()
        for word in words:
            if word[len(word)-1]=='j':
                count+=1
```

```
print("No of words ending with 'j':",  
count)
```

**g. COUNT WORDS STARTING WITH
SMALL/CAPITAL VOWELS**

```
def fun():  
    count = 0  
    with open("nowfile.txt","r") as f:  
        data = f.read()  
        words = data.split()  
        for word in words:  
            if word[0] in 'aeiouAEIOU':  
                count+=1  
        print("No of words starting with vowels:",  
count)
```

**h. COUNT WORDS ENDING WITH
SMALL/CAPITAL VOWELS**

```
def fun():  
    count = 0  
    with open("nowfile.txt","r") as f:  
        data = f.read()  
        words = data.split()  
        for word in words:  
            if word[-1] in 'aeiouAEIOU':  
                count+=1  
        print("No of words ending with vowels:",  
count)
```

**i. COUNT/PRINT WORDS HAVING
SECOND LAST LETTER AS VOWEL**

```
def fun():
```

```
    count = 0
```

```
    with open("nowfile.txt","r") as f:
```

```
        data = f.read()
```

```
        words = data.split()
```

```
        for word in words:
```

```
            if word[-2] in 'aeiouAEIOU':
```

```
                count+=1
```

```
        print("No of words having second last letter  
as vowel:", count)
```

j. COUNT PARTICULAR SIZED WORD

```
def fun():
```

```
    count = 0
```

```
    with open("f:\\nowfile.TXT","r") as f:
```

```
        data = f.read()
```

```
        words = data.split()
```

```
        for word in words:
```

```
            if len(word)==2:
```

```
                count+=1
```

```
    print(count)
```

C) LINE BASED

k. COUNT/PRINT NUMBER OF LINES

MODEL 1

```
def fun():
```

```
    count = 0
```

```
    with open("f:\\nowfile.txt","r") as f:
```

```
        data = f.readlines()
```

```
        for line in data:
```

```
            count+=1
```

```
    print("No of lines present:",count)
```

MODEL 2

```
def fun():  
  
    count = 0  
  
    f = open("ths.txt","r")  
  
    for line in f:  
  
        count+=1  
  
    print("No. of lines:", count)  
  
    f.close()
```

I. COUNT NUMBER OF LINES STARTING WITH A PARTICULAR LETTER

MODEL 1

```
def fun():  
  
    count = 0  
    with open("f:\\nowfile.txt","r") as f:  
        data = f.readlines()  
        for line in data:  
            if line[0]=='b':  
                count+=1  
        print("No of lines starting with  
'b':",count)
```

MODEL 2

```
def fun():  
  
    count = 0  
  
    f = open("ths.txt","r")  
  
    for line in f:  
  
        if line[0]=='b':
```

```
        count+=1
```

```
        print("No. of lines:", count)
```

```
        f.close()
```

m. COUNT NUMBER OF LINES ENDING WITH A PARTICULAR LETTER

```
def fun():  
  
    #assuming each line is ending with full stop  
    count = 0  
    with open("f:\\nowfile.txt","r") as f:  
        data = f.readlines()  
        for line in data:  
            if line[len(line)-2]=='b':  
                count+=1  
        print("No of lines ending with 'b':",count)
```

USING “w” or “a” MODE

1.ADD NEW WORD/LINE ERASING ALL CONTENT

a) WORD/LINE received as string argument (ex: str = “school of management”)

```
def fun(str):  
  
    with open("f:\\nowfile.txt","w") as f:  
  
        f.write(str)
```

b)WORD/LINE received as list argument (ex: str = ['school', 'of', 'management'])

```
def fun(str):  
  
    with open("f:\\nowfile.txt","a") as f:  
  
        f.writelines(str)
```


2.ADD NEW WORD/LINE KEEPING OLD CONTENT

a) WORD/LINE received as string argument (ex: str = "school of management")

```
def fun(str):  
    with open("f:\\nowfile.txt","a") as f:  
        f.write("\n")  
        f.write(str)
```

b)WORD/LINE received as list argument (ex: str = ['school', 'of', 'management'])

```
def fun(str):  
    with open("f:\\nowfile.txt","a") as f:  
        f.write("\n")  
        f.writelines(str)
```

USING "r+" MODE

ADD A NEW LINE ONLY IF PARTICULAR WORD IS PRESENT

```
def fun():  
    with open("myflie.txt", "r+") as f:  
        data = f.read()  
        words = data.split()  
        for word in words:  
            if word=='rama':  
                f.write("Rama is present")
```

TRANSFER PARTICULAR LETTER/WORD/LINE TO ANOTHER FILE

(Transfer words containing 'abcde' to another file)

```
def fun():  
    f1 = open("oldfile.txt", "r")  
    f2 = open("newfile.txt", "w")  
    data = f1.read()  
    words = data.split()  
    for words in data:  
        for word in words:  
            for letter in word:  
                if letter in 'abcde':  
                    f2.write(word)  
                    break  
    f1.close()  
    f2.close()
```

~~~~~

*"There are no limits. There are only plateaus, and you must not stay there — you must go beyond them." – Bruce Lee*

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS – See Yes!*

### 9. CSV FILE HANDLING

**1) Write a Program in Python that defines and calls the following user defined functions:**

(i) ADD() – To accept and add data of an employee to a CSV file 'record.csv'. Each record consists of a list with field elements as empid, name and mobile to store employee id, employee name and employee salary respectively.

(ii) COUNTR() – To count the number of records present in the CSV file named 'record.csv'. (CBSE)

```
import csv
```

```
def ADD():
```

```
    fout=open("record.csv","a",newline="\n")
```

```
    wr=csv.writer(fout)
```

```
    empid=int(input("Enter Employee id :: "))
```

```
    name=input("Enter name :: ")
```

```
    mobile=int(input("Enter mobile number :: "))
```

```
    lst=[empid,name,mobile] -----1/2 mark
```

```
    wr.writerow(lst) -----1/2 mark
```

```
    fout.close()
```

```
def COUNTR():
```

```
    fin=open("record.csv","r",newline="\n")
```

```
    data=csv.reader(fin)
```

```
    d=list(data)
```

```
    print(len(d))
```

```
    fin.close()
```

```
ADD()
```

```
COUNTR()
```

**2) Write a Program in Python that defines and calls the following user defined functions:**

a) add() – To accept and add data of an employee to a CSV file 'furddata.csv'. Each record consists of a list with field elements as fid, fname and fprice to store furniture id, furniture name and furniture price respectively.

b) search()- To display the records of the furniture whose price is more than 10000.(CBSE)

```
import csv
```

```
def add():
```

```
    fout=open("furddata.csv","a",newline='\n')
```

```
    wr=csv.writer(fout)
```

```
    fid=int(input("Enter Furniture Id :: "))
```

```
    fname=input("Enter Furniture name :: ")
```

```
    fprice=int(input("Enter price :: "))
```

```
    rec=[fid,fname,fprice]
```

```
    wr.writerow(rec)
```

```
    fout.close()
```

```
def search():

    fin=open("furdata.csv","r",newline='\n')

    data=csv.reader(fin)

    found=False

    print("The Details are")

    for rec in data:

        if int(rec[2])>10000:

            found=True

            print(rec[0], rec[1], rec[2])

    if found==False:

        print("Record not found")

    fin.close()
```

add()

search()

### 3)Write a Program in Python that defines and calls the following user defined functions:

a) addstud() – To accept and add data of a student to a CSV file ‘stud.csv’. Each record consists of a list with field elements as sid, sname and score to store student id, student name and student score respectively.

b) searchstud()- To display only the names of the students who scored more than 450.

```
import csv
```

#### def addstud():

```
f=open("stud.csv","a",newline='\n')

mycsv=csv.writer(f)

sid=int(input("Enter student Id: "))

sname=input("Enter student name: ")

score=int(input("Enter score : "))

rec=[sid, sname, score]

mycsv.writerow(f)

f.close()
```

#### def searchstud():

```
f=open("stud.csv","r",newline='\n')

mycsv=csv.reader(f)

found = False

print("Students who scored more than 450:")

for rec in data:

    if int(rec[2])>450:

        found=True

        print(rec[1])

    if found==False:

        print("Record not found")

    fin.close()

addstud()

searchstud()
```

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS - See Yes!*

### 10. BINARY FILE HANDOUT

#### LIST-BASED

**1) Write a Program in Python that defines and calls the following user defined functions:**

adddata() - to add student details such as id, name and score to STUD. DAT

readdata() – to search for the records of students who have scored more than 450

#### **Model 1: with while continuity**

```
import pickle
def adddata():
    f = open("stud.dat", "ab")
    while True:
        id = int(input("Enter your id:"))
        name = input("Enter name:")
        tot = int(input("Enter total:"))
        rec = [id, name, tot]
        pickle.dump(record,f)
        opt = input("Add more (y/n):")
        if opt.lower()=='n':
            break
    f.close()
```

#### **Model 2: with no . of records from user**

```
def adddata():
    f = open("stud.dat", "ab")
    n = int(input("Enter no. of records:"))
    for i in range(n):
        id = int(input("Enter your id:"))
        name = input("Entername:")
        tot = int(input("Enter total:"))
        rec = [id, name, tot]
        pickle.dump(record,f)
    f.close()
```

#### **Model 3: for a single record**

```
def adddata():
    f = open("stud.dat", "ab")
    id = int(input("Enter your id:"))
    name = input("Enter name:")
    tot = int(input("Enter total:"))
    rec = [id, name, tot]
    pickle.dump(record,f)
    f.close()
```

```

def readdata():
    f = open("stud.dat", "rb")
    flag=0
    try:
        while True:
            data = pickle.load(f)
            for rec in data:
                if rec[2] >450:
                    print(rec[0], rec[1], rec[2])
                    flag=1
                    break
            if flag==0:
                print("Record not found!!!")
    except EOFError:
        f.close()

```

adddata()

readdata()

## 2) Write a Program in Python that defines and calls the following user defined functions:

adddata() - to add a product detail such as id, name and price to PROD. DAT

readdata(id) – to search for the id of the product received from the user, passed as argument and print its details. If there are no record with the id, then print appropriate message.

```

def adddata():
    f = open("prod.dat", "ab")
    id = int(input("Enter product id:"))
    name = input("Enter name:")

```

```

price = float(input("Enter price:"))
rec = [id, name, tot]
pickle.dump(record,f)
f.close()

```

```

def readdata(id):
    f = open("prod.dat", "rb")
    flag=0
    try:
        while True:
            data = pickle.load(f)
            for rec in data:
                if rec[0]==id:
                    print(rec[0], rec[1], rec[2])
                    flag=1
                    break
            if flag==0:
                print("Record not found!!!")
    except EOFError:
        f.close()

```

adddata()

id = int(input("Enter ID to search:"))

readdata(id)

## Model to count no. of products with price more than 1000

```

def readdata():
    f = open("prod.dat", "rb")
    count = 0
    try:

```

```

while True:
    data = pickle.load(f)
    for rec in data:
        if rec[2]>1000:
            count+=1
except EOFError:
    f.close()
print("No. of records with price >1000:", count)

```

**Model to update a student name and score for the id received from the user. Assume each record as [id, name, score]**

```

def update():
    f1=open("stud.dat",'rb+')
    flag = 0
    id=int(input("Enter the id to update:"))
    f1.seek(0)
    try:
        while True:
            pos=f1.tell()
            data=pickle.load(f)
            for rec in data:
                if(rec[0]==id):
                    rec[1]=input("Enter new name:")
                    rec[2]=int(input("Enter new score:"))
                    f1.seek(rpos)
                    pickle.dump(s, f)
                    flag = 1
                    break
            if flag==0:
                print("Record not found!!!")

```

```

except EOFError:
    f1.close()

```

## DICTIONARY-BASED

**2) Write a Program in Python that defines and calls the following user defined functions:**

insertdata() - to add a student detail such as roll no, name and marks to STUDENT. DAT using dictionary

readdata(id) – to print id and name of students who have scored less than or equal to 300. Print appropriate message if the record is not found.

```

import pickle
rec = {}
def insertRec():
    fout = open("student.dat", "ab")
    rec['Roll'] = int(input('Enter roll number:'))
    rec['Name'] = input('Enter Name:')
    rec['Marks'] = int(input('Enter Marks'))
    pickle.dump(rec,f)
    fout.close()

```

```

def readRec():
    f = open('student.dat','rb')
    flag = 0
    try:
        while True:
            rec = pickle.load(f)
            if rec['Marks']<=300:
                print(rec['Roll'], rec['Name'])
                flag = 1
            if flag==0:

```

```

        print("Record not found!")

except EOFError:
    f.close()

```

### #Searching a record based on Rollno

```

def searchRollNo(r):
    f = open('student.dat', 'rb')
    flag = 0
    while True:
        try:
            rec = pickle.load(f)
            if rec['Roll'] == r:
                print('Roll Num:',rec['Roll'])
                print('Name:',rec['Name'])
                print('Marks:',rec['Marks'])
                flag = True
        except EOFError:
            break
    if flag == 0:
        print('No Records found')
    f.close()

```

### #Marks Modification for a RollNo

```

def searchRollNo(r):
    f = open('student.dat', 'rb+')
    flag = 0
    while True:
        try:
            pos = f.tell()
            rec = pickle.load(f)
            if rec['Roll'] == r:
                flag = 1
                rec['Name'] =input("New name:")
                rec['Marks']=int(input(New Score:"))

```

```

        f.seek(pos)
        pickle.dump(rec)
    except EOFError:
        break
    f.close()
if flag==0:
    print("No record found!")

```

~~~~~

XII COMPUTER SCIENCE

Towards Getting Centum in CS - See Yes!

11. LIST MANIPULATION (DIGITS)

VARIATIONS IN CBSE SAMPLE PAPER QUESTION (2022-23)

1) Write a function INDEX_LIST(L), where L is the list of elements passed as argument to the function. The function returns another list named indexList that stores the indices of all Non-Zero Elements of L. (CBSE)

```
def INDEX_LIST(L):
    indexList=[]
    for i in range(len(L)):
        if L[i]!=0:
            indexList.append(i)
    return indexList
```

2) Write a function INDEX_LIST(L), where L is the list of elements passed as argument to the function. The function returns another list named indexList that stores the indices of all Zero Elements of L.

```
def INDEX_LIST(L):
    indexList=[]
    for i in range(len(L)):
        if L[i]==0:
            indexList.append(i)
    return indexList
```

3) Write a function CHANGE_LIST(L), where L is the list of elements passed as argument to the function. The function returns another list named indexList that stores all Non-Zero Elements of L.

```
def CHANGE_LIST(L):
    indexList=[]
    for i in range(len(L)):
        if L[i]!=0:
            indexList.append(L[i])
    return indexList
```

4) Write a function CHANGE_LIST(L), where L is the list of elements passed as argument to the function. The function prints another list named indexList that stores all Non-Zero Elements of L.

```
def CHANGE_LIST(L):
    indexList=[]
    for i in range(len(L)):
        if L[i]!=0:
            indexList.append(L[i])
    print(indexList)
```

5) Write a function CHANGE_LIST() to create a list L of 10 integers. The function returns another list named indexList that stores all Non-Zero Elements of L.

```
def CHANGE_LIST():
    L=[]
    for i in range(10):
        n = int(input("Enter the integer:"))
        L.append(n)

    indexList=[]
```



```

for i in range(len(L)):
    if L[i]!=0:

        indexList.append(L[i])
return indexList

```

MODEL 1: PRINT ADDITION/PRODUCT OF SPECIFIC ELEMENTS

- Direct traversal
- Check and Add/product specific integers
- Print/return output

1) Write a function LIST_FUN() to add all the even integers of a list L passed as argument to the function and print and return the addition.

#If L = [1,4,6,2,3,9,8], then it should print 'Addition of even integers:20'

```

def LIST_FUN(L):
    addi=0
    for i in L:
        if i%2==0:
            addi+=i
    print("Addition:",addi)
    return addi

```

2) Write a function LIST_FUN() to multiply all the odd integers of a list L passed as argument to the function and print and return the addition.

#If L = [1,4,6,2,3,9,8], then it should print 'Product of odd integers:27'

```

def LIST_FUN(L):
    prod=1
    for i in L:
        if i%2!=0:
            prod*=i

```

```

        print("Product of odd of
integers:",prod)
    return addi

```

3) Write a function LIST_FUN() to multiply all the POSITIVE integers of a list L passed as argument to the function and print and return the addition.

#If L = [1,-4,6,2,-3,9,-8], then it should print 'Product of positive integers:18'

```

def LIST_FUN(L):
    prod=1
    for i in L:
        if i>=0:
            prod*=i
    print("Product of positive
integers:",prod)
    return addi

```

MODEL 2: JUST PRINT SPECIFIC ELEMENTS

- Direct traversal
- Check and Print odd/even/divisible/unit/negative/positive/negative/non-zero integers
- Print/return output

1)PRINTING EVEN INTEGERS

```

def LIST_FUN(L):
    for i in L:
        if i%2==0:
            print(i, end=' ')

```

2)PRINTING ODD INTEGERS

```

def LIST_FUN(L):
    for i in L:
        if i%2!=0:
            print(i, end=' ')

```

3)PRINTING NUMBERS DIVISIBLE BY 5

```
def LIST_FUN(L):  
    for i in L:  
        if i%5==0:  
            print(i, end=' ')
```

4)PRINTING NUMBERS ENDING WITH 8

```
def LIST_FUN(L):  
    for i in L:  
        if i%10==8:  
            print(i, end=' ')
```

5)PRINTING NUMBERS ENDING WITH 6

```
def LIST_FUN(L):  
    for i in L:  
        if i%10==6:  
            print(i, end=' ')
```

6)PRINTING NEGATIVE NUMBERS

```
def LIST_FUN(L):  
    for i in L:  
        if i<0:  
            print(i, end=' ')
```

7)PRINTING POSITIVE NUMBERS

```
def LIST_FUN(L):  
    for i in L:  
        if i>=0:  
            print(i, end=' ')
```

8)PRINTING NON-ZERO NUMBERS

```
def LIST_FUN(L):  
    for i in L:  
        if i!=0:  
            print(i, end=' ')
```

9)PRINTING POSITIVE AND WITHIN 45

```
def LIST_FUN(L):  
    for i in L:  
        if i>=0 and i<=45:  
            print(i, end=' ')
```

10) PRINTING NEGATIVE OR ODD NUMBERS

```
def LIST_FUN(L):  
    for i in L:  
        if i<0 or i%2!=0:  
            print(i, end=' ')
```

MODEL 3: CREATE A NEW LIST WITH SPECIFIC ELEMENTS

- Direct traversal
- Create empty list
- Check and append odd/even/divisible/unit/negative/positive/negative/non-zero integers to the list created
- Print/return output

1)CREATING LIST WITH EVEN INTEGERS

```
def LIST_FUN(L):  
    NLIST = []  
    for i in L:  
        if i%2==0:  
            NLIST.append(i)
```

```
print(NLIST, end=' ')
```

```
return NLIST
```

2) CREATING LIST WITH ODD INTEGERS

```
def LIST_FUN(L):
```

```
    NLIST = []
```

```
    for i in L:
```

```
        if i%2!=0:
```

```
            NLIST.append(i)
```

```
    print(NLIST, end=' ')
```

```
    return NLIST
```

3) CREATING LIST WITH INTEGERS DIVISIBLE BY 5

```
def LIST_FUN(L):
```

```
    NLIST = []
```

```
    for i in L:
```

```
        if i%5==0:
```

```
            NLIST.append(i)
```

```
    print(NLIST, end=' ')
```

```
    return NLIST
```

4) CREATING LIST WITH NUMBERS ENDING WITH 8

```
def LIST_FUN(L):
```

```
    NLIST = []
```

```
    for i in L:
```

```
        if i%10==8:
```

```
            NLIST.append(i)
```

```
    print(NLIST, end=' ')
```

```
    return NLIST
```

5) CREATING LIST WITH NUMBERS ENDING WITH 6

```
def LIST_FUN(L):
```

```
    NLIST = []
```

```
    for i in L:
```

```
        if i%10==6:
```

```
            NLIST.append(i)
```

```
    print(NLIST, end=' ')
```

```
    return NLIST
```

6) CREATING LIST WITH NEGATIVE NUMBERS

```
def LIST_FUN(L):
```

```
    NLIST = []
```

```
    for i in L:
```

```
        if i<0:
```

```
            NLIST.append(i)
```

```
    print(NLIST, end=' ')
```

```
    return NLIST
```

7) CREATING LIST WITH POSITIVE NUMBERS

```
def LIST_FUN(L):
```

```
    NLIST = []
```

```
    for i in L:
```

```
        if i>=0:
```

```
            NLIST.append(i)
```

```
    print(NLIST, end=' ')
```

```
    return NLIST
```

8) CREATING LIST WITH NON-ZERO NUMBERS

```
def LIST_FUN(L):
```

```
    NLIST = []
```

```
    for i in L:
```

```
        if i!=0:
```

```
            NLIST.append(i)
```

```
    print(NLIST, end=' ')
```

```
    return NLIST
```

9) CREATING LIST WITH POSITIVE AND WITHIN 45

```
def LIST_FUN(L):  
    NLIST = []  
    for i in L:  
        if i>=0 and i<=45:  
            NLIST.append(i)  
    print(NLIST, end=' ')  
    return NLIST
```

10) CREATING LIST WITH NEGATIVE OR ODD NUMBERS

```
def LIST_FUN(L):  
    NLIST = []  
    for i in L:  
        if i<0 or i%2!=0:  
            NLIST.append(i)  
    print(NLIST, end=' ')  
    return NLIST
```

MODEL 4: CREATE A NEW LIST WITH INDEX OF SPECIFIC ELEMENTS

- Range traversal
- Create empty list
- Check and append index of odd/even/divisible/unit/negative/positive/negative/non-zero integers to the list created
- Print/return output

1) CREATING LIST WITH INDICES OF EVEN INTEGERS

```
def LIST_FUN(L):  
    NLIST = []  
    for i in range(len(L):  
        if L[i]%2==0:  
            NLIST.append(i)  
    print(NLIST, end=' ')  
    return NLIST
```

2) CREATING LIST WITH INDICES OF ODD INTEGERS

```
def LIST_FUN(L):  
    NLIST = []  
    for i in range(len(L):  
        if L[i]%2!=0:  
            NLIST.append(i)  
    print(NLIST, end=' ')  
    return NLIST
```

3) CREATING LIST WITH INDICES OF INTEGERS DIVISIBLE BY 5

```
def LIST_FUN(L):  
    NLIST = []  
    for i in range(len(L):  
        if L[i]%5==0:  
            NLIST.append(i)  
    print(NLIST, end=' ')  
    return NLIST
```

4) CREATING LIST WITH INDICES OF NUMBERS ENDING WITH 8

```
def LIST_FUN(L):  
    NLIST = []  
    for i in range(len(L):  
        if L[i]%10==8:  
            NLIST.append(i)  
    print(NLIST, end=' ')  
    return NLIST
```

5) CREATING LIST WITH INDICES OF NUMBERS ENDING WITH 6

```
def LIST_FUN(L):  
    NLIST = []  
    for i in range(len(L):  
        if L[i]%10==6:  
            NLIST.append(i)
```

```
print(NLIST, end=' ')
```

```
return NLIST
```

6) CREATING LIST WITH INDICES OF NEGATIVE NUMBERS

```
def LIST_FUN(L):
```

```
    NLIST = []
```

```
    for i in range(len(L):
```

```
        if L[i]<0:
```

```
            NLIST.append(i)
```

```
    print(NLIST, end=' ')
```

```
    return NLIST
```

7) CREATING LIST WITH INDICES OF POSITIVE NUMBERS

```
def LIST_FUN(L):
```

```
    NLIST = []
```

```
    for i in range(len(L):
```

```
        if L[i]>=0:
```

```
            NLIST.append(i)
```

```
    print(NLIST, end=' ')
```

```
    return NLIST
```

8) CREATING LIST WITH INDICES OF NON-ZERO NUMBERS

```
def LIST_FUN(L):
```

```
    NLIST = []
```

```
    for i in range(len(L):
```

```
        if L[i]!=0:
```

```
            NLIST.append(i)
```

```
    print(NLIST, end=' ')
```

```
    return NLIST
```

9) CREATING LIST WITH INDICES OF POSITIVE AND WITHIN 45

```
def LIST_FUN(L):
```

```
NLIST = []
```

```
for i in range(len(L):
```

```
    if L[i]>=0 and L[i]<=45:
```

```
        NLIST.append(i)
```

```
print(NLIST, end=' ')
```

```
return NLIST
```

10) CREATING LIST WITH INDICES OF NEGATIVE OR ODD NUMBERS

```
def LIST_FUN(L):
```

```
    NLIST = []
```

```
    for i in range(len(L):
```

```
        if L[i]<0 or L[i]%2!=0:
```

```
            NLIST.append(i)
```

```
print(NLIST, end=' ')
```

```
return NLIST
```

MODEL 5: CHANGE/MODIFY SPECIFIC ELEMENTS IN THE LIST

- Range traversal
- Check odd/even/divisible/unit/negative/positive/negative/non-zero elements of the list and change accordingly
- Print/return output

1) DIVIDE EVEN DIGITS BY 2 AND MULTIPLY ODD DIGITS BY 3

```
def LIST_FUN(L):
```

```
    for i in range(len(L):
```

```
        if L[i]%2==0:
```

```
            L[i]/=2
```

```
        else:
```

```
            L[i]*=3
```

```
    print(L)
```

```
    return L
```

**2) DIVIDE 5 DIVISIBLE DIGITS BY 5
AND MULTIPLY 3 DIVISIBLE DIGITS BY
4**

```
def LIST_FUN(L):  
    for i in range(len(L):  
        if L[i]%5==0:  
            L[i]/=5  
        else L[i]%3==0:  
            L[i]*=4  
  
    Print(L)  
    return L
```

**3) CHANGE 7 DIVISIBLE DIGITS AS 0
OTHERS AS 1**

```
def LIST_FUN(L):  
    for i in range(len(L):  
        if L[i]%7==0:  
            L[i]=0  
        else:  
            L[i]=1  
  
    Print(L)  
    return L
```

**4) CHANGE POSITIVE INTEGERS AS True
and NEGATIVE INTEGERS AS False**

```
def LIST_FUN(L):  
    for i in range(len(L):  
        if L[i]>=0:  
            L[i]=True  
        else:  
            L[i]=False  
  
    Print(L)  
    return L
```

LIST SWAPPING

#FULL SWAP

```
a = [1,2,3,4,5,6,7,8]
```

```
a = [8,7,6,5,4,3,2,1]
```

```
def list_swap():
```

```
    l = len(a)-1
```

```
    for i in range(len(a)//2):
```

```
        a[i],a[l] = a[l],a[i]
```

```
        l=l-1
```

```
    print(a)
```

#HALF SWAP

```
a = [1,2,3,4,5,6,7,8]
```

```
a = [5,6,7,8,1,2,3,4] #ans
```

```
def list_swap():
```

```
    m = len(a)//2
```

```
    for i in range(len(a)//2):
```

```
        a[i],a[m] = a[m],a[i]
```

```
        m = m+1
```

```
    print(a)
```

#ALTERNATE SWAP

```
a = [1,2,3,4,5,6,7,8]
```

```
a = [2,1,4,3,6,5,8,7] #ans
```

```
def list_swap():
```

```
    for i in range(0,len(a),2):
```

```
        a[i],a[i+1] = a[i+1],a[i]
```

```
    print(a)
```

#SELECT AND SWAP

```
#SWAP MULTIPLES OF 10
```

```
a = [1,2,3,20,5,31,40,8]
```

```
a = [1,2,20,3,5,31,8,40] #ans
```

```
def list_swap():
```

```
    i =0
```

```
    while i<len(a)-1:
```

```
        if a[i]==14:
```

```
            a[i],a[i+1]=a[i+1],a[i]
```

```
            i+=1
```

```
        i+=1
```

```
    print(a)
```

~~~~~

*"Let us sacrifice our today so that our children can have a better tomorrow." - A. P. J. Abdul Kalam*

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS - See Yes!*

### 12. LIST MANIPULATION (STRINGS)

#### LENGTH-BASED

##### (DIRECT TRAVERSAL)

1) Create a List of 10 student names. Pass the list to a function CHAR\_LIST(L). Create another list of strings NLIST with strings having EVEN length. Print and return the newly created list. (CBSE)

```
L = []
for i in range(10):
    name = input("Enter a name:")
    L.append(name)
```

```
def CHAR_LIST(L):
    NEWLIST=[]
    for i in L:
        if len(L)%2==0:
            NLIST.append(i)
    print(NEWLIST)
    return NEWLIST
```

2) Create a List of 10 student names. Pass the list to a function CHAR\_LIST(L). Create another list of strings NLIST with strings having ODD length. Print and return the newly created list.

```
L = []
for i in range(10):
    name = input("Enter a name:")
    L.append(name)
```

```
def CHAR_LIST(L):
    NEWLIST=[]
    for i in L:
        if len(L)%2!=0:
            NLIST.append(i)
    print(NEWLIST)
    return NEWLIST
```

3) Create a List of 10 student names. Pass the list to a function CHAR\_LIST(L). Create another list of strings NLIST with strings having length greater than or equal to 5. Print and return the newly created list.

```
L = []
for i in range(10):
    name = input("Enter a name:")
    L.append(name)
```

```
def CHAR_LIST(L):
    NEWLIST=[]
    for i in L:
        if len(L)>=5:
            NLIST.append(i)
    print(NEWLIST)
    return NEWLIST
```

#### CHARACTER-BASED

##### (DIRECT TRAVERSAL)

4) Create a List of 10 student names. Pass the list to a function CHAR\_LIST(L). Create another list of strings NLIST with only the first character of the names. Print and return the newly created list.

```
L = []
for i in range(10):
    name = input("Enter a name:")
    L.append(name)
```



```
def CHAR_LIST(L):
    NEWLIST=[]
    for i in L:
        NLIST.append(i[0])
    print(NEWLIST)
    return NEWLIST
```

5) Create a List of 10 student names. Pass the list to a function CHAR\_LIST(L). Create another list of strings NLIST with only the last character of the names . Print and return the newly created list.

```
L = []
for i in range(10):
    name = input("Enter a name:")
    L.append(name)
```

```
def CHAR_LIST(L):
    NEWLIST=[]
    for i in L:
        NLIST.append(i[-1])
    print(NEWLIST)
    return NEWLIST
```

6) Create a List of 10 student names. Pass the list to a function CHAR\_LIST(L). Create another list of strings NLIST with names containing vowels lower or upper case as the first letter. Print and return the newly created list.

```
L = []
for i in range(10):
    name = input("Enter a name:")
    L.append(name)
```

```
def CHAR_LIST(L):
    NEWLIST=[]
```

```
    for i in L:
        if i[0] in "aeiouAEIOU":
            NLIST.append(i)
    print(NEWLIST)
    return NEWLIST
```

7) Create a List of 10 student names. Pass the list to a function CHAR\_LIST(L). Create another list of strings NLIST with only middle character of the names with odd length. Print and return the newly created list.

```
L = []
for i in range(10):
    name = input("Enter a name:")
    L.append(name)
```

```
def CHAR_LIST(L):
    NEWLIST=[]
    for i in L:
        if len(i)%2!=0:
            mid = len(i)//2
            # get the index of the mid character
            NLIST.append(i[mid])
            # add the character to the list
    print(NEWLIST)
    return NEWLIST
```

## MODIFYING BASED ON LENGTH/FIRST LETTER/LAST LETTER ETC.

### (RANGE TRAVERSAL)

8) Create a List of 10 student names. Pass the list to a function CHAR\_LIST(L). Change the elements whose length is even as character @. Print and return the modified list

```
L = []
for i in range(10):
    name = input("Enter a name:")
    L.append(name)

def CHAR_LIST(L):
    for i in range(len(L)):
        if len(L[i])%2==0:
            L[i] = '@'

    print(L)
    return L
```

9) Create a List of 10 student names. Pass the list to a function CHAR\_LIST(L). Change the elements whose length is odd as character \*. Print and return the modified list

```
L = []
for i in range(10):
    name = input("Enter a name:")
    L.append(name)

def CHAR_LIST(L):
    for i in range(len(L)):
        if len(L[i])%2!=0:
            L[i] = '*'

    print(L)
    return L
```

10) Create a List of 10 student names. Pass the list to a function CHAR\_LIST(L). Change the elements whose first letter is H as False. Print and return the modified list

```
L = []
for i in range(10):
    name = input("Enter a name:")
    L.append(name)

def CHAR_LIST(L):
    for i in range(len(L)):
        if L[i][0]=='H':
            L[i] = False

    print(L)
    return L
```

11) Create a List of 10 student names. Pass the list to a function CHAR\_LIST(L). Change the elements whose last letter is N as #. Print and return the modified list

```
L = []
for i in range(10):
    name = input("Enter a name:")
    L.append(name)

def CHAR_LIST(L):
    for i in range(len(L)):
        if L[i][-1]=='N':
            L[i] = '#'

    print(L)
    return L
```

~~~~~

XII COMPUTER SCIENCE

Towards Getting Centum in CS - See Yes!

13. STACK HANDOUT - PUSH

A) NUMBER-BASED

1) PUSHING FINITE NUMBER OF INTEGERS

Write a function PUSHSTACK() to get 10 integers from user and add to a stack S.

```
S=[]
def PUSHSTACK():
    for i in range range(10):
        n = int(input("Enter your digit:"))
        S.append(n)
```

2) PUSHING SINGLE INTEGER RECEIVED FROM USER

Write a function PUSHSTACK() to get an integer from user and add to a stack S.

```
S=[]
def PUSHSTACK():
    n = int(input("Enter your digit:"))
    S.append(n)
```

3) PUSHING SINGLE INTEGER PASSED TO FUNCTION

Write a function PUSHSTACK() to pass an integer n and add to a stack S.

```
S=[]
```

```
def PUSHSTACK(n):
    S.append(n)
```

4) CREATING A LIST OF INTEGERS FROM USER AND ADDING SPECIFIC ELEMENTS

a) Create a list L of 10 integers from user. Pass the list to a function PUSHSTACK() and add only even digits to a stack S.

```
L = [] #list for adding 10 integers
```

```
for i in range range(10):
    n = int(input("Enter your digit:"))
    L.append(n)
```

```
S = [] #list to create stack
```

```
def PUSHSTACK(L):
    for i in L:
        if i%2==0:
            S.append(i)
```

b) Create a list L of 10 integers from user. Pass the list to a function PUSHSTACK() and add only odd digits to a stack S.

```
L=[] #list for adding 10 integers
```

```
for i in range range(10):
    n = int(input("Enter your digit:"))
    L.append(n)
```

```
S=[] #list to create stack
```

```
def PUSHSTACK(L):
    for i in L:
        if i%2!=0:
            S.append(i)
```

c) Create a list L of 10 integers from user. Pass the list to a function PUSHSTACK() and add digits ending with 7 to a stack.

```
L = []    #list for adding 10 integers
for i in range range(10):
    n = int(input("Enter your digit:"))
    L.append(n)

S = []    #list to create stack

def PUSHSTACK(L):
    for i in L:
        if i%10==7: #if the digit is ending with 7
            S.append(i)
```

B) STRING BASED

1) PUSHING FINITE NUMBER OF STRINGS

Write a function PUSHSTACK() to get 10 student names from user and add to a stack S.

```
S=[]

def PUSHSTACK():
    for i in range range(10):
        name = input("Enter your name:")
        S.append(name)
```

2)PUSHING SINGLE STRING RECEIVED FROM USER

Write a function PUSHSTACK() to get a name from user and add to a stack S.

```
S=[]

def PUSHSTACK():
    name = input("Enter your digit:")
    S.append(name)
```

3)PUSHING SINGLE STRING PASSED TO FUNCTION

Write a function PUSHSTACK() to pass a name and add to a stack S.

```
S=[]

def PUSHSTACK(name):
    S.append(name)
```

4)CREATING A LIST OF STRINGS FROM USER AND ADDING SPECIFIC STRINGS

a) Create a list L of 10 names from user. Pass the list to a function PUSHSTACK() and add only strings with even length to a stack S.

```
L = []    #list for adding 10 strings

for i in range range(10):
    name = input("Enter your digit:")
    L.append(name)

S = []    #list to create stack

def PUSHSTACK(L):
    for i in L:
        if len(i)%2==0:    #only even length
            S.append(i)
```

b)Create a list L of 10 names from user. Pass the list to a function PUSHSTACK() and add only strings with odd length to a stack S.

```
L = []    #list for adding 10 strings

for i in range range(10):
    name = input("Enter your digit:")
    L.append(name)
```

```
S = []    #list to create stack
```

```
def PUSHSTACK(L):  
    for i in L:  
        if len(i)%2!=0:    #only odd  
            S.append(i)
```

c)Create a list L of 10 names from user. Pass the list to a function PUSHSTACK() and add only strings with starting letter is 'A' to a stack S.

```
L = []    #list for adding 10 strings
```

```
for i in range range(10):  
    name = input("Enter your digit:")  
    L.append(name)
```

```
S=[]    #list to create stack
```

```
def PUSHSTACK(L):  
    for i in L:  
        if i[0]=='A' :  
            S.append(i)
```

d)Create a list L of 10 names from user. Pass the list to a function PUSHSTACK() and add only strings with ending with vowel to a stack S.

```
L = []    #list for adding 10 strings
```

```
for i in range range(10):  
    name = input("Enter your digit:")  
    L.append(name)
```

```
S=[]    #list to create stack
```

```
def PUSHSTACK(L):  
    for i in L:  
        if i[-1] in "aeiouAEIOU":  
            S.append(i)
```

C) LIST- BASED

1) ADDING PARTICULAR LIST OF RECORDS FROM THE MAIN LIST

a)Create a list LIBRARY containing 10 records of BOOKS with BOOK ID, NAME, AUTHOR, PRICE as fields. Write a function PUSHSTACK() that receives the list LIBRARY and push only book's name and author whose price is more than 500 to a Stack BOOK500.

```
LIBRARY = []    #main list
```

```
for i in range range(10):  
    id = int(input("Enter ID:"))  
    bn = input("Enter book name:")  
    ba = input("Enter book author name:")  
    bp = float(input("Enter price:"))  
    rec = [id, bn, ba, bp]  
    LIBRARY.append(rec)
```

```
BOOK500 = []    #list to create stack
```

```
def PUSHSTACK(LIBRARY):  
    for rec in LIBRARY:  
        if rec[3]>500:    # books with price > 500  
            brec = [rec[1], rec[2]]    # name,author  
            BOOK500.append(brec)
```

b) Create a list FOREST containing 10 records of trees with TREE ID, NAME, PLACE, HEIGHT as fields. Write a function PUSHSTACK() that receives the list FOREST and push only tree's id and place whose height is more than 100 Feet to a Stack TREE_100.

```
FOREST = []      #main list

for i in range range(10):
    id = int(input("Enter tree ID:"))
    tn = input("Enter tree name:")
    pl = input("Enter tree place:")
    ht = float(input("Enter height:"))
    rec = [id, tn, pl, ht]
    FOREST.append(rec)

TREE_100 = []    #list to create stack

def PUSHSTACK(FOREST):
    for rec in FOREST:
        if rec[3]>100:    #books with price > 500
            trec = [rec[0], rec[2]]    # list with id,place
            TREE_100.append(trec)
```

2) ADDING A WHOLE LIST OF RECORDS

c) Write a function to get 10 records of STUDENTS with STUDENT ID, NAME, PLACE, SCORE as fields and push them to a stack SCHOOL.

```
SCHOOL = []      #stack
def PUSHSTACK():
    for i in range range(10):
        id = int(input("Enter student ID:"))
        tn = input("Enter name:")
```

```
pl = input("Enter place:")
sc = float(input("Enter score:"))
rec = [id, tn, pl, sc]
SCHOOL.append(rec)
```

D) DICTIONARY- BASED

1) ADDING ALL KEYS OF A DICTIONARY TO A STACK

a) Create a dictionary STUDENT of 10 students' name and age {name:age}. Write a function PUSHSTACK() that receives the dictionary STUDENT and pushes all the names alone to a stack S.

```
STUDENT = {}

for i in range(10):
    name = input("Enter student name:")
    STUDENT[name] = int(input(" Student age:"))

S = []    #list to create stack of student names

def PUSHSTACK(STUDENT):
    for k in STUDENT:
        S.append(k)    #appending only keys(names)
```

b) Create a dictionary STUDENT of 10 students' name and age {name:age}. Write a function PUSHSTACK() that receives the dictionary STUDENT and pushes all the names whose age is more than 15 to a stack S.

```
STUDENT = {}

for i in range(10):
    name = input("Enter student name:")
    STUDENT[name] = int(input("Student age:"))
```

S = [] #list to create stack of student names

```
def PUSHSTACK(STUDENT):
    for k in STUDENT:
        if STUDENT[k] > 15:
            S.append(k) #only keys (names)
```

c) Create a dictionary STUDENT of 10 students' name and age {name:age}. Write a function PUSHSTACK() that receives the dictionary STUDENT and pushes all the names whose length is more than 10 to a stack S.

```
STUDENT = {}

for i in range(10):
    name = input("Enter student name:")
    STUDENT[name] = int(input("Enter age:"))
```

S = [] #list to create stack of student names

```
def PUSHSTACK(STUDENT):
    for k in STUDENT:
        if len(k) > 10:
            S.append(k) #only keys (names)
```

d) Create a dictionary STUDENT of 10 students' name and age {name:age}. Write a function PUSHSTACK() that receives the dictionary STUDENT and pushes all the ages that are more than or equal to 12 to a stack S.

```
STUDENT = {}

for i in range(10):
    name = input("Enter student name:")
    STUDENT[name] = int(input("Enter age:"))
```

S=[] #list to create stack of student names

```
def PUSHSTACK(STUDENT):
    for k in STUDENT:
        if STUDENT[k]>=12: #checking values of the
            dictionary
            S.append(STUDENT[k])
```

e) Create a dictionary STUDENT of 10 students' name and age {name:age}. Write a function PUSHSTACK() that receives the dictionary STUDENT and pushes all the names whose age is more than 12 to a stack S.

```
STUDENT = {}

for i in range(10):
    name = input("Enter student name:")
    STUDENT[name] = int(input("Enter age:"))
```

S = [] #list to create stack of student names

```
def PUSHSTACK(STUDENT):
    for k in STUDENT:
        if STUDENT[k]>=12: #checking values
            S.append(k) #only keys (name)
```

11. STACK HANDOUT - POP

1)POP 1 ELEMENT

Write a function POPSTACK() to pop an element from a stack S and print.

```
def POPSTACK(S):  
    if S==[]:  
        print("UNDERFLOW")  
    else:  
        print("Element popped is:",S.pop())
```

2)POP & DISPLAY ALL THE ELEMENTS 1 BY 1

Write a function POPSTACK() to pop all the elements from a stack S and print. Call the function.

If the stack S contains [1,53,2,7,8,9], then the output should be,

9 8 7 2 53 1

```
def POPSTACK(S):  
    if S==[]:  
        print("UNDERFLOW")  
    else:  
        while len(S)!=0:  
            print(S.pop(), end=' ')
```

POPSTACK()

3)POP AND DISPLAY ALL THE ELEMENTS 1 BY 1, IF STACK IS OVER, PRINT 'STACK EMPTY'

Write a function POPSTACK() to pop all the elements from a stack S and print. Call the function.

If the stack S contains [1,53,2,7,8,9], then the output should be,

9 8 7 2 53 1 STACK EMPTY

```
def POPSTACK(S):  
    if S==[]:  
        print("UNDERFLOW")  
    else:  
        while len(S)!=0:  
            print(S.pop(), end=' ')  
    else:  
        print("STACK EMPTY")
```

POPSTACK(S)

PREVIOUS YEAR MODELS

1) Julie has created a dictionary containing names and marks as key value pairs of 6 students. Write a program, with separate user defined functions to perform the following operations:

- Push the keys (name of the student) of the dictionary into a stack, where the corresponding value (marks) is greater than 75.
- Pop and display the content of the stack.

For example:

If the sample content of the dictionary is as follows:

R={"OM":76, "JAI":45, "BOB":89, "ALI":65, "ANU":90, "TOM":82}

The output from the program should be:

TOM ANU BOB OM

#DECLARE GIVEN

```
R={"OM":76, "JAI":45, "BOB":89, "ALI":65,  
"ANU":90, "TOM":82}
```

#PUSH

```
def PUSH(S,N):
```

```
    S.append(N)
```

#POP

```
def POP(S):
```

```
    if S!=[]:
```

```
        return S.pop()
```

```
    else:
```

```
        return None
```

#CREATE STOCK

```
ST=[]
```

#TRAVERSE N PUSH

```
for k in R:
```

```
    if R[k]>=75:
```

```
        PUSH(ST,k)
```

#POP ALL

```
while True:
```

```
    if ST!=[]:
```

```
        print(POP(ST), end=" ")
```

```
    else:
```

```
        break
```

2) Alam has a list containing 10 integers. You need to help him create a program with separate user defined functions to perform the following operations based on this list.

- Traverse the content of the list and push the even numbers into a stack.

- Pop and display the content of the stack.

For Example:

If the sample Content of the list is as follows:

```
N=[12, 13, 34, 56, 21, 79, 98, 22, 35, 38]
```

Sample Output of the code should be:

```
38 22 98 56 34 12
```

Answer

```
N=[12, 13, 34, 56, 21, 79, 98, 22, 35, 38]
```

```
def PUSH(S,N):
```

```
    S.append(N)
```

```
def POP(S):
```

```
    if S!=[]:
```

```
        return S.pop()
```

```
    else:
```

```
        return None
```

```
ST=[]
```

```
for k in N:
```

```
    if k%2==0:
```

```
        PUSH(ST,k)
```

```
while True:
```

```
    if ST!=[]:
```

```
        print(POP(ST),end=" ")
```

```
    else:
```

```
        break
```

3) Rahul has created a tuple containing names of his friends. Write a program, with separate user defined functions to perform the following operations:

- Push the names that are odd in length dictionary into a stack
- Pop and display the content of the stack.

For example:

If the sample content of the dictionary is as follows:

Names = ('Ranga', 'Seetha', 'John', 'Sairam', 'Sagar')

The output from the program should be:

Ranga Sagar

Answer:

```
names = ('Ranga', 'Seetha', 'John', 'Sairam', 'Sagar')
```

```
def PUSH(S,N):
```

```
    S.append(N)
```

```
def POP(S):
```

```
    if S!=[]:
```

```
        return S.pop()
```

```
    else:
```

```
        return None
```

```
ST=[]
```

```
for i in names:
```

```
    if len(i)%2!=0:
```

```
        PUSH(ST,i)
```

```
while True:
```

```
    if ST!=[]:
```

```
        print(POP(ST),end=" ")
```

```
    else:
```

```
        break
```

4) Suman has a list containing 10 integers. You need to help him create a program with separate user defined functions to perform the following operations based on this list.

- Traverse the content of the list and push the numbers ending with 3 into a stack.
- Pop and display the content of the stack.

For Example:

If the sample Content of the list is as follows:

N=[12, 13, 33, 56, 21, 73, 98, 35, 3]

Sample Output of the code should be:

13

33

73

3

Answer:

N=[12, 13, 33, 56, 21, 73, 98, 35, 3]

```
def PUSH(S,N):
```

```
    S.append(N)
```

```
def POP(S):
```

```
    if S!=[]:
```

```
        return S.pop()
```

```
    else:
```

```
        return None
```

```
ST=[]
```

```
for k in N:
```

```
    if k%10==3:
```

```
        PUSH(ST,k)
```

```
while True:
```

```
    if ST!=[]:
```

```
        print(POP(ST))
```

```
    else:
```

```
        break
```

~~~~~

*"Success isn't a result of spontaneous combustion. You must set yourself on fire." - Arnold H. Glasow*

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS - See Yes!*

### 14. SQL MADE EASY

#### DDL (DATA DEFINITION LANGUAGE) - CAD

1. **CREATE TABLE** *table\_name*
2. **ALTER TABLE** *table\_name*
3. **DROP TABLE** *table\_name*

#### DML (DATA MANIPULATION LANGUAGE) - SUDI

4. **SELECT** *column name(s)* **FROM** *table\_name*
5. **UPDATE** *table\_name* **SET**
6. **DELETE FROM** *table\_name*
7. **INSERT INTO** *table\_name* **VALUES**

#### SAMPLE TABLE

FACULTY

| <b>F_ID</b> | <b>Fname</b> | <b>Lname</b> | <b>Hire_date</b> | <b>Salary</b> |
|-------------|--------------|--------------|------------------|---------------|
| 102         | Amit         | Mishra       | 12-10-1998       | 12000         |
| 103         | Nitin        | Vyas         | 24-12-1994       | 8000          |
| 104         | Rakshit      | Soni         | 18-5-2001        | 14000         |
| 105         | Rashmi       | Malhotra     | 11-9-2004        | 11000         |
| 106         | Sulekha      | Srivastava   | 5-6-2006         | 10000         |

COURSES

| <b>C_ID</b> | <b>F_ID</b> | <b>Cname</b>      | <b>Fees</b> |
|-------------|-------------|-------------------|-------------|
| C21         | 102         | Grid Computing    | 40000       |
| C22         | 106         | System Design     | 16000       |
| C23         | 104         | Computer Security | 8000        |
| C24         | 106         | Human Biology     | 15000       |
| C25         | 102         | Computer Network  | 20000       |
| C26         | 105         | Visual Basic      | 6000        |

## DDL COMMANDS

### 1. CREATE TABLE COMMAND

#### i) CREATING NEW TABLE WITH NEW ATTRIBUTES

➤ CREATE TABLE FACULTY

```
( F_ID      INTEGER PRIMARY KEY,  
  Fname     CHAR(20),  
  Lname     CHAR(20),  
  Hire_date DATE,  
  Salary    DECIMAL);
```

- Primary key is a constraint applied on the attribute F\_ID. We can use any no. of constraints with any attribute like NOT NULL, UNIQUE, PRIMARY KEY, CHECK, DEFAULT etc.
- Do not give comma in between attribute name, data type and constraints.

FACULTY

| F_ID | Fname   | Lname      | Hire_date  | Salary |
|------|---------|------------|------------|--------|
| 102  | Amit    | Mishra     | 12-10-1998 | 12000  |
| 103  | Nitin   | Vyas       | 24-12-1994 | 8000   |
| 104  | Rakshit | Soni       | 18-5-2001  | 14000  |
| 105  | Rashmi  | Malhotra   | 11-9-2004  | 11000  |
| 106  | Sulekha | Srivastava | 5-6-2006   | 10000  |

COURSES

| C_ID | F_ID | Cname             | Fees  |
|------|------|-------------------|-------|
| C21  | 102  | Grid Computing    | 40000 |
| C22  | 106  | System Design     | 16000 |
| C23  | 104  | Computer Security | 8000  |
| C24  | 106  | Human Biology     | 15000 |
| C25  | 102  | Computer Network  | 20000 |
| C26  | 105  | Visual Basic      | 6000  |

#### ii) CREATING NEW TABLE WITH a) NOT NULL b) UNIQUE CONSTRAINT

➤ CREATE TABLE FACULTY

```
( F_ID      INTEGER PRIMARY KEY,  
  Fname     CHAR(20) NOT NULL UNIQUE,  
  Lname     CHAR(20) NOT NULL,  
  Hire_date DATE UNIQUE,  
  Salary    DECIMAL);
```

## 2. ALTER TABLE COMMAND

### a) To add a new column/attribute

- ALTER TABLE FACULTY  
ADD( Places CHAR(20));
- ALTER TABLE COURSES  
ADD(Duration INT);
- ALTER TABLE COURSES  
MODIFY (Duration CHAR(4));

### b) To remove a column/attribute

- ALTER TABLE FACULTY  
DROP COLUMN PLACES;
- ALTER TABLE COURSES  
DROP COLUMN DURATION;

### c) To Modify a column/attribute

- ALTER TABLE FACULTY  
MODIFY(Lname char(30));

### d) To add PRIMARY KEY

- ALTER TABLE Persons  
ADD PRIMARY KEY (ID);
- ALTER TABLE FACULTY  
ADD PRIMARY KEY (F\_ID);

### d) To remove PRIMARY KEY

- ALTER TABLE FACULTY  
DROP PRIMARY KEY;
- ALTER TABLE COURSES  
DROP PRIMARY KEY;

## 3. DROP TABLE COMMAND

- To completely delete the table from memory.
- DROP TABLE FACULTY;
- DROP TABLE COURSES;

## DML COMMANDS

## 4. INSERT INTO COMMAND

### i) To insert new values fresh (to all the columns)

- INSERT INTO FACULTY  
VALUES(107, 'Ranga', 'Rajan', '2-3-1998', 15000.00);
- INSERT INTO FACULTY  
VALUES(108, 'Seetha', 'Raman', '4-5-1997', 16000.00);

## ii) To insert new values fresh (to particular columns)

- INSERT INTO FACULTY (F\_ID, Fname, Lname)  
VALUES(109, 'Rahul', 'Kumar');
- INSERT INTO FACULTY (F\_ID, Fname, Salary)  
VALUES(110, 'Johan', '20000.00');

## 5. SELECT COMMAND (to display anything from the table)

### i) To display all rows present in the table

- SELECT \* FROM FACULTY;

### ii) To display all columns/particular columns without duplicates (SELECT DISTINCT)

- SELECT DISTINCT \* FROM FACULTY;
- SELECT DISTINCT Fname FROM FACULTY;
- If there are more null records present, only 1 null record will be displayed when we use DISTINCT. We can use only one DISTINCT in a query.

### iii) Using RELATIONAL(<, >, <=, >=, !=, <>, /LOGICAL OPERATORS (AND/OR/NOT)

- SELECT \* FROM FACULTY WHERE Salary >=10000;
- SELECT \* FROM FACULTY WHERE Salary >=9000 AND Salary <=12000;
- SELECT \* FROM FACULTY WHERE Salary >=12000 OR Fname LIKE "R%";

#### Using NOT operation

- SELECT \* FROM FACULTY WHERE (NOT Salary >=12000 OR Fname LIKE "R%");
- SELECT \* FROM FACULTY WHERE (NOT Fname LIKE "R%");
- SELECT \* FROM FACULTY WHERE (NOT F\_ID=106);
- SELECT \* FROM FACULTY WHERE (NOT Grade='E1');
- SELECT \* FROM FACULTY WHERE NOT Fname = 'Amit';

➤ SELECT \* FROM FACULTY WHERE Saslary <> 12000;

➤ SELECT \* FROM FACULTY WHERE Fname <> 'Amit';

➤ SELECT \* FROM FACULTY WHERE (Salary >=12000 AND Fname <> 'Amit');

#### **iv) SCALAR EXPRESSIONS USING ARITHMETIC OPERATORS (ONLY DURING DISPLAY)**

Display Course name and 'NEW FEES' (calculated by adding 3% increase in the current fees.

➤ SELECT CNAME, FEES+FEES\*0.03 AS 'NEW FEES' FROM COURSE

Display First name of the faculty and 'NEW SALARY' (calculated by deducting 2% in the current salary for members between ID 103 and 105.

➤ SELECT FNAME, SALARY-SALARY\*0.02 AS 'NEW SALARY' FROM FACULTY  
WHERE FID>=103 AND FID<=105

➤ SELECT FNAME, SALARY-SALARY\*0.02 AS 'NEW SALARY' FROM FACULTY  
WHERE FID BETWEEN 103 AND 105

Display First name of the Faculty and 'NEW SALARY' (calculated by adding 100 Rs extra with the current salary).

➤ SELECT FNAME, SALARY+100AS 'NEW SALARY' FROM FACULTY  
WHERE FID BETWEEN 103 AND 105

#### **v) To display tuples based on a range (BETWEEN, NOT BETWEEN)**

➤ SELECT \* FROM FACULTY WHERE Salary BETWEEN 8000 AND 10000;

➤ SELECT \* FROM FACULTY WHERE Salary NOT BETWEEN 8000 AND 10000;

➤ SELECT \* FROM FACULTY WHERE Salary BETWEEN 8000 AND 10000 AND Fname <> 'Amit';

(you can have as many AND/OR/NOT in a query as you want)

#### **vi) To display tuples based on a list (IN, NOT IN)**

➤ SELECT \* FROM FACULTY WHERE Fname IN('Amit', 'Nitin', 'Rakshit');

➤ SELECT \* FROM FACULTY WHERE Fname NOT IN('Amit', 'Nitin', 'Rakshit');

- SELECT \* FROM FACULTY WHERE F\_ID IN(102,103);
- SELECT \* FROM FACULTY WHERE F\_ID NOT IN(102,103);

#### **vii) To display tuples based Pattern Matches (LIKE “%”, “\_\_\_”)**

- SELECT \* FROM FACULTY WHERE Fname LIKE “R%”; (ex: Raju, Rangan, Rahul)
- SELECT \* FROM FACULTY WHERE Fname LIKE “Ran%”; (ex: Rani, Rangan,)
- SELECT \* FROM FACULTY WHERE Fname LIKE “%an%”; (ex: Kanthan, Rangan, Johan)
- SELECT \* FROM FACULTY WHERE Fname LIKE “\_\_\_”; (ex: Rani, John, Arun, Sugu – exactly 4)
- SELECT \* FROM FACULTY WHERE Fname LIKE “\_\_\_%”; (at least 4 characters or more but not less than that)

#### **viii) Use of IS NULL/ IS NOT NULL**

- SELECT \* FROM FACULTY WHERE Fname IS NULL; (to display NULL VALUES)
- SELECT \* FROM FACULTY WHERE Fname IS NOT NULL;

(IMPORTANT: Don’t use IS with BETWEEN/LIKE etc. It is wrong.)

#### **ix) SORTING THE TUPLES (ORDER BY)**

##### **a) ASCENDING**

- SELECT \* FROM FACULTY ORDER BY Fname;
- SELECT \* FROM FACULTY ORDER BY Fname ASC; (both are same)
- SELECT \* FROM FACULTY WHERE Salary>=12000 ORDER BY Fname ASC;

##### **b) DESCENDING**

- SELECT \* FROM FACULTY ORDER BY Fname DESC;
- SELECT \* FROM FACULTY WHERE Salary>=12000 ORDER BY Fname DESC;

##### **b)ASCENDING AND DESCENDING**



- SELECT \* FROM FACULTY ORDER BY Fname ASC, F\_ID DESC;
- SELECT \* FROM FACULTY ORDER BY Fname ASC, Salary DESC;

## xi) GROUPING THE COLUMNS WITH AGGREGATE FUNCTIONS (count avg, sum, max, min)

### a) GROUP BY

**Keywords clue in question:** each/wise

Table: FLIGHT

| FL_NO | DEPARTURE | ARRIVAL    | NO_FLIGHTS | NOOFSTOPS |
|-------|-----------|------------|------------|-----------|
| IC301 | MUMBAI    | DELHI      | 8          | 0         |
| IC799 | BANGALORE | DELHI      | 2          | 1         |
| MC101 | INDORE    | MUMBAI     | 3          | 0         |
| IC302 | DELHI     | MUMBAI     | 8          | 0         |
| AM812 | KANPUR    | BANGALORE  | 3          | 1         |
| IC899 | MUMBAI    | KOCHI      | 1          | 4         |
| AM501 | DELHI     | TRIVANDRUM | 1          | 5         |
| MU499 | MUMBAI    | MADRAS     | 3          | 3         |
| IC701 | DELHI     | AHMEDABAD  | 4          | 0         |

Table: FARE

| FL_NO | AIRLINES        | FARE  | TAX |
|-------|-----------------|-------|-----|
| 1C701 | Indian Airlines | 6500  | 10  |
| MU499 | Sahara          | 9400  | 5   |
| AM501 | Jet Airways     | 13450 | 8   |
| IC899 | Indian Airlines | 8300  | 4   |
| 1C302 | Indian Airlines | 4300  | 9   |
| 1C799 | Indian Airlines | 10500 | 10  |
| MC101 | Deccan Airlines | 3500  | 4   |

### Displaying columns without duplicates using GROUP BY

- SELECT DEPARTURE FROM FLIGHT GROUP BY DEPARTURE;
- SELECT AIRLINES FROM FARE GROUP BY AIRLINES; (both just avoids duplicates)

### b) COUNT WITHOUT GROUP BY

#### COUNT(\*)

- SELECT COUNT(\*) FROM FLIGHT; (displays no. of flights including duplicates, nulls)
- SELECT COUNT(\*) FROM FARE;

#### COUNT(DISTINCT col name)

- SELECT COUNT(DISTINCT DEPARTURE) FROM FLIGHT;  
(displays no. of departures(5), avoids duplicates, includes null)

- SELECT COUNT(DISTINCT AIRLINES) FROM FLIGHT;  
(displays no. of airlines(4), avoids duplicates, includes null)

### **COUNT(ALL col name)**

- SELECT COUNT(ALL DEPARTURE) FROM FLIGHT;  
(displays no. of departures(5), includes duplicates, does not include null)
- SELECT COUNT(ALL AIRLINES) FROM FLIGHT;  
(displays no. of airlines(5), includes duplicates, does not include null)

## **c) COUNT WITH GROUP BY**

Display no. of flights departure place wise

- SELECT DEPARTURE, COUNT(DEPARTURE) FROM FLIGHT GROUP BY DEPARTURE;  

|           |   |
|-----------|---|
| MUMBAI    | 3 |
| BANGALORE | 1 |
| INDORE    | 1 |
| DELHI     | 3 |
| KANPUR    | 1 |

Display no. of flights for each airlines

- SELECT AIRLINES, COUNT(AIRLINES) FROM FARE GROUP BY AIRLINES;  

|                 |   |
|-----------------|---|
| Indian Airlines | 4 |
| Sahara          | 1 |
| Jet Airways     | 1 |
| Deccan Airlines | 1 |

## **d) COUNT WITH WHERE CONDITION AND GROUP BY**

- SELECT DEPARTURE, COUNT(DEPARTURE) FROM FLIGHT WHERE NO\_FLIGHTS > 5  
GROUP BY DEPARTURE;  

|        |   |
|--------|---|
| MUMBAI | 1 |
| DELHI  | 1 |
- SELECT AIRLINES, COUNT(AIRLINES) FROM FARE WHERE FARE > 5000  
GROUP BY AIRLINES;  

|                 |   |
|-----------------|---|
| Indian Airlines | 3 |
| Sahara          | 1 |
| Jet Airways     | 1 |

## e) SUM

Display total no. of flights for **each** departure place

- SELECT DEPARTURE, SUM(NO\_FLIGHTS) FROM FLIGHT GROUP BY DEPARTURE;

|           |    |
|-----------|----|
| MUMBAI    | 12 |
| BANGALORE | 2  |
| INDORE    | 3  |
| DELHI     | 13 |
| KANPUR    | 3  |

Display total tax details airlines- wise with fare above 5000

- SELECT AIRLINES, SUM(TAX) FROM FARE WHERE FARE > 5000  
GROUP BY AIRLINES;

|                 |    |
|-----------------|----|
| Indian Airlines | 24 |
| Sahara          | 5  |
| Jet Airways     | 8  |

## e) AVG

Display average no. of flights for **each** departure place

- SELECT DEPARTURE, AVG(NO\_FLIGHTS) FROM FLIGHT GROUP BY DEPARTURE;

|           |     |
|-----------|-----|
| MUMBAI    | 4   |
| BANGALORE | 2   |
| INDORE    | 3   |
| DELHI     | 4.3 |
| KANPUR    | 3   |

Display average tax airlines- wise with fare above 5000

- SELECT AIRLINES, AVG(TAX) FROM FARE WHERE FARE > 5000  
GROUP BY AIRLINES;

|                 |   |
|-----------------|---|
| Indian Airlines | 6 |
| Sahara          | 5 |
| Jet Airways     | 8 |

## **f) SIMPLE MIN/MAX**

- SELECT MAX(NO\_FLIGHTS), DEPARTURE FROM FLIGHT;  
  
8     MUMBAI  
8     DELHI
- SELECT MIN(NO\_FLIGHTS), ARRIVAL FROM FLIGHT;  
  
1     KOCHI  
1     TRIVANDRUM
- SELECT MAX(TAX), AIRLINES FROM FARE WHERE FARE >10000;  
  
10     INDIAN AIRLINES
- SELECT MIN(TAX), AIRLINES FROM FARE WHERE FARE >10000;  
  
8     JET AIRWAYS
- SELECT MAX(HIRE\_DATE) FROM FACULTY  
  
5-6-2006
- SELECT MIN(HIRE\_DATE) FROM FACULTY  
  
24-12-1994

## **g) MIN/MAX WITH GROUP BY**

Display Maximum number of flights for each arrival place

SELECT ARRIVAL, MAX(NO\_FLIGHTS) FROM FLIGHT GROUP BY ARRIVAL;

Display Minimum tax collected airlines-wise / for each airlines

SELECT AIRLINES, MIN(TAX) FROM FARE GROUP BY AIRLINES;

## h) GROUP BY WITH HAVING

### To apply conditions over aggregate functions after grouping

Display total no. of flights for each departure whose no. of flights is more than 2

- SELECT DEPARTURE, COUNT(\*) FROM FLIGHT GROUP BY DEPARTURE HAVING COUNT(\*)>2;

|        |   |
|--------|---|
| MUMBAI | 3 |
|--------|---|

|       |   |
|-------|---|
| DELHI | 3 |
|-------|---|

Display total tax for each airlines that have average tax more than or equal to 6%

SELECT AIRLINES, SUM(TAX) FROM FARE GROUP BY AIRLINES HAVING AVG(TAX)>=6;

|             |   |
|-------------|---|
| Jet Airways | 8 |
|-------------|---|

|                 |    |
|-----------------|----|
| Indian Airlines | 33 |
|-----------------|----|

## xii) FORMATTING THE OUTPUT (SIMILAR TO SCALAR EXPRESSIONS)

- SELECT DEPARTURE, NO\_FLIGHTS+3 AS 'NEW FLIGHTS FOR DELHI' FROM FLIGHT WHERE ARRIVAL = 'DELHI';

| DEPARTURE | NEW FLIGHTS FOR DELHI |
|-----------|-----------------------|
|-----------|-----------------------|

|        |    |
|--------|----|
| MUMBAI | 11 |
|--------|----|

|           |   |
|-----------|---|
| BANGALORE | 5 |
|-----------|---|

- SELECT FL\_NO, FARE+2000 AS 'NEW FARE' FROM FARE WHERE AIRLINES = 'SAHARA';

| FL_NO | NEW FARE |
|-------|----------|
|-------|----------|

|       |       |
|-------|-------|
| MU499 | 10400 |
|-------|-------|

SELECT 'Rs.', FARE+2000 AS 'NEW FARE', '/- only for IA' FROM FARE WHERE AIRLINES = 'INDIAN AIRLINES';

|              |                |
|--------------|----------------|
| Rs. NEW FARE | /- only for IA |
|--------------|----------------|

|          |                |
|----------|----------------|
| Rs. 8500 | /- only for IA |
|----------|----------------|

|           |                |
|-----------|----------------|
| Rs. 10300 | /- only for IA |
|-----------|----------------|

|          |                |
|----------|----------------|
| Rs. 6300 | /- only for IA |
|----------|----------------|

|           |                |
|-----------|----------------|
| Rs. 12500 | /- only for IA |
|-----------|----------------|

### xiii) JOINS - TAKING VALUES FROM 2 TABLES

- SELECT DEPARTURE, AIRLINES FROM FLIGHT, FARE WHERE FLIGHT.FL\_NO=FARE.FL\_NO AND FARE>10000;

DELHI          JET AIRWAYS  
BANGALORE INDIAN AIRLINES

- SELECT f1.DEPARTURE, f2.AIRLINES FROM FLIGHT f1, FARE f2 WHERE f1.FL\_NO=f2.FL\_NO AND f2.FARE>10000; (f1, f2 are alias names)

DELHI          JET AIRWAYS  
BANGALORE INDIAN AIRLINES

- SELECT AIRLINES, SUM(FARE), COUNT(DEPARTURE) FROM FLIGHT, FARE WHERE FLIGHT.FL\_NO=FARE.FL\_NO AND FARE<7000 GROUP BY AIRLINES;

INDIAN AIRLINES 10000 2  
DECCAN AIRLINES 3500 1

### NATURAL JOIN

(Automatically, the duplicate column will be removed and all the rows that have same foreign key column will be displayed. All the columns will be displayed)

#### EMPLOYEE TABLE

| EmployeeID | FirstName | LastName | DeptID |
|------------|-----------|----------|--------|
| E62549     | John      | Doe      | D1001  |
| E82743     | Priya     | Sharma   | D3002  |
| E58461     | Raj       | Kumar    | D1002  |
| E95462     | Ravi      |          | D1001  |
| E25947     | Shreya    | P        | D3000  |
| E42650     | Jane      | Scott    | D3001  |

#### DEPARTMENT TABLE

| DeptID | DeptName   | Location  |
|--------|------------|-----------|
| D1001  | Technology | Bangalore |
| D1002  | Technology | Hyderabad |
| D3001  | Sales      | Gurugram  |
| D3002  | Operations | Hyderabad |
| D4002  | Finance    | Mumbai    |

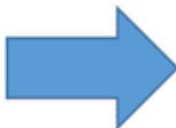
SELECT \* FROM EMPLOYEE NATURAL JOIN DEPARTMENT

OUTPUT:

| EmployeeID | FirstName | LastName | DeptID | DeptName   | Location  |
|------------|-----------|----------|--------|------------|-----------|
| E62549     | John      | Doe      | D1001  | Technology | Bangalore |
| E82743     | Priya     | Sharma   | D3002  | Operations | Hyderabad |
| E58461     | Raj       | Kumar    | D1002  | Technology | Hyderabad |
| E95462     | Ravi      |          | D1001  | Technology | Bangalore |
| E42650     | Jane      | Scott    | D3001  | Sales      | Gurugram  |

#### xiv) CARTESIAN PRODUCT OR CROSS JOIN

- SELECT \* FROM FLIGHT, FARE;  
(gives combination each FLIGHT tuple with all the tuples in FARE);
- SELECT \* FROM FARE, FLIGHT;  
(gives combination each FARE tuple with all the tuples in FLIGHT);
- SELECT \* FROM COLOR, SIZE

| Tables                                                                                                                                   |             | Query Result |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
|------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------|--------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|-----|-------|------|-------|-----|--------|------|--------|-----|-------|------|-------|-----|-------------|------|-------------|
| <table><tr><th>Color</th></tr><tr><td>Red</td></tr><tr><td>Blue</td></tr></table>                                                        | Color       | Red          | Blue   |  | <table><tr><th>Color</th><th>Size</th></tr><tr><td>Red</td><td>Small</td></tr><tr><td>Blue</td><td>Small</td></tr><tr><td>Red</td><td>Medium</td></tr><tr><td>Blue</td><td>Medium</td></tr><tr><td>Red</td><td>Large</td></tr><tr><td>Blue</td><td>Large</td></tr><tr><td>Red</td><td>Extra Large</td></tr><tr><td>Blue</td><td>Extra Large</td></tr></table> | Color | Size | Red | Small | Blue | Small | Red | Medium | Blue | Medium | Red | Large | Blue | Large | Red | Extra Large | Blue | Extra Large |
| Color                                                                                                                                    |             |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Red                                                                                                                                      |             |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Blue                                                                                                                                     |             |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Color                                                                                                                                    | Size        |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Red                                                                                                                                      | Small       |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Blue                                                                                                                                     | Small       |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Red                                                                                                                                      | Medium      |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Blue                                                                                                                                     | Medium      |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Red                                                                                                                                      | Large       |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Blue                                                                                                                                     | Large       |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Red                                                                                                                                      | Extra Large |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Blue                                                                                                                                     | Extra Large |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| <table><tr><th>Size</th></tr><tr><td>Small</td></tr><tr><td>Medium</td></tr><tr><td>Large</td></tr><tr><td>Extra Large</td></tr></table> | Size        | Small        | Medium | Large                                                                               | Extra Large                                                                                                                                                                                                                                                                                                                                                   |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Size                                                                                                                                     |             |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Small                                                                                                                                    |             |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Medium                                                                                                                                   |             |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Large                                                                                                                                    |             |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |
| Extra Large                                                                                                                              |             |              |        |                                                                                     |                                                                                                                                                                                                                                                                                                                                                               |       |      |     |       |      |       |     |        |      |        |     |       |      |       |     |             |      |             |

## SELECT COMMAND FULL SYNTAX:

**SELECT** column name(s)

**FROM** <table name>

**WHERE** <criteria/predicate>

**GROUP BY** <column name(s)>

**HAVING** <search condition>

**ORDER BY** col\_name;

## 6. UPDATE COMMAND (to make permanent change on the table)

Increase 10 with number of flights where the departure is MUMBAI

- UPDATE FLIGHT SET NO\_FLIGHTS=NO\_FLIGHTS+10 WHERE DEPARTURE = 'MUMBAI';

Increase 3 % in the tax for all the INDIAN AIRLINES

- UPDATE FARE SET TAX=TAX+(TAX\*0.03) WHERE AIRLINES = 'INDIAN AIRLINES';

Decrease 2 % in the tax for records with fare is less than 7000

- UPDATE FARE SET TAX=TAX-(TAX\*0.02) WHERE FARE < 7000;

To modify all tax as 7

UPDATE FARE SET TAX=7;

- After UPDATE command, give table name
- Then don't forget to use SET keyword
- Finally WHERE Clause (if condition is specified)

## 7. DELETE FROM COMMAND

- DELETE FROM FLIGHT; (deletes all tuples/records in FLIGHT. It doesn't drop the table.)
- DELETE FROM FARE WHERE AIRLINES = 'INDIAN AIRLINES';
- DELETE FROM FACULTY WHERE FNAME LIKE "%N%"



## 8. SIMPLE SQL COMMANDS

### **To create database**

```
CREATE DATABASE SCHOOL;  
CREATE DATABASE HOSPITAL;
```

### **To use the database**

```
USE SCHOOL;  
USE HOSPITAL;
```

### **To list all the databases in the server**

```
SHOW DATABASES;
```

### **To list all the tables in a database**

```
USE SCHOOL;  
  
SHOW TABLES;
```

```
USE HOSPITAL;  
  
SHOW TABLES;
```

### **To view the structure of a table in a database**

```
USE SCHOOL;  
  
DESC STUDENT;
```

```
USE HOSPITAL  
  
DESC PATIENT;
```

~~~~~

KEY-BASED QUESTION MODELS

1) Consider the table, MOVIEDETAILS given below: (CBSE SP)

Table: MOVIEDETAILS

MOVIEID	TITLE	LANGUAGE	RATING	PLATFORM
M001	Minari	Korean	5	Netflix
M004	MGR Magan	Tamil	4	Hotstar
M010	Kaagaz	Hindi	3	Zee5
M011	Harry Potter and the Chamber of Secrets	English	4	Prime Video
M015	Uri	Hindi	5	Zee5
M020	Avengers: Endgame	English	4	Hotstar

a) Identify the degree and cardinality of the table.

b) Which field should be made the primary key? Justify your answer.

a) Degree: 5 Cardinality: 6

b) MOVIEID should be made the primary key as it uniquely identifies each record of the table.

2) (CBSE SP)

(a) Identify the candidate key(s) from the table MOVIEDETAILS.

(b) Consider the table SCHEDULE given below:

Table: SCHEDULE

SLOTID	MOVIEID	TIMESLOT
S001	M010	10 AM to 12 PM
S002	M020	2 PM to 5 PM
S003	M010	6 PM to 8 PM
S004	M011	9 PM to 11 PM

Which field will be considered as the foreign key if the tables MOVIEDETAILS and SCHEDULE are related in a database?

(a) MOVIEID and TITLE

(b) MOVIEID

3) Give suitable example of a table with sample data and illustrate primary and candidate keys in it.

Candidate Key: It is a set of all attributes that uniquely identifies records in a table. Each table may have one or more candidate keys.

Table : Student

AdmNo	RollNo	Name	Class	Marks
2715	1	Rame	12	90
2816	2	Shyeam	11	95
2404	3	Ajay	10	92
2917	4	Tarun	12	94

e.g. In Student table, AdmNo and RollNo both can identify records uniquely. So, both are candidate key.

Primary Key: It is a set of one or more attributes that can uniquely identify each tuple of a relation. A relation can have only one primary key.

e.g. In Student table, AdmNo of all students are different. So, we have created AdmNo as primary key.

4) Anita has created a table “Players” to store the details of players who play in her sports academy . She has planned to create the following table with columns : PlayerId, PlayerName , Game, Type , AadharNo

Table : Players

PlayerId	PlayerName	Game	Type	AadharNo
P01	Becker	Tennis	Indoor	333444657
P02	Robin	Tennis	Indoor	192900877
P03	Sunetra	Football	Outdoor	214567432
P04	Rakhi	Cricket	Outdoor	111231896

(i) Which column can she make the primary key?

The PlayerId column stores unique and non-blank values , hence it can serve as primary key.

(ii) Which column(s) can act as candidate key?

All the field combinations that can serve as primary key for unique identification of records in a table are called candidate keys. Here PlayerId and AadharNo can serve as primary key. So, these fields can act as candidate keys.

(iii) Which column is the foreign key in the table?

The concept of foreign key is relevant only when there are multiple tables.

(iv) What is the degree of the table?

Degree of a table is the total number of columns, i.e. 5

v) What will be the cardinality of the table, if two columns are added to the table?

The total number of rows in a table is its cardinality, i.e. 4. Adding columns to table increases its degree not cardinality

5) Explain the use of ‘Foreign Key’ in a Relational Database Management System. Give example to support (2022-23).

Ans: A foreign key is used to set or represent a relationship between two relations (or tables) in a database. Its value is derived from the primary key attribute of another relation. For example: In the tables TRAINER and COURSE given below, TID is primary key in TRAINER table but foreign key in COURSE table.

TRAINER

TID	TNAME	CITY	HIREDATE	SALARY
101	Ritu	Nagpur	1998-10-15	56700
102	Navin	Goa	1994-12-24	80000
103	Murugan	Chandigarh	2001-12-21	82000
104	Jyothi	Guwahati	2002-12-25	68000
105	Chanu	Mumbai	1996-01-12	95000
106	Arbaaz	Delhi	2001-12-12	69000

COURSE

CID	CNAME	FEES	STARTDATE	TID
C201	Deepa	12000	2018-07-02	101
C202	Sindhu	15000	2018-07-15	103
C203	Neeraj	10000	2018-10-01	102
C204	Prakash	9000	2018-09-15	104
C205	Nikhat	20000	2018-08-01	101

Consider the following tables **GAMES** and **PLAYER** and answer (b) and (c) parts of this question:

Table: GAMES					
GCode	GameName	Type	Number	Prize Money	Schedule Date
101	Carom Board	Indoor	2	5000	23-Jan-2004
102	Badminton	Outdoor	2	12000	12-Dec-2003
103	Table Tennis	Indoor	4	8000	14-Feb-2004
105	Chess	Indoor	2	9000	01-Jan-2004
108	Lawn Tennis	Outdoor	4	25000	19-Mar-2004

Table: PLAYER		
PCode	Name	Gcode
1	Nabi Ahmad	101
2	Ravi Sahai	108
3	Jatin	101
4	Nazneen	103

Write SQL commands for the flowing statements:

- (i) To display the name of all GAMES with their GCodes
- (ii) To display details of those GAMES which are having PrizeMoney more than 7000.
- (iii) To display the content of the GAMES table in ascending order of ScheduleDate.
- (iv) To display sum of PrizeMoney for each Type of GAMES

(c) Give the output of the following SQL queries:

- (i) SELECT COUNT(DISTINCT Number) FROM GAMES;
- (ii) SELECT MAX(ScheduleDate),MIN(ScheduleDate) FROM GAMES;
- (ii) SELECT Name, GameName FROM GAMES G, PLAYER PWHERE G.Gcode=P.Gcode AND G.PrizeMoney>10000;
- (iv) SELECT DISTINCT Gcode FROM PLAYER;

Consider the following tables **STOCK** and **DEALERS** and answer (b1) & (b2) parts of this question:

Table : STOCK					
Itcode	Itname	Dcode	Qty	unitpr	Stkdate
444	Drawing Copy	101	110	21	31-July-2010
445	Sharpener Camlin	102	235	3	01-Aug-2010
450	Eraser Natraj	101	40	2	17-Aug-2010
452	Gel Pen Montex	103	50	5	30-Dec-2009
457	Geometry Box	101	35	45	15-Nov-2009
467	Parker Premium	102	60	205	27-Oct-2009
469	Office File	103	32	25	13-Sep-2010

Table : DEALERS	
Dcode	Dname
101	Vikash Stationers
102	Bharat Drawing Emporium
103	Banaras Books Corporation

Write SQL commands for the following statements :

- (i) to display details of all items in the stock table in descending order of Stkdate.
- (ii) to increase unitpr (unit price) by 20% in the stock table for the items which has been purchased before 2010.
- (iii) to display the details of those items whose Dcode (Dealer Code) is 102 or Qty (Quantity) is more than 50 from stock.
- (iv) to display minimum unit price of items for each dealer individually as per Dealer Code from the table stock.

Give the output of the following SQL queries :

- (i) SELECT COUNT(DISTINCT Dcode)FROM STOCK;
- (ii) SELECT Qty*unitpr FROM STOCK WHERE Itcode=457;
- (iii) SELECT Itname, Dname FROM STOCK S, DEALERS D WHERE S.Dcode=D.Dcode AND Itcode=450;
- (iv) SELECT MAX(Stkdate) FROM STOCK;

HAPPY SQL! JAI SAIRAM!

XII COMPUTER SCIENCE

Towards Getting Centum in CS - See Yes!

15. CONNECTIVITY HANDOUT

1) A resultset is extracted from the database using the cursor object (that has been already created) by giving the following statement.

Mydata=cursor.fetchone()

(a) How many records will be returned by fetchone() method?

(b) What will be the datatype of Mydata object after the given command is executed?

Ans: (a) One record (b) tuple

2) Name the module used for connecting MySQL with Python. Write statement to import it.

import mysql.connector

3) Write the code to create a table Product in database Inventory with following fields :

Fields	Datatype
PID	varechar(5)
PName	char(30)
Price	float
Rank	varechar(2)

import mysql.connector as mc

mycon = mc.connect (

host = "localhost", user = "system",

passwd = "hello", database = "Inventory")

cur = mycon.cursor ()

db = cur.execute (

"CREATE TABLE Production(

PID varchar (5) Primary key,

PName char (30),

Price float,

Rank varchar(2))"

)

mycon.close()

4) Consider the table Student whose fields are

SCODE	Name	Age	stredc	Points	Grade
101	Amit	16	1	6	NULL
102	Arjun	13	3	4	NULL
103	Zaheer	14	2	1	NULL
104	Gagan	15	5	2	NULL
105	Kumar	13	6	8	NULL

Write the Python code to update grade to 'A' for all these students who are getting more than 8 as points.

import mysql.connector as mydb

mycon = mydb.connect (host = "localhost",

user = "Admin",

passwd = "Admin@123",

database = "system")

cursor = mycon.cursor ()

sql = "UPDATE Student SET Grade = 'A' WHERE Points > 8"

cursor. execute (sql)

mycon.commit ()

mycon.close ()

5) Consider the following table Traders with following fields

TCode	TName	City
T01	Electronic Sales	Mumbai
T03	Busy Store Corp	Delhi
T02	Disp House Inc	Chennai

Write Python code to display the names of those traders who are either from Delhi or from Mumbai.

```
import mysql.connector as mc

mycon = mc.connect(host = "localhost", user =
"root", passwd = "system", database = "Admin")

cursor = mycon.cursor ()

sql = "SELECT * FROM Traders WHERE
City = 'Mumbai' OR City = 'Delhi'"

cursor.execute(sql)

dis = cursor.fetchall ()

for rec in disp:

    print (rec[1])

mycon.close ()
```

6) Differentiate between fetchall() and fetchmany().

- fetchall() fetches all the rows of a query result.
- It returns all the rows as a list of tuples.
- An empty list is returned if there is no record to fetch.

```
mycur = mycon.cursor();

mycur.exeucte("SELECT * FROM student");

data = mycur.fetchall()

fetchone() method returns a single record as a
tuple or None if no more rows are available.

mycur = mycon.cursor();

mycur.exeucte("SELECT * FROM student");

data = mycur.fetchone()
```

7) Give example code for inserting a data of student (Name and Roll no) through python connectivity, using data received from user. Also, Write the number of records present in the table.

Old Style parametrizing Query

```
import mysql.connector as mc

mydb = mc.connect(

    host = "localhost",

    user = "root",

    password = "tiger",

    database = "school"

)

mycursor = mydb.cursor()

nam = input("Enter name:")

roll = int(input("Enter roll number:"))

sql = "INSERT INTO student(Name, Roll_no)
VALUES ('%s', %s)" %(nam, roll)

mycursor.execute(sql)

mydb.commit()

count = mycursor.rowcount,

print(No. of records", count)

mydb.close()
```

Modern style Parametrizing Query

```
import mysql.connector as mc

mydb = mc.connect(

    host = "localhost",

    user = "root",

    password = "tiger",

    database = "school"

)

mycursor = mydb.cursor()

nam = input("Enter name:")

roll = int(input("Enter roll number:"))
```

```
sql = "INSERT INTO student (Name, Roll_no)
VALUES ({}, {})" .format(nam, roll)
mycursor.execute(sql)
mydb.commit()
count = mycursor.rowcount,
print("No. of records", count)
mydb.close()
```

8) Write the Python code for (i) and (ii) on the basis of table College.

No	Name	Age	Department	DateofJoin	Basic	Sex
1	Aren	51	Mathematics	22/01/91	8500	M
2	Reeta	27	Chemistry	14/02/94	9000	F
3	Viren	49	Mathematics	03/01/88	9000	M
4	Prakash	22	Physics	17/02/92	8000	M
5	Shalaz	45	Biology	13/02/88	10500	M
6	Urvashi	29	Biology	10/02/93	8500	F

(i) To insert a new row in the table College with the following data

15, "Atin", 27, "Physics", "15/05/02", 8500, "M"

(ii) To delete a row from table in which name is Viren.

```
i) import mysql.connector as mydb
con = mydb.connect (
host = "localhost",
user = "root",
passwd = "admin@123",
database = "Management")
cursor = con.cursor ( )
sql = "INSERT INTO College (No, Name, Age,
Department, DateofJoin, Basic, Sex)
VALUES
(15, 'Atin', 27, 'Physics', '15/05/02', 8500, 'M')
cursor.execute (sql)
con.commit()
con.close ( )
```

```
ii) import mysql.connector as mydb
con = mydb.connect
(host = "localhost",
user = "root",
passwd = "admin@123",
database = "Management")
cursor = con.cursor ( )
cursor.execute ("DELETE FROM College WHERE
Name = 'Viren' ")
con.commit ( )
con.close ( )
```

8) Write the command to get no. or records using a cursor object mycursor.

```
>>>mycursor.rowcount
```

9) Write a function sql_data() to add details received from the user to a mysql table named Student, using python connectivity.

Fields to be inserted:

RollNo – integer

Name – string

Clas – integer

Marks – integer

Note the following to establish connectivity between Python and MYSQL:

- Username is root
- Password is tiger
- The table exists in a MYSQL database named school.

```
import mysql.connector as mysql
```

```
def sql_data():
```

```
con1=mysql.connect(host="localhost",
user="root",password="tiger",
database="school")
mycursor=con1.cursor()
```

```

rno=int(input("Enter Roll Number :: "))
name=input("Enter name :: ")
clas=int(input("Enter class :: "))
marks=int(input("Enter Marks :: "))
query="insert into student
values({}, '{}', {}, {})".format(
                                rno,name,clas,marks)
mycursor.execute(query)
con1.commit()
print("Data Added successfully")
con1.close()

```

```

database="school")
mycursor=con1.cursor()
print("Students with marks more than 75: ")
query = "SELECT * FROM STUDENT WHERE
MARKS > 75"
mycursor.execut(query)
data=mycursor.fetchall()
count = mycursor.rowcount

for rec in data:
    print(rec[1])          #printing names alone

```

10) Write a function read_sql_data() to read names of students who have got marks more than 75 and print in Python using connectivity. The records are present in a mysql table named Student. Also, print the number of records.

```

print("Total Number of records:", count)
con1.close()

```

~~~~~

#### **Fields:**

RollNo – integer

Name – string

Clas – integer

Marks – integer

Note the following to establish connectivity between Python and MYSQL:

- Username is root
- Password is tiger
- The table exists in a MYSQL database named school.

```
import mysql.connector as mysql
```

```
def read_sql_data():
```

```

    con1=mysql.connect(host="localhost",
    user="root",password="tiger",

```

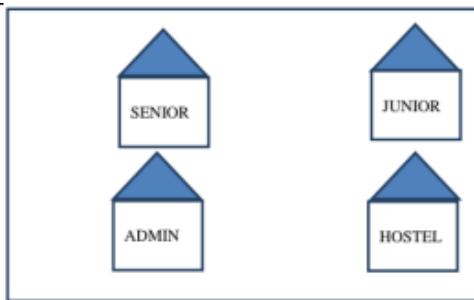


## XII COMPUTER SCIENCE

*Towards Getting Centum in CS – See Yes!*

### 16. NETWORKING CASE STUDY

1) Multipurpose Public School, Bengaluru is Setting up the network between its Different Wings of school campus. There are 4 wings named as SENIOR(S), JUNIOR(J), ADMIN(A) and HOSTEL(H).



Distance between various wings are given below:

|                  |      |
|------------------|------|
| Wing A to Wing S | 100m |
| Wing A to Wing J | 200m |
| Wing A to Wing H | 400m |
| Wing S to Wing J | 300m |
| Wing S to Wing H | 100m |
| Wing J to Wing H | 450m |

Number of Computers installed at various wings are as follows:

| Wings  | Number of Computers |
|--------|---------------------|
| Wing A | 20                  |
| Wing S | 150                 |
| Wing J | 50                  |
| Wing H | 25                  |

a. Suggest the best wired medium and draw the cable layout to efficiently connect various wings of Multipurpose Public School, Bengaluru.

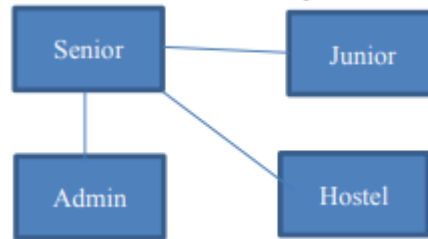
b. Name the most suitable wing where the Server should be installed. Justify your answer.

c. Suggest a device/software and its placement that would provide data security for the entire network of the School.

d. Suggest a device and the protocol that shall be needed to provide wireless Internet access to all smartphone/laptop users in the campus of Multipurpose Public School, Bengaluru.

answers:

a)



b) The server should be installed at Wing S (Senior) as per 80-20 rule i.e. maximum traffic should be local and minimum traffic should pass over backbone.

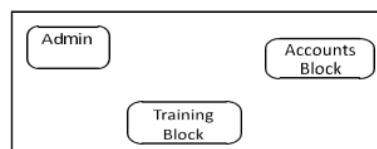
c) Firewall.

d) Device: Wireless Access Point or Router or WiFi hotspot device or Wifi Dongle  
Protocol: IEEE 802.11x or TCP/IP

2) Hi Standard Tech Training Ltd. is a Mumbai based organization which is expanding its office set-up to Chennai. At Chennai office compound, they are planning to have 3 different blocks for Admin, Training and Accounts related activities. Each block has a number of computers, which are required to be connected in a network for communication, data and resource sharing. As a network consultant, you have to suggest the best network related solutions for the more issues/problems raised by them in (i) to (iv), as per the distances between various blocks/locations and other given parameters.

# CHENNAI OFFICE

# MUMBAI Head Office



Shortest distances between various blocks/locations :

|                                      |            |
|--------------------------------------|------------|
| Admin Block to Accounts Block        | 300 Metres |
| Accounts Block to TrainingBlock      | 150 Metres |
| Admin Block to Training Block        | 200 Metres |
| MUMBAI Head Office to CHENNAI Office | 1300 Km    |

Number of computers installed at various blocks are as follows :

|                |     |
|----------------|-----|
| Training Block | 150 |
| Accounts Block | 30  |
| Admin Block    | 40  |

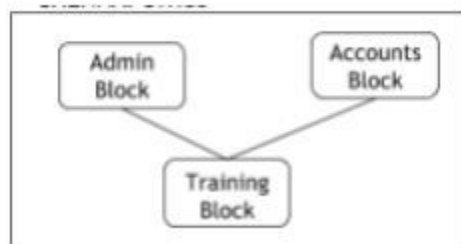
a) Suggest the most appropriate block/location to house the SERVER in the CHENNAI office (out of the 3 blocks) to get the best and effective connectivity. Justify your answer.

Training Block – As per 80:20 rule, 80% of the traffic is local, so to minimize/reduce the local traffic, we have keep the server in the Training Block or Because it has maximum number of computers.

b) Suggest the best wired medium and draw the cable layout (Block to Block) to efficiently connect various blocks within the CHENNAI office compound.

Best wired medium: Optical Fibre OR CAT5 OR CAT6 OR CAT7 OR CAT8 OR Ethernet Cable

# CHENNAI OFFICE



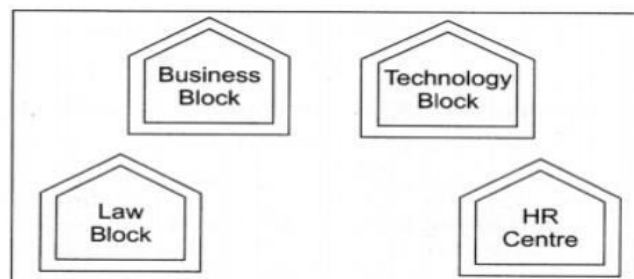
c) Suggest a device/software and its placement that would provide data security for the entire network of the CHENNAI office.

Firewall - Placed with the server at the Training Block

d) Suggest a device and the protocol that shall be needed to provide wireless Internet access to all smartphone/laptop users in the CHENNAI office.

Device Name: WiFi Router OR WiMax OR RF Router OR Wireless Modem OR RF Transmitter Protocol : WAP OR 802.16 OR TCP/IP OR VOIP OR 802.11

3) Quick Learn University is setting up its academic blocks at Prayag Nagar and planning to set up a network. The university has three academic blocks and one human resource centre as shown in the diagram below:



Centre to centre distance between various blocks/centre is as follows:

|                                    |       |
|------------------------------------|-------|
| Law Block to Business Block        | 40 m  |
| Law Block to Technology Block      | 80 m  |
| Law Block to HR Centre             | 105 m |
| Business Block to Technology Block | 30 m  |
| Business Block to HR Centre        | 35 m  |
| Technology Block to HR Centre      | 15 m  |

Number of computers in each of the blocks/centre are as follows:

|                  |     |
|------------------|-----|
| Law Block        | 15  |
| Technology Block | 40  |
| HR Centre        | 115 |
| Business Block   | 25  |

1. Suggest the most suitable place (i.e. block/centre) to install the server of this university with a suitable reason.

2. Suggest an ideal layout for connecting these block/centre for a wired connectivity.

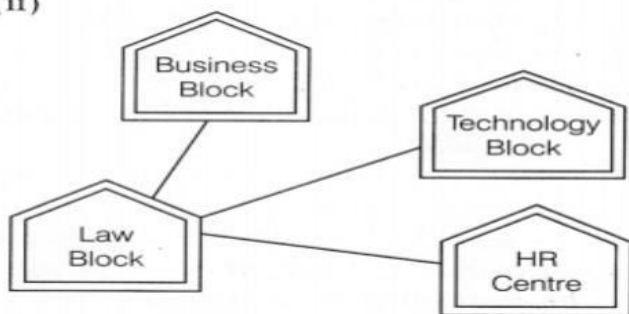
3. Which device you will suggest to be placed/installed in each of these blocks/centre to efficiently connect all the computers with in these blocks/centre ?

4. The university is planning to connect its admission office in the closest big city, which is more than 250 km from university, which type of network out of LAN, MAN or WAN will be formed? Justify your answer.

#### Answers:

1) HR centre as it has maximum number of computers.

2)  
(ii)

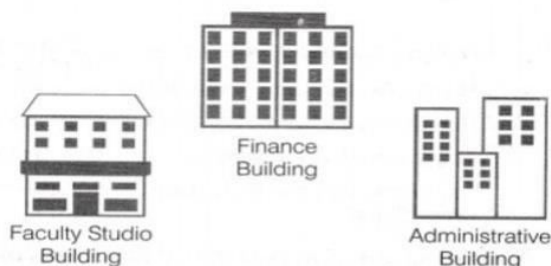


3) Switch.

4) WAN as it is another city.

4) Expertia Professional Global (EPG) is an online corporate training provider company for IT related courses. The company is setting up their new campus in Mumbai. You as a network expert have to study the physical locations of various buildings and the number of computers to be installed. In the planning phase, provide the best possible answers for the queries (i) to (iv) raised by them.

Physical locations of the buildings of EPG



| Building to building distance (in m.) |                         |          |
|---------------------------------------|-------------------------|----------|
| From                                  | To                      | Distance |
| Administrative Building               | Finance Building        | 60       |
| Administrative Building               | Faculty Studio Building | 120      |
| Finance Building                      | Faculty Studio Building | 70       |

Expected computers to be installed in each building

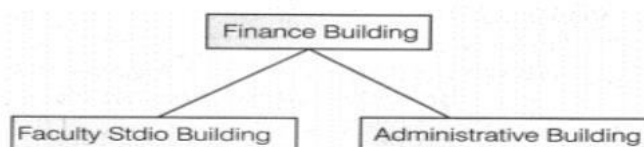
| Buildings               | Computers |
|-------------------------|-----------|
| Administrative Building | 20        |
| Finance Building        | 40        |
| Faculty Studio Building | 120       |

1. Suggest the most appropriate building, where EPG should plan to install the server.
2. Suggest the most appropriate building to building cable layout to connect all three buildings for efficient communication.
3. Which type of network out of the following is formed by connection the computers of these three buildings?  
(a) LAN  
(b) MAN  
(c) WAN
4. Which wireless channel out of the following should be opted by EPG to connect to students of all over the world?  
(a) Infrared  
(b) Microwave  
(c) Satellite

#### Answers:

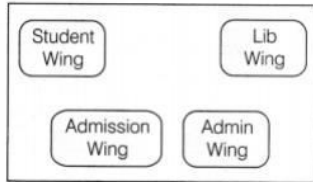
1. EPG should install the server in the Faculty Studio Building as it is having maximum number of computers.

2.



3. (a) LAN (Local Area Network).
4. (c) Satellite.

53. Institute of Distance Learning is located in Pune and is planning to go in for networking of four wings for better interaction. The details are shown below.



**The distance between various wings**

|                                |       |
|--------------------------------|-------|
| Student Wing to Admin Wing     | 150 m |
| Student Wing to Admission Wing | 100 m |
| Student Wing to Lib Wing       | 325 m |
| Admission Wing to Admin Wing   | 100 m |
| Admission Wing to Lib Wing     | 125 m |
| Admin Wing to Lib Wing         | 90 m  |

**Number of computers**

|                |     |
|----------------|-----|
| Student Wing   | 225 |
| Admission Wing | 50  |
| Admin Wing     | 10  |
| Lib Wing       | 25  |

~~~~~

the server is Student Wing.

3. (a) Repeater should be installed between Student Wing and Admin Wing as distance is more than 70m.

(b) Switch should be installed in each wing to connect several computers.

4. Broadband connection as it is between economical and speedy.

1. Suggest the type of networking (LAN, MAN, WAN) for connecting Lib Wing to Admin

Wing. Justify your answer.

2. Suggest the most suitable place (i.e. wing) to house the server, with a suitable reason.

3. Suggest and placement of the following devices with reasons.

(a) Repeater

(b) Switch

4. The Institute is planning to link its study centre situated in Delhi. Suggest an economic way to connect it with reasonably high speed. Justify your answer.

Answers:

1. Since, the distance between Lib Wing and Admin Wing is small. So type of networking

is small, i.e. LAN.

2. Since, maximum number of computers are in Student Wing, so suitable place to house

*"Dream, Dream, Dream, Dream transform into thoughts
and thoughts result in action."*

- Dr. A.P.J. Abdul Kalam

XII COMPUTER SCIENCE

Towards Getting Centum in CS - See Yes!

VIVA HANDOUT - 1

REVIEW OF PYTHON FROM CLASS XI

1) What does IDLE stand for? What is Python IDLE?

IDLE is an acronym of *Integrated Development Environment* and is the standard, most popular Python development environment.

To run a program, we basically need an editor to write it, an interpreter/compiler to execute it and a debugger to catch and remove the errors. Python IDLE provides all these tools as a bundle. It lets edit, run, browse and debug Python Programs from a single interface and makes it easy to write programs.

2) What is the difference between Interactive mode and Script mode?

Python IDLE can be used in two modes: Interactive mode and Script mode. **Python shell** is an **interactive interpreter**. **Python editor** allows us to work in **script mode** i.e. we can create and edit python source file.

3) What are tokens?

Tokens- Smallest individual unit of a program is called token. Example keywords, identifiers, literals, operators and punctuators.

4) What are Keywords?

They are the words used by Python interpreter to recognize the structure of program. As these words have specific meaning for interpreter, they cannot be used for any other purpose.

5) What are identifiers and literals?

IDENTIFIERS: These are the names given to identify any memory block, program unit or program objects, function, or module. Examples of identifiers are: num, roll_no, name etc.

LITERALS: A fixed numeric or non-numeric value is called a literal. Examples of literals are: 502, -178.76, "Rajan" etc.

6. What is a variable?

Variables are like containers which are used to store the values for some input, intermediate result of calculations or the final result of a mathematical operation in computer memory.

7) What are the naming conventions of identifiers/variables/literals.

- > Variable names are **case sensitive**. For eg. num and NUM are treated as two different variable names.
- **Keywords** or words, which have special meaning, should **not** be used as the variable names.
- Variable names should be **short** and **meaningful**.
- All variable names must begin with a **letter** or an **underscore** (**_**).
- After the first initial letter, variable names may contain **letters** and **digits** (0 to 9) and an **underscore** (**_**), but no spaces or special characters are allowed.

Examples of valid variable names: sum, marks1, first_name, _money

Examples of invalid variables names: marks%, 12grade, class, last-name

8. What do you understand by the terms L-value and R-value?

L-value is a value which has an address. Thus, all variables are l-values since variables have addresses. As the name suggests l-value appears on left hand side but it can appear on right hand side of an assignment operator(=). l-value often represents as identifier.

R-value refers to data value that is stored at some address in memory. An **r-value** is a value which is assigned to an l-value. An r-value appears on right but not on left hand side of an assignment operator(=).

9) What is the default separator and terminator in print() function in Python? How can we change it?

By default, print uses a single space as a separator and a \n as a terminator (appears at the end of the string). We can use the parameters **sep** and **end** to change the separator and terminator respectively.

10) Differentiate between list, tuple and dictionary.

A list is an ordered collection of comma-separated values (items) within square brackets.	A tuple is an ordered collection of commas separated values. The comma separated values can be enclosed in parenthesis but parenthesis are not mandatory.	A dictionary is an unordered collection of items where each item is a key: value pair. We can also refer to a dictionary as a mapping between a set of keys/indices and a set of values. The entire dictionary is enclosed in curly braces.
it is mutable	it is immutable.	it is mutable
<code>list2=["Raman", 100, 200, 300, "Ashwin"]</code>	<code>tup1=('Sunday', 'Monday', 10, 20)</code>	<code>dict1={'R':'RAINY', 'S':'SUMMER', 'W':'WINTER', 'A':'AUTUMN'}</code>

11) What are the different types of errors in a Python program?

SYNTAX ERROR: An error in the syntax of writing code that does not conform to the syntax of the programming language is called a syntax error.

LOGICAL ERROR: It is an error in a program's source code that results in incorrect or unexpected result. It is a type of runtime error that may simply produce the wrong output or may cause a program to crash while running.

RUN TIME ERROR: A runtime error is an error that occurs during execution of the program and causes abnormal termination of program.

12) What are operators?

Operators performs some action on data. Various types of operators are:

- Arithmetic (+, -, *, /, %)
- Assignment Operator (=)
- Relational/comparison (<, >, <=, >=, ==, !=)
- Logical (AND, OR, NOT)
- Identity Operators (is, is not)
- Membership Operators (in, not in)

13) What is the difference between mutable and immutable data types?

Mutable data types	immutable data types
Variables of mutable data types can be changed after creation and assignment of values.	Variables of immutable data types can not be changed or modified once they are created and assigned the value. If an attempt is made to change the value of variable of immutable datatype, the new is stored in some other memory location and the variable starts pointing to that location.
Some of the mutable data types in Python are list and dictionary	some of the immutable data types are int, float, decimal, bool, string and tuple.

14) What is difference between /, //, %?

/	//	%
Divides two operands and gives quotient	Integer division	Divides two operands and gives remainder
10/5=2 10/3.0 = 3.3 10.0/3=3.3 10/3=3	5//2=2 5.0//2=2.0	10%5=0

15) What is the difference between * and **?

*	**
Multiplies two operands	Exponentiation
10*5=50	2**4= 16

16) What is the difference between = and ==?

=	==
Assigns the value	Checks if the value of left operand is equal to the value of right operand, if yes then condition becomes true.
x=50	15= = 15, true 16 = = 15, false

17) What is the difference between concatenation and repetition?

Concatenation (+)	Repetition (*)
Joins the objects (strings, lists etc.) on either side of the operator	Concatenates multiple copies of the same object (strings, lists etc.) given number of times
<pre>>>> str1="Tech" >>> str2="Era" >>> str1+str2 'TechEra' >>> str3=str2+str1</pre>	<pre>>>> str1*2 'TechTech'</pre>

18) What is the difference between is and in operators?

is, is not - Identity Operators	in, not in - Membership Operators
Identity operators are used to compare two variables whether they are of same type with the same memory location.	Membership operators are used to validate the membership of a value. It tests for membership in a sequence, such as strings, lists, or tuples.
The is operator evaluates to true if the variables on either side of the operator point to the same memory location and false otherwise. <pre>>>>x = 5 >>>type(x) is int True</pre>	The in operator is used to check if a value exists in a sequence or not. Evaluates to true if it finds a variable in the specified sequence and false otherwise. <pre>>>>x = [1,2,3,4,5] >>>3 in x True</pre>
The is not operator evaluates to true if the variables on either side of the operator point to the different memory location and false otherwise. <pre>>>>x = 5 >>>type(x) is not int False</pre>	The not in operator evaluates to true if it does not find a variable in the specified sequence and false otherwise. <pre>>>>x = [1,2,3,4,5] >>>7 not in x True</pre>

19) What are the major differences between key and value in a dictionary?

key	value
Keys are unique	Values may not be unique
The keys must be of an immutable data type such as strings, numbers, or tuples.	The values of a dictionary can be of any type

20) Compare random(), randint() and randrange().

Function name	Purpose	Example
<u>random()</u>	Generates a random float number between 0.0 to 1.0.	>>> <u>random.random()</u> 0.21787519275240708
<u>randint()</u>	Returns a random integer between the specified integers.	>>> <u>random.randint(1,50)</u> 21 >>> <u>random.randint(1,50)</u> 35
<u>randrange()</u>	Returns a randomly selected element from the range created by the start, stop and step arguments. The value of start is 0 by default. Similarly, the value of step is 1 by default.	>>> <u>random.randrange(1,20)</u> 1 >>> <u>random.randrange(1,20,5)</u> 16

21) Compare break and continue.

BREAK	CONTINUE
Break statement causes the loop to break/ terminate immediately.	Continue statement causes the current iteration of the loop to skip and go to the next iteration.
The loop of which, break statement is a part, stops.	All the statements following the continue statement in the loop will not be executed and loop control goes to the next iteration.
<pre>s=10; for i in range (10, 20, 3): s+=i if(i==16): break print(s); print("end");</pre> OUTPUT: <pre>20 33 end</pre>	<pre>s=10; for i in range (10, 20, 3): s+=i if(i==16): continue print(s); print("end");</pre> OUTPUT: <pre>20 33 68 end</pre>

22) What is the difference among insert, append and extend?

<u>insert()</u>	<u>append()</u>	<u>extend()</u>
It is used to insert data at the given index position.	It is used to add an element at the end of the list.	is used to add one or more element at the end of the list.
<pre>z1=[121, "RAMESH", 890, 453.90] z1.insert(1, "KIRTI")</pre> OUTPUT <pre>[121, 'KABIR', 'RAMESH', 890, 453.9]</pre>	<pre>list1=[100, 200, 300, 400, 500] list2=["Raman", 100, "Ashwin"] list1.extend(list2) print(list1)</pre> OUTPUT <pre>[100, 200, 300, 400, 500, 'Raman', 100, 'Ashwin']</pre> here list2 is added at the end of the list list1.	<pre>z1=[121, "RAMESH", 890, 453.90] z1.insert(1, "KIRTI") z1.append(10) print(z1)</pre> OUTPUT <pre>[121, 'KABIR', 'RAMESH', 890, 453.9, 10]</pre> here list2 is added at the end of the list list1.

23) What is the difference among del, remove or pop methods to delete elements from a List?

<u>pop()</u>	<u>del</u>	<u>remove()</u>
It removes the element from the specified index, and also return the element which was removed. The last element is deleted if no index value is provided in pop ().	The del method removes the specified element from the list, but it does not return the deleted value.	The remove method is used when the index value is unknown and the element to be deleted is known.
<pre>>>> list1=[100, 200, 90, 'Raman', 100, 'Ashwin'] >>> list1.pop(2)</pre> OUTPUT <pre>90</pre>	<pre>>>> list1=[100, 200, 'Raman', 100] >>> del list1[2] >>> print(list1)</pre> OUTPUT <pre>[100, 200, 100]</pre>	<pre>>>> list1=[100, 200, 50, 400, 500, 'Raman', 100, 'Ashwin'] >>> list1.remove(400) >>> print(list1)</pre> OUTPUT <pre>[100, 200, 50, 500, 'Raman', 100, 'Ashwin']</pre>

24) What is the difference between is and in operators?

is, is not - Identity Operators	in, not in - Membership Operators
Identity operators are used to compare two variables whether they are of same type with the same memory location.	Membership operators are used to validate the membership of a value. It tests for membership in a sequence, such as strings, lists, or tuples.
The is operator evaluates to true if the variables on either side of the operator point to the same memory location and false otherwise. <pre>>>>x = 5 >>>type(x) is int True</pre>	The in operator is used to check if a value exists in a sequence or not. Evaluates to true if it finds a variable in the specified sequence and false otherwise. <pre>>>>x = [1,2,3,4,5] >>>3 in x True</pre>
The is not operator evaluates to true if the variables on either side of the operator point to the different memory location and false otherwise. <pre>>>>x = 5 >>>type(x) is not int False</pre>	The not in operator evaluates to true if it does not find a variable in the specified sequence and false otherwise. <pre>>>>x = [1,2,3,4,5] >>>7 not in x True</pre>

25) Give the difference between upper() and isupper() functions of string.

The upper() function returns the copy of the string with all the letters in uppercase.

The isupper() function returns True if the string is in uppercase.

Example

```
>>> print(str3.upper())  
ERATECH  
>>> str5='tech era *2018*'  
>>> print(str5.isupper())  
False
```

26) What are the comments? Declare a single line comment and a multiline comment.

Comments are the line that compiler ignores to compile or execute. There are two types of comments in Python

- **Single line comment:** This type of comment deactivates only that line where comment is applied. Single line comments are applied with the help of “ #”.
For e.g.
This program calculates the sum of 2 numbers

```
>>>n1=10  
>>>n2=20  
>>>sum= n1 + n2 # sum of numbers 10 and 20  
>>>print ("sum=", sum)
```

- **Multi line Comment:** This Type of comment deactivates group of lines when applied. This

type of comments is applied with the help of triple quoted string.

```
"""This is a  
multiline comment"""
```

or

```
"""This is a  
multiline comment"""
```

27) Write functions for the following:

To convert string to int	int()
To convert string to float	float()
To convert numeric data to string	str()

28) What is the use of indentation in Python?

- Indentation in Python is used to form block of statements also known as suite.
- It is required for indicating what block of code, a statement belongs to.
- Indentation is typically used in flow of control statements (like if, for etc.), function, class etc.
- Although we can give any number of spaces for indentation but all the statements in a block must have the same indent.
- Unnecessary indentation results in syntax error and incorrect indentation results in logical error.

29) What is the difference between indexing and slicing?

Indexing is used to retrieve an element from an ordered data type like string, list, tuple etc.

Slicing is used to retrieve a subset of values from an ordered data type. A slice of a list is basically its sub-list.

30) When is the else clause executed in looping statements?

The else clause in looping statements is executed when the loop terminates.

~~~~~

## LET US REVISE FOR VIVA (PYTHON BASICS)

### 1) Who developed Python and in which year?

Python was developed by Guido Van Rossum in February 1991.

### 2) What are the advantages/pluses of Python?

- easy to use
- expressive
- complete
- cross-platform
- free and open source

### 3) What are limitations/minuses of Python?

- Not the fastest language
- Lesser libraries than C, Java, Perl
- Not strong on Type-binding.

### 4) What are the 2 modes of Python?

(i) Interactive mode (ii) Script mode

### 5) Name some popular python distributions.

Popular Python distributions are CPython (default), Anaconda distribution etc.

### 6) Name some popular Python IDEs.

Popular Python IDEs are Python IDLE, Spyder IDE, Jupyter Notes, Pycharm IDE etc.

### 7) Say about interactive mode of Python.

- Interactive mode does not save commands in form of a program and also, output is sandwiched between commands.
- Interactive mode is suitable for testing code.

### 8) Say about script mode of Python.

Script mode is useful for creating programs and then run the programs later and get complete output

### 9) Python is compiled or interpreted?

Python is an interpreted language

### 10) What is Python shell?

Python's interactive interpreter is also called Python Shell.

### 11) What is a token in Python?

A token is the smallest individual unit in a program

### 12) Name the tokens of Python.

Python provides following tokens: Keywords, identifiers (names), Values (literals), punctuators, operators and comments.

### 13) What is a keyword? Give example.

A Keyword is a reserved word carrying special meaning and purpose. ex: while, pass

### 14) What is an identifier?

Identifiers are the user-defined names for different parts of the program.

### 15) Say the rules for making Python identified.

In Python, an identifier may contain **letters** (a-z, A-Z) **digits** (0-9) and a symbol underscore (\_). However, an identifier must begin with a letter or underscore; all letters / digits in an identifier are significant.

### 16) What are literals? Give example.

Literals are the fixed values. ex: 45, 'dog'

### 17) What are the literals in Python?

Python allows following literals: string literals, numeric (integer, floating-point literals, Boolean literals, special literal **None** and literal collections).

### 18) What is an operator?

Operators are tokens that trigger some computation / action when applied to variables and other objects in an expression.

### 19) What are punctuators?

Punctuators are symbols used to organize programming – sentence structures and indicate the rhythm and emphasis of expressions, statements and program-structure.

### 20) What is an expression?

An expression is a legal combination of symbols that represents a value.

### 21) What is a statement?

A statement is a programming instruction.

### 22) What are comments?

Comments are non-executable, additional information added in program for readability.

### 23) Comments in python begin with?

In Python, comments begin with a # character or triple quotes ''' ''' for multiline comments

### 24) What are the types of comments in python?

Comments can be single-line comment, multi-line comments and inline comments.

**25) What is a function?**

Function is a named code that can be reused with a program.

**26) What is a block?**

A block/ suite/ code-block is a group of statements that are part of another statement. Blocks are represented through indentation.

**27) When does indentation appear in Python?**

When a block is created, indentation appears.

**28) What is a variable in Python?**

A variable in Python is defined only when some value is assigned to it.

**29) What is dynamic typing?**

Python supports dynamic typing i.e., a variable can hold values of different types at different times.

ex:

```
a = 2      # a is integer
```

```
a = 'sai'  # now a is a string
```

**30) What is the use of input() function? What is the type of data returned by input()?**

The input() is used to obtain input from user ; it always returns a string type of value.

**31) What is the use of print() function?**

Output is generated through print() (by calling print function) statement.

**31)What is an expression?**

An expression is composed of one or more operations. It is a valid combination of operators, literals and variables.

**32) What is an atom in python?**

An atom in Python is something that has a value. Example of atoms are variables, literals, strings, lists, tuples, sets etc.

**33) Say some types of expression.**

Expression can be arithmetic, relational or logical, compound etc.

**34) How do you determine the type of an expression?**

Types of operators used in an expression determine its type. For instance, use of arithmetic operators makes it arithmetic expression.

**35) What are the types of arithmetic expressions?**

Arithmetic expression can either can be integer expressions or real expressions or complex number operations or mixed-mode expressions.

**36) What will be the result of an arithmetic expression?**

An arithmetic expression always results in a number (integer or floating-point number or a complex number);

ex: 5+6.4

**37) What will be the result of a relational expression?**

Relational expression always results in a Boolean value i.e., either True or False

ex:>>> 8 > 4

**38) What will be the result of a logical expression?**

Logical expression results into a number or a string or a Boolean value, depending upon its operands.

ex: 8 or 5 and 2

**39) What is type conversion?**

In a mixed-mode expression, different types of variables / constants are converted to one same type. This process is called type conversion.

#### 40) What are the 2 types of type conversion?

Type conversion can take place into two forms : implicit (that is performed by compiler without programmers's intervention) and explicit (that is defined by the user)

#### 41) What is type promotion or coercion?

In implicit conversion, all operands are converted up to the type of the largest operand, which is called type promotion or coercion.

#### 42) What is type casting?

The explicit conversion of an operand to a specific type is called type casting and it is done using type conversion functions that is used as

<type conversion function> (<expression>)

e.g. to convert to float, one may write

```
float (<expression>) >>>float(2)
```

#### 43) What do you mean by debugging?

Debugging refers to the process of locating the place of error, cause of error, and correcting the code accordingly.

#### 44) What are compile-time errors?

Compile-time errors (syntax error and semantics errors) refer to the errors that violate the grammatical rules and regulations of a programming language.

#### 45) What is a run-time error?

Runtime errors occur during the execution of a program.

#### 46) What is a logical error?

Logical errors occur due to mistaken analysis of the problem.

#### 47) What do you mean by exception?

Irregular unexpected situations occurring during runtime are called Exceptions. Exceptions may occur even if the program is free from all types of errors.

#### 48) What do you mean by immutable type in python?

Immutable Types in Python mean that their values cannot be changed in place.

#### 49) What do you mean by mutable type in python?

Mutable Types Python mean that their values can be changed in place.

#### 50) Which are the mutable types in Python?

Python has only 3 mutable type : **lists**, **dictionaries** and **sets**.

#### 51) Which are the arithmetic operators in python?

Arithmetic operators carry out arithmetic for Python. These are unary +, unary -, +, -, \*, /, //, % and \*\*.

#### 52) What is the use of Unary + and Unary - operators?

Unary + gives the value of its operand, unary - changes the sign of its operand's value.

```
ex: >>>+7
7
>>>-8
-8
```

#### 53) What is the use of / operator?

The operator / divides the first operand by second and returns a float result

#### 54) What is the use of // operator?

// performs the floor division, % gives the remainder after dividing first operand by second

#### 55) What is the use of \*\* operator?

\*\* is the exponentiation operator, i.e., it gives base raised to power.

**56) Which are the relational operators in Python?**

Relational operators compare the values of their operands. These are >, <, ==, <=, >= and != i.e., less than, greater than, equal to, less than or equal to, greater than or equal to and not equal to respectively.

**57) What is the use of identity operators?**

Identify operators (is, is not) compare the memory two objects are referencing.

**58) What is the use of logical operators?**

Logical operators perform comparisons on the basis of the truth-ness of an expression or value. These are 'or', 'and', and 'not'.

**59) What is the use of boolean operators?**

Boolean or relational expressions truth value depends on their Boolean result True or False.

**60) How do we determine the boolean value of an item?**

Values' truth value depends on their emptiness or non-emptiness. Empty numbers (such as 0, 0.0, 0j, etc.) and empty sequences (such as "", [], ()) and None have truth-value as **false** and all others (non-empty ones) have truth-value as **true**.

**61) What are statements?**

Statements are the instructions given to the computer to perform any kind of action.

**62) What are the types of python statements?**

Python statements can be on one of these types : empty statement, single statement and compound statement.

**63) What is the use of pass statement in python?**

An empty statement is the statement that does nothing. Python offers **pass** statement as empty statement.

**64) Which is a simple statement in python?**

Single executable statement forms a simple statement.

**65) What is a compound statement?**

A compound statement represents a group of statement executed as a unit.

Every compound statement of Python has a header and an indented body below the header.

**66) Give examples for compound statement?**

Some examples of compound statements are : if statement, while statement etc.

**67) What are the 3 types of constructs in Python?**

The flow of control in a program can be in three ways : sequentially (the sequence construct), selectively (the selection construct), and iteratively (the iteration construct).

**68) What is a sequence construct in Python?**

The sequence construct means statements get executed sequentially.

**69) What is a selection construct in Python?**

The selection construct means the execution of statement(s) depending upon a condition-test.

**70) What is the use of iteration construct in Python?**

The iteration (looping) constructs mean repetition of a set-of-statements depending upon a condition-test.

**71) Which is the selection statement provided by Python?**

Python provides one selection statement if in many forms – if..else and if..elif..else.



### 72) What is the use of if...else statement?

The **if..else** statement tests an expression and depending upon its truth value one of the two sets-of-action is executed.

### 73) What is a sequence?

A sequence is a succession of values bound together by a single name. Some Python sequences are strings, lists and tuples.

### 74) What is the use of range() function?

The `range()` function generates a sequence of list type.

### 75) What are called iteration statements?

The statements that allow a set of instructions to be performed repeatedly are iteration statements.

### 76) What are the looping statements or constructs in Python?

Python provides two looping constructs – **for** and **while**. The **for** is a counting loop and **while** is a conditional loop.

### 77) Say about while loop in Python.

The while loop is an entry-controlled loop as it has a control over entry in the loop in the form of test condition.

### 78) What is loop else?

Loops in Python can have **else** clause too. The **else** clause of a loop is executed in the end of the loop only when loop terminates normally.

```
for i in range(9):
    print('Sairam')
else:
    print('Loop over')
```

### 79) What is the use of break statement?

The **break** statement can terminate a loop immediately and the control passes over to the statement following the statement containing break. In nested loops, a **break** terminates the very loop it appears in.

### 80) What is the use of continue statement?

The **continue** statement abandons the current iteration of the loop by skipping over the rest of statements in the loop-body. It immediately transfers control to the beginning of the next iteration of the loop.

### 81) How does python store string in memory?

Python strings are stored in memory by storing individual characters in contiguous memory locations.

### 82) What is an index or subscript?

The memory locations of string characters are given indexes or indices. The index (also called subscript sometimes) is the numbered position of a letter in the string.

### 83) What do you mean by 2 way indexing in Python?

In Python, indices begin 0 onwards in the forward direction up to **length -1** and

-1, -2, ... up to **-length** in the backward direction. This is called two-way indexing.

### 84) Which operator is used for concatenation and replication in strings?

For strings, **+** means concatenation, **\*** means replication.

### 85) What is use of membership operators on strings?

The **in** and **not in** are membership operators that test for a substring's presence in the string.

### 86) How does Python compare strings?

Comparison in Python strings takes place in dictionary order by applying character-by-character comparison rules for ASCII or Unicode.

**87) Which function is used to get the ASCII value of a given character?**

The ord( ) function returns the ASCII value of a given character.

```
>>>ord('A')  
65
```

**88) Which function is used to get equivalent character of an ASCII code?**

The chr( ) function returns the character value of a given ASCII code.

```
>>>chr(65)  
'A'
```

**89) What is a string slice?**

The string slice refers to a part of the string **s[start:end]** is the element beginning at **start** and extending up to but not including **end**.

If the string slice has the syntax **s[start : end : n]**, then it is the slice beginning at **start** and extending up to but not including **end**, taking every **nth** character.

**90) What are lists?**

Lists are **mutable sequences** of Python i.e., you can change elements of a list in place.

**91) How to create a list?**

To create a list, put a number of comma-separated expressions in square brackets.

**92) What is an empty list?**

The empty square brackets i.e., [ ] indicate an empty list.

**93) List follows indexing, same that of?**

Lists index their elements just like strings, i.e., two way indexing.

**94) How are lists stored in memory?**

Lists are stored in memory exactly like strings, except that because some of their objects are larger than others, they store a reference at each index instead of single character as in strings.

**95) Compare list and strings?**

Lists are similar to strings in many ways like **indexing**, **slicing** and **accessing individual elements** but they are different in the sense that Lists are **mutable** while strings are not.

What is the use of len() function on lists?

Function **len(L)** returns the number of items (count) in the List **L**.

**96) What is the use of + and \* operators on lists?**

The + operator adds one list to the end of another. The \* operator repeats a list.

**97) What is a list slice?**

List slice is an extracted part of a list; list slice is a list in itself.

**98) How do you create a list slice?**

L[start : stop] creates a list slice out of list L with elements falling between indexes start and stop, not including stop.

**99) What are tuples?**

Tuples are **immutable sequences** of Python i.e., you cannot change elements of a tuple in place.

**100) How do you create a tuple?**

To create a tuple, put a number of comma-separated expressions in round brackets.

**101) What is an empty tuple?**

The empty round brackets i.e., ( ) indicate an empty tuple.

**102) Difference between tuples and strings?**

Tuples are stored in memory exactly like strings, except that because some of their objects are larger than others, they store a reference at each index instead of single character as in strings.

### 103) Similarity between tuples and strings?

Tuples are similar to strings in many ways like indexing, slicing and accessing individual elements and they are immutable just like strings are.

### 104) What is the use of + and \* operators on a tuple?

The + operator adds one tuple to the end of another. The \* operator repeats a tuple.

### 105) What is type of a tuple slice?

Tuple slice is an extracted part of tuple ; tuple slice is a tuple in itself.

### 106) How to create a tuple slice?

T[start : stop] creates a tuple slice out of tuple T with elements falling between indexes start and stop, not including stop.

### 107) What are dictionaries?

Dictionaries are mutable with elements in the form of a **key : value** pair that associate keys to values.

### 108) Which part of dictionary is mutable and which one is immutable?

The values of a dictionary can be mutable or immutable. But, the **keys** must be of immutable types.

### 109) Other name for dictionaries?

Dictionaries are also called **associative arrays** or **mappings** or **hashes**.

### 110) Is dictionary ordered?

In Python dictionaries, the elements (**key : value** pairs) are unordered ; one cannot access element as per specific order.

### 111) Which part of the dictionary could be duplicate?

Keys of a dictionary must be unique. Values could be duplicate.

ex: d = {1:'sai', 2:'sai'}

### 112) What are the ways to create a dictionary?

Dictionaries can be created in multiple ways – through ( ) and through dict( ) constructor.

### 113) Differentiate between updation and addition of elements in dictionary?

In Dictionaries, the updation and addition of elements are similar in syntax. But for addition, the key must not exist in the dictionary and for updation, the key must exist in the dictionary.

### 114) What are the ways to delete element in a dictionary?

To delete an element, you can either use **del** statement or use **pop( )** or **popitem( )** method.

### 115) Can membership operators work on values?

No. The membership operators **in** and **not in** only work with dictionary keys.

### 116) What is the use of clear() function?

The clear( ) function removes all the items from dictionary but the dictionary object exists as an empty dictionary.

~~~~~

*"Keep your face always toward the sunshine - and
shadows will fall behind you."*

- Walt Whitman

XII COMPUTER SCIENCE

Towards Getting Centum in CS - See Yes!

VIVA HANDOUT – 2

FUNCTIONS

1) What is a function?

Function is a named group of related programming statements which perform a specific task. Functions are used to break the large code into smaller manageable modules which are easy to manage and debug.

2) What are the needs of functions?

Some of the major advantages of using functions in a program are:

Modularity: Functions provide modularity to a program. It is difficult to manage large programs. Thus, a large program is usually broken down into small units called functions. This makes the program easy to understand, more organized and manageable.

Code Reusability: Functions save space and time. Sometime a part of the program code needs to be executed repeatedly. If this part of program code is stored in a function then we do not need to write the same code again and again.

Instead, function can be called repeatedly. This saves time and space. We can also store the frequently used functions in the form of modules which can easily be imported in other programs when needed.

Easy debugging: It is easy to debug small units of a program than a long program.

3) What are the types of functions?

Basically, we can divide functions into the following three types:

- Built-in Functions
- Python Modules
- User-defined Functions

4) What are the types of functions based upon return statement?

- * Returning functions are fruitful/non-void functions
- * Non returning functions are non-fruitful/void functions

5) What is the difference between global and local variables?

A variable declared outside the function is known as global variable. The global variable can be accessed inside or outside of the function.

A variable declared inside the function is known as local variable. The local variable can be accessed only inside the function in which it is declared.

6) How can we access a global variable inside a function if a local variable exists inside the function by the same name?

The keyword global may be used to access a global variable inside a function if a local variable exists inside the function by the same name.

7) What is the purpose of a return statement? How can a function return more than one value?

The return statement is used to return either a single value or multiple values from a function. More than values are returned by a function in the form of a tuple.

8) What is the difference between parameters and arguments?

Information can be passed to functions using parameters. They are specified inside the parenthesis after the function name and are separated by a comma. The values provided in the function call are called arguments.

Arguments and parameters have one to one correspondence.

9) Name the keyword used for defining a function.

Ans: def

10) What are the uses of functions?

- Functions help to achieve modularity, readability and reusability.
- Functions make program handling easier as only a small part of the program is dealt with at a time, thereby avoiding ambiguity.
- Once defined, a function can be called repeatedly from any part of the program.

11) What are the rules for creating a function?

- A function definition should begin with the keyword def (short for define).
- A function may or may not have parameters. Also, a function may or may not return a value.
- Function header always ends with a colon (:)
- Function name should be unique. Rules for naming identifiers also applies for function naming.
- The statements outside the function indentation are not considered as part of the function.

ex:

```
def fun():  
    print("Hello")  
print("Welcome")
```

Here, print("Welcome") is not part of the function fun()

12) What is __main__?

By default, Python names the segment with top-level statements (main program) as `s__main__`.

13) What are arguments?

An argument is a value passed to the function during the function call which is received in corresponding parameter defined in function header.

ex:

```
def fun(n):  
    print(n)  
  
fun(n)
```

Here, in this function call-statement, n is called as an argument.

14) What are parameters?

The values received in the function header are called parameters (or formal parameters or formal arguments)

ex:

```
def fun(n):  
    print(n)
```

Here, n is called as a parameter.

15) What are the 3 types of formal arguments?

- Positional arguments
- Default arguments
- Keyword (or named) arguments

16) What is positional argument matching?

When the function call statement must match the number and order of arguments as defined in the function definition, this is called the positional argument matching and the arguments are called positional arguments.

ex:

```
def fun(a,b):  
    print(a+b)
```

x=1

y =2

fun(x,y)

Here, x will match with a, and y will match with b.
These are called positional arguments.

17) What is a default parameter or default argument?

A parameter having default value in the function header is known as default parameter.

ex:

```
def fun(a, b = 4):  
    print(a+b)
```

fun(3)

Here, a will get 3. Value of b will be taken from function header. b is a default argument.

18) When are the default arguments considered?

The default values for parameters are considered only if no value is provided for that parameter in the function call statement.

19) Which one rule must be satisfied for default arguments?

- Default arguments must not be followed by non-default arguments
- any parameter cannot have a default value unless all parameters appearing on its right side have their default values.

20) What are keyword arguments?

Named arguments with assigned values being passed in the function call statement.

ex:

```
def fun(a,b):  
    print(a+b)
```

fun(b=3, a=1)

Here, a will get 1 and b will get 3, though the order is changed and sent.

21) Differentiate between positional and keyword arguments with a suitable example.

Positional arguments

- When arguments are passed as positional arguments, the order in which the arguments are sent or their position is very important.
- It is mandatory to pass arguments in the same order as the parameters.
- The arguments should have one to one correspondence/mapping with parameters present in the function header.
- The number and type of arguments passed should match the number and type of parameters present in the function header.

```
def fun(a,b):  
    print(a+b)
```

x=1

y =2

fun(x,y)

Here, x will match with a, and y will match with b.
These are called positional arguments.

22) Keyword arguments

- When arguments are passed as keyword arguments, the name of the argument is given along with the value. In this case the position of arguments does not matter.
- They can be written in random order.

```
def fun(a,b):  
    print(a+b)
```

```
fun(b=3, a=1)
```

Here, a will get 1 and b will get 3, though the order is changed and sent.

23) What is a void function?

A function that does not return a value is a void function.

ex:

```
def fun(a,b):  
    print(a+b)
```

24) What is a non-void function?

A function that returns a non-empty value is a non-void function.

ex:

```
def fun(a,b):  
    return (a+b)
```

25) What are fruitful functions?

Functions returning values are known as fruitful functions

ex: ref 13)

26) What is a non-fruitful function?

A function that does not return a value is known as a non-fruitful function. They are also called as a void function.

ex: ref 12)

27) A void function internally returns legal empty value _____

Ans: None

28) What is the default return value for a function that does not return any value explicitly?

Ans: None

29) State the use of return keyword in a function. or

What is the purpose of a return statement? How can a function return more than one value?

- A function may or may not return a value when called. The return statement returns the values from the function.
- The return statement helps to return a single value or multiple values or an expression.
- Multiple values are returned as a tuple.

30) What is variable scope?

The program part(s) in which a particular piece of code or a variable can be accessed is known as variable scope.

31) What are the 2 scopes in Python?

Global scope and local scope

32) What is global scope?

A name declared in top level segment (`__main__`) of a program is said to have global scope.

```
def fun():  
    a = 8  
    b = 2  
    c = a+b  
    return c
```

```
ans = fun()
```

Here, ans is said to have global scope and it is a global variable.

33) What is local scope?

A name declared in a function-body is said to have local scope.

```
def fun():
```

```
    a = 8
```

```
    b = 2
```

```
    c = a+b
```

Here, a,b and c are said to have local scope and are local variables.

34) What are global variables?

A global variable is a variable defined in the 'main' program. Such variables are said to have global scope.

35) What are local variables?

A local variable is a variable defined within a function. Such variables are said to have local scope.

36) State the use of global keyword in a function with the help of a suitable example.

- In Python, global keyword allows the programmer to use a global variable in the local context.
- A variable declared inside a function is by default local and a variable declared outside the function is global by default. The keyword global is written inside the function to use its global value.
- Outside the function, global keyword has no effect

ex:

```
def add():
```

```
    global c
```

```
    c = c + 2
```

```
    print("Inside add():", c)    #output:2
```

```
c = 0    # global variable
```

```
add()
```

```
print("In main:", c)    #output:2
```

37) How name or scope resolution is done in Python?

Through LEGB rule

38) What is LEGB Rule?

Python resolves the scope of a name using LEGB rule. i.e., it checks environments in the order: **Local**, **Enclosing**, **Global** and **Built-in**

39) What do you mean by flow of execution?

The order in which the statements in a program are executed is called the flow of execution

40) When will the statements inside a function are executed?

When the interpreter encounters a function definition, the statements inside the function are not executed until the function is called.

41) What is composition?

An expression as part of a larger expression; or a statement as a part of larger statement.

ex:

```
greater((4+5), (3+4))    #arithmetic
```

```
test( a or b)            #logical
```

```
int(str(52)+str(10))    # function composition
```

42) What is the memory area which stores the parameters and local variables of a function call?

Ans: Stack

43) Differentiate between local and global variables.

Global Variables	Local Variables
In Python, a variable that is defined outside any function or any block is known as a global variable	A variable that is defined inside any function or a block is known as a local variable.
It can be accessed in any functions defined onwards.	It can be accessed only in the function or a block where it is defined.
Any change made to the global variable will impact all the functions in the program where that variable can be accessed.	Any change made to the global variable will impact only the function where it is present.

```
def fun():  
    a = 8  
    b = 2  
    c = a+b  
    return c
```

```
ans = fun()
```

Here a,b and c are local variables whereas ans is a global variable.

44) What is lifetime of a variable?

- It is the time for which a variable lives in memory
- The lifetime of Global variables is entire program run
- The lifetime of local variables is only during their function's run

45) Differentiate between actual and formal parameters.

The values being sent through a function-call statement to the corresponding parameters in the function header are called actual parameters.

The values received in the function header, after sent from a function-call statement are called formal parameters

```
def fun(n):  
    print(n)  
  
fun(n)
```

Here, the parameter n present in the function call-statement is called actual argument.

The parameter n received in the function header is called formal argument.

46) Can keyword arguments first before positional arguments in function call statement?

No. First, the positional arguments should be placed before keyword arguments.

47) What is the purpose of a return statement? How can a function return more than one value?

The return statement is used to return either a single value or multiple values from a function. More than values are returned by a function in the form of a tuple.

~~~~~

### LET US REVISE FOR VIVA - FUNCTIONS

#### 1) What is a function?

A **Function** is a subprogram that acts on data and often returns a value.

#### 2) What is the use of functions?

Functions make program handling easier as only a small part of the program is dealt with at a time, thereby avoiding ambiguity. A function in a program can invoke any other function of that program.

### 3) What is main program in python?

By default, Python names the segment with top-level statements (main program) as `_main_`.

### 4) What are arguments or actual parameters or actual arguments?

The values being passed through a function-call statement are called **arguments** (or **actual parameters** or **actual arguments**).

### 5) What are parameters or formal parameters or formal arguments?

The values received in the function definition/header are called **parameters** (or **formal parameters** or **formal arguments**).

### 6) What are the 3 types of arguments supported by Python?

Python supports three types of formal arguments/parameters (i) Positional arguments (Required arguments), (ii) Default arguments and (iii) Keyword (or named) arguments.

### 7) What is positional argument matching?

When the function call statement must match the number and order of arguments as defined in the function definition, this is called the **positional argument matching**.

### 8) What is a default parameter?

A parameter having default value in the function header is known as a **default parameter**.

### 9) Which argument can be skipped in the function call statement?

A default argument can be skipped in the function call statement.

### 10) When will the default parameters be considered?

The default values for parameters are considered only if no value is provided for that parameter in the function call statement.

### 11) What are keyword arguments?

**Keyword arguments** are the named arguments with assigned values being passed in the function call statement.

### 12) Are all the functions return a value?

A function may or may not return a value.

### 13) What happens when a function return multiple values?

A function may also return multiple values that can either be received in a tuple variable or equal number of individual variables.

### 14) What is a non-void function?

A function that returns a non-empty value is a non-void function.

### 15) What are fruitful functions?

Functions returning value are also known as **fruitful functions**.

**16) What do you mean by variable scope?**

The program part(s) in which a particular piece of code or a data value (e.g., variable) can be accessed is known as Variable Scope.

**17) What are the 2 types of scopes?**

In Python, broadly scopes can either be global scope or local scope.

**18) What is LEGB rule?**

Python resolves, the scope of a name using LEGB rule, i.e., it check environments in the order : Local, Enclosing, Global and Built-in.

**19) What is the use of global keyword statement?**

A local variable having the same name as that of a global variable, hides the global variable in its function. The global statement tells a function that the mentioned variable is to be used from global environment. The global statement cannot be undone in a code-block. i.e., once an identifier is declared global, it cannot be reverted to local namespace.

**20) What happens to mutable arguments passed to a function?**

Mutability of arguments / parameter affects the change of value in caller function.

---

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS – See Yes!*

### VIVA HANDOUT – 3

#### FILE HANDLING

**1. What is the difference between files and variables of other data types (like list, dictionary etc.) in terms of data storage ?**

Variables store data in RAM which remains in memory as long as the program is running and gets erased the moment our program gets over. This happens because RAM is volatile in nature.

Files store the data on secondary storage devices like hard disk. The data stored on secondary storage devices is permanent in nature and does not get erased when the program gets over. When input/output operations are performed on files data gets transferred from secondary storage to RAM and vice versa.

**2. What is the difference in opening a file using open() function or using with statement?**

Using with ensures that all the resources allocated to file objects gets deallocated automatically once we stop using the file.

**3. What is the need of closing files?**

Closing a file frees up the resources that were tied with the file. Before closing a file, any material which is not written in file, is flushed off i.e., written to file. So, it is a good practice to close the file once we have finished using it.

**4. What is the difference between the write() and writelines() methods used to write data in a file?**

The write() function write a single string at a time and writelines() methods can be used to write a sequence of strings.

**5. What is the difference between the read() and readlines() methods used to read data from a file?**

The **readline()** method reads one line (i.e. till newline) at a time from a file and returns that line. It reads the file till newline including the newline character.

The file is read sequentially from top i.e. first call to readline() method returns first line, second call returns second line till end of file.

The readline() method returns an empty string when the end of file is reached.

The **readlines()** method reads the entire content of the file in one go and returns a list of lines of the entire file. This method returns an empty value when an end of file (EOF) is reached.

**6. What is the purpose of read(n) method?**

This method reads a string of size (here n) from the specified file and returns it. If size parameter is not given or a negative value is specified as size, it reads and returns up to the end of the file. At the end of the file, it returns an empty string.

**7. What do you understand by the terms pickling and unpickling?**

To store data in binary files, Pickle module is used. Pickle module is used to store any kind of object in file as it allows us to store python objects with their structure. Pickle module supports two major operations: Pickling and Unpickling.

- **Pickling** is the process whereby a Python object is converted into a byte stream.

- **Unpickling** is the process by which a byte stream is converted back into the desired object.

The process of pickling and unpickling a Python data object is known as object **serialization**.

**8. Name the methods which are used to write and read into/from a binary file.**

The dump method of pickle module is used to write objects to binary file. The load method of pickle module is used to read the object from the binary file.

## 9. Compare Text, Binary and CSV files.

| Text files                                                                                                                                                                                                                                                                                    | Binary files                                                                                                                                                                                                                                                                            | CSV files                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Text files</b> are sequence of lines, where each line includes a sequence of characters. All lines are terminated by a special character, called the EOL or <b>End of Line</b> character. Text files are stored in human readable form and they can also be created using any text editor. | <b>Binary files</b> deal with non-text files such as images or exe and store data in binary format i.e. in the form of 0's and 1's which is understandable by the machine. So when we open the binary file in our machine, it decodes the data and displays in a human-readable format. | <b>CSV</b> (Comma Separated Values) is a simple file format used to store tabular data. They are a convenient way to export data from spreadsheets and databases as well as import or use it in other programs. These files are saved with the <b>.csv</b> file extension. In general, the separator character is called a delimiter, and other popular delimiters include the tab (\t), colon (:) and semi-colon (;) characters. |
| <b>Examples of text files are:</b> Web standards: html, XML, CSS, JSON etc.<br><b>Source code:</b> c, app, js, py, java etc.<br><b>Documents:</b> txt, tex, RTF etc.                                                                                                                          | <b>Binary files</b> can range from image files like JPEGs or GIFs, audio files like MP3s or <b>binary</b> document formats like Word or PDF.                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## 10) Name two important functions of CSV module which are used for reading and writing.

|              |                                                                                                                                                                                        |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| csv.reader() | returns a reader object which iterates over lines of a CSV file                                                                                                                        |
| csv.writer() | returns a writer object that converts the user's data into a delimited string. This string can later be used to write into CSV files using the writerow() or the writerows() function. |

## 11) What is the difference between 'a' and 'w' modes?

'w' : Open a file for writing. Creates a new file if it does not exist or truncates the file if it exists.

'a' Open for appending at the end of the file without truncating it. Creates a new file if it does not exist.

## LET US REVISE FOR VIVA - FILE HANDLING

### 1) What is a file?

A file in itself is a bunch of bytes stored on some storage devices like hard-disk, thumb-drive etc.

### 2) What are the 3 types of data files?

The data files can be stored in three ways : (i) Text files  
(ii) Binary files (iii) CSV files

### **3) Differentiate between text file and binary file.**

A text file stores information in ASCII or Unicode characters, where each line of text is terminated, (delimited) with a special character known as EOL (End of Line) character. In text files some internal manipulations take place when this EOL character is read or written.

A binary file is just a file that contains information in the some format in which the information is held in memory, i.e., the file content that is returned to you is raw (with no translation or no specific encoding).

### **4) What is the use open() function?**

The open( ) function is used to open a data file in a program through a file-object (or a file-handle).

### **5) What is the use of file mode?**

A file-mode governs the type of operations (e.g., read / write / append) possible in the opened file i.e., it refers to how the file will be used once it's opened.

### **6) Name the text and csv file modes.**

A text file and a csv file can be opened in these file modes : 'r', 'w', 'a', 'r+', 'w+', 'a+'

### **7) Name the binary file modes?**

A binary file can be opened in these file modes : 'rb', 'wb', 'ab', 'r+b' ('rb+'), 'w+b'('wb+'), 'a+b'('ab+')

### **8) What are the 3 functions for reading a text file?**

The three file reading functions of text files are : read( ), readline( ), readlines( )

### **9) Differentiate between read(), readline() and readlines().**

While read( ) reads some bytes from the file and returns it as a string, readline( ) reads a line at a time and readlines( ) reads all the lines from the file and returns it in the form of a list.

### **10) Differentiate between write() and writelines()**

The two writing functions for Python text files are write( ) and writelines( ). While write( ) writes a string in file, writelines( ) writes a list in a file.

### **11) What is the use of dump() and load() functions?**

Pickle module's dump( ) and load( ) functions write and read into binary files.

### **12) How are the input and output devices treated in file handling?**

The input and output devices are implemented as files, also called standard streams.

### **13) What are the 3 standard streams in file handling?**

There are three standard streams : stdin (standard input), stdout (standard output) and stderr (standard error)

### **14) Differentiate between absolute and relative paths.**

The absolute paths are from the topmost level of the directory structure. The relative paths are relative to current working directory denoted as a dot(.) while its parent directory is denoted with two dots(..)

**15) What is a file-pointer in file handling?**

Every open file maintains a file-pointer to determine and control the position of read or write operation in file.

**16) Differentiate between seek() and tell() functions.**

The seek( ) function places the file pointer at specified position.

The tell( ) function returns the current position of file-pointer in open file.

**17) What are CSV files?**

CSV files are delimited files that store tabular data (data stored in rows and columns as we see in spreadsheets or databases) where comma delimits every value.

**18) What is the use of csv.writer()?**

For writing onto a csv file, user data is written on a csv.writer object which converts the user data into delimited form and writes it on to the csv file.

**19) What is the use of csv.reader()?**

For reading from a csv file, **csv.reader** loads data from the csv file, parses it and makes it available in the form of an iterator wherefrom the records can be fetched row by row.

**20) Why should the CSV files be opened with newline argument?**

CSV files should be opened with newline argument to suppress EOL translation.

**21) Name the functions used for writing onto a CSV file associated with csv.writer object.**

writerow() for single list of data

writerows() for nested list of data

**22) Expand CSV.**

Comma Separated Values

~~~~~

XII COMPUTER SCIENCE

Towards Getting Centum in CS - See Yes!

VIVA HANDOUT - 4

DATA STRUCTURE & STACK

1) What is a stack? Give some real-life examples.

Stack is a data structure that provides temporary storage of data in such a way that the element stored last will be retrieved first.

This method is also called LIFO – Last In First Out. In real life we can think of stack as a stack of copies, stack of plates, stack of rotis etc.

2) Which function is used to push an element in stack?

The append() function.

3) Which function is used to pop an element in stack?

The pop() function

4) Which data structure is used to implement stacks in Python?

List

5) Give applications of stack.

Converting an infix expression to postfix operation and to evaluate the postfix expression.

Reversing strings.

6) Stack is a data structure that follows _____ order

LIFO (Last In First Out)

7) Queue is a data structure that follows _____ order

FIFO (First In First Out)

8) In stack, elements can be added or removed from_____

Top only

9) What is a data structure?

Is a named group of data of different data types which is stored in a specific way and can be processed as a single unit.

10) How data structure is different from data type?

Data type is used to identify type of data. ex: string

Data structure is a group of data of ex: stack

11) Say about the types of data structure

Data structure can be classified into following two types:

Simple Data structure : these are built from primitive data types like integer, float, string or Boolean. Example: Linear List or Array

Complex Data structure : simple data structure can be combined in various way to form more complex data structure. It is of two types:

Linear : are single level data structure i.e. in linear fashion to form a sequence. Example : STACK, QUEUE, LINKED LIST

Non-Linear: are multilevel data structure. Example: TREE and GRAPH.

12) _____ allows to insert and delete items from one end only called TOP.

Stack

13) _____ allows insertion and deletion from different end i.e. insertion from REAR and deletion from FRONT

Queue

14) What is traversal?

TRAVERSAL Processing all data elements one by one

15) Say about the features of STACK.

- + A stack is a collection of data items that can be accessed at only one end, called top.
- + Items can be inserted and deleted in a stack only at the top.
- + The last item inserted in a stack is the first one to be deleted.
- + Therefore, a stack is called a Last-In-First-Out (LIFO) data structure.
- + Two main operations on Stack is PUSH & POP
- + PUSH means inserting new item at top and POP means deleting item from top.

16) STACK TERMS:

Peek : getting the most recent value of stack i.e value at TOP

Overflow : a situation when we are Pushing item in Stack that is full.

Underflow : a situation when we are Popping item from empty stack

17) Name some operations on data structures.

INSERTION, DELETION, TRAVERSAL, SEARCH, MERGE

~~~~~

### **LET US REVISE FOR VIVA – DATA STRUCTURE**

**1) What is a data structure?**

A data structure is a named group of data of different data types which can be processed as a single unit.

**2) How compound data structures are formed? What are its types?**

Simple data structures can be combined in various ways to form compound data structures. The compound data structures may be linear (whose elements form a sequence) and non-linear (which are multi-level).

**3) What is a structure?**

Structure refers to a named collection of variables of different data types.

**4) Differentiate between stacks and queues.**

Stacks are LIFO (Last In First Out) lists where insertions and deletions take place only at one end.

Queues are FIFO (First In First Out) lists where insertions take place at “rear” end and deletions take place at the “front” end.

**5) How linear search is performed?**

In linear search, each element of the array is compared with the given Item to be searched for, one by one.

### **6) What is a 2D list?**

A regular two dimensional list is a list having lists as its elements and each element-list has the same shape i.e., same number of elements (length).

### **7) What is pushing and popping in stack?**

An insertion in a stack is called pushing and a deletion from a stack is called popping.

### **8) Differentiate between OVERFLOW and UNDERFLOW.**

When a stack, implemented as an array, is full and no new element can be accommodated, it is called an OVERFLOW.

When a stack is empty and an element's deletion is tried from the stack, it is called an UNDERFLOW.

~~~~~

XII COMPUTER SCIENCE

Towards Getting Centum in CS - See Yes!

VIVA HANDOUT – 5

SQL

1) What are DDL and DML? Give one command of each.

DDL stands for Data Definition Language. DDL commands are used to create table, changing the structure of table or dropping the table. Example: CREATE, ALTER & DROP

DML stands for Data Manipulation Language. DML commands are used to manipulate the information stored in a table. Like adding new records, changing existing records or deleting the records. Example: INSERT, UPDATE & DELETE

2) What is RDBMS?

RDBMS: A DBMS used to manage Relational Databases is called an RDBMS (Relational Data Base Management System). Some popular RDBMS software available are: Oracle, MySQL,

3) What is data redundancy?

Presence of duplicate data in a database. It leads to data inconsistency.

4) What is data inconsistency?

Data inconsistency occurs when same data present in two different tables does not match.

5) What is a relation?

Table consisting rows and columns is called a relation.

6) What is cardinality and degree?

Cardinality: Total number of rows in a table is called the cardinality of the table. Degree: Total

number of columns of a table is called the Degree of the table.

7) Say the cardinality and degree of the following table:

ANO	ANAME	ADDRESS
101	Nirja Singh	Bangalore
102	Rohan Gupta	Chennai
103	Ali Reza	Hyderabad
104	Rishabh Jain	Chennai
105	Simran Kaur	Chandigarh

Degree: 3, Cardinality: 5

8) What is a Primary Key?

The group of one or more columns used to uniquely identify each row of a relation is called its Primary Key.

9) What is a Candidate Key?

A column or a group of columns which can be used as the primary key of a relation is called a candidate key.

10) What is an Alternate Key? A candidate key of a table which is not made as its primary key is called its Alternate Key.

11) What is a Foreign key?

It is a column in one table which is the primary key of another table. The foreign key links these two tables.

12) What is selection and projection?

The SELECTION operation is a horizontal subset of table/relation and is used to choose the tuples(rows) from a relation that satisfies a given condition. It is denoted by the symbol σ (sigma).

The PROJECTION operation is a horizontal subset of table/relation and used to choose the attributes (columns) from a relation that are given in the attribute list. It is denoted by the symbol Π (pi).

13) What are the different types of clauses used in where command?

CLAUSE/KEYWORD	USAGE
DISTINCT	Used to <i>display distinct values (removes duplicate values)</i> from a column of a table.
WHERE	Used to <i>specify the condition</i> based on which rows of a table are displayed.
BETWEEN	Used to define the <i>range of values</i> within which the column values must fall to make a condition true. Range includes both the upper and the lower values.
IN	Used to select values that match any value in a <i>list of specified values</i> .
LIKE	Used for <i>pattern matching</i> of string data using wildcard characters % and _. % (percentage): It is used to represent any sequence of zero or more characters. _ (underscore): It is used to represent a single character.
IS NULL	Used to select rows in which the specified column is NULL (or is NOT NULL).
ORDER BY	Used to display the selected rows in ascending or in descending order of the specified column/expression.
GROUP BY	Group by clause is used when we need <i>to group the data of the table</i> based on certain type.

14) What is cartesian product?

A Cartesian product is the result of joining every row in one table with every row in another table. It is also called cross join. It has,

columns = sum of the columns of both the tables
rows = product of the rows of both the tables.
It is represented by the symbol X.

ex: SELECT * FROM TABLE1, TABLE2

15) What is union operation in sql?

It is used to merge two different tables with same number and type of columns. It removes duplicate tuples. It is represented with U symbol.

ex: SELECT * FROM TABLE 1

UNION

SELECT * FROM TABLE 2

(UNION ALL includes duplicates)

16) What is a difference between database and table?

Database is a collection of related tables. Table is a collection of related data in the form of rows and columns

17) Give a command to view records/tuples from a table. SELECT

18) Give a command to view only particular columns from a table. SELECT

19) Give a command to insert a new column into a table. ALTER TABLE

20) Give a command to delete a column from a table. ALTER TABLE

21) Give a command to delete a row from a table. DELETE FROM

22) Give a command to insert a new row into a table. INSERT INTO

23) Which command is used to change the existing information of table? UPDATE

24) Which command is used to remove the table from database? DROP TABLE

25) Which command is used to see information like name of columns, data type, size etc. of a table? DESCRIBE OR DESC

26) Which command is used to create i) a table ii) a database?

i) CREATE TABLE student

ii) CREATE DATABASE school

27) Command used to display the list of tables in a database? SHOW TABLES

28) Command used to display the list of databases in SQL.

SHOW DATABASES

29) In mysql, database is called as _____ (SCHEMA)

30) What is the default order of sorting using ORDER BY? Ans. Ascending

31) What is the difference between CHAR and VARCHAR?

Ans. CHAR is fixed length data type. For example if the column „name“ is of CHAR(20) then all name will occupy 20 bytes for each name irrespective of actual data.

VARCHAR is variable length data type i.e. it will occupy size according to the actual length of data

32) What is the Difference between ALTER Table command and UPDATE command?

Ans. ALTER is DDL command and is used for modifying the schema of table like adding new column, modifying column definition, dropping column.

UPDATE is DML command and is used for modifying the existing data of table like changing the mobile number, changing the salary etc.

33) DIFFERENCE BETWEEN

i. HAVING AND WHERE

ii. % AND _

i. HAVING – this clause is used with GROUP BY to filter the group of records. We can use aggregate functions with HAVING.

WHERE – this clause is used to apply condition on all the rows of table. We cannot use aggregate functions with WHERE.

ii. % is a wildcard character used with LIKE and it is used for substituting multiple characters while matching the pattern. Matching text can be of any length

EX: SELECT * FROM STUDENT WHERE NAME LIKE “%s%”

_ (underscore) is also a wildcard character used with LIKE but it substitutes only a single character at a given position while matching the pattern. Length will be fixed.

EX: SELECT * FROM STUDENT WHERE NAME LIKE “s_”

34. What happens when the drop table command is executed?

All the records as well as the entire structure of the table is deleted.

35. What are aggregate functions?

Aggregate functions work on multiple rows. There are 5 types of aggregate functions:

Aggregate function	Purpose
MAX()	To find the maximum value under the specified column
MIN()	To find the minimum value under the specified column
AVG()	To find the average of values under the specified column
SUM()	To find the sum of values under the specified column
COUNT()	To count the values under the specified column

36) What is the difference between COUNT() and COUNT(*) function?

Ans. COUNT() function will count number of values in any column excluding the NULLs

COUNT(*) will count number of rows in query output including NULLs

37) Expand SQL: Structured Query Language

38) What is a join?

It is used to combine rows from two or more tables, based on a related column between them. It could be of equi-join, natural join, cross join.

39) What is the difference between Equi-Join and Natural Join?

Ans. In Equi join we compare value of any column from two tables and it will return matching rows. In Equi-join common column appears twice in output because we fetch using (*) not by specifying column name. for e.g.

SELECT * FROM TABLE1, TABLE2 WHERE TABLE1.ID=TABLE2.ID

In natural join, the matching rows of the query will be returned. In natural join column will appear only once in output.

```
SELECT NAME, TOTAL FROM TABLE1, TABLE2  
WHERE TABLE1.ID=TABLE2.ID
```

40) What is a constraint?

Constraint can be used to specify the limit on the data type of table. Constraint can be specified while creating or altering the table statement. Sample of constraint are.

NOT NULL, CHECK, DEFAULT, UNIQUE,
PRIMARY KEY,

FOREIGN KEY

41) What is a query?

A query is a question or enquiry or request about a set of data.

42) Which operator is used in query for pattern matching?

LIKE operator is used for pattern matching, and it can be used as -.

% – Matches zero or more characters.

_(Underscore) – Matching exactly one character.

43) What is data integrity?

Data Integrity defines the accuracy and consistency of data stored in a database.

44) What is referential integrity?

Referential Integrity is a constraint in the database that enforces the relationship between two tables. It is done through foreign key.

45) What is DBMS?

A Database Management System ([DBMS](#)) is a program that controls creation, maintenance and use of a database.

46) What is RDBMS?

RDBMS stands for Relational Database Management System. RDBMS store the data into

the collection of tables, which is related by common fields between the columns of the table.

47) Which option of ORDER BY clause is used to arrange the output in descending order?

Ans. DESC

48) What is a tuple in SQL? a record

49) Column names must be unique in SQL?

Yes.

50) Does SQL maintain sort the records while inserting?

No. It maintains the order as they are inserted.

51) Horizontal subset of table is known as _____

Tuple

52) Vertical subset of table is known as _____

Column

53) For each attribute of a relation, there is a set of permitted values, called the _____ of that attribute.

Domain

54) Give any 2 advantages of using Database/SQL/RDBMS?

i) Sharing of Data ii) Consistency of Data

iii). Eliminate Redundancy of data

iv) Security of Data

~~~~~

### LET US REVISE FOR VIVA – SQL

#### 1) What is a database? What is database management system?

A Collection of data is referred to as database and a database management system is basically a computer based record keeping system.

## 2) What is the use of DBMS?

Database systems (DBMS) help reduce data redundancy, data inconsistency and facilitate sharing of data, standardization of data and data security.

## 3) What are relations?

A relational data model organizes the data into tables known as relations.

## 4) What is MySQL?

MySQL is a free, open-source Relational Database Management System.

## 5) What is the use of DDL commands?

The DDL commands are used to define or redefine schema objects.

## 6) What is the use of DML commands?

The DML commands are used to manipulate data in existing schema objects.

---

## 7) Short Definitions:

|                         |                                                                                              |
|-------------------------|----------------------------------------------------------------------------------------------|
| <b>Alternate Key</b>    | : A candidate key which is not primary key                                                   |
| <b>Attribute</b>        | : A column of a relation.                                                                    |
| <b>Base Table</b>       | : A table having an independent existence; represented in storage by a distinct stored file. |
| <b>Candidate Keys</b>   | : Attribute combinations in a relation that can serve as a primary key.                      |
| <b>Cardinality</b>      | : Number of tuples in a relation.                                                            |
| <b>Data Redundancy:</b> | Duplication of data.                                                                         |

**Database** : Collection of interrelated data.

**Database System** : A Combination of a computer based recordkeeping system and a database

**Degree** : Number of attributes in a relation.

**Domain** : A pool of values from which the actual values appearing in a given column are drawn.

**Primary Key** : Set of one or more attributes that can uniquely tuples within a relation.

**Relation** : Data arranged in rows and columns and having certain properties.

**Tuple** : A row of a relation.

**Data Dictionary** : A file containing metadata i.e., data about data.

**SQL** : Structured Query Language. A Language to operate upon RDBMSs.

---

## 8) What is a NULL value?

Column having no value is said to have NULL value.

## 9) What is the use of SELECT command?

The SELECT command of SQL lets you make queries on the database.

## 10) What is the use of DISTINCT keyword?

The DISTINCT keyword eliminates redundant data.

## 11) What is the use of dummy table?

A small work table **Dummy** can be used for making simple calculations.

## 12) What is the use of WHERE clause?

The WHERE clause is used to select specific rows.

### 13) What is the use of logical operators?

The logical operators OR, AND and NOT are used for making range checks in queries.

### 14) What is the use of IN operator?

The IN operator is used for selecting values from a list of values.

### 15) What is the use of LIKE operator?

The LIKE operator is used for making character comparisons using strings.

### 16) What is the use of IS NULL operator?

The null value in a column can be searched for in a table using IS NULL in the WHERE clause.

### 17) What is the use of ORDER BY clause?

The ORDER BY clause is used to sort the results of a query.

### 18) What is the use of CREATE TABLE command?

**CREATE TABLE** command is used to create tables in database.

### 19) How do we insert data in a table?

**INSERT INTO** command is used to insert data in the table.

### 20) How do we insert NULL values?

Nulls are inserted in the table by typing NULL without quotes.

### 21) What is the use of UPDATE command?

Existing data in tables can be changed with UPDATE command.

### 22) What is the use of DELETE command?

Tuples or rows in a table can be deleted using **DELETE** command.

### 26) What is the use of ALTER TABLE command?

**ALTER TABLE** command is used to alter the definition of already created tables. With **ALTER TABLE**, new columns can be added, existing columns can be redefined.

### 27) What is the DROP TABLE command?

**DROP TABLE** command drops a table from a database.

### 28) What is the use of GROUP BY clause?

The GROUP BY clause combines all those records that have identical value in a particular field or a group of fields.

### 29) How does GROUP BY react on NULL VALUES?

All fields containing a NULL value are considered to have a value and are grouped to have a value

### 30) What is the use of HAVING clause?

The HAVING clause is used to specify filtering condition for groups.

### 31) Differentiate between WHERE and HAVING clauses?

The difference between WHERE and HAVING clause is that WHERE conditions are applicable on individual rows whereas HAVING conditions are applicable on groups as formed by GROUP BY clause.



## XII COMPUTER SCIENCE

*Towards Getting Centum in CS - See Yes!*

### VIVA HANDOUT - 6

#### CONNECTIVITY

##### 1) Name of connector to establish bridge between Python and MySQL

mysql.connector

##### 2) Which function of connection is used to check whether connection to mysql is successfully done or not? `con.is_connected()`

##### 3) What is a cursor?

The component act as a container to hold all the data returned from the query and from there we can fetch data one at a time.

##### 4) What is the difference in `fetchall()` and `fetchone()`?

Ans `fetchall()` function is used to fetch all the records from the cursor in the form of tuple.

`fetchone()` is used to fetch one record at a time. Subsequent `fetchone()` will fetch next records.

If no more records to fetch it return None

##### 5) Which attribute of of cursor is used to get number of records stored in cursor?

`mycursor.rowcount`,

`%` and `{}`

##### 7) Which function is used to fetch n number of records from cursor? `fetchmany()`

##### 8) Which cursor function is used to send query to connection? `execute()`

##### 9) What are the five major steps for connecting MySQL and Python?

- Import MySQL connector
- Open a connection to a database
- Create a cursor object
- Execute a query
- Close the connection

##### 10) How do we create a connection object?

Connection object is created with the help of `connect()` function. The `connect()` function of `mysql.connector` package is used to establish a connection to MySQL database. For example:

```
conn1 =  
mysql.connector.connect(host='localhost',  
database='test', user='root', password='tiger')
```

##### 11) How do we execute SQL query through Python?

To execute SQL queries through Python, cursor object is used along with `execute()` method. For example: `cursor1.execute("select * from emp;")`

##### 12) What is the purpose of rowcount parameter?

It returns the number of rows affected by the last `execute` method for the same cur object.

##### 13) How do we create a cursor object?

You can create a cursor object using the `cursor()` method of the connection object/class. The cursor object is used to execute statements to perform database operations. For example:

```
cursor1 = conn1.cursor()
```

Here, `cursor1` is a cursor object created using connection object `conn1` using `cursor()` method

##### 14) The result of `fetchone()`, `fetchall()` etc. methods will be in the form of 6) Which Symbols are used for passing parameterized query for execution to cursor? Tuple

##### 15) Method To see the effects of INSERT/UPDATE/DELETE FROM commands: `mycursor.commit()`

## **LET US REVISE FOR VIVA – CONNECTIVITY**

### **1)How do we connect python program with DBMS?**

To connect to a database from within a programming application, you need a framework that facilitates communication between two different genres of software (programming application and DBMS). This framework is created through the `mysql.connector` module.

### **2)What is the use of `mysql.connector` module?**

To connect from Python to MySQL, you need a library called **`mysql connector`**.

### **3)What are the basic steps of a database connectivity?**

Steps to create a database connectivity Python application are :

**Step 1** Start Python

**Step 2** Import the packages required for database programming

**Step 3** Open a connection

**Step 4** Create a cursor instance

**Step 5** Execute a query

**Step 6** Extract data from result set

**Step 7** Clean up the environment

### **4) What is a database connection?**

A database Connection object controls the connection to the database. It represents a unique session with a database connected from within a script / program.

### **5) What is a database cursor?**

A Database Cursor is a special control structure that facilitates the row by row processing of records in the resultset, i.e., the set of records retrieved as per query.

### **6) What is a result set?**

The result set refers to a logical set of records that are fetched from the database by executing an SQL query and made available to the application program.

### **7) What are the basic methods from `mysql.connector` module used in connectivity?**

**`connect( )`** method for establishing database connection, **`cursor( )`** to create a cursor and **`execute( )`** to execute an SQL query.

### **8) How do we fetch records from a result set?**

To fetch records from a result set, you can use **`fetchone( )`**, **`fetchmany( )`** and **`fetchall( )`** methods.

### **9) When should we use `commit()` method?**

For INSERT, UPDATE and DELETE queries, you must run **`commit( )`** with the connection object.

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS - See Yes!*

### VIVA HANDOUT - 7

#### COMMUNICATION & NETWORKING

##### 1) Say about evolution of networking:

i) ARPANET: In 1969, The US govt. formed an agency named ARPANET (Advanced Research Projects Agency NETwork) to connect computers at various universities and defense agencies. The main objective of ARPANET was to develop a network that could continue to function efficiently even in the event of a nuclear attack.

ii) Internet (INTERconnection NETwork): The Internet is a worldwide network of computer networks. It is not owned by anybody.

iii) Interspace: InterSpace is a client/server software program that allows multiple users to communicate online with real - time audio, video and text chat in dynamic 3D environments.

##### 2) Expand the following Terms:

1. ARPANET: Advanced Research Projects Agency NETwork
2. NSFnet: National Science Foundation network
3. Internet: Inter networking
4. TCP/IP: Transmission Control Protocol / Internet Protocol
5. NIU: Network Interface Unit
6. TAP: Terminal Access Point
7. NIC: Network Interface Card
8. MAC Address: Media Access Control Address
9. bps: bits per second
10. Bps: Bytes per second
11. kbps: kilo bits per second
12. Kbps: Kilo bytes per second

13. mbps: mega bits per second
14. Mbps: mega bytes per second
15. kHz: kilohertz
16. MHz: megahertz
17. GHz: gigahertz
18. THz: terahertz
19. LAN: Local Area Network
20. VGM: Voice Grade Medium
21. DGM: Data Grade Medium
22. UTP: Unshielded Twisted Pair Cable
23. CAT1: Category 1 UTP cable (same acronym for other CATs)
24. STP: Shielded Twisted Pair Cable
25. LEDs: Light Emitting Diodes
26. LDs: Laser Diodes
27. PDAs: Personal Digital Assistants / Personal Data Assistant
28. WAN: Wide Area Network
29. MAN: Metropolitan Area Network
30. PAN: Personal Area Network
31. P-P link: Point to Point Link
32. Modem: Modulator - Demodulator
33. AM: Amplitude Modulation
34. FM: Frequency Modulation
35. PM: Phase Modulation
36. A/F: Audio Frequency
37. RTS: Request To Send
44. RJ-45: Registered Jack-45
46. RJ-11: Registered Jack-11
51. ISP: Internet Service Provider
52. PCI-Express: Peripheral Component Interconnect Express
53. PCMCIA WiFi Card: Personal Computer Memory Card International Association
54. OSI model: Open Systems Interconnection Model
55. IP address: Internet Protocol Address
58. ISDN: Integrated Services Digital Network
64. NFS: Network File System
65. HTTP: Hypertext Transfer Protocol
66. WWW: World Wide Web
67. URI: Uniform Resource Identifier
68. URL: Uniform Resource Locator
69. URN: Uniform Resource Name
70. MIME: Multipurpose Internet Mail Extensions
71. CPU: Central Processing Unit
72. SMTP: Simple Mail Transfer Protocol

73. NNTP: Network News Transfer Protocol  
 74. FTP: File Transfer Protocol  
 75. WAIS: Wide Area Information Server  
 76. ASCII: American Standard Code for Information Interchange  
 77. SLIP: Serial Line Internet Protocol  
 78. PPP: Point to Point Protocols  
 82. GUI: Graphical User Interface  
 83. GSM: Global System for Mobile communications  
 84. SIM: Subscriber Identification Module / Subscriber Identity Module  
 85. TDMA: Time Division Multiple Access  
 89. CDMA: Code-Division Multiple Access  
 90. WLL/WiLL: Wireless in Local Loop  
 92. PSTN: Public Switched Telephone Network  
 93. GPRS: General Packet Radio Service  
 95. 4G: Fourth Generation (same for 1G, 2G and 3G)  
 96. 4G wireless technology is also referred to as 'MAGIC': Mobile multimedia, Any-where, Global mobility solutions over, Integrated wireless and Customized services  
 97. HDTV: High Definition Television  
 98. LTE: Long Term Evolution  
 99. SMS: Short Message Service  
 100. SMSC: Short Message Service Center  
 102. IRC: Internet Relay Chat  
 103. SIP: Session Initiation Protocol  
 105. VoIP: Voice over Internet Protocol  
 106. Wi-Fi: Wireless Fidelity  
 108. WiMAX: Worldwide Interoperability for Microwave Access  
 109. EDGE: Enhanced Data rates for GSM Evolution  
 110. Telnet: Teletype Network  
 111. HTML: Hypertext Markup Language  
 112. DNS: Domain Name System  
 113. POP: Post Office Protocol  
 115. XML: eXtensible Markup Language  
 116. DHTML: Dynamic HTML  
 117. VBScript: Visual Basic Script  
 118. ASP: Active Server Pages  
 119. PHP: Hypertext Preprocessor  
 120. PERL: Practical Extraction and Reporting Language  
 121. JSP: Java Server Pages

123. LINUX: Linus' Unix (after the inventor of Linux, Linus)  
 124. SSL: Secure Socket Layer  
 125. IE: Internet Explorer  
 126. IT Act 2000: Information Technology Act, 2000  
 127. UNCITRAL: United Nation's Commission for International Trade related laws  
 128. IT Act 2008: The Information Technology Amendment Act, 2008  
 133. C.D.ROM: Compact Disc Read-Only Memory  
 134. IPR: Intellectual Property Rights  
 135. FAT: File Allocation Tables  
 136. E-mail: Electronic Mail  
 137. GNU: GNU's Not Unix  
 138. Cc: Carbon Copy  
 139. Bcc: Blind Carbon Copy  
 141. IMAP: Internet Mail Access Protocol  
 143. UMTS: Universal Mobile Telephone System  
 144. CSS: Cascading Style Sheets  
 145. WAP: Wireless Application Protocol  
 146. IIS: Internet Information Services  
 150. W3C: World Wide Web Consortium

### 3) Differentiate between Star Topology and Bus Topology.

| STAR                                                                        | BUS                                                         |
|-----------------------------------------------------------------------------|-------------------------------------------------------------|
| Each node is connected to each other through central node i.e. Switch / Hub | Each node is connected to each other through backbone cable |
| If Switch/Hub fails entire network will be stopped                          | If Backbone cable fails entire network will be stopped      |
| Data transmission speed is fast                                             | Data transmission speed is slow as compare to Star          |
| More cable is required                                                      | Less cable is required                                      |
| Fault Diagnosis is easy                                                     | Fault Diagnosis is difficult                                |
| Adding/Removal of node is easy                                              | Difficult as compare to Star to Add/Remove nodes            |
| High cost setup                                                             | Low cost setup as compare to Star                           |
| Signal transmission is not unidirectional                                   | Signal transmission is unidirectional                       |

### 4) Say about switching techniques:

Switching techniques are used for transmitting data across networks.

1. Circuit Switching: In the Circuit Switching technique, first, the complete end-to-end transmission path between the source and the destination computers is established and then the message is transmitted through the path. The main advantage of this technique is guaranteed delivery of the message. Mostly used for voice communication.

2. Message Switching: In the Message switching technique, no physical path is established between sender and receiver in advance. This technique follows the store and forward mechanism.

3. Packet Switching: In this switching technique fixed size of packet can be transmitted across the network.

### 5) Define the following:

**Data channel :-** The information / data carry from one end to another in the network by channel.

**Baud & bits per second (bps) :-** It's used to measurement for the information carry of a communication channel.

#### Measurement Units :- bit

1 Byte= 8 bits

1 KBPS ( Kilo Byte Per Second)= 1024 Bytes

1 Kbps (kilobits Per Second) = 1024 bits

1 Mbps ( Mega bits Per Second )=1024 Kbps

**Bandwidth :-** It is amount of information transmitted or receives per unit time.

### 6) Two types of transmission media:

wired media(guided media) : twisted pair cable, co-axial cable, optical fiber cable

wireless media(unguided media): radio wave, micro wave, infrared, satellite, laser etc.

### 7) Say about twisted pair cable.

Twisted pair cable: - It consists of two identical 1 mm thick copper wires insulated and twisted together. The twisted pair cables are twisted in order to reduce crosstalk and electromagnetic induction.

Advantages:

(i) It is easy to install and maintain.

(ii) It is very inexpensive

Disadvantages:

(i) It is incapable to carry a signal over long distances without the use of repeaters.

(ii) Due to low bandwidth, these are unsuitable for broadband applications.

### 8) Say about co-axial cable:

It consists of a solid wire core surrounded by one or more foil shields, each separated from the other by some kind of plastic insulator. It is mostly used in the cable wires.

Advantages:

(i) Data transmission rate is better than twisted pair cables.

(ii) It provides a cheap means of transporting multi-channel television signals around metropolitan areas.

Disadvantages:

(i) Expensive than twisted pair cables.

(ii) Difficult to manage and reconfigure.

### 9) Say about optical fiber cable.

An optical fiber consists of thin glass fibers that can carry information in the form of visible light.

Advantages:

(i) Transmit data over long distance with high security.

(ii) Data transmission speed is high

(iii) Provide better noise immunity

(iv) Bandwidth is up to 10 Gbps.

Disadvantages:

(i) Expensive as compared to other guided media.

(ii) Need special care while installation

### 10) Say about radio waves.

Radio Wave an electromagnetic wave with a wavelength between 0.5 cm and 30,000m. The transmission making use of radio frequencies is termed as radio-wave transmission

Advantages:

- (i) Radio wave transmission offers mobility.
- (ii) It is cheaper than laying cables and fibers.
- (iii) It offers ease of communication over difficult terrain.

Disadvantages:

- (i) Radio wave communication is insecure communication.
- (ii) Radio wave propagation is susceptible to weather effects like rains, thunder storms etc.

### **11) Say about micro waves.**

The Microwave transmission is a line of sight transmission. Microwave signals travel at a higher frequency than radio waves and are popularly used for transmitting data over long distances.

Advantages:

- (i) It is cheaper than laying cable or fiber.
- (ii) It has the ability to communicate over oceans.

Disadvantages:

- (i) Microwave communication is an insecure communication.
- (ii) Signals from antenna may split up and transmitted in different way to different antenna which leads to reduce to signal strength.

### **12) Say about infrared waves.**

The infrared light transmits data through the air and can propagate throughout a room, but will not penetrate walls. It is a secure medium of signal transmission. The infrared transmission has become common in TV remotes, automotive garage doors, wireless speakers etc.

### **13) What is a Modem?**

A MODEM (MODulator DEModulator) is an electronic device that enables a computer to transmit data over telephone lines. There are two types of modems, namely, internal modem and external modem.

### **14) What is RJ45 connector?**

The RJ-45(Registered Jack) connectors are the plug-in devices used in the networking and telecommunications applications. They are used primarily for connecting LANs, particularly Ethernet.

### **15) What is Ethernet Card?**

It is a hardware device that helps in connection of nodes within a network.

### **16) What is Hub?**

A hub is a hardware device used to connect several computers together. Hubs can be either active or i passive.

### **17) What is Switch?**

A switch (switching hub) is a network device which is used to interconnect computers or devices on a network. It filters and forwards data packets across a network. The main difference between hub and switch is that hub replicates what it receives on one port onto all the other ports while switch keeps a record of the MAC addresses of the devices attached to it.

### **18) What is a Gateway?**

A gateway is a device that connects dissimilar networks.

### **19) What is a Repeater?**

A repeater is a network device that amplifies and restores signals for long distance transmission.

### **20) What is topology?**

Topology refers to the way in which the workstations attached to the network are interconnected.

### **21) Say about Types of Networks:**

**LAN (Local Area Network):** A Local Area Network (LAN) is a network that is confined to a relatively small area. It is generally limited to a geographic area such as writing lab, school or building. It is generally privately owned networks over a distance not more than 5 Km.

**MAN (Metropolitan Area Network):** MAN is the networks cover a group of nearby corporate offices or a city and might be either private or public.

**WAN (Wide Area Network):** These are the networks spread over large distances, say across countries or even continents through cabling or satellite uplinks are called WAN.

**PAN (Personal Area Network):** A Personal Area Network is computer network organized around an individual person. It generally covers a range of less than 10 meters. Personal Area Networks can be constructed with cables or wirelessly.

## 22) What is a protocol?

A protocol means the rules that are applicable for a network.

## 23) What is use of HTTP?

- Hypertext Transfer Protocol (HTTP) is a communications protocol for the transfer of information on the intranet and the World Wide Web. HTTP is a request/response standard between a client and a server. A client is the end-user; the server is the web site.

## 24) What is use of FTP?

- FTP (File Transfer Protocol) is the simplest and most secure way to exchange files over the Internet.

## 25) What is use of TCP/IP?

- TCP/IP (Transmission Control Protocol / Internet Protocol) TCP - is responsible for verifying the correct delivery of data from client to server. Data can be lost in the intermediate network. TCP adds support to detect errors or lost data and to trigger retransmission until the data is correctly and completely received.

IP - is responsible for moving packet of data from node to node. IP forwards each packet based on a four byte destination address (the IP number). The Internet authorities assign ranges of numbers to different organizations. The organizations assign groups of their numbers to departments. IP

operates on gateway machines that move data from department to organization to region and then around the world.

## 26) What is Telnet?

It is an older internet utility that lets us log on to remote computer system.

## 27) What is email?

Electronic mail is sending and receiving messages by computer.

## 28) What is chatting?

Online textual talk in real time, is called Chatting.

## 29) What is SMS?

Short Message Service: SMS is the transmission of short text messages to and from a mobile phone, fax machine and or IP address.

## 30) What is EDGE?

Enhanced Data rates for Global Evolution : is a radio based high speed mobile data standard.

## 31) What is WLL(Wireless in Local Loop) ?

WLL is a system that connects subscribers to the public switched telephone network using radio signals as a substitute for other connecting media.

## 32) Define the following:

**Cookies:** Cookies are the text messages sent by a web server to the web browser primarily for identifying the user.

**Firewall:** A firewall is used to control the traffic between computer networks. It intercepts the packets between the computer networks and allows only authorized packets to pass.

**Cyber Law:** Cyber law refers to all the legal and regulatory aspects of Internet and the World Wide Web.

**Cyber Crimes:** Cyber crime involves the usage of the computer system and the computer network for criminal activity.

**Hacking:** Hacking is an unauthorized access to computer in order to exploit the resources.

**WWW:** The World Wide Web or W3 or simply the Web is a collection of linked documents or pages, stored on millions of computers and distributed across the Internet.

**HTML (Hyper Text Markup Language):-** HTML is a computer language that describes the structure and behavior of a web page. This language is used to create web pages.

**XML (eXtensible Markup Language):-** Extensible Markup Language (XML) is a meta language that helps to describe the markup language.

**HTTP (Hyper Text Transfer Protocol):-** A protocol to transfer hypertext requests and information between servers and browsers.

**Domain Names:** A domain name is a unique name that identifies a particular website and represents the name of the server where the web pages reside.

**URL:-** The Uniform Resource Locator is a means to locate resources such as web pages on the Internet.

**Website:** A collection of related web pages stored on a web server is known as a website.

**Web browser:** A software application that enables to browse, search and collect information from the Web is known as Web browser.

**Web Servers:** The web pages on the Internet are stored on the computers that are connected to the Internet. These computers are known as web servers.

**Web Hosting:** - Web Hosting or website hosting is the service to host, store and maintain the websites on the World Wide Web.

### **33) Where the following devices be placed:**

**HUB / SWITCH:-** In all the wings

**REPEATER:** It is used if the distances higher than 70 m. It regenerates data and voice signals.

**ROUTER:** When one LAN will be connected to the other LAN.

**Server:** Server should be placed in the building where the number of computers is maximum.

### **34) What are the main functions of TCP?**

The TCP does the following activities

- It breaks the data into packets that the network
- Verifies that all the packets arrived at the destination
- Reassembles the data

### **35) Say about tree topology.**

**The TREE Topology:** - The tree topology combines the characteristics of the linear bus and the star topologies. It consists of groups of star — configured workstations connected to a bus backbone cable.

### **36) Say some video conferencing protocols:**

SLIP/PPP, H.323, VoIP

### **37) Explain about web hosting.**

**Web hosting** is a type of internet hosting service by companies that allows organizations and individuals to post a website or web page onto the server, which can be viewed by everyone on the internet.

**There are four main types of web hosting:**

**Shared hosting** – share by multiple domains/web sites.

**VPS (virtual private server) hosting** - The main server is split into multiple virtual servers—hence the name. These virtual servers can be customized by individual websites.

**Dedicated hosting** - Websites being hosted on a dedicated server have complete technical control over the server settings. We choose the software, configurations, and anything else we need.



**Cloud hosting** - Cloud hosting plans come with multiple remote servers. Each server has different responsibilities. If one of the servers is compromised or has a problem, the other servers on the network will take over those responsibilities and pick up the slack.

**Free hosting:** web hosting services free of charge. Such free web hosting sites may place certain restrictions on the files such as size of the files may be limited, only certain formats allowed etc.

~~~~~

LET US REVISE FOR VIVA – NETWORKING

1) What is a network?

A network is a collection of interlined computers by means of a communication system.

2) What are the advantages of networking?

The networks facilitate resource sharing, increased reliability, reduced costs, and increased and fast communication.

3) Say about the evolution of internet.

Today's Internet has evolved from ARPANET (Advanced Research Projects NETwork) of U.S. Department of Defense along with some other networks such as NSFnet and other private networks.

4) What is internet?

Internet is a worldwide network of computer networks.

5) What is interspace?

InterSpace is said to be the future of Internet. It is a client/server software program that allows multiple users to communicate online with real-time audio, video and text chat in dynamic 3D environments.

6) What is the use of switching techniques?

Switching techniques are used for transmitting data across networks.

7) What are the different types of switching techniques?

Various switching techniques are : **circuit switching** (where complete physical connection is setup prior to communication), **message switching** (which follows store and forward principle for complete messages), and **packet switching** (which follows store and forward principle for packets – a message is divided into fixed sized packets).

8) Give example for transmission media:

A network can have any of these transmission media or connecting media : Wired: twisted pair cable, coaxial cable, optical fibre Wireless: microwave, radiowave, satellite, etc.

9) What are the types of network?

On the basis of geographical spread, networks can be classified into LAN (Local Area Network), WAN (Wide Area Network), and PAN (Personal Area Network)

10) What is LAN?

Small computer networks that are confined to a localised area e.g. an office, a building etc., are called LANs.

11) What is MAN?

MANs are the networks spread over a city.

12) What is WAN?

A WAN is a group of computers that are separated by large distances and tied together. It can even be a group of LANs that are spread across several locations and connected together to look like a big LAN.

13) What do you mean by topology?

The pattern of interconnection of nodes in a network is called topology.

14) Give example for topologies.

Most popular topologies are star, bus, ring and tree.

15) What is a modem?

A modem is a computer peripheral that allows you to connect and communicate with other computers via telephone lines. Its expansion is Modulator Demodulator

16) What is the use of RJ-45?

RJ-45 (Registered Jack – 45) is an eight-wire connector, which is commonly used to connect computers on LANs especially Ethernets.

17) What is Ethernet?

Ethernet is a LAN architecture developed by Xerox Corp along with DEC and intel. It uses either a bus or star topology and supports data transfer rates of upto 10 Mbps.

18) What is Ethernet Card?

Computers part of Ethernet are connected through a special card called Ethernet card.

19) What is a hub? What are its types?

A hub is a hardware device used to connect several computers together. Hubs can be either active hubs or passive hubs.

20) What is a switch?

A switch is a device that is used to segment networks into different subnetworks called subnets or LAN segments

21) What is a backbone?

A backbone network is a network that is used as a backbone to connect LANs together to form a WAN.

22) What is a repeater?

A repeater is a device that amplifies a signal being transmitted on the network.

23) What is a bridge?

A bridge is a device that links two networks together.

24) What is a router?

A router is a device that works like a bridge but can handle different protocols.

25) What is a gateway?

A gateway is a device that connects dissimilar networks.

26) What is a protocol?

A protocol is a set of standardized rules for data packets, techniques for detecting and correcting errors and so on.

27) Expand: HTTP, FTP, TCP/IP

HTTP (Hyper Text Transfer Protocol), FTP (File Transfer Protocol), TCP/IP (Transmission Control Protocol / Internet Protocol)

28) What is wireless communication?

Wireless communication is simply data communication without the use of landlines.

29) What is WLL?

WLL (Wireless in Local Loop) is a system that connects subscribers to the Public switched telephone network (PSTN) using radio signal as a substitute for other connecting media.

30) What is SMS?

SMS (Short Message Service) is the transmission of short text message to and from a mobile phone, fax machine and/or IP address.

31) What is e-mail?

Email (Electronic Mail) is sending and receiving messages by computer. Protocols used SMTP, POP3, IMAP

32) What is chatting?

Online textual talk, in real time, is called chatting. Protocols used: IRC

33) What is video conferencing?

A two-way videophone conversation among multiple participants is called video-conferencing.

Protocols used: H.323, VoIP etc.

34) What is Telnet?

Remote login (Telnet) is the process of accessing a network from a remote place without actually being at the actual place of working.

35) What is WWW?

The World Wide Web (WWW) is a set of protocols that allows you to access any document on the net (short for internet) through a naming system based on URL's (Uniform Resource Locators).

36) Differentiate between web browser and web server.

A Web Browser is a WWW client that navigates through the World Wide Web and displays web pages. A Web Server is a WWW Server that responds to the requests made by web browsers.

37) What are URL and domain name?

A URL (Uniform Resource Locator) specifies the distinct address for each resource on the Internet. An Internet address which is character based is called a Domain Name.

38) Differentiate between web site and web page.

A location on a net server is called a Web Site. A document that uses HTTP is called a Web Page.

39) What is Firewall?

The system designed to prevent unauthorized access to or from a private network is called Firewall.

40) What is a database? What is database management system?

A Collection of data is referred to as database and a database management system is basically a computer based record keeping system.

41) What is the use of DBMS?

Database systems (DBMS) help reduce data redundancy, data inconsistency and facilitate sharing of data, standardization of data and data security.

42) What are relations?

A relational data model organizes the data into tables known as relations.

43) What is MySQL?

MySQL is a free, open-source Relational Database Management System.

44) What is the use of DDL commands?

The DDL commands are used to define or redefine schema objects.

45) What is the use of DML commands?

The DML commands are used to manipulate data in existing schema objects.

~~~~~

*"Our greatest weakness lies in giving up. The most certain way to succeed is always to try just one more time." – Thomas A. Edison*

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS – See Yes!*

*Easy Theory*

### NCERT FOCUS – GETTING STARTED WITH PYTHON

#### 1) Write some features of Python.

- Python is a high level language.
- It is a free and open source language.
- It is an interpreted language,
- Python programs are easy to understand as they have a clearly defined syntax and relatively simple structure.
- Python is case-sensitive. For example, NUMBER and number are not same in Python.
- Python is portable and platform independent, means it can run on various operating systems and hardware platforms.
- Python has a rich library of predefined functions.
- Python is also helpful in web development. Many popular web services and applications are built using Python.
- Python uses indentation for blocks and nested blocks.

#### 2) What are the 2 ways to use the Python interpreter?

There are two ways to use the Python interpreter:

- a) Interactive mode
- b) Script mode

Interactive mode allows execution of individual statement instantaneously. Whereas, Script mode

allows us to write more than one instruction in a file called Python source code file that can be executed.

But in the interactive mode, we cannot save the statements for future use and we have to retype the statements to run them again.

#### 3) How python files are saved?

Python scripts are saved as files where file name has extension ".py". By default, the Python scripts are saved in the Python installation folder.

#### 4) Write about Python keywords.

Keywords are reserved words. Each keyword has a specific meaning to the Python interpreter, and we can use a keyword in our program only for the purpose for which it has been defined. As Python is case sensitive, keywords must be written exactly as given

|        |          |         |          |        |
|--------|----------|---------|----------|--------|
| False  | class    | finally | is       | return |
| None   | continue | for     | lambda   | try    |
| True   | def      | from    | nonlocal | while  |
| and    | del      | global  | not      | with   |
| as     | elif     | if      | or       | yield  |
| assert | else     | import  | pass     |        |
| break  | except   | in      | raise    |        |

#### 5) Where are identifiers? State the rules to define them.

Identifiers are names used to identify a variable, function, or other entities in a program. The rules for naming an identifier in Python are as follows:

- The name should begin with an uppercase or a lowercase alphabet or an underscore sign (\_).
- This may be followed by any combination of characters a–z, A–Z, 0–9 or underscore (\_). Thus, an identifier cannot start with a digit.
- It can be of any length. (However, it is preferred to keep it short and meaningful).
- It should not be a keyword or reserved word given
- We cannot use special symbols like !, @, #, \$, %, etc., in identifiers

## 6) What are variables?

- A variable in a program is uniquely identified by a name (identifier).
- Variable in Python refers to an object — an item or element that is stored in the memory.
- Value of a variable can be a string (e.g., 'b', 'Global Citizen'), numeric (e.g., 345) or any combination of alphanumeric characters (CD67).
- In Python we can use an assignment statement to create new variables and assign specific values to them.

ex:

```
gender = 'M'  
message = "Keep Smiling"  
price = 987.9
```

## 7) Write about Python variable declaration.

- Variable declaration is implicit in Python, means variables are automatically declared and defined when they are assigned a value the first time.
- Variables must always be assigned values before they are used in expressions as otherwise it will lead to an error in the program.
- Wherever a variable name occurs in an expression, the interpreter replaces it with the value of that particular variable.

## 8) Where are comments? State their use.

- Comments are used to add a remark or a note in the source code. Comments are not executed by interpreter.
- They are added with the purpose of making the source code easier for humans to understand.
- They are used primarily to document the meaning and purpose of source code and its input and output requirements, so that we can remember later how it functions and how to use it.
- In Python, a comment starts with # (hash sign).

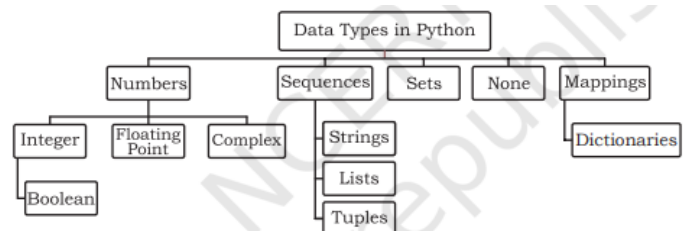
- Everything following the # till the end of that line is treated as a comment and the interpreter simply ignores it while executing the statement.

## 9) What is an object in python?

- Python treats every value or data item whether numeric, string, or other type (discussed in the next section) as an object in the sense that it can be assigned to some variable or can be passed to a function as an argument.
- Every object in Python is assigned a unique identity (ID) which remains the same for the lifetime of that object. This ID is akin to the memory address of the object. The function id() returns the identity of an object.

## 10) What is the use of data types? What are the data types present in Python?

Every value belongs to a specific data type in Python. Data type identifies the type of data values a variable can hold and the operations that can be performed on that data.



Python also comes under the category of \_\_\_\_\_  
programming Ans: Object oriented

## 11) What are the number types in Python?

Number data type stores numerical values only. It is further classified into three different types: int, float and complex.

## 12) Is boolean an integer data type?

Boolean data type (bool) is a subtype of integer. It is a unique data type, consisting of two constants, True and False. Boolean True value is non-zero, non-null and non-empty. Boolean False is the value zero.

### 13) Which data types store single values?

Variables of simple data types like integers, float, boolean, etc., hold single values.

### 14) What is a sequence?

A Python sequence is an ordered collection of items, where each item is indexed by an integer. The three types of sequence data types available in Python are Strings, Lists and Tuples.

### 15) What are strings?

String is a group of characters. These characters may be alphabets, digits or special characters including spaces. String values are enclosed either in single quotation marks (e.g., 'Hello') or in double quotation marks (e.g., "Hello").

The quotes are not a part of the string, they are used to mark the beginning and end of the string for the interpreter.

We cannot perform numerical operations on strings, even when the string contains a numeric value,

ex:

```
>>> str1 = 'Hello Friend'
```

```
>>> str2 = "452"
```

### 16) What is a list?

List is a sequence of items separated by commas and the items are enclosed in square brackets [ ].

```
list1 = [5, 3.4, "New Delhi"]
```

### 17) What is a tuple?

Tuple is a sequence of items separated by commas and items are enclosed in parenthesis ( ). This is unlike list, where values are enclosed in brackets [ ]. Once created, we cannot change the tuple.

```
tuple1 = (10, 20, "Apple")
```

### 19) What is a set?

Set is an unordered collection of items separated by commas and the items are enclosed in curly brackets { }. A set is similar to list, except that it cannot have duplicate entries. Once created, elements of a set cannot be changed.

```
>>> set1 = {10,20,3.14,"New Delhi"}
```

### 20) What is None data type?

None is a special data type with a single value. It is used to signify the absence of value in a situation. None supports no special operations, and it is neither same as False nor 0 (zero).

```
>>> myVar = None
```

### 21) What is mapping?

Mapping is an unordered data type in Python. Currently, there is only one standard mapping data type in Python called dictionary

### 22) What is dictionary? Write about its structure.

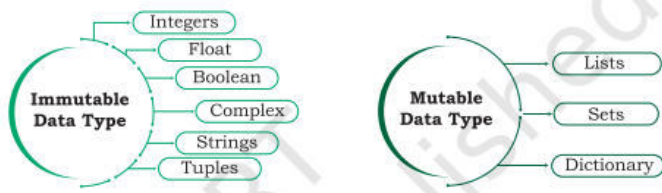
- Dictionary in Python holds data items in key-value pairs.
- Items in a dictionary are enclosed in curly brackets { }.
- Dictionaries permit faster access to data.
- Every key is separated from its value using a colon (:) sign.
- The key : value pairs of a dictionary can be accessed using the key.
- The keys are usually strings and their values can be any data type.
- In order to access any value in the dictionary, we have to specify its key in square brackets [ ]

### 23) Differentiate between mutable and immutable data types.

Variables whose values can be changed after they are created and assigned are called mutable.

Variables whose values cannot be changed after they are created and assigned are called immutable.

When an attempt is made to update the value of an immutable variable, the old variable is destroyed and a new variable is created by the same name in memory.



#### 24) How strings are compared in Python?

Python compares strings lexicographically, using ASCII value of the characters. If the first character of both the strings are same, the second character is compared, and so on.

When do we need to use list, tuple and dictionary data types? or How the usage of the data types are decided?

##### List

It is preferred to use lists when we need a simple iterable collection of data that may go for frequent modifications. For example, if we store the names of students of a class in a list, then it is easy to update the list when some new students join or some leave the course.

##### Tuple

Tuples are used when we do not need any change in the data. For example, names of months in a year. When we need uniqueness of elements and to avoid duplicacy it is preferable to use sets, for example, list of artefacts in a museum.

##### Dictionary

If our data is being constantly modified or we need a fast lookup based on a custom key or we need a logical association between the key : value pair, it is advised to use dictionaries. A mobile phone book is a good application of dictionary

#### 25) What are operators?

An operator is used to perform specific mathematical or logical operation on values. The values that the operators work on are called operands. For example, in the expression  $10 + \text{num}$ , the value 10, and the variable num are operands and the + (plus) sign is an operator.

#### 26) What is the use of floor division operator //?

Divides the operand on the left by the operand on the right and returns the quotient by removing the decimal part. It is sometimes also called integer division.

```
>>> num1 = 13
>>> num2 = 4
>>> num1 // num2    3
>>> num2 // num1    0
```

#### 27) What is the use of \*\*?

Performs exponential (power) calculation on operands. That is, raise the operand on the left to the power of the operand on the right.

```
>>> num1 = 3
>>> num2 = 4
>>> num1 ** num2    81
```

#### 28) What is the use of relational operators?

Relational operator compares the values of the operands on its either side and determines the relationship among them.

What is the use of assignment operators?

Assignment operator assigns or changes the value of the variable on its left.

#### 29) What are logical operators?

There are three logical operators supported by Python. These operators (and, or, not) are to be written in lower case only.

The logical operator evaluates to either True or False based on the logical operands on either side.



Every value is logically either True or False. By default, all values are True except None, False, 0 (zero), empty collections "", (), [], {}, and few other special values.

### 30) What are identity operators?

Identity operators are used to determine whether the value of a variable is of a certain type or not. Identity operators can also be used to determine whether two variables are referring to the same object or not. There are two identity operators.

**is**

Evaluates True if the variables on either side of the operator point towards the same memory location and False otherwise.

var1 is var2 results to True if id(var1) is equal to id(var2).

**is not**

Evaluates to False if the variables on either side of the operator point to the same memory location and True otherwise.

var1 is not var2 results to True if id(var1) is not equal to id(var2)

### 31) What are membership operators?

Membership operators are used to check if a value is a member of the given sequence or not.

**in**

Returns True if the variable/value is found in the specified sequence and False otherwise

**not in**

Returns True if the variable/value is not found in the specified sequence and False otherwise

### 32) What are expressions?

An expression is defined as a combination of constants, variables, and operators. An expression always evaluates to a value. A value or a standalone variable is also considered as an expression but a standalone operator is not an expression. Some examples of valid expressions are given below

(i) 100 (ii) 3.0 + 3.14 (iii) num (iv) 23/3 - 5 \* 7(14 - 2)  
(v) num - 20.4 (vi) "Global" + "Citizen"

### 33) What do you mean by precedence of operators?

- Evaluation of the expression is based on precedence of operators.
- When an expression contains different kinds of operators, precedence determines which operator should be applied first.
- Higher precedence operator is evaluated before the lower precedence operator.
- Parenthesis can be used to override the precedence of operators.
- The expression within () is evaluated first. b) For operators with equal precedence, the expression is evaluated from left to right.

| Operators              | Description                                 |
|------------------------|---------------------------------------------|
| **                     | Exponentiation (raise to the power)         |
| ~, +, -                | Complement, unary plus and unary minus      |
| *, /, %, //            | Multiply, divide, modulo and floor division |
| +, -                   | Addition and subtraction                    |
| <=, <, >, >=, ==, !=   | Relational and Comparison operators         |
| =, %=, /=, //=, -=, += | Assignment operators                        |
| is, is not             | Identity operators                          |
| in, not in             | Membership operators                        |
| not                    | Logical operators                           |
| and                    |                                             |
| or                     |                                             |

### 34) What is a statement?

In Python, a statement is a unit of code that the Python interpreter can execute.

### 35) Write about input() function in python.

- We have the input() function for taking the user input.
- The input() function prompts the user to enter data.
- It accepts all user input as string.
- The user may enter a number or a string but the input() function treats them as strings only.

- The syntax for input() is: input ([Prompt]), Prompt is the string we may like to display on the screen prior to taking the input, and it is optional.
- When a prompt is specified, first it is displayed on the screen after which the user can enter data.
- The input() takes exactly what is typed from the keyboard, converts it into a string and assigns it to the variable on left-hand side of the assignment operator (=).
- Entering data for the input function is terminated by pressing the enter key.
- Type casting is done to change the type of the input data

### 36) Write about print() function.

- Python uses the print() function to output data to standard output device - the screen.
- The function print() evaluates the expression before displaying it on the screen.
- The print() outputs a complete line and then moves to the next line for subsequent output.
- The syntax for print() is: print(value [, ..., sep = ' ', end = '\n'])
- sep: The optional parameter sep is a separator between the output values. We can use a character, integer or a string as a separator. The default separator is space.
- end: This is also optional and it allows us to specify any string to be appended after the last value. The default is a new line.

### 37) What is type conversion? What are its 2 types?

As and when required, we can change the data type of a variable in Python from one type to another.

Such data type conversion can happen in two ways: either explicitly (forced) when the programmer specifies for the interpreter to convert a data type to another type; or implicitly, when the interpreter understands such a need by itself and does the type conversion automatically.

### 38) What do you mean by explicit type conversion? What is the risk in it?

Explicit conversion, also called type casting happens when data type conversion takes place because the programmer forced it in the program.

The general form of an explicit data type conversion is: (new\_data\_type) (expression)

#### Risk of explicit type conversion

With explicit type conversion, there is a risk of loss of information since we are forcing an expression to be of a specific type. For example, converting a floating value of x = 20.67 into an integer type, i.e., int(x) will discard the fractional part .67.

```
pricelcecream = 25
```

```
priceBrownie = 45
```

```
totalPrice = pricelcecream + priceBrownie
```

```
print("The total in Rs." + str(totalPrice))
```

Output:

The total in Rs.70

### 39) What is the use chr() and ord() functions?

chr(x) Converts ASCII value of x to character

ord(x) returns the character associated with the ASCII code x

### 40) What is implicit conversion or coercion?

Implicit conversion, also known as coercion, happens when data type conversion is done automatically by Python and is not instructed by the programmer.

```
num1 = 10 #num1 is an integer
```

```
num2 = 20.0 #num2 is a float
```

```
sum1 = num1 + num2 #sum1 is sum of a float
```

and an integer

```
print(sum1, end=' ')
```

```
print(type(sum1))
```

Output:

30.0 <class 'float'>

#### **41) What do you mean by debugging?**

A programmer can make mistakes while writing a program, and hence, the program may not execute or may generate wrong output. The process of identifying and removing such mistakes, also known as bugs or errors, from a program is called debugging

#### **42) What are syntax errors?**

Python has its own rules that determine its syntax. The interpreter interprets the statements only if it is syntactically (as per the rules of Python) correct. If any syntax error is present, the interpreter shows error message(s) and stops the execution there

#### **43) What are logical errors? Why are they difficult to get identified?**

A logical error is a bug in the program that causes it to behave incorrectly. A logical error produces an undesired output but without abrupt termination of the execution of the program.

Since the program interprets successfully even when logical errors are present in it, it is sometimes difficult to identify these errors. The only evidence to the existence of logical errors is the wrong output.

While working backwards from the output of the program, one can identify what went wrong.

#### **44) What are runtime errors?**

A runtime error causes abnormal termination of program while it is executing. Runtime error is when the statement is correct syntactically, but the interpreter cannot execute it. Runtime errors do not appear until after the program starts running or executing.

~~~~~

"Challenges are what make life interesting and overcoming them is what makes life meaningful."

- Joshua J. Marine

XII COMPUTER SCIENCE

Towards Getting Centum in CS - See Yes!

Easy Theory

NCERT FOCUS FLOW OF CONTROL

1) What is flow of control?

The order of execution of the statements in a program is known as flow of control. The flow of control can be implemented using control structures. Python supports two types of control structures—selection and repetition.

2)When is selection statement used?

A decision involves selecting from one of the two or more possible options. In programming, this concept of decision making or selection is implemented with the help of if..else statement.

There is no limit on the number of statements that can appear as a block under the if statement.

3)Write about indentation in python.

- In most programming languages, the statements within a block are put inside curly brackets. However, Python uses indentation for block as well as for nested block structures.
- Leading whitespace (spaces and tabs) at the beginning of a statement is called indentation.
- In Python, the same level of indentation associates statements into a single block of code.
- The interpreter checks indentation levels very strictly and throws up syntax errors if indentation is not correct.

- It is a common practice to use a single tab for each level of indentation.

4)What is iteration statement? How looping constructs are used to create iteration?

- Iteration is repetition a set of statements in a program is made possible using looping constructs.
- Looping constructs provide the facility to execute a set of statements in a program repetitively, based on a condition.
- The statements in a loop are executed again and again as long as particular logical condition remains true. This condition is checked based on the value of a variable called the loop's control variable.
- When the condition becomes false, the loop terminates.

5)What is the use of for loop?

The for statement is used to iterate over a range of values or a sequence.

The for loop is executed for each of the items in the range. These values can be either numeric or a sequence.

6)What is range() function in Python?

- The range() is a built-in function in Python. Syntax of range() function is: range([start], stop[, step])
- It is used to create a list containing a sequence of integers from the given start value upto stop value (excluding stop value), with a difference of the given step value.
- The start and step parameters are optional.
- If start value is not specified, by default the list starts from 0.
- If step is also not specified, by default the value increases by 1 in each iteration.
- All parameters of range() function must be integers. The step parameter can be a positive or a negative integer excluding zero.

- The function range() is often used in for loops for generating a sequence of numbers.

ex:

#start and step not specified

```
>>> list(range(10))
```

```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

#default step value is 1

```
>>> list(range(2, 10))
```

```
[2, 3, 4, 5, 6, 7, 8, 9]
```

#step value is 5

```
>>> list(range(0, 30, 5))
```

```
[0, 5, 10, 15, 20, 25]
```

#step value is -1. Hence, decreasing

#sequence is generated

```
>>> list (range (0, -9, -1))
```

```
[0, -1, -2, -3, -4, -5, -6, -7, -8]
```

7)How does while loop function?

The while statement executes a block of code repeatedly as long as the control condition of the loop is true. The control condition of the while loop is executed before any statement inside the loop is executed.

8)What is the use of break statement?

The break statement alters the normal flow of execution as it terminates the current loop and resumes execution of the statement following that loop.

9)What is the use of continue statement?

When a continue statement is encountered, the control skips the execution of remaining statements inside the body of the loop for the current iteration and jumps to the beginning of the loop for the next iteration.

10)What is nested loop?

A loop may contain another loop inside it. A loop inside another loop is called a nested loop.

Python does not impose any restriction on how many loops can be nested inside a loop or on the levels of nesting. Any type of loop (for/while) may be nested within another loop (for/while).

~~~~~

*"In most things success depends on knowing how long it takes to succeed." – Montesquieu*

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS – See Yes!*

*Easy Theory*

### NCERT FOCUS – FUNCTIONS

#### 1) What is a function? State its uses.

- In programming, the use of function is one of the means to achieve modularity and reusability.
- Function can be defined as a named group of instructions that accomplish a specific task when it is invoked.
- Once defined, a function can be called repeatedly from different places of the program without writing all the codes of that function everytime, or it can be called from inside another function, by simply writing the name of the function and passing the required parameters, if any

#### 2) State some advantages of a function.

- Increases readability
- Reduces code length
- Increases reusability

#### 3) What is a user defined function?

A function defined to achieve some task as per the programmer's requirement is called a user defined function.

#### 4) Write about the syntax of a function.

A function definition begins with def (short for define). The syntax for creating a user defined function is as follows:

```
def<Function name> ([parameter 1, parameter 2,...]):
    set of instructions to be executed
    [return <value>]
```

Function Header

Function Body (Should be indented within the function header)

- The items enclosed in "[ ]" are called parameters and they are optional. Hence, a function may or may not have parameters. Also, a function may or may not return a value.
- Function header always ends with a colon (:).
- Function name should be unique. Rules for naming identifiers also applies for function naming.
- The statements outside the function indentation are not considered as part of the function.

#### 5) What are arguments and parameters?

An argument is a value passed to the function during the function call which is received in corresponding parameter defined in function header.

#### 6) What is a default parameter?

Python allows assigning a default value to the parameter. A default value is a value that is predecided and assigned to the parameter when the function call does not have its corresponding argument.

The default parameters must be the trailing parameters in the function header that means if any parameter is having default value then all the other parameters to its right must also have default values.

For example,

```
def mixedFraction(num,deno = 1)
```

```
def mixedFraction(num = 2,deno = 1)
```

#### 7) What is a void function?

A function not returning anything is called a void function?

What does the return statement do?

The return statement does the following:

- returns the control to the calling function.
- return value(s) or None.

### 8)What do you mean by flow of execution?

Flow of execution can be defined as the order in which the statements in a program are executed.

### 9)What is scope of a variable? What are its 2 types?

A variable defined inside a function cannot be accessed outside it. Every variable has a well-defined accessibility. The part of the program where a variable is accessible can be defined as the scope of that variable.

A variable can have one of the following two scopes: A variable that has global scope is known as a global variable and a variable that has a local scope is known as a local variable.

#### Global Variable

In Python, a variable that is defined outside any function or any block is known as a global variable. It can be accessed in any functions defined onwards. Any change made to the global variable will impact all the functions in the program where that variable can be accessed.

#### Local Variable

A variable that is defined inside any function or a block is known as a local variable. It can be accessed only in the function or a block where it is defined. It exists only till the function executes.

ex:

```
a = 2 # global a

def calc():
    a = 4    #local a
    print(a)
```

### 10)What is python standard library?

Python has a very extensive standard library. It is a collection of many built in functions that can be called in the program as and when required, thus saving programmer's time of creating those commonly used functions everytime.

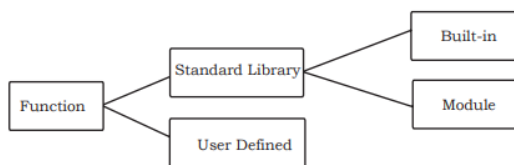


Figure 7.7: Types of functions

### 11) What do you mean by built-in functions?

Built-in functions are the ready-made functions in Python that are frequently used in programs.

Let us inspect the following Python program:

```
a = int(input("Enter a number: "))
b = a * a
print(" The square of ",a ,"is", b)
```

In the above program input(), int() and print() are the built-in functions. The set of instructions to be executed for these built-in functions are already defined in the python interpreter.

In the above code, input() is a return function.

### 12)What is a module?

Other than the built-in functions, the Python standard library consists of a number of modules. While a function is a grouping of instructions, a module is a grouping of functions.

A module is created as a python (.py) file containing a collection of function definitions.

### 13)How to import a module?

To use a module, we need to import the module. Once we import a module, we can directly use all the functions of that module. The syntax of import statement is as follows:

```
import modulename1 [,modulename2, ...]
```

To call a function of a module, the function name should be preceded with the name of the module with a dot(.) as a separator. The syntax is as shown below: modulename.functionname()

ex: math.pow(3,2)

### 14)What is the use of From statement while importing a module?

Instead of loading all the functions into memory by importing a module, from statement can be used to access only the required functions from a module. It loads only the specified function(s) instead of all the functions in a module. Its syntax is

```
>>> from modulename import functionname [,
functionname,...]
```

ex: from math import sqrt

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS - See Yes!*

*Easy Theory*

### NCERT FOCUS – STRINGS

#### 1) What is a string?

String is a sequence which is made up of one or more UNICODE characters. Here the character can be a letter, digit, whitespace or any other symbol. A string can be created by enclosing one or more characters in single, double or triple quote.

#### 2) Write about indexing in strings.

- Each individual character in a string can be accessed using a technique called indexing.
- The index specifies the character to be accessed in the string and is written in square brackets ([ ]).
- The index of the first character (from left) in the string is 0 and the last character is n-1 where n is the length of the string.
- If we give index value out of this range then we get an IndexError.
- The index must be an integer (positive, zero or negative).
- The index can also be an expression including variables and operators but the expression must evaluate to an integer. a[2+3, 4+9]

#### 3) Write about negative indexing in python.

- Negative indices are used when we want to access the characters of the string from right to left.

- Starting from right hand side, the first character has the index as -1 and the last character has the index -n where n is the length of the string

#### 4) What is concatenation?

To concatenate means to join. Python allows us to join two strings using concatenation operator plus which is denoted by symbol +.

c= 'sai' + ' ram'

#### 5) How replication or repetition of strings is done?

Python allows us to repeat the given string using repetition operator which is denoted by symbol \*.

ex: 'sai' \*2

#### 6) What is slicing?

- In Python, to access some part of a string or substring, we use a method called slicing.
- This can be done by specifying an index range. Given a string str1, the slice operation str1[n:m] returns the part of the string str1 starting from index n (inclusive) and ending at m (exclusive).
- In other words, we can say that str1[n:m] returns all the characters starting from str1[n] till str1[m-1].
- The numbers of characters in the substring will always be equal to difference of two indices m and n, i.e., (m-n).

ex: a[5:10]

#### 7) What is the use of title()?

Returns the string with first letter of every word in the string in uppercase and rest in lowercase.

#### 8) What is the use of lower()?

Returns the string with all uppercase letters converted to lowercase

#### 9) What is the use of upper()?

Returns the string with all lowercase letters converted to uppercase



**10) What is the use of endswith()?**

Returns True if the given string ends with the supplied substring otherwise returns False

**11) What is the use of startswith()?**

Returns True if the given string starts with the supplied substring otherwise returns False

**12) What is the use of isalnum()?**

Returns True if characters of the given string are either alphabets or numeric. If whitespace or special symbols are part of the given string or the string is empty it returns False

**13) What is the use of islower()?**

Returns True if the string is non-empty and has all lowercase alphabets, or has at least one character as lowercase alphabet and rest are non-alphabet characters

**14) What is the use of isupper()?**

Returns True if the string is non-empty and has all uppercase alphabets, or has at least one character as uppercase character and rest are non-alphabet characters

**15) What is the use of isspace()?**

Returns True if the string is non-empty and all characters are white spaces (blank, tab, newline, carriage return)

**16) What is the use of lstrip()?**

Returns the string after removing the spaces only on the left of the string

**17) What is the use of rstrip()?**

Returns the string after removing the spaces only on the right of the string

**18) What is the use of strip()?**

Returns the string after removing the spaces both on the left and the right of the string

**19) What is the use of replace()?**

Replaces all occurrences of old string with the new string

**20) What is the use of join()?**

Returns a string in which the characters in the string have been joined by a separator

**21) What is the use of partition()?**

Partitions the given string at the first occurrence of the substring (separator) and returns the string partitioned into three parts.

1. Substring before the separator
2. Separator
3. Substring after the separator

If the separator is not found in the string, it returns the whole string itself and two empty strings

**22) What is the use of split()?**

Returns a list of words delimited by the specified substring. If no delimiter is given then words are separated by space.

~~~~~

XII COMPUTER SCIENCE

Towards Getting Centum in CS - See Yes!

Easy Theory

NCERT FOCUS – LISTS

1) Compare list and strings.

- The data type list is an ordered sequence which is mutable and made up of one or more elements.
- Unlike a string which consists of only characters, a list can have elements of different data types, such as integer, float, string, tuple or even another list.
- A list is very useful to group together elements of mixed data types.
- Elements of a list are enclosed in square brackets and are separated by comma.
- Like string indices, list indices also start from 0.
- The elements of a list are accessed in the same way as characters are accessed in a string.
- Like strings, the slicing operation can also be applied to lists.

2) What is the use of append()?

Appends a single element passed as an argument at the end of the list. The single element can also be a list.

3) What is the use of extend()?

Appends each element of the list passed as argument to the end of the given list.

4) What is the use of insert()?

Inserts an element at a particular index in the list.

5) What is the use of count()?

Returns the number of times a given element appears in the list.

6) What is the use of index()?

Returns index of the first occurrence of the element in the list. If the element is not present, ValueError is generated.

7) What is the use of remove()?

Removes the given element from the list. If the element is present multiple times, only the first occurrence is removed. If the element is not present, then ValueError is generated.

8) What is the use of pop()?

Returns the element whose index is passed as parameter to this function and also removes it from the list. If no parameter is given, then it returns and removes the last element of the list.

9) What is the use of reverse()?

Reverses the order of elements in the given list.

10) What is the use of sort()?

Sorts the elements of the given list in-place.

11) What is the use of sorted()?

It takes a list as parameter and creates a new list consisting of the same elements arranged in sorted order.

12) What is the use of min()?

Returns minimum or smallest element of the list.

13) What is the use of max()?

Returns maximum or largest element of the list.

14) What is the use of sum()?

Returns sum of the elements of the list.

15) What is a nested list?

When a list appears as an element of another list, it is called a nested list.

16) What are the different ways to copy a list?

We can also create a copy or clone of the list as a distinct object by three methods. The first method uses slicing, the second method uses built-in function list() and the third method uses copy() function of python library copy.

Using Slicing:

```
>>> list1 = [1,2,3,4,5]
>>> list2 = list1[:]
>>> list2
[1, 2, 3, 4, 5]
```

Using list()

```
>>> list1 = [10,20,30,40]
>>> list2 = list(list1)
>>> list2
[10, 20, 30, 40]
```

Using copy module

```
>>> import copy
>>> list1 = [1,2,3,4,5]
>>> list2 = copy.copy(list1)
>>> list2 [1, 2, 3, 4, 5]
```

(B) If the list is assigned a new value inside the function, then a new list object is created and it becomes the local copy of the function.

Any changes made inside the local copy of the function are not reflected back to the calling function.

def change(x):

 x = [4,7,9] #new list assigned

x = [2,5,7,9]

print(x) #prints [2, 5, 7,9]

change(x)

print(x) #prints [2, 5, 7,9]

~~~~~

## 17) What happens when a list is passed as argument to a function?

Whenever a list is passed as an argument to a function, we have to consider two scenarios:

(A) Elements of the original list may be changed, i.e. changes made to the list in the function are reflected back in the calling function.

def change(x):

    x[2]= 33

x = [2,5,7,9]

print(x)       #prints [2, 5, 7,9]

change(x)

print(x)       #prints [2, 5, 33,9]

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS – See Yes!*

*Easy Theory*

### NCERT FOCUS – TUPLES & DICTIONARIES

#### TUPLE

##### 1) What is a tuple? Write about its features.

- A tuple is an ordered sequence of elements of different data types, such as integer, float, string, list or even a tuple.
- Elements of a tuple are enclosed in parenthesis (round brackets) and are separated by commas.
- Like list and string, elements of a tuple can be accessed using index values, starting from 0
- Elements of a tuple cannot be changed as it is an immutable type.

ex: a = {1,3,'exam', (5,6)}

- If there is only a single element in a tuple then the element should be followed by a comma.  
ex: a = 4,     #a tuple
- If we assign the value without comma it is treated as integer.  
ex: a = (4)     #an integer
- It should be noted that a sequence without parenthesis is treated as tuple by default

##### 2) When should we prefer tuple over lists?

- List is mutable but tuple is immutable. So iterating through a tuple is faster as compared to a list.
- If we have data that does not change then storing this data in a tuple will make sure that it is not changed accidentally.

##### 3) Can the elements of a tuple be changed?

The elements of a tuple cannot be altered except a list. If a tuple contains a list, the elements of the list can be changed.

#4th element of the tuple2 is a list

```
>>> tuple2 = (1,2,3,[8,9])
```

#modify the list element of the tuple tuple2

```
>>> tuple2[3][1] = 10
```

#modification is reflected in tuple2

```
>>> tuple2
```

```
(1, 2, 3, [8, 10])
```

##### 4) What are the operations allowed on a tuple?

Concatenation using +, Repetition or Replication using \* and Membership operation

##### 5) Name some methods that can be applied on a tuple.

len(), tuple(), count(), index(), sorted(), min(), max() and sum()

##### 6) What is tuple assignment?

- Assignment of tuple is a useful feature in Python.
- It allows a tuple of variables on the left side of the assignment operator to be assigned respective values from a tuple on the right side.
- The number of variables on the left should be same as the number of elements in the tuple.

### Example

```
>>> record = ( "Pooja",40,"CS")
>>> (name,rollNo,subject) = record
>>> name
'Pooja'
>>> rollNo
40
>>> subject
'CS'
```

## DICTIONARY

### 1) What is a dictionary? Write about its features.

- The data type dictionary fall under mapping.
- It is a mapping between a set of keys and a set of values.
- The key-value pair is called an item.
- A key is separated from its value by a colon(:) and consecutive items are separated by commas.
- Items in dictionaries are unordered, so we may not get back the data in the same order in which we had entered the data initially in the dictionary.
- The keys in the dictionary must be unique and should be of any immutable data type, i.e., number, string or tuple.
- The values can be repeated and can be of any data type.

```
d = {1:'ram', 2:'krish'}
```

### 2) How are the items in a dictionary accessed?

The items of a dictionary are accessed via the keys rather than via their relative positions or indices. Each key serves as the index and maps to a value.

```
dict3 = {'Mohan':95,'Ram':89,'Suhel':92}
print(dict3['Mohan'])    #prints 95
```

In the above examples the key 'Mohan' always maps to the value 95 and key 'Sangeeta' always maps to the value 85. So the order of items does not matter. If the key is not present in the dictionary we get KeyError.

### 2) What are the operations allowed on a dictionary?

Only Membership operation is allowed on a dictionary.

```
>>> dict1 = {'Mohan':95,'Ram':89,'Suhel':92}
```

```
>>> 'Suhel' in dict1
```

True

### 12) What is the use of dict()?

Creates a dictionary from a sequence of key-value pairs

### 13) What is the use of keys()?

Returns a list of keys in the dictionary

### 13) What is the use of values()?

Returns a list of values in the dictionary

### 13) What is the use of items()?

Returns a list of tuples(key – value) pair

### 13) What is the use of get()?

Returns the value corresponding to the key passed as the argument. If the key is not present in the dictionary it will return None.

### 14) What is the use of update()?

appends the key-value pair of the dictionary passed as the argument to the key-value pair of the given dictionary

### 14) What is the use of del statement?

Deletes the item with the given key. To delete the dictionary from the memory we write:

```
>>>del Dict_name
```

### 14) What is the use of clear()?

Deletes or clear all the items of the dictionary

~~~~~

XII COMPUTER SCIENCE

Towards Getting Centum in CS – See Yes!

Easy Theory

NCERT FOCUS – FILE HANDLING

1) What is a file?

A file is a named location on a secondary storage media where data are permanently stored for later access.

Technically, they are a series of bytes.

2) What are the 2 types of files?

- There are mainly two types of data files — text file and binary file.
- A text file consists of human readable characters, which can be opened by any text editor.
- On the other hand, binary files are made up of non-human readable characters and symbols, which require specific programs to access its contents.

3) Write about text files.

- A text file can be understood as a sequence of characters consisting of alphabets, numbers and other special symbols.
- Files with extensions like .txt, .py, .csv, etc. are some examples of text files.
- When we open a text file using a text editor (e.g., Notepad), we see several lines of text. However, the file contents are not stored in such a way internally. Rather, they are stored in sequence of bytes consisting of 0s and 1s.

- In ASCII, UNICODE or any other encoding scheme, the value of each character of the text file is stored as bytes. So, while opening a text file, the text editor translates each ASCII value and shows us the equivalent character that is readable by the human being.
- For example, the ASCII value 65 (binary equivalent 1000001) will be displayed by a text editor as the letter 'A'.
- Each line of a text file is terminated by a special character, called the End of Line (EOL). For example, the default EOL character in Python is the newline (\n). However, other characters can be used to indicate EOL.
- **Contents in a text file are usually separated by whitespace, but comma (,) and tab (\t) are also commonly used to separate values in a text file.**

4) Write about binary files.

- Binary files are stored in terms of bytes (0s and 1s), but unlike text files, these bytes do not represent the ASCII values of characters.
- Rather, they represent the actual content such as image, audio, video, compressed versions of other files, executable files, etc.
- These files are not human readable. Thus, trying to open a binary file using a text editor will show some garbage values.
- We need specific software to read or write the contents of a binary file.
- Binary files are stored in a computer in a sequence of bytes.
- Even a single bit change can corrupt the file and make it unreadable to the supporting application.
- Also, it is difficult to remove any error which may occur in the binary file as the stored contents are not human readable.
- We can read and write both text and binary files through Python programs.

5) Which module is used by python for handling files?

Python has the **io module** that contains different functions for handling files.

6) What is the use of open()? What is a file handle?

To create and open a file in Python, we use the open() function. The syntax of open() is as follows:

```
file_object= open(file_name, access_mode)
```

This function returns a file object called file handle which is stored in the variable file_object. We can use this variable to transfer data to and from the file (read and write) by calling the functions defined in the Python's io module.

7) What is use of fileclosed()?

file.closed returns true if the file is closed and false otherwise.

8) What is the of file.mode?

returns the access mode in which the file was opened.

9) What is the of file.name?

returns the name of the file.

10) What is an access_mode?

- The access_mode is an optional argument that represents the mode in which the file has to be accessed by the program. It is also referred to as processing mode.
- Here mode means the operation for which the file has to be opened like for reading, for writing, for both reading and writing, for appending at the end of an existing file.
- The default is the read mode.
- In addition, we can specify whether the file will be handled as binary () or text mode. By default, files are opened in text mode that means strings can be read or written.

- Files containing non-textual data are opened in binary mode that means read/write are performed in terms of bytes.

11) What is an file offset?

The file offset position refers to the position of the file object when the file is opened in a particular mode.

12) What is the use of r mode?

Opens the file in read-only mode. If the file does not exist, error arises. The file offset position is at the beginning of the file.

13) What is the use of r+ or +r mode?

Opens the file in both read and write mode. If the file does not exist, error arises. The file offset position is at the beginning of the file.

14) What is the use of rb mode?

Opens the file in binary and read-only mode. If the file does not exist, error arises. The file offset position is at the beginning of the file.

15) What is the use of rb+ or +rb mode?

Opens the file in binary and read and write mode. If the file does not exist, error arises. The file offset position is at the beginning of the file.

16) What is the use of w mode?

Opens the file in write mode. If the file already exists, all the contents will be overwritten. If the file doesn't exist, then a new file will be created. The file offset position is at the beginning of the file.

17) What is the use of w+ or +w mode?

Opens the file in write and read mode. If the file already exists, all the contents will be overwritten. If the file

doesn't exist, then a new file will be created. The file offset position is at the beginning of the file.

18) What is the use of wb mode?

Opens the binary file in write mode. If the file already exists, all the contents will be overwritten. If the file doesn't exist, then a new file will be created. The file offset position is at the beginning of the file.

19) What is the use of wb+ or +wb mode?

Opens the binary file in write mode. Opens the file in read, write and binary mode. If the file already exists, the contents will be overwritten. If the file doesn't exist, then a new file will be created. The file offset position is at the beginning of the file.

20) What is the use of a mode?

Opens the file in append mode. If the file already exists, old contents will be retained and new contents will be added in the end. If the file doesn't exist, then a new file will be created. The file offset position is at the end of the file.

21) What is the use of a+ or +a mode?

Opens the file in append and read mode. If the file already exists, old contents will be retained and new contents will be added in the end. If the file doesn't exist, then a new file will be created. The file offset position is at the end of the file.

22) What is the use of ab mode?

Opens the binary file in append mode. If the file already exists, old contents will be retained and new contents will be added in the end. If the file doesn't exist, then a new file will be created. The file offset position is at the end of the file.

23) What is the use of ab+ or +ab mode?

Opens the binary file in append and read mode. If the file already exists, old contents will be retained and new contents will be added in the end. If the file doesn't exist, then a new file will be created. The file offset position is at the end of the file.

24) Why should we close a file?

- Once we are done with the read/write operations on a file, it is a good practice to close the file. Python provides a close() method to do so. While closing a file, the system frees the memory allocated to it.
- The syntax of close() is: `file_object.close()`
- Python makes sure that any unwritten or unsaved data is flushed off (written) to the file before it is closed. Hence, it is always advised to close the file once our work is done.
- Also, if the file object is re-assigned to some other file, the previous file is automatically closed.

25) What is the use of with clause in opening a file?

- In Python, we can also open a file using with clause.
- The advantage of using with clause is that any file that is opened using this clause is closed automatically, once the control comes outside the with clause.
- In case the user forgets to close the file explicitly or if an exception occurs, the file is closed automatically.

- Also, it provides a simpler syntax.

with open("myfile.txt","r+") as myObject:

`content = myObject.read()`

- Here, we don't have to close the file explicitly using close() statement. Python will automatically close the file.

26) How should we write data onto a text file?

- For writing to a file, we first need to open it in write or append mode.

- If we open an existing file in write mode, the previous data will be erased, and the file object will be positioned at the beginning of the file.
- On the other hand, in append mode, new data will be added at the end of the previous data as the file object is at the end of the file.
- After opening the file, we can use the following methods to write data in the file.
- write() - for writing a single string
- writelines() - for writing a sequence of strings

27) What is the use of write() method?

- write() method takes a string as an argument and writes it to the text file.
- It returns the number of characters being written on single execution of the write() method.
- Also, we need to add a newline character (\n) at the end of every sentence to mark the end of line.

```
>>> myobject=open("myfile.txt",'w')
```

```
>>> myobject.write("Hey I have started using files in Python\n")
```

```
41
```

```
>>> myobject.close()
```

- On execution, write() returns the number of characters written on to the file. Hence, 41, which is the length of the string passed as an argument, is displayed. \n is also considered as a character.
- The write() actually writes data onto a buffer. When the close() method is executed, the contents from this buffer are moved to the file located on the permanent storage.

28) How can you write numeric data onto a text file?

If numeric data are to be written to a text file, the data need to be converted into string before writing to the file. For example:

```
>>>myobject=open("myfile.txt",'w')
```

```
>>> marks=58
```

```
>>> myobject.write(str(marks))
```

```
2
```

```
>>>myobject.close()
```

29) What is the use of writelines() method?

This method is used to write multiple strings to a file. We need to pass an iterable object like lists, tuple, etc. containing strings to the writelines() method. Unlike write(), the writelines() method does not return the number of characters written in the file. The following code explains the use of writelines().

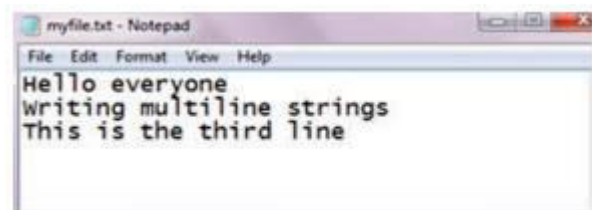
```
>>> myobject=open("myfile.txt",'w')
```

```
>>> lines = ["Hello everyone\n", "Writing multiline strings\n", "This is the third line"]
```

```
>>> myobject.writelines(lines)
```

```
>>>myobject.close()
```

On the file:



30) What are modes available for reading a text file?

“r”, “r+”, “w+” or “a+” modes.

31) What is the use of read() method?

This method is used to read a specified number of bytes of data from a data file.

The syntax of read() method is: file_object.read(n)

If no argument or a negative number is specified in read(), the entire file content is read.

```
>>>myobject=open("myfile.txt",'r')
```

```
>>> myobject.read(10)
```

```
'Hello ever'
```

```
>>> myobject.close()
```

32) What is the use of readline(n) method?

- This method reads one complete line from a file where each line terminates with a newline (\n) character.
- It can also be used to read a specified number (n) of bytes of data from a file but maximum up to the newline character (\n).
- In the following example, the second statement reads the first ten characters of the first line of the text file and displays them on the screen

```
>>> myobject=open("myfile.txt",'r')
```

```
>>> myobject.readline(10)
```

```
'Hello ever'
```

```
>>> myobject.close()
```

- If no argument or a negative number is specified, it reads a complete line and returns string.

```
>>>myobject=open("myfile.txt",'r')
```

```
>>> print (myobject.readline()) 'Hello  
everyone\n'
```

- An empty string is displayed when the readline() method reaches EOF.

33) What is the use of readlines() method?

The method reads all the lines and returns the lines along with newline as a list of strings. The following example uses readlines() to read data from the text file myfile.txt.

```
>>> myobject=open("myfile.txt", 'r')
```

```
>>> print(myobject.readlines())
```

```
['Hello everyone\n', 'Writing multiline
```

```
strings\n', 'This is the third line']
```

```
>>> myobject.close()
```

when we read a file using readlines() function, lines in the file become members of a list, where each list element ends with a newline character ('\n').

34) What is the general use of seek() and tell()?

read(), readline() etc. methods access a file sequentially. if we want to access data in a random fashion, then Python gives us seek() and tell() functions to do so.

35) What is the use of tell() method?

- This function returns an integer that specifies the current position of the file object in the file.
- The position so specified is the byte position from the beginning of the file till the current position of the file object.
- The syntax of using tell() is:

file_object.tell()

36) What is the use of seek() method? Write about its reference point in seek() method.

seek() method is used to position the file object at a particular position in a file.

The syntax of seek() is:

```
file_object.seek(offset [, reference_point])
```

In the above syntax, offset is the number of bytes by which the file object is to be moved.

reference point indicates the starting position of the file object. That is, with reference to which position, the offset has to be counted. It can have any of the following values:

0 - beginning of the file

1 - current position of the file

2 - end of file

By default, the value of reference_point is 0, i.e. the offset is counted from the beginning of the file.

For example, the statement fileObject.seek(5,0) will position the file object at 5th byte position from the beginning of the file.

37) What happens when readline() method reaches End of the File?

The readline() will return an empty string on reaching EOF.

38) What is the use of pickle module?

- Python considers everything as an object. So, all data types including list, tuple, dictionary, etc. are also considered as objects.
- During execution of a program, we may require to store current state of variables so that we can retrieve them later to its present state. We may like to store a Python dictionary as an object, to be able to retrieve later in the same format.
- To save any object structure along with data, Python provides a module called Pickle.
- The module Pickle is used for serializing and de-serializing any Python object structure.

39) What is the actual meaning of the term pickling?

Pickling is a method of preserving food items by placing them in some solution, which increases the shelf life. In other words, it is a method to store food items for later consumption.

40) Differentiate between serialization or pickling and de-serialization or unpickling in python?

Serialization is the process of transforming data or an object in memory (RAM) to a stream of bytes called byte streams. These byte streams in a binary file can then be stored in a disk or in a database or sent through a network. Serialization process is also called pickling.

De-serialization or unpickling is the inverse of pickling process where a byte stream is converted back to Python object. This is also called unpickling.

41) Instead of reading or writing, how the files are processed using pickle module?

The pickle module deals with binary files. Here, data are not written but dumped using dump() method, also called as pickling and similarly, data are not read but loaded using load() method, also called unpickling.

42) What is the use of dump() method?

This method is used to convert (pickling) Python objects for writing data in a binary file. The file in which data are to be dumped, needs to be opened in binary write mode (wb).

Syntax of dump() is as follows:

```
dump(data_object, file_object)
```

where data_object is the object that has to be dumped to the file with the file handle named file_object.

43) What is the use of load() method?

This method is used to load (unpickling) data from a binary file. The file to be loaded is opened in binary read (rb) mode.

Syntax of load() is as follows:

```
Store_object = load(file_object)
```

Here, the pickled Python object is loaded from the file having a file handle named file_object and is stored in a new file handle called store_object.

~~~~~

*"They succeed, because they think they can."*

– Virgil

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS – See Yes!*

*Easy Theory*

### NCERT FOCUS – STACK

#### What do you mean by the term data structure?

- A data structure defines a mechanism to store, organise and access data along with operations (processing) that can be efficiently performed on the data.
- For example, string is a data structure containing a sequence of elements where each element is a character.
- On the other hand, list is a sequence data structure in which each element may be of different types.
- We can apply different operations like reversal, slicing, counting of elements, etc. on list and string.
- Hence, a data structure organises multiple elements in a way so that certain operations on each element as well as the collective data unit could be performed easily.

#### What is linear data structure?

A data structure in which elements are organised in a sequence is called linear data structure

#### What is a stack?

Stack is a data structure in which insertion and deletion is done from one end only, usually referred to as TOP. It follows LIFO, means Last In First Out. The item added in the last will be the item which is removed in the first.

#### Give real life examples of stack.

- Pile of clothes in an almirah
- Multiple chairs in a vertical pile
- Bangles worn on wrist
- Pile of boxes of eatables in pantry or on a kitchen shelf

#### What are the application of stack in programming?

- When we need to **reverse a string**, the string is traversed from the last character till the first character. i.e. characters are traversed in the reverse order of their appearance in the string. This is very easily done by putting the characters of a string in a stack.
- We use text/image editor for editing the text/image where we have options to **redo/undo** the editing done. When we click on the redo /undo icon, the most recent editing is redone/undone. In this scenario, the system uses a stack to keep track of changes made.
- While browsing the web, we move from one web page to another by accessing links between them. In order to go back to the last visited web page, we may use the back button on the browser. In this case, the **history of browsed pages** is maintained as stack.
- While writing any arithmetic expression in a program, we may use parentheses to **order the evaluation of operators**. While executing the program, the compiler checks for matched parentheses i.e. each opening parenthesis should have a corresponding closing parenthesis and the pairs of parentheses are properly nested. In case of parentheses are mismatched, the compiler needs to throw an error. To handle matching of parentheses, stack is used.

### What are the operations performed on a stack?

- A stack is a mechanism that implements LIFO arrangement hence elements are added and deleted from the stack at one end only.
- The end from which elements are added or deleted is called TOP of the stack.
- Two fundamental operations performed on the stack are PUSH (insertion) and POP(deletion).
- Other than these, PEEK (viewing the topmost element), TRAVERSAL (checking every element from top to bottom), DISPLAY are the other operations performed on a stack.

### What do you mean by PUSH on stack?

PUSH adds a new element at the TOP of the stack. It is an insertion operation. We can add elements to a stack until it is full.

A stack is full when no more elements can be added to it. Trying to add an element to a full stack results in an exception called 'overflow'.

### What do you mean by POP on stack?

- POP operation is used to remove the top most element of the stack, that is, the element at the TOP of the stack. It is a delete operation.
- We can delete elements from a stack until it is empty i.e. there is no element in it.
- Trying to delete an element from an empty stack results in an exception called 'underflow'

### How stack is implemented in Python? Which side is fixed as top in python stack?

In python, Stack is implemented through List.

We can fix either of the sides of the list as TOP to insert/remove elements.

### Why does stack in python not get OVERFLOW?

As there is no limit on the size of list in Python, the implemented stack will never be full unless there is no more space available in memory. Hence, we will never face 'overflow' (no space for new element) condition for stack.

### Why is it not required to fix an element as TOP in Python stack explicitly?

We are using built-in methods append() and pop() of the list for implementation of the stack. As these built-in methods insert/delete elements at the rightmost end of the list, hence explicit declaration of TOP is not needed.

### Which method is used to insert an element on the top in python stack?

append() of list

### Which method is used to delete an element on the top in python stack?

pop() of list

~~~~~

XII COMPUTER SCIENCE

Towards Getting Centum in CS – See Yes!

Easy Theory

NCERT FOCUS – DATABASE CONCEPTS

1. What do you mean by database schema?

- Database Schema is the design of a database.
- It is the skeleton of the database that represents the structure (table names and their fields/columns), the type of data each column can hold, constraints on the data to be stored (if any), and the relationships among the tables.
- Database schema is also called the visual or logical architecture as it tells us how the data are organised in a database.

2. What are data constraints? Mention their use.

- Sometimes we put certain restrictions or limitations on the type of data that can be inserted in one or more columns of a table. This is done by specifying one or more constraints on that column(s) while creating the tables.
- column mobile number can only have non-negative integer values of exactly 10 digits. Since each student shall have one unique roll number, we can put the NOT NULL and UNIQUE constraints on the RollNumber column.
- Constraints are used to ensure accuracy and reliability of data in the database.

3. What is Meta data or data dictionary?

The database schema along with various constraints on the data is stored by DBMS in a database catalog or dictionary, called meta-data.

A meta-data is data about the data.

4. What is database instance?

- When we define database structure or schema, state of database is empty i.e. no data entry is there. After loading data, the state or snapshot of the database at any given time is the database instance.
- We may then retrieve data through queries or manipulate data through updation, modification or deletion.
- Thus, the state of database can change, and thus a database schema can have many instances at different times.

5. What do you understand by Database engine?

Database engine is the underlying component or set of programs used by a DBMS to create database and handle various queries for data retrieval and manipulation.

6. What is relational data model? Give example.

- A data model describes the structure of the database, including how data are defined and represented, relationships among data, and the constraints. The most commonly used data model is Relational Data Model.
- In relational model, tables are called relations that store data for different columns.
- Each table can have multiple columns where each column name should be unique.
- Relations in a database are not independent tables, but are associated with each other.

ex: any meaningful table.

7. Mention properties of a relation.

- Each attribute in a relation has a unique name.
- Sequence of attributes in a relation is immaterial.
- Each tuple in a relation is distinct.
- Sequence of tuples in a relation is immaterial.
- All data values in an attribute must be from the same domain (same data type).

- Each data value associated with an attribute must be atomic (cannot be further divisible)
- No attribute can have many data values in one tuple.
- A special value “NULL” is used to represent values that are unknown or non-applicable to certain attributes.

8. What are keys? Mention their use.

The tuples within a relation must be unique.

Relational data model imposes some restrictions constraints on the values of the attributes and how the contents of one relation be referred through another relation. These restrictions are specified at the time of defining the database through different types of keys.

Primary key, candidate key, alternate key, foreign key are used in relational data model.

9. What are fields?

The columns of a relation are the attributes which are also referred as fields.

10. What do you mean by domain in a relational data model?

- It is a set of values from which an attribute can take a value in each row.
- Usually, a data type is used to specify domain for an attribute.
- For example, in STUDENT relation, the attribute RollNumber takes integer values and hence its domain is a set of integer values.

11. Differentiate between primary key and candidate key with a suitable example.

Primary Key:

Primary Key is a unique and non-null key which identify a record uniquely in table. A table can have only one primary key.

Candidate key:

Candidate key is also a unique key to identify a record uniquely in a table but a table can have multiple candidate keys.

12. Differentiate between Candidate Key and alternate Key in context of RDBMS. Give a suitable example.

Candidate Key: All attribute combinations inside a relation that can serve as primary key are Candidate Keys as they are candidates for the primary key position.

Alternate Key: A candidate key that is not the primary key is called an Alternate Key. (Where Candidate Key: All attribute combinations inside a relation that can serve primary key(uniquely identifies a row in a relation) are Candidate Keys as they are candidates for the primary key position.)

13. Differentiate between primary key and alternate key. Give a suitable example.

Primary Key: A primary key is a set of one or more attributes that can uniquely identify tuples within the relations.

Alternate Key: A candidate key that is not the primary key is called an Alternate Key.

STUDENTS:

SID	NAME	CLASS	SUBJECT
101	RAHUL	XI	MATHS
102	JOHN	XII	PHYSICS
103	SUGU	VI	CHEMSITRY
104	RAHUL	VII	BIOLOGY

In the table STUDENTS above, SID, CLASS are candidate keys.

When SID is selected as the primary key, CLASS becomes the alternate key.

14. What is a foreign key? What is its use? Give a suitable example.

- A foreign key is the one that is used to link two tables together or create connectivity between the two.

- It means the columns of one table points to the primary key attribute of the other table and establishes a link.

Consider two tables **Student (Parent Table)** and **Department (Child Table)** having their respective attributes as shown in the below table structure:

Primary Key

Student

Stud_Id	Name	Course
101	John	Computer
105	Merry	AI
107	Sheero	Biology
108	Bisle	Maths

Foreign Key

Department

Dept_name	Stud_Id
CS_Department	105
CS_Department	101
Science_Department	101
Maths_Department	108

Here, Stud_Id which is primary key of Student table, is present in Department table as a Foreign key and establishes a link between the 2 tables.

Use of Foreign key:

The **FOREIGN KEY** constraint prevents invalid data from being inserted into the foreign key column, because it has to be one of the values contained in the parent table.

A table can have many foreign keys.

15. What is Cartesian Product? Give a suitable example.

- A Cartesian product is the result of joining every row in one table with every row in another table.
- It is denoted by symbol X.

- The degree of the resulting relation is calculated as the sum of the degrees of both the relations.
- The cardinality of the resulting relation is calculated as the product of the cardinality of relations.

If Name and Department are 2 relations, then Name X Department:

SELECT * FROM NAME, DEPARTMENT

Name		Name	Department
John		John	IT
Paul		Paul	IT
Sara		Sara	IT
		John	Marketing
		Paul	Marketing
		Sara	Marketing

CROSS JOIN

16. Differentiate between Cartesian Product and Joins.

Cartesian product which make all possible combinations of tuples.

SELECT * FROM NAME, DEPARTMENT

JOIN operation combines tuples from two tables on specified conditions.

SELECT * FROM NAME, DEPARTMENT WHERE NAME.ID = DEPARTMENT.ID

17. State the use of JOIN Clause in query.

In The JOIN clause of SQL, we specify conditions on the related attributes of two tables within the FROM clause. Usually, such an attribute is the primary key in one table and foreign key in another table.

Example (using JOIN explicitly) :

SELECT * FROM UNIFORM U JOIN COST C ON U.Ucode=C.Ucode;

This is similar to,

SELECT * FROM UNIFORM U, COST C
WHERE U.UCode = C.UCode;

18. State the use of NATURAL JOIN in query.

NATURAL JOIN works similar to JOIN clause in SQL but removes the redundant (duplicate) attribute. This operator can be used to join the contents of two tables implicitly if there is one common attribute in both the tables.

```
SELECT * FROM UNIFORM NATURAL JOIN COST;
```

19. Differentiate between natural join and equi join.

A NATURAL JOIN is a JOIN operation that creates an implicit join clause based on the common columns in the 2 tables being joined. In natural join, the common column will appear only once in output.

```
SELECT * FROM UNIFORM NATURAL JOIN COST;
```

In Equi join we compare value of any column from two tables using = operator, and it will return matching rows. In Equi-join common column appears twice in output because we fetch using (*) not by specifying column name.

```
SELECT * FROM UNIFORM U, COST C
```

```
WHERE U.UGrade = C.UGrade;
```

20. What is NATURAL JOIN? Give a suitable example.

A NATURAL JOIN is a JOIN operation that creates an implicit join clause based on the common columns in the 2 tables being joined. In natural join, the common column will appear only once in output.

If student and Course are 2 tables, then

```
SELECT * FROM STUDENT NATURAL JOIN COURSE
```

will result as follows:

Student

S_id	Name	Class	Age	C_id
1	Andrew	5	25	11
2	Angel	10	30	11
3	Anamika	8	35	22

Course

C_id	C_name
11	Foundation C
21	C++

Student $\bowtie$ Course

S_id	Name	Class	Age	C_id	C_name
1	Andrew	5	25	11	Foundation C
2	Angel	10	30	11	Foundation C

21. What is equi-join? Give a suitable example.

In Equi join we compare value of any column from two tables using = operator, and it will return matching rows. In Equi-join common column appears twice in output because we fetch using (*) not by specifying column name.

If R and S are 2 tables, then

```
SELECT * FROM R, S WHERE R.ID=S.ID
```

will result as follows:

Equijoin vs Natural Join

R

Id	Name	Sex
1	John	M
2	Mary	F
3	Bob	M

S

Id	Test	Status
1	Eye	Fail
2	Hearing	Pass
4	Eye	Pass

Equijoin

R $\bowtie$ R.Id=S.Id S

R.Id	Name	Sex	S.Id	Test	Status
1	John	M	1	Eye	Fail
2	Mary	F	2	Hearing	Pass

Natural Join

R * S

Id	Name	Sex	Test	Status
1	John	M	Eye	Fail
2	Mary	F	Hearing	Pass

22. What are the mathematical operators in SQL?

Arithmetic (Mathematical) Operators in SQL are:

+	[Addition]
-	[Subtraction]
/	[Division]
*	[Multiplication]
%	[Modulus]

23. What are the relational operators in SQL? State their use. Give example.

Relational operators <, >, <=, >=, != are used to specify conditions.

```
SELECT * FROM STUDENTS WHERE MARKS >=490
```

24. What are the logical operators in SQL? State their use.

The logical operators AND, OR, and NOT are used to combine multiple conditions.

```
SELECT * FROM STUDENTS WHERE MARKS >=490 AND  
REMARKS = 'GOOD'
```

25. Give example for column alias and table alias.

Column alias:

```
SELECT Ename AS Name, Salary*12 AS 'Annual Income'
```

In the above query, Ename will be printed as 'Name' and Salary multiplied by 12 will be printed as 'Annual Income'

Table alias:

Just like column aliases, table aliases are used to refer tables by shortened names.

```
SELECT * FROM UNIFORM U, COST C  
WHERE U.UGCode = C.UGCode;
```

26. How an attribute can be added or removed in an existing table or relation?

An attribute can be added or removed from a table using ALTER TABLE command

Adding an attribute:

It is done with ADD keyword. To add a new attribute Income of data type INT in the table GUARDIAN:

```
ALTER TABLE GUARDIAN ADD income INT;
```

Removing an attribute:

To remove attribute INCOME from table GUARDIAN. DROP keyword is used.

```
ALTER TABLE GUARDIAN DROP income;
```

27. How Primary Key can be added or removed in an existing table or relation?

Adding Primary Key:

It is done with ADD keyword. To add a PRIMARY KEY to the attribute GUID in the table GUARDIAN:

```
ALTER TABLE GUARDIAN ADD PRIMARY KEY (GUID);
```

Removing Primary Key:

It is done with DROP keyword.

To remove primary key from GUARDIAN table.

```
ALTER TABLE GUARDIAN DROP PRIMARY KEY;
```

28. State use of CREATE DATABASE, USE DATABASE, SHOW DATABASES, DROP DATABASE, SHOW TABLES commands. (refer viva handout)

29. What is the use of NULL operator? Give a suitable example.

- SQL supports a special value called NULL to represent a missing or unknown value. For example, the Name column in the GUARDIAN table below can have missing value for certain records. Hence, NULL is used to represent such unknown values.
- It is important to note that NULL is different from 0 (zero).
- Also, any arithmetic operation performed with NULL value gives NULL. For example: 5 + NULL = NULL because NULL is unknown hence the result is also unknown.
- In order to check for NULL value in a column, we use IS NULL operator.

GUARDIAN

ID	NAME
101	SEKAR
102	NULL
103	KUMAR
104	NULL

30) Differentiate between ALTER and UPDATE commands in SQL. (NCERT)

Ans. Differences between ALTER and UPDATE commands in SQL are

ALTER command	UPDATE command
It belongs to DDL category.	It belongs to DML category.
It changes the structure of the table.	It modifies data of the table.
Columns can be added, modified, deleted etc.	Data can be changed, updated with values and expressions.

31) How is char data type different from varchar data type? (NCERT)

Ans: Differences between char data type and varchar data type are

Char	Varchar
It is fixed length.	It is variable length.
Wastage of memory.	Memory usage only as per data size.
Less useful.	Better data type.

32) What is the differences between HAVING clause and WHERE clause?

Ans. Differences between HAVING clause and WHERE clause are

WHERE clause	HAVING clause
WHERE clause is used to filter the records from the table based on the specified condition.	HAVING clause is used to filter record from the groups based on the specified condition.
WHERE clause implements in row operation.	HAVING clause implements in column operation.
WHERE clause cannot contain aggregate function.	HAVING clause can contain aggregate function.
WHERE clause can be used with SELECT, UPDATE, DELETE statement.	HAVING clause can only be used with SELECT statement.
WHERE clause is used with single row function like UPPER, LOWER etc.	HAVING clause is used with multiple row function like SUM, COUNT etc.

33) Is it compulsory to provide values for all columns of a table while adding records? Give an example.

No, it is not compulsory to provide values for all columns of a table while adding records. We can use NULL values wherever values are missing.

e.g. INSERT INTO Employee VALUES (1,NULL,"Sales",89000);

34) What do you mean by an operator? Name any four operators used in queries.

Ans.An operator is a component of an expression that represents the action that should be taken over a set of values.

Four operators used in queries are

- (i) Arithmetic operators (ii) Comparison operators
- (iii) Boolean/Logical operators (iv) Between operator

35. Rakesh wants to increase the price of some of the products by 20% , of his store whose price is less than 200. Assuming the following structure , what will be the query?

PNo PName Quality Price

UPDATE ITEM SET Price=Price + Price * 0.2
WHERE Price<200 ;

36. State the use of UPDATE command.

The UPDATE command updates data of a table . While updating, the expression for update value can be assigned to the updating field. The records to be updated can be specified as WHERE condition.

UPDATE ITEM SET Price=Price + Price * 0.2

WHERE Price<200 ;

37. Explain the use of ORDER BY clause.

- The ORDER BY clause is used to arrange the records in ascending or descending order.
- Data present in a table can be arranged as per requirement on a specific field in ascending or descending order.
- The default is ascending order. To arrange in descending order the DESC clause is to be used.
- To arrange in ascending order ASC may be used.
e.g.
SELECT * FROM Employee ORDER BY
EMP_SALARY DESC;
- The above command arranges the records in descending order of salary.

38) What is the purpose of GROUP BY clause in MySQL? How is it different from ORDER BY clause?

The GROUP BY clause can be used to combine all those records that have identical value in a particular field or a group of fields.

Whereas, ORDER BY clause is used to display the records either in ascending or descending order based on a particular field. For ascending order ASC is used and for descending order, DESC is used. The default order is ascending order

39) What are the aggregate functions in SQL? Give example.

Ans: Aggregate function is a function where the values of multiple-rows are grouped together as input on certain criteria to form a single value of more significant meaning.

Some aggregate functions used in SQL are SUM (), AVG(), MIN(), etc.

40) If the value in the column is repeatable, how do you find out the unique values?

The DISTINCT clause in SQL is used to display only distinct values in a column of a table. Hence, if the column allows duplicate values the unique values can be extracted using the DISTINCT clause.

SELECT DISTINCT CLASS FROM Student ; This displays only the unique classes.

41) Write the use of LIKE clause and a short explanation on the two characters used with it.

This operator is used to search a specified pattern in a column. It is useful when you want to search rows to match a specific pattern or when you do not know the entire value.

The SQL LIKE clause is used to compare a value to similar values using wildcard characters. We describe patterns by using two special wildcard characters, given below:

(i) The per cent sign (%) is used to match any substring.

(ii) The underscore (_) is used to match any single character.

The symbols can also be used in combinations.

~~~~~

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS – See Yes!*

*Easy Theory*

### NCERT FOCUS – NETWORKING

#### 1) What do you mean by data communication?

Data communication refers to the exchange of data between two or more networked or connected devices.

#### 2) What are the components of communication medium?

Sender, receiver, message, communication media and protocols are the components of communication medium.

**Sender:** A sender is a computer or any such device which is capable of sending data over a network. It can be a computer, mobile phone, smartwatch, walkietalkie, video recording device, etc.

**Receiver:** A receiver is a computer or any such device which is capable of receiving data from the network. It can be any computer, printer, laptop, mobile phone, television, etc. In computer communication, the sender and receiver are known as nodes in a network.

**Message:** It is the data or information that needs to be exchanged between the sender and the receiver. Messages can be in the form of text, number, image, audio, video, multimedia, etc.

**Communication media:** It is the path through which the message travels between source and destination. It is also called medium or link which is either wired or wireless. For example, a television cable, telephone cable, ethernet cable, satellite link, microwaves, etc.

**Protocols:** It is a set of rules that need to be followed by the communicating parties in order to have successful and reliable data communication.

#### 3) What is a channel?

In data communication, the transmission medium is known as channel.

#### 4) How the capacity of a communication medium is measured?

It is measured in terms of bandwidth and data transfer rate.

#### 5) What do you mean by data transfer rate?

Data travels in the form of signals over a channel. One signal carries one or more bits over the channel. Data transfer rate is the number of bits transmitted between source and destination in one second. It is also known as bit rate. It is measured in terms of bits per second (bps).

$$1 \text{ Kbps} = 2^{10} \text{ bps} = 1024 \text{ bps}$$

$$1 \text{ Mbps} = 2^{20} \text{ bps} = 1024 \text{ Kbps}$$

$$1 \text{ Gbps} = 2^{30} \text{ bps} = 1024 \text{ Mbps}$$

$$1 \text{ Tbps} = 2^{40} \text{ bps} = 1024 \text{ Gbps}$$

#### 6) What is the use of switching techniques?

To avoid establishing a dedicated connection between the sender and the receiver, in switching techniques data is routed through various nodes in a network. This switching process forms a temporary route for the data to be transmitted, avoiding high cost needed to establish dedicated connection.

#### 7) Explain circuit switching.

- In circuit switching, before a communication starts, a dedicated path is identified between the sender and the receiver.
- This path is a connected sequence of links between network nodes.
- All packets follow the same path established during the connection.
- In earlier days, when we placed a telephone call, the switching equipment within the telephone system finds out a physical path or channel all the way from our telephone at home to the receiver's telephone. This is an example of circuit switching.

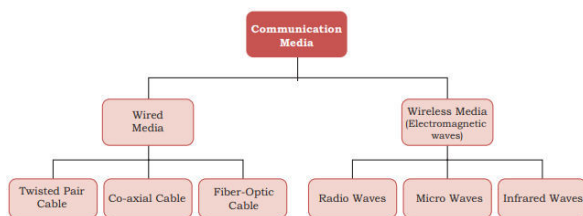
### 8) Explain packet switching.

- In packet switching, each information or message to be transmitted between sender and receiver is broken down into smaller pieces, called packets.
- These packets are then transmitted independently through the network.
- Different packets of the same message may take different routes depending on availability.
- Each packet has two parts — a header containing the address of the destination and other information, and the main message part.
- When all the packets reach the destination, they are reassembled and the complete message is received by the receiver.
- Unlike circuit switching, a channel is occupied in packet switching only during the transmission of the packet. On completion of the transmission, the channel is available for transfer of packets from other communicating parties.

### 9) Differentiate between guided and unguided media.

- Transmission can be classified as guided or unguided.
- In guided transmission, there is a physical link made of wire/cable through which data in terms of signals are propagated between the nodes. These are usually metallic cable, fiber-optic cable, etc. They are also known as wired media.
- In unguided transmission, data travels in air in terms of electromagnetic waves using an antenna. They are also known as wireless media.

### 10) Illustrate the types of transmission media.

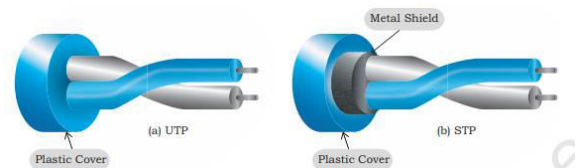


### 11) What is a transmission medium?

A transmission medium can be anything that can carry signals or data between the source (transmitter) and destination (receiver).

### 12) Describe about twisted pair cable and its uses.

- A twisted-pair consists of two copper wires twisted like a DNA helical structure.
- Both the copper wires are insulated with plastic covers.
- Usually, a number of such pairs are combined together and covered with a protective outer wrapping.
- Each of the twisted pairs act as a single communication link.
- The use of twisted configuration minimises the effect of electrical interference from similar pairs close by.
- Twisted pairs are less expensive and most commonly used in telephone lines and LANs.
- These cables are of two types: Unshielded twisted-pair (UTP) and Shielded twisted-pair (STP).



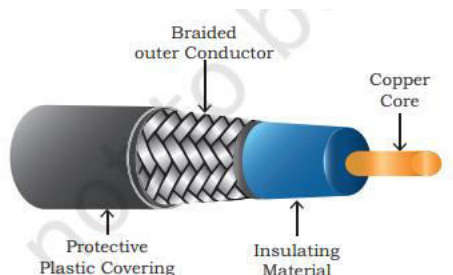
### 13) Describe about co-axial cable and its use.

- Coaxial cable is another type of data transmission medium.
- It is better shielded and has more bandwidth than a twisted pair.
- It has a copper wire at the core of the cable which is surrounded with insulating material.
- The insulator is further surrounded with an outer conductor (usually a copper mesh). This outer conductor is wrapped in a plastic cover.
- The key to success of coaxial cable is its shielded design that allows the cable's copper core to



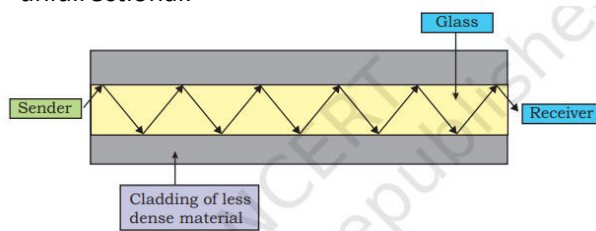
transmit data quickly, without interference of environmental factors.

- These types of cables are used to carry signals of higher frequencies to a longer distance.



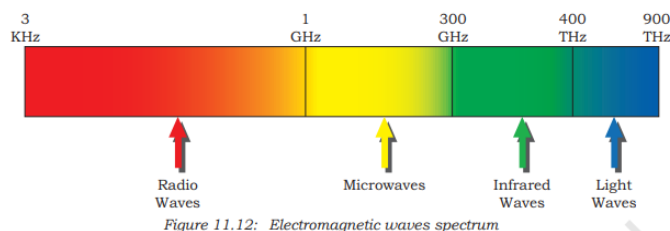
#### 14) Describe about optical fibre cable and its use.

- The optical fiber cable carries data as light, which travels inside a thin fiber of glass.
- Optic fiber uses refraction to direct the light through the media.
- A thin transparent strand of glass at the centre is covered with a layer of less dense glass called cladding.
- This whole arrangement is covered with an outer jacket made of PVC or Teflon. Such types of cables are usually used in backbone networks.
- These cables are of light weight and have higher bandwidth which means higher data transfer rate.
- Signals can travel longer distances and electromagnetic noise cannot affect the cable.
- However, optic fibers are expensive and unidirectional.



#### 15) How does information or data travel in wireless media?

- In wireless communication technology, information travels in the form of electromagnetic signals through air.
- Electromagnetic spectrum of frequency ranging from 3 KHz to 900 THz is available for wireless communication.
- Wireless technologies allow communication between two or more devices in short to long distance without requiring any physical media. There are many types of wireless communication technologies such as Bluetooth, WiFi, WiMax etc.



#### 16) What are the features of radio waves?

1. Waves of frequency range 3 KHz - 1 GHz
2. Omni-directional, these waves can move in all directions
3. Radio waves of frequency 300KHz-30MHz can travel long distance
4. Susceptible to interference
5. Radio waves of frequency 3-300KHz can penetrate walls
6. These waves are used in AM and FM radio, television, cordless phones.

#### 17) What are the features of micro waves?

1. Electromagnetic waves of frequency range 1GHz - 300GHz.
2. Unidirectional, can move in only one direction.
3. Cannot penetrate solid objects such as walls, hills or mountains.

4. Needs line-of-sight propagation i.e. both communicating antenna must be in the direction of each other.
5. Used in point-to-point communication or unicast communication such as radar and satellite.
6. Provide very large information-carrying capacity.

### 18) What are the features of infrared waves?

1. Electromagnetic waves of frequency range 300GHz - 400THz.
2. Very high frequency waves.
3. Cannot penetrate solid objects such as walls.
4. Used for short-distance point-to-point communication such as mobile-to-mobile, mobile-to-printer, remote-control-to-TV, and Bluetooth enabled devices to other devices like mouse, keyboards etc.

### 19) What is Protocol?

Protocol is a set of standard rules that the communicating parties the sender, the receiver, and all other intermediate devices need to follow.

### 20) What are the needs for protocols?

We need protocol,

#### For Flow Control

Flow control is required when the sender and receiver have different speeds of sending and receiving the data.

#### For Access Control

Access control is required to decide which nodes in a communication channel will access the link shared among them at a particular instant of time. Otherwise, the transmitted data packets will collide if computers are sending data simultaneously through the same link resulting in the loss or corruption of data.

### 21) What are the functions of protocols?

Protocols define,

- how computers identify one another on a network.
- the form to which the data should be converted for transit.

- how to decide whether the data received is for that node or to be forwarded to another node.
- ensuring that all the data have reached the destination without any loss.
- how to rearrange the packets and process them at the destination.

### 22) Briefly write about HTTP and its uses.

- HTTP stands for HyperText Transfer Protocol. It is the primary protocol used to access the World Wide Web.
- HTTP is a request-response (also called clientserver) protocol that runs over TCP.
- The common use of HTTP is between a web browser (client) and a web server (server).
- HTTP facilitates access of hypertext from the World Wide Web by defining how information are formatted and transmitted, and how the Web servers and browsers should respond to various commands.
- A web page is written using a markup language like HTML and is stored on a web server for access via its URL. Once a user opens a web browser and types in the URL of the intended web page, a logical communication link between the user machine (client) and the web server is created using HTTP. Then, the webpage is displayed on the browser.

### 23) Briefly describe about FTP and its uses.

File Transfer Protocol (FTP) is the protocol used for transferring files from one machine to another.

Like HTTP, FTP also works on a client-server model. When a user requests for a file transfer with another system, FTP sets up a connection between the two nodes for accessing the file.

File transfer between two systems seems simple and straightforward because FTP takes care of issues between two communicating devices,

such as,

- use of different conventions while naming files.
- representation of text and data in different formats.
- having different directory structure



**24) Briefly describe about PPP and its uses.**

- PPP is a communication protocol which establishes a dedicated and direct connection between two communicating devices.
- This protocol defines how two devices will authenticate each other and establish a direct link between them to exchange data.
- For example, two routers with direct connection communicate using PPP. The Internet users who connect their home computers to the server of an Internet Service Provider (ISP) through a modem also use PPP.
- This protocol maintains data integrity ensuring that the packets arrive in order.
- It intimates the sender about damage or lost packets and asks to resend it.

**25) Briefly describe about SMTP and its uses.**

- SMTP is a protocol used for email services.
- It uses information written on the message header (like an envelope on a letter sent by post), and is not concerned with the content of the email message.
- Each email header contains email addresses of recipients. The email containing header and body are entered into a queue of outgoing mails.
- The SMTP sender program takes mails from the outgoing queue and transmits them to the destination(s).
- When the SMTP sender successfully delivers a particular mail to one or more destinations, it removes the corresponding receiver's email address from the mail's destination list.
- When that mail is delivered to all the recipients, it is removed from the outgoing queue.
- The SMTP receiver program accepts each mail that has arrived and places it in the appropriate user mailbox

**26) Briefly describe about HTTP and its uses.**

- TCP/IP stands for Transmission Control Protocol/Internet Protocol.
- It is a set of standardised rules that uses a client-server model of communication in which a user

or machine (a client) requests a service by a server in the network.

- The IP protocol ensures that each computer or node connected to the Internet is assigned an IP address, which is used to identify each node independently.
- TCP ensures that the message or data is broken into smaller chunks, called IP packets. Each of these packets are routed (transmitted) through the Internet, along a path from one router to the next, until it reaches the specified destination.
- TCP guarantees the delivery of packets on the designated IP address. It is also responsible for ordering the packets so that they are delivered in sequence.
- When all the packets finally reach the destination machine, they are reassembled into the original message at the receiver's end.

**27) What are the advantages of networking**

- Interconnectivity of computing devices in a network allows us to exchange information simultaneously with many parties through email, websites, audio/video calls, etc.
- Network allows sharing of resources. For example, a printer can be made available to multiple computers through a network;
- People often connect their devices through hotspot, thus forming a small personal network.

**28) What is ARPANET?**

- In the 1960s a research project was commissioned by Advanced Research Projects Agency Network (ARPANET) in the U.S. Department of Defence to connect the academic and research institutions located at different places for scientific collaborations.
- Slowly but gradually, more and more organisations joined the ARPANET, and many independent smaller networks were formed.

**29) What are the different types of networks?**

Based on the geographical area covered and data transfer rate, computer networks are broadly categorised as:

- PAN ( Personal Area Network)
- LAN (Local Area Network)
- MAN (Metropolitan Area Network)
- WAN (Wide Area Network)

### 30) Explain about PAN.

- It is a network formed by connecting a few personal devices like computers, laptops, mobile phones, smart phones, printers etc.,
- All these devices lie within an approximate range of 10 metres.
- A personal area network may be wired or wireless.
- For example, a mobile phone connected to the laptop through USB forms a wired PAN while two smartphones communicating with each other through Bluetooth technology form a wireless PAN or WPAN.

### 31) Briefly describe about LAN.

- It is a network that connects computers, mobile phones, tablet, mouse, printer, etc., placed at a limited distance.
- The geographical area covered by a LAN can range from a single room, a floor, an office having one or more buildings in the same premise.
- The connectivity is done by means of wires, Ethernet cables, fibre optics, or Wi-Fi.
- LANs provide the short range communication with the high speed data transfer rates.
- These types of networks can be extended up to 1 km.
- Data transfer in LAN is quite high, and usually varies from 10 Mbps (called Ethernet) to 1000 Mbps (called Gigabit Ethernet).

### 32) Briefly describe about MAN.

- Metropolitan Area Network (MAN) is an extended form of LAN which covers a larger geographical area like a city or a town.
- Data transfer rate in MAN also ranges in Mbps, but it is considerably less as compared to LAN.

- Cable TV network or cable based broadband internet services are examples of MAN.
- This kind of network can be extended up to 30-40 km.
- Sometimes, many LANs are connected together to form MAN.

### 33) What are the features of WAN?

- Wide Area Network connects computers and other LANs and MANs, which are spread across different geographical locations of a country or in different countries or continents.
- Large business, educational and government organisations connect their different branches in different locations across the world through WAN.
- The Internet is the largest WAN that connects billions of computers, smartphones and millions of LANs from different continents.

### 34) Write about the range for different networks.

- PAN – 10 metres
- LAN – upto 1 km
- MAN – upto 30-40 km
- WAN – distance in hundreds, thousands

### 35) What is MODEM? How does it function?

- Modem stands for 'MOdulator DEModulator'.
- It refers to a device used for conversion between analog signals and digital bits.
- The modem at the sender's end acts as a modulator that converts the digital data into analog signals. The modem at the receiver's end acts as a demodulator that converts the analog signals into digital data for the destination node to understand.

**36) Write about the features and uses of Ethernet Card.**

- Ethernet card, also known as Network Interface Card (NIC card in short) is a network adapter used to set up a wired network.
- It acts as an interface between computer and the network.
- It is a circuit board mounted on the motherboard of a computer.
- The Ethernet cable connects the computer to the network through NIC.
- Ethernet cards can support data transfer between 10 Mbps and 1 Gbps (1000 Mbps). Each
- NIC has a MAC (Media Access Control) address, which helps in uniquely identifying the computer on the network.

**37) What is the use of RJ45?**

- RJ 45 or Registered Jack-45 is an eight-pin connector that is used exclusively with Ethernet cables for networking.
- It is a standard networking interface that can be seen at the end of all network cables.
- Basically, it is a small plastic plug that fits into RJ-45 jacks of the Ethernet cards present in various computing devices.

**38) What is the use of Repeaters?**

- Data are carried in the form of signals over the cable. These signals can travel a specified distance (usually about 100 m).
- Signals lose their strength beyond this limit and become weak. In such conditions, original signals need to be regenerated.
- A repeater is an analog device that works with signals on the cables to which it is connected.
- The weakened signal appearing on the cable is regenerated and put back on the cable by a repeater.

**39) How does a hub function? Write one of its limitations.**

- An Ethernet hub is a network device used to connect different devices through wires.
- Data arriving on any of the lines are sent out on all the others.
- The limitation of Hub is that if data from two devices come at the same time, they will collide.

**40) What is a Switch? How does it function?**

- A switch is a networking device that plays a central role in a Local Area Network (LAN).
- Like a hub, a network switch is used to connect multiple computers or communicating devices.
- When data arrives, the switch extracts the destination address from the data packet and looks it up in a table to see where to send the packet. Thus, it sends signals to only selected devices instead of sending to all.
- It can forward multiple packets at the same time. A switch does not forward the signals which are noisy or corrupted. It drops such signals and asks the sender to resend it.

**41) What do you mean by Gateway? What are its uses?**

- As the term "Gateway" suggests, it is a key access point that acts as a "gate" between an organisation's network and the outside world of the Internet.
- Gateway serves as the entry and exit point of a network, as all data coming in or going out of a network must first pass through the gateway in order to use routing paths.
- Besides routing data packets, gateways also maintain information about the host network's internal connection paths and the identified paths of other remote networks.
- If a node from one network wants to communicate with a node of a foreign network, it will pass the data packet to the gateway, which then routes it to the destination using the best possible router.
- For simple Internet connectivity at homes, the gateway is usually the Internet Service Provider

(BSNL, airtel etc.) that provides access to the entire Internet.

- A gateway can be implemented completely in software, hardware, or a combination of both.



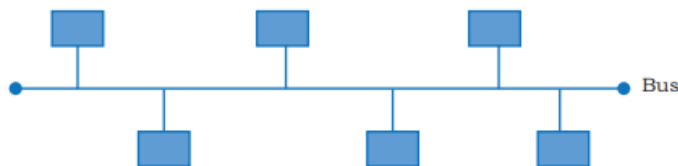
#### 42) What is a topology?

Arrangement of computers and other peripherals in a network is called its topology.

Common network topologies are Mesh, Ring, Bus, Star and Tree.

#### 43) Write about bus topology.

- In bus topology, each communicating device connects to a transmission medium, known as bus.
- Data sent from a node are passed on to the bus and hence are transmitted to the length of the bus in both directions. That means, data can be received by any of the nodes connected to the bus.



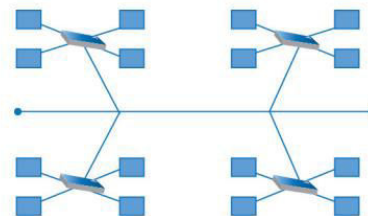
- In this topology, a single backbone wire called bus is shared among the nodes, which makes it cheaper and easier to maintain.
- Both ring and bus topologies are considered to be less secure and less reliable.

#### 44. How does star topology function?

- In star topology, each communicating device is connected to a central node, which is a networking device like a hub or a switch.
- Star topology is considered very effective, efficient and fast as each device is directly connected with the central device.
- Although disturbance in one device will not affect the rest of the network, any failure in a central networking device may lead to the failure of complete network

#### 45. What are the features of tree topology?

- It is a hierarchical topology, in which there are multiple branches and each branch can have one or more basic topologies like star, ring and bus.
- Such topologies are usually realised in WANs where multiple LANs are connected. Those LANs may be in the form of a ring, bus or star.
- In this type of network, data transmitted from source first reaches the centralised device and from there the data passes through every branch where each branch can have links for more nodes.



#### 46) What is IP Address? How is it useful in networking?

- IP address, also known as Internet Protocol address, is also a unique address that can be used to uniquely identify each node in a network.
- The IP addresses are assigned to each node in a network that uses the Internet Protocol for communication. Thus, if we know a computer's IP address, we can communicate with that computer from anywhere in the world.
- However, unlike MAC address, IP address can change if a node is removed from one network and connected to another network.
- The initial IP Address called version 4 (IPv4 in short), is a 32 bit numeric address, written as four numbers separated by periods, where each number is the decimal (base-10) representation for an 8-bit binary (base-2) number and each can take any value from 0 - 255.
- A sample IPV4 address looks like: 192:168:0:178
- With more and more devices getting connected to the Internet, it was realised that the 32-bit IP

address will not be sufficient as it offers just under 4.3 billion unique addresses.

- Thus, a 128 bits IP address, called IP version 6 (IPv6 in short) was proposed. An IPv6 address is represented by eight groups of hexadecimal (base-16) numbers separated by colons.
- A sample IPV6 address looks like: 2001:CDBA:0000:0000:0000:0000:3257:9652

#### 47) What is WWW? What are its features?

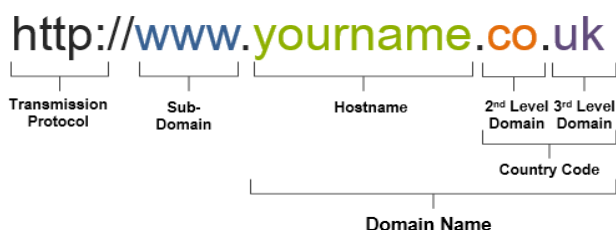
Earlier, to access files residing in different computers, one had to login individually to each computer through the Internet. Besides, files in different computers were sometimes in different formats, and it was difficult to understand each other's files and documents. Sir Tim Berners-Lee — a British computer scientist invented the revolutionary World Wide Web in 1990 by defining three fundamental technologies that lead to creation of web:

- HTML – HyperText Markup Language. It is a language which is used to design standardised Web Pages so that the Web contents can be read and understood from any computer. Basic structure of every webpage is designed using HTML.
- URI – Uniform Resource Identifier. It is a unique address or path for each resource located on the web. It is also known as Uniform Resource Locator (URL). Every page on the web has a unique URL. ex: [www.ncert.nic.in](http://www.ncert.nic.in)
- HTTP – The HyperText Transfer Protocol is a set of rules which is used to retrieve linked web pages across the web. The more secure and advanced version is HTTPS.

#### 48) What is a domain name? Give Example.

- Domain names are unique names given to websites to help humans remember them. Computers don't use the domain name at all; it's designed to make internet browsing easier for human beings.

Example:



#### 49) What are domain names?

- It is very difficult to remember the IP addresses of different websites as they are in terms of numbers or strings.
- It is easier to remember names, and therefore, each computer server hosting a website or web resource is given a name against its IP address. These names are called the Domain names.

#### 50) What do you mean by domain name resolution? What is a DNS Server?

- To access a web resource, a browser needs to find out the IP address corresponding to the domain name entered.
- Conversion of the domain name of each web server to its corresponding IP address is called domain name resolution. It is done through a server called DNS server.
- Thus, when we enter a URL on a web browser, the HTTP protocol approaches a computer server called DNS server to obtain the IP address corresponding to that domain name.
- After getting the IP address, the HTTP protocol retrieves the information and loads it in our browser.

#### 51) Describe about VoIP.

- Voice over Internet Protocol (VoIP), is a technology that allowing you to make voice calls over a broadband Internet connection instead of an analog (regular) phone line.
- Some VoIP services allow you to call people using the same service, but others may allow you to call anyone.
- They can have a telephone number – including local, long-distance, mobile, and international numbers or not.
- Some VoIP services only work over your computer or a special VoIP phone while other services allow you to use a traditional phone connected to a VoIP adapter.
- Voice is converted into a digital signal by VoIP services that travel over the Internet.
- If the regular phone number is called, the signal is converted to a regular telephone signal i.e. an analog signal before it reaches the destination.

### 52) Write about POP3.

- Post Office Protocol 3, or POP3, is the most commonly used protocol for receiving email over the internet.
- This standard protocol, which most email servers and their clients support, is used to receive emails from a remote server and send to a local client.
- POP3 is a one-way client-server protocol in which email is received and held on the email server. The "3" refers to the third version of the original POP protocol.
- A recipient or their email client can download mail periodically from the server using POP3. Thus, POP3 offers a means of downloading email from a server to the client so the recipient can view the email offline.
- Once the email is on the client, POP3 then deletes it from the server.

### 53) What is Telnet?

Telnet is a network protocol used to virtually access a computer and to provide a two-way, collaborative and text-based communication channel between two machines.

It follows a user command Transmission Control Protocol/Internet Protocol (TCP/IP) networking protocol for creating remote sessions.

### 54) Differentiate between HTTP and HTTPS.

- Both the HTTP (Hyper Text Transfer Protocol) and its variant HTTPS (Hyper Text Transfer Protocol Secure) are a set of rules (protocol) that govern how data can be transmitted over the WWW (World Wide Web).
- HTTP sends information over the network as it is. It does not scramble the data to be transmitted, leaving it vulnerable to attacks from hackers. Hence, HTTP is sufficient for websites with public information sharing like news portals, blogs, etc.
- However, when it comes to dealing with personal information, banking credentials and passwords, we need to communicate data more securely over the network using HTTPS.

- HTTPS encrypts the data before transmission. At the receiver end, it decrypts to recover the original data.
- The HTTPS based websites require SSL Digital Certificate.

### 55) Write about XML.

- XML stands for Extensible Markup Language. It is a text-based markup language.
- XML tags identify the data and are used to store and organize the data, rather than specifying how to display it like HTML tags, which are used to display the data.
- XML allows you to create your own self-descriptive tags that suits your application.

### 56) Define website, web browser, web servers, web hosting. refer viva handout

#### NCERT Book back questions

1. What is data communication? What are the main components of data communication?
2. Which communication mode allows communication in both directions simultaneously?
3. Among LAN, MAN, and WAN, which has the highest speed and which one can cover the largest area?
4. What are three categories of wired media? Explain them.
5. Compare wired and wireless media.
6. Which transmission media carries signals in the form of light?
7. List out the advantages and disadvantages of optical fiber cable.
8. What is the range of frequency for radio waves?
10. HTTP stands for?
11. Write short note on the following: a) HTTP b) Bandwidth c) Bluetooth d) DNS e) Data transfer rate
12. What is protocol in data communication? Explain with an example.
13. Expand the following: a) ARPANET b) MAC c) ISP d) URI
14. What do you understand by the term network?
15. Mention any two main advantages of using a network of computing devices.
16. Differentiate between LAN and WAN. Write down the names of few commonly used networking devices.

18. Two universities in different States want to transfer information. Which type of network they need to use for this?

19. Define the term topology. What are the popular network topologies?

20. How is tree topology different from bus topology?

21. Identify the type of topology from the following: a) Each node is connected with the help of a single cable. b) Each node is connected with central switching through independent cables.

22. What do you mean by a modem? Why is it used?

23. Explain the following devices: a) Switch b) Repeater c) Router d) Gateway e) NIC

24. Draw a network layout of star topology and bus topology connecting five computers.

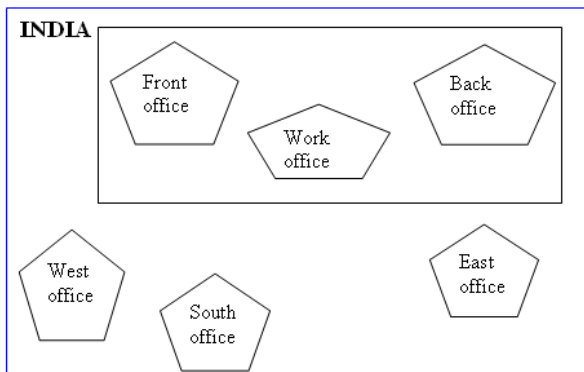
25. What is the significance of MAC address?

26. How is IP address different from MAC address? Discuss briefly.

27. What is DNS? What is a DNS server?

28. Sahil, a class X student, has just started understanding the basics of Internet and web technologies. He is a bit confused in between the terms "World Wide Web" and "Internet". Help him in understanding both the terms with the help of suitable examples of each.

- Many people confuse the web with the Internet. The Internet as we know is the huge global network of interconnected computers, which may or may not have any file or webpage to share with the world.
- The web on the other hand is the interlinking of collection of Webpages on these computers which are accessible over the Internet. WWW today gives users access to a vast collection of information created and shared by people across the world. It is today the most popular information retrieval system.



"SAMAR & SEER TELELINK ASSOCIATION" is Planning to spread their offices in four major cities in India to provide regional IT infrastructure support in the field of Education & Culture.

The company has planned to setup their head office in New Delhi in three locations and have named their New Delhi offices as "Front Office ","Back Office "and "Work Office ".The company has three more regional offices as "South Office ","East Office"and "West Office " located in other three major cities of India. A rough layout of the same is given here. (Picture shown above)

Approximate distance between these offices as per network survey team is as follows

| Place From  | Place To     | Distance |
|-------------|--------------|----------|
| Back Office | Front Office | 10 KM    |
| Back Office | Work Office  | 70 KM    |
| Back Office | East Office  | 1291KM   |
| Back Office | West Office  | 790 KM   |
| Back Office | South Office | 1952KM   |

In continuation of the above, the company experts have planned to install the following number of computers in each of their offices:

|                     |     |
|---------------------|-----|
| <b>Front Office</b> | 100 |
| <b>Work Office</b>  | 20  |
| <b>East Office</b>  | 501 |
| <b>West Office</b>  | 50  |
| <b>SouthOffice</b>  | 50  |

a) Suggest network type(out of LAN,MAN,WAN) for connecting each of the following set of their offices:

Back Office and Work Office

Back Office and south Office

b) Which device you will suggest to be produced by the company for connecting all the computers with in each of their offices out of the following devices?

Switch/Hub

Modem

Telephone

*“When something is important enough, you do it even if the odds are not in your favor.”*

– Elon Musk

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS – See Yes!*

### OUTSTANDING OUTPUTS

#### 1) Find output

```
a = (1,6,5,4,7)
b = list('unittest')
b.sort()
c = []
for i in a:
    if i%2==0:
        c.append(i)
        c.extend(b[i])
```

```
c.pop()
```

```
print(c)
```

#### 2) Choose the correct output:

```
t = 5
for x in range(10):
    if x%2==0:
        continue
    else:
        t+=x
x+=1
```

```
print(t)
```

#### 3) Find the output:

```
a='sathyasaiinstitute'
b = a[3:10:2]
c = a[-3:-10:-2]
d = b+c
print(d)
```

#### 4) Find the output:

```
s = 'Love'
t = s*2
u = s+'2'
print(u[:2]+u[2:])
```

#### 5) Find the output:

```
a = 'ReVislon@ExaM'
b = ""
for i in range(len(a)):
    if a[i].islower():
        b+='a'
    elif a[i].isupper():
        b+='b'
    else:
        b+='c'
print(b)
```

#### 6) Find the output:

```
x = 'God'
y = 2
z = 'knows'
w = x*y
x = x+z
y = w+x
z = len(y)//2
print(z)
```

#### 7) Find the output:

```
s = 'exam' < 'Test' and 'test' >= 'Test'
t = 'I' != 'we' or not 'Me' and 'Mine'
print(s or t)
```



**8) Find the output:**

```
a = 2
b = -4
c = 6
d = a+b*c**1+c//a-c%a
print(d)
```

**9) Find the output:**

```
a = 2
for i in range(1,10,3):
    a+=(i**1)
print(a)
```

**10) Find the output:**

```
n = bool(int(int(len(str('sathya')))*0.2))
print(n)
```

**11) Find the output:**

```
m = 'I am' is 'Aham'
n = 'm' not in 'Aham'
p = 'i am' < 'I am' or 'i am' > 'I am'
print(not (m or n and p))
```

**12) Find the output:**

```
h = 1
j = 1
while h<20:
    if h%10==5:
        j+=5
    else:
        j=j
    h+=5

print(j)
```

```
13)
a = 30%9 + 3//2 * 7%108 - 2
b = a/2 + a//2
print(a)
```

**14) Find the output:**

```
v = 'VicTorY#ouRs'
w = ''
for i in range(len(v)):
    if v[i].isupper():
        w+=v[i].lower()
    if v[i].islower():
        w+='#'
    else:
        w+='$'
print(w)
```

**15) Find the output:**

```
c = 1
while c<10:
    if c%3==0:
        c+=2
    else:
        c+=1
    continue
print(c)
```

**16) Find the output:**

```
b = 'Indiaismycountry'
c = b[::2]
d = c[::2]
e = d[1::1]
print(e)
```

**17) Find the output:**

```
s = 'DiwaliholiDays'
t = 'Daddies'
u = s.strip(t)
v = u.lstrip('Hawaii')
print(v)
```

**18) Find the output:**

```
a = 'AvoidCrackers'
b = a.lower()
c = a.count('a')
d = a.index('r')
e = a.find('s')
print(c+d+e)
```

**19) Find the output:**

```
import math
a = math.sqrt(16)
b = math.fabs(4)
c = math.floor(4.8)
print(a+b+c)
```

**20) Find the output:**

```
a = 8+2*8**1
b = 1**8*2+8
print(a-b)
```

**21) Find the output:**

```
x = 10
for y in range(-10,-20,-1):
    if y%2!=0:
        x += (y//2)
print(x)
```

**22) Find the output:**

```
x, y, z = int(44//30), int(44/3), int(44%3)
x, y, z = x/x, y/y, z/z
x, y, z = z, y, x
x, y, z = x**1, y*2, z+3
print(x,y,z)
```

**23) Find the output:**

```
e, f, g = bool(0), bool([0]), bool("")
e, f, g = g, f, e
e, f, g = str(g), str(f), str(e)
e, f, g = e+f, f+g, g+e
print(e, g)
```

**24) Find the output:**

```
a, b, c, d, e = 6, 7, 8, 9, 10
a, b, c, d, e = e, d, c, b, a
a, b, c, d = a+e, b-e, c*e, d//e
a, b, c, d = d, c, b, a
print(a+b, b+c, c+d, d+e)
```

**25) Find the output:**

```
x, y, z = len('sri'), len('sai'), len('baba')
x,y,z = z, y, z
x, y = x+y+z, x-y-z
x, y = y, x
print(x,y,z)
```

**26) Find the output:**

```
z = 5.6+2j
a = len(str(float(int(z.real))))
print(a)
```

**27) Find the output:**

```
i, j, k = True, False, True
i, j, k = k, j, i
i, j, k = k>j, j==i, i<k
i,j, k = i+j, j+k, k+i
print(i,j,k)
```

**28) Find the output:**

```
a, b = 34.2//3, 45%7
a, b = a/b, b+(a-b)
a, b = b, a
print(a, a)
```

**29) Find the output:**

```
x = 'Examination'
y = x.strip('Extension')
z = y.split()
print(len(z)+len(x))
```

**30) Find the output:**

```
import math
import statistics
a = [1,2,3,4,1,6.2]
b = math.floor(a[-1])
c = statistics.mode(a)
d = b+c
print(d)
```

**31) Find the output:**

```
a = int(float(4/2))
b = float(int(4/2))
print(a//b)
```

32)

```
print("I" "am" "Good", end=' ')
print('and strong', end='*', sep='#')
```

**33) Find the output:**

```
def ChangeVal(A,B):
    for i in range(1,B):
        if A[i]%3 == 0:
            A[i]//=3
        if A[i]%5 == 0:
            A[i]//=5
        else:
            A[i]+=1
```

```
K = [3,6,7,10,11,12]
ChangeVal(K,5)
print(K)
```

**34) Find the output:**

```
p = [1,3,4,6,7,8,9]
q = 'listmethods'
p.insert(len(p)-1,len(q))
p.sort(reverse = True)
r = p[::-2]
r.append(max(q))
print(r[2], r[-1])
```

**35) Find the output:**

```
a = "getcentum"
b = ""get
centum""
c = (((1,2),3),4)
print(len(a)+len(b)+len(c))
```

**36) Find the output:**

```
a = 'allthebest'
b = {}
```

for i in a:

```
    if i>'d' and i < 'j':
        b[i] = 1
```

```
    elif i>'p' and i<'z':
        b[i] = 2
```

```
    else:
        pass
```

print(b)

**37) Find the output:**

```
import math as m
a = m.floor(4.5)
b = m.ceil(3.8)
c = a+b
print(c)
```

**38) Find the output:**

```
def findcode(y):
    msg = ""
    for i in range(len(y)):
        if y[i].isalnum():
            msg+='*'
        elif y[i].isspace():
            msg+=y[i+1]
        else:
            msg+=y[i-1]
    print(msg)
```

```
t = 'Term I Exam@21'
findcode(t)
```

**39) Find the output:**

```
h = 'SweeTHomE'
txt = ""
for i in h:
    if i.isupper():
        txt+=i.lower()
    elif i.islower():
        if h.index(i)%2==0:
            txt+=i.upper()
        else:
            txt+='#'
print(txt)
```

**40) Find the output:**

```

a = 1
b = 2
c = 3
d = 4
e = 5
k = d//b*a**b+(e%c)-int(e/c)
print(k)

```

**41) Find the output:**

```

m = 9 > 8 and 'Baba' < 'Baba' or 12.5 < 11.9 and not
(1,2)
print(m)

```

**42) Find the output:**

```

a = 'Truth'
b = (1,2)
c = ((b*2),b*1,len(a),)
print(len(c))

```

**43) Find the output:**

```

a='India2022'
k=0
for i in a:
    if i.isalnum():
        k+=2
print(k)

```

**44) Find the output:**

```

x = "writing"
y = "exam"
print(x+y*(int(len(y)/2)))

```

**45) Find the output:**

```

a = 'I am'
b = 'fine'
print(a, b, sep='@',end='*')

```

**46) Find the output:**

```

x,y,z = 2,4,6
x,y,z = x<y, y>z, z!=x
x,y,z = z,y,x
x,y,= x+x,y+z
print(x,y,z)

```

**47) Find the output:**

```

a = 3
for b in range(2):
    for c in range(3):
        if b%2!=0:
            a+=(c*2)
print(a)

```

**48) Find the output:**

```

names1 = ['Amir', 'Bear', 'Charlton', 'Daman']
names2 = names1
names3 = names1[:]
names2[0] = 'Alice'
names3[1] = 'Bob'
sum = 0
for ls in (names1, names2, names3):
    if ls[0] == 'Alice':
        sum += 1
    if ls[1] == 'Bob':
        sum += 10
print (sum)

```

**49) Find the output:**

```

a=list((45,)*4)
print((45)*4)
print(a)

```

**50) Find the output:**

```

values = [[3, 4, 5, 1], [33, 6, 1, 2]]
v = values[0][0]
for lst in values:
    for element in lst:
        if v > element:
            v = element
print(v)

```

**51) Find the output:**

Assume somefile.txt contains:

```

This is line 1
This is line 2
This is line 3
This is line 4

```

Then, what is the output of the following

```
f1 = open("f:\somefile.txt", "r+")
print(f1.read(5))
print(f1.readline())
f1.readline()
print(f1.readline())
f1.readline()
f1.close()
```

## 52) Find the output:

```
def display(s):
    l = len(s)
    m=""
    for i in range(0,l):
        if s[i].isupper():
            m=m+s[i].lower()
        elif s[i].isalpha():
            m=m+s[i].upper()
        elif s[i].isdigit():
            m=m+"$"
        else:
            m=m+"*"
    print(m)

display("SSSIE.org2022")
```

## 53) Find the output:

```
X = 100
def Change(P=10, Q=25):
    global X
    if P%6==0:
        X+=100
    else:
        X+=50
    Sum=P+Q+X
    print(P,'#',Q,'$',Sum)

Change()
Change(18,50)
Change(30,100)
```

## 54) Find the output:

```
st = ("one", "two", "three")
nt = "$".join(st)
print(nt)
```

## 55) Find the output:

```
sum = 0
for i in range(1,12):
    if i%2==0:
        continue
    elif i%5==0:
        break
    sum+=i

print(sum)
```

## 56) Find the output:

```
d = {"rama":1, "krishna":2, "baba":3}
for x in d.keys():
    print(x[2], end='%')
```

57)

```
TXT=["10","20","30","40"]
CNT = 3
TOTAL = 0
for C in [7,5,4,6]:
    T = TXT[CNT]
    TOTAL = float(T) + C
    print(TOTAL)
    CNT-=1
```

## 58) Find the output:

```
def Changer(P,Q=10):

    P=P+Q

    Q=P+Q

    return P

A=200
B=20

A=Changer(A,B)

print(A,"$",B)

B=Changer(B)

print(A,"$",B)

A=Changer(A)

print(A,"$",B)
```

**59) Find the output:**

```
def Display(str):
    m=""
    for i in range(0,len(str)):
        if(str[i].isupper()):
            m=m+str[i].lower()
        elif str[i].islower():
            m=m+str[i].upper()
        else:
            if i%2==0:
                m=m+str[i-1]
            else:
                m=m+"#"
    print(m)
```

Display('Fun@Python3.0')

**60) Find the output:**

```
value = 50
def display(N):
    global value
    value = 25
    if N%7==0:
        value = value + N
    else:
        value = value - N
    print(value, end="#")
```

```
display(20)
print(value)
```

**61) Find the output:**

```
D1={1: "One",2: "Two", 3: "C"}
D2={4: "Four",5: "Five"}
D1.update(D2)
print (D1)
```

**62) Find the output:**

```
a = "ReviSioN1Exam"
b = a[::-2]
for i in b:
    print(i, end="")
```

**63) Find the output:**

```
x = {"Rahul":450, "Raju":340, "Karthi":490}
for k,v in x.items():
    if v >= 450:
        print(k, v)
```

**64) Find the output:**

```
a = (4,9,10,13,34,20)
b = []
for i in a:
    if i%2==0:
        b.append(i)
b.pop()
b.append(a[-1])
b.extend([6,1])
b.sort(reverse=True)
c = b[1::-1]
print(c)
```

**65) Find the output:**

```
a = [4,3,2,7,6]
for i in a:
    if i%2!=0:
        print(str(i)+str(i))
```

**66) Find the output:**

```
a = "sai"
b = "baba"
c = "krish"
d = {}
d[a] = 1
d[b] = "1"
d[a] = 2
d[b] = "3"
d[c] = [1,2,3]
print(d[a])
```

**67) Find the output:**

```
j = [0,1,2,3,4]
a = j[0]
b= j[1]
print(len(str(float(bool(a)+bool(b)))))
```

**68) Find the output: 7 - 7//2 + 4\*\*2\*2**

**69) Find the output:**

```
not False and "CBSE" >= "STATE" or bool('False')
```

**70) Find the output:**

```
8 or 9 and 2 or 0
```

**71) Find the output:**

```
a) 1<5>2<3
```

```
b) 7<4>8
```

**72) Find the output:**

```
def funsend(P ,Q=10):
```

```
    P=P+Q
```

```
    Q=P-Q
```

```
    print( P,"#",Q)
```

```
    return P
```

```
R=4
```

```
S=30
```

```
R=funsend(R,S)
```

```
print(R,"@",S)
```

```
S=funsend(S)
```

**73) Find the output:**

```
def Changelt(Text,C):
```

```
    T=""
```

```
    for K in range(len(Text)):
```

```
        if Text[K]>='F' and Text[K]<='L':
```

```
            T=T+Text[K].lower();
```

```
        elif Text[K]=='E' or Text[K]=='e':
```

```
            T=T+C;
```

```
        elif K%2==0:
```

```
            T=T+Text[K].upper()
```

```
        else:
```

```
            T=T+T[K-1]
```

```
    print("New TEXT:",T)
```

```
oldText="FrEeExamKioSK"
```

```
Changelt(oldText,"%")
```

**74) Find the output:**

```
Data = ["P",20,"R",10,"S",30]
```

```
Times = 0
```

```
Alpha = ""
```

```
Add = 0
```

```
for C in range(1,6,2):
```

```
    Times= Times + C
```

```
    Alpha= Alpha + Data[C-1]+"$"
```

```
    Add = Add + Data[C]
```

```
print(Times, Add, Alpha)
```

**75) Find the output:**

```
def div(lst,n):
```

```
    for i in range(0,n):
```

```
        if lst[i]%5==0:
```

```
            lst[i]+=5
```

```
        else:
```

```
            lst[i]=lst[i]/2
```

```
lt=[45,20,23,54,5]
```

```
div(lt,len(lt))
```

```
for i in lt:
```

```
    print(i,end='#')
```

**76) Find the output:**

```
def short(lst,n):
```

```
    for i in range(0,n):
```

```
        if len(lst[i])>4:
```

```
            lst[i]=lst[i][0:4]
```

```
        else:
```

```
            lst[i]=lst[i]
```

```
sub=['IP','HINDI','CS','Mathematics','Chemistry','Phy  
sics']
```

```
short(sub,6)
```

```
for i in sub:
```

```
    print(i, end=',')
```

**77) Find the output:**

```
def A_func(x=10,y=20):
```

```
    x=x+1
```

```
    y=y-2
```

```
    return(x+y)
```

```
print(A_func(5),A_func())
```

**78) Find the output:**

```
def fun3(num1,num2):
    for x in range(num1,num2):
        if x%4==0:
            print(x,end=' ')
fun3(10,20)
```

```
elif S[i].isupper():
    m=m+S[i].lower()
else:
    m=m+'@'
print(m)

func('CSWORKSHOP2023')
```

**79) Find the output:**

```
val=100
def display(N):
    global val
    val=50
    if N%14==0:
        val=val+N
    else:
        val=val-N
    print(val,end="@")

display(40)
print(val)
```

**83) Find the output:**

```
n=7
c=0
while(n):
    if(n>5):
        c=c+n-1
        n=n-1
    else:
        break
print(n)
print(c)
```

**80) Find the output:**

```
i=1
while i<6:
    print(i)
    if i<2:
        break
    i+=1
```

**84) Find the output:**

```
val = 25
def Fun1():
    val=100
    print(val,end='#')

def Fun2():
    global val
    val=val%6
```

**81) Find the output:**

```
tupl=([2,3],"abc",0,9)
tupl[0][1]=1
print(tupl)
```

```
print(val,end='#')
Fun1()
print(val,end='#')
Fun2()
print(val,end='#')
```

**82) Find the output:**

```
def func(S):
    k=len(S)
    m=""
    for i in range(0,k):
        if S[i].islower():
            m=m+S[i].upper()
        elif S[i].isdigit():
            m=m+'*'
```

**85) Find the output:**

```
def Manipulate(M):
    for i in range(len(M)):
        if M[i]%10==0:
            M[i]-=10
        elif M[i]//10==6:
            M[i]+=M[i-1]
```



```
else:
```

```
    M[i]+=1
```

```
MyList = [10,5,17,62,37,65,80]
```

```
Manipulate(MyList)
```

```
print(MyList)
```

#### 86) Find the output:

```
msg="Pre-Board 2023!"
```

```
m = "
```

```
for i in range(len(msg)):
```

```
    if msg[i].isupper():
```

```
        m+=msg[i].lower()
```

```
    elif msg[i].islower():
```

```
        m+=msg[i].upper()
```

```
    elif msg[i].isdigit():
```

```
        m+=msg[i+1]
```

```
    else:
```

```
        m+=msg[i-1]
```

```
print(m)
```

#### 87) Find the output:

```
def alpha(num):
```

```
    return num+2
```

```
def beta(num):
```

```
    return alpha(num)
```

```
def gamma(num):
```

```
    return beta(num*2)
```

```
print(gamma(10))
```

#### 88) Find the output:

```
str1="hello"
```

```
c=0
```

```
for x in str1:
```

```
    if(x!="l"):
```

```
        c=c+1
```

```
    else:
```

```
        pass
```

```
print(c)
```

#### 89) Find the output:

```
def Change(L):
```

```
    k=len(L)-1
```

```
    for i in range(len(L)//2):
```

```
        L[i] = L[k]
```

```
    k-=1
```

```
    if i>=3:
```

```
        break
```

```
L=[24,78,90,66,45,12,77,120]
```

```
Change(L)
```

```
print(L)
```

#### 90) Find the output:

```
def display(s):
```

```
    l = len(s)
```

```
    m=""
```

```
    for i in range(0,l):
```

```
        if s[i].isupper():
```

```
            m=m+s[i].lower()
```

```
        elif s[i].isalpha():
```

```
            m=m+s[i].upper()
```

```
        elif s[i].isdigit():
```

```
            m=m+"$"
```

```
        else:
```

```
            m=m+"*"
```

```
    print(m)
```

```
display("EXAM23@cbse.com")
```

#### 91) Find the output:

```
def fun3(a,b,c):
```

```
    return a+1 ,b+2,c+3
```

```
t=fun3(10,20,30)
```

```
print(t)
```

#### 92) Find the output:

```
def foo(i, x=[ ]):
```

```
    x.append(i)
```

```
    return x
```

```
for i in range(3):
```

```
    print(foo(i),end=" ")
```

**93) Find the output:**

```
def func(x, y=2):
    num = 1
    for i in range(y):
        num = num * x
    return num

print (func(4))
print (func(4, 4))
```

**94) Find the output:**

```
lst1 = [10, 15, 20, 25, 30]
lst1.insert( 3, 4)
lst1.insert( 2, 3)
print (lst1[-5])
```

**95) Find the output:**

```
def my_func(var1=100, var2=200):
    var1+=10
    var2 = var2 - 10
    return var1+var2

print(my_func(50),my_func())
```

**96) Find the output:**

```
def ChangeVal(M,N):
    for i in range(N):
        if M[i]%5 == 0:
            M[i]//=5
        if M[i]%3 == 0:
            M[i]//=3

L = [25,8,75,12]
ChangeVal(L,4)

for i in L:
    print(i,end="##")
```

**97) Find the output:**

```
g=0
def fun1(x,y):
    global g
    g=x+y
    return g
```

```
def fun2(m,n):
```

```
    global g
    g=m-n
    return g
```

```
k=fun1(2,3)
fun2(k,7)
print(g)
```

**98) Find the output:**

```
def fun1(m=8,n=10):
    global x
    x=4
    y=6
    print (m+n-x)
```

```
def fun2():
    x=5
    return x
```

```
print(fun2())
a,b=5,10
x=20
fun1(b)
print(x)
```

**99) Find the output:**

```
def f3(a,b):
    global x,y
    x=a+b
    a,y=a+x,a*x
    print(a,b,x,y)
```

```
f3(5,10)
print(f3(b=x,a=y))
```

**100) Find the output:**

```
def check(a):
    for i in range(len(a)):
        a[i]=a[i]+5
    return a
```

```
b=[1,2,3,4]
c=check(b)
print(c)
```

**101) Find the output:**

```
x = ('key1', 'key2', 'key3')
thisdict = dict.fromkeys(x)
print(thisdict)
```

**102) Find the output:**

```
def power(x, y=2):
    r = 1
    for i in range(y):
        r = r * x
    return r
```

```
print(power(3))
print(power(3, 3))
```

**103) Find the output:**

```
a=100
def show():
    global a

a=200
def invoke():
    global a

a=500
show()
invoke()
print(a)
```

**104) Find the output:**

```
def Updater(A,B=5):
    A = A // B
    B = A % B
    print(A,'$',B)
    return A + B

A=100
B=30
A = Updater(A,B)
print(A,'#',B)
B = Updater(B)
```

```
print(A,'#',B)
A = Updater(A)
print(A,'$',B)
```

**105) Find the output:**

```
def Modify(X,Y):
    for i in range(Y):
        if X[i]%5==0:
            X[i] //= 5
        if X[i]%3==0:
            X[i] //= 3
```

```
MyList = [35, 11, 105, 27]
Modify(MyList,4)
for i in MyList:
    print(i,end='$')
```

**106) Find the output:**

```
def Total(Number=10):
    Sum=0
    for C in range(1,Number+1):
        if C%2==0:
            continue
        Sum+=C
    return Sum

print(Total(4))
print(Total(7))
print(Total())
```

**107) Find the output:**

```
X = 50
def Alpha(num1):
    global X
    num1 += X
    X += 20
    num1 = Beta(num1)
    return num1

def Beta(num1):
    global X
    num1 += X
    X += 10
```

```
num1 = Gamma(num1)
return num1
```

```
def Gamma(num1):
    X = 200
    num1 += X
    return num1
```

```
num = 100
num = Alpha(num)
print(num,X)
```

**108) Find the output:**

```
a = {}
a[1] = 1
a['1'] = 2
a[1]=a[1]+1
count = 0
for i in a:
    count += a[i]
```

```
print(count)
```

**109) Find the output:**

```
Val = [20,"A",40,"K",10,"H"]
Freq = 0
Sum = 0
Cat = ""
for l in range(1,6,2):
    Freq = Freq + l
    Sum = Sum + Val[l-1]
    Cat = Cat + Val[l] + "*"
print(Freq,Sum,Cat)
```

**110) Find the output:**

```
Data = ["P",20,"R",10,"S",30]
Times = 0
Alpha = ""
Add = 0
for C in range(1,6,2):
    Times = Times + C
    Alpha = Alpha + Data[C-1]+"$"
    Add = Add + Data[C]
    print (Times,Add,Alpha)
```

**111) Find the output:**

```
Str1="EXAM2023"
Str2=""
l=0
while l<len(Str1):
    if Str1[l]>="A" and Str1[l]<="M":
        Str2=Str2+Str1[l+1]
    elif Str1[l]>="0" and Str1[l]<="9":
        Str2=Str2+ (Str1[l-1])
    else:
        Str2=Str2+"*"
    l=l+1

print(Str2)
```

**112) Find the output:**

```
def Alter(P=15,Q=10):
    P=P*Q
    Q=P/Q
    return Q
```

```
A=100
B=200
A=Alter(A,B)
print(A,"$",B)
B=Alter(B)
print (A,"$",B)
A=Alter(A)
print (A,"$",B)
```

**113) Find the output:**

```
Msg1="WeLcOME"
Msg2="GUeSTs"
Msg3=""
for l in range(0,len(Msg2)+1):
    if Msg1[l]>="A" and Msg1[l]<="M":
        Msg3=Msg3+Msg1[l]
    elif Msg1[l]>="N" and Msg1[l]<="Z":
        Msg3=Msg3+Msg2[l]
    else:
        Msg3=Msg3+"*"
print(Msg3)
```

**114) Find the output:**

```
def Changer(P,Q=10):
    P=P/Q
    Q=P%Q
    return P
```

```
A=200
B=20
B=Changer(A,B)
print(A,"$",B)
A=Changer(B)
print (A,"$",B)
B=Changer(A)
print (A,"$",B)
ans:
```

**115) Find the output:**

```
def Call(P=40,Q=20):
    P=P+Q
    Q=P-Q
    print(P,'@',Q)
    return P
```

```
R=200
S=100
R=Call(R,S)
print (R,'@',S)
S=Call(S)
print(R,'@',S)
```

**116) Find the output:**

```
Txt="Some2Thing"
STxt=""
Fold=0
for C in range(0,len(Txt)):
    if Txt[C]>="0" and Txt[C]<="9":
        Fold=1
        STxt = STxt + "#"
    elif Fold==0 and Txt[C]>="A" and Txt[C]<="S":
        STxt = STxt + "@"
    elif Fold==1 and Txt[C]>="T" and Txt[C]<="Z":
        STxt = STxt + "*"
    else:
        STxt = STxt + Txt[C]
print(STxt)
```

**117) Find the output:**

```
def Change(A):
    S=0
    for i in range(len(A)//2):
        S+=(A[i]*2)
    return S
```

```
B = [10,11,12,30,32,34,35,38,40,2]
C = Change(B)
print('Output is',C)
```

**118) Find the output:**

```
string="aabbcc"
count=3
while True:
    if string[0]=='a':
        string=string[2:]
    elif string[-1]=='b':
        string=string[:2]
    else:
        count+=1
        break
print(string)
print(count)
```

**119) Find the output:**

```
x="hello world"
print(x[:2],x[:-2],x[-2:])
print(x[6],x[2:4])
print(x[2:-3],x[-4:-2])
```

**120) Find the output:**

```
a=(2 + 3) ** 3 - 6 / 2
b=(2 + 3) * 5// 4 + (4 + 6) / 2
c=12 + ( 3 * 4 - 6 ) / 3
d=12 % 5 * 3 + (2 * 6) // 4
print(a, b, c, d)
```

**121) Find the output:**

```
a = "Year 2022 at All the best"
a = a.split('2')
b = a[0] + ". " + a[1] + ". " + a[3]
print (b)
```

## 122) Connectivity output Question (CBSE)

Consider the following SQL table MEMBER in a SQL Database CLUB :

Table : **MEMBER**

| M_ID  | NAME   | ACTIVITY |
|-------|--------|----------|
| M1001 | Amina  | GYM      |
| M1002 | Pratik | GYM      |
| M1003 | Simon  | SWIMMING |
| M1004 | Rakesh | GYM      |
| M1005 | Avneet | SWIMMING |

Assume that the required library for establishing the connection between Python and MYSQL is already imported in the given Python code. Also assume that DB is the name of the database connection for table MEMBER stored in the database CLUB.

Predict the output of the following code :

```
MYCUR = DB.cursor()
MYCUR.execute ("USE CLUB")
MYCUR.execute ("SELECT * FROM MEMBER WHERE ACTIVITY= 'GYM' ")
R=MYCUR.fetchone()
for i in range (2) :
    R=MYCUR.fetchone()
    print (R[0], R[1], sep = "#")
```

### 123) Find the output:

```
x = "abcdef"
i = "a"
while i in x:
    print(i, end = " ")
```

### 124) Find the output:

```
d = {'x': 1, 'y': 2, 'z': 3}
a = d.setdefault('z',4)
print(a)
a = d.setdefault('w',0)
print(a)
```

### 125) Find the output:

```
d = {'x': 1, 'y': 2, 'z': 3}
for v in d.items():
    print (v[0], ":", v[1])
```

~~~~~

OUTPUTS ANSWER KEY

1	[6, 't', 4]
2	30
3	haaiuisi
4	Love2
5	babaabaacbaab
6	7
7	True
8	-19
9	14
10	True
11	True
12	1
13	8
14	v\$##t\$##y\$##r\$#
15	11
16	ayn
17	liholiDay
18	19
19	12.0
20	14
21	-30
22	1.0 2.0 4.0
23	FalseTrue FalseFalse
24	49 51 19 22
25	-3 11 4
26	3
27	1 0 1
28	11.0 11.0
29	12
30	7
31	1.0

32	IamGood and strong*
33	[3, 3, 8, 2, 12, 12]
34	6 t
35	21
36	{'t': 2, 'h': 1, 'e': 1, 's': 2}
37	8
38	**** *E*****m**
39	s#EEthO#e
40	3
41	False
42	3
43	18
44	writingexamexam
45	I am@fine*
46	2 1 True
47	9
48	12
49	180 [45, 45, 45, 45]
50	1
51	This is line 1 This is line 3
52	sssie*ORG\$\$\$\$
53	10 # 25 \$ 185 18 # 50 \$ 318 30 # 100 \$ 480
54	one\$two\$three
55	4
56	m%i%b%
57	47.0 35.0 24.0 16.0
58	220 \$ 20 220 \$ 30 230 \$ 30
59	fUN#pYTHONn#.
60	5#5

61	{1: 'One', 2: 'Two', 3: 'C', 4: 'Four', 5: 'Five'}
62	mx1oSvR
63	Rahul 450 Karthi 490
64	[20, 34]
65	33 77
66	2
67	3
68	36
69	True
70	0
71	a) True b) False
72	34 # 4 34 @ 30 40 # 30
73	New TEXT: ff%%%%%%%%AAkkOOk
74	9 60 P\$R\$S\$
75	50#25#11#27#10#
76	IP,HIND,CS,Math,Chem,Phys,
77	24 29
78	12 16
79	10@10
80	1
81	([2, 1], 'abc', 0, 9)
82	csworkshop****
83	5 11
84	25#100#25#1#
85	[0, 6, 18, 80, 38, 103, 70]
86	pREebOARDd023!3
87	22
88	3
89	[120, 77, 12, 45, 45, 12, 77, 120]
90	exam\$\$*CBSE*COM
91	(11, 22, 33)
92	[0] [0, 1] [0, 1, 2]

93	4 4
94	3
95	250 300
96	5#8#5#4#
97	-2
98	5 16 4
99	20 10 15 75 165 15 90 6750 None
100	[6, 2, 3, 4]
101	{'key1': None, 'key2': None, 'key3': None}
102	9 27
103	500
104	3 \$ 3 6 # 30 6 \$ 1 6 # 7 1 \$ 1 2 \$ 7
105	7\$11\$7\$9\$
106	4 16 25
107	420 80
108	4
109	9 70 A*K*H*
110	1 20 P\$ 4 30 P\$R\$ 9 60 P\$R\$S\$
111	X*M2M202
112	100.0 \$ 200 100.0 \$ 200.0 100.0 \$ 200.0
113	G*L*TME
114	200 \$ 10.0 1.0 \$ 10.0 1.0 \$ 0.1
115	300 @ 200

	300 @ 100 120 @ 100 300 @ 120
116	@ome#*hing
117	Output is 20
118	bbcc 4
119	('he', 'hello wor', 'ld') ('w', 'll') ('llo wo', 'or')
120	(122, 11, 14, 9)
121	Year . 0. at All the best
122	M1002#Pratik M1004#Rakesh
123	Infinite loop :aaaaaaa....
124	3 0
125	x : 1 y : 2 z : 3

~~~~~

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS – See Yes!*

### REJOICE & REASON

**1) Assertion** - close () method closes the file and frees the memory space acquired by that file.

**Reason** - one should always close their files, in some cases, due to buffering, changes made to a file may not show until the file is closed.

A - Both A and R are true and R is the correct explanation of A.

B - Both A and R are true but R is not the correct explanation of A.

C - A is true but R is false.

D - A is false but R is true.

**2) Assertion** - The with statement simplifies the working with files but does not support exception handling.

**Reason** - The with statement will automatically close the file after the nested block of code.

A - Both A and R are true and R is the correct explanation of A.

B - Both A and R are true but R is not the correct explanation of A.

C - A is true but R is false.

D - A is false but R is true.

**3) Assertion** - The open a file one should use the open() function, open() is a built in function

**Reason** - The open () functions returns a file object, it is used to read or modify the file accordingly.

A - Both A and R are true and R is the correct explanation of A.

B - Both A and R are true but R is not the correct explanation of A.

C - A is true but R is false.

D - A is false but R is true.

**4) Assertion:** The mode a+ allows one to open a binary file in append mode.

**Reason:** Append mode allows to add contents to existing file.

A - Both A and R are true and R is the correct explanation of A.

B - Both A and R are true but R is not the correct explanation of A.

C - A is true but R is false.

D - A is false but R is true.

**5) Assertion :** Pickle module is to be imported to work with a binary file.

**Reason :** It retains the structure of the mutable data types and allows effective manipulations over the data.

A - Both A and R are true and R is the correct explanation of A.

B - Both A and R are true but R is not the correct explanation of A.

C - A is true but R is false.

D - A is false but R is true.

**6) Assertion** - A CSV file is a type of plain text file that uses specific structuring to arrange tabular data.

**Reason** - A CSV file is a simple text file where each line contains a list of values delimited by commas.

A - Both A and R are true and R is the correct explanation of A.

B - Both A and R are true but R is not the correct explanation of A.

C - A is true but R is false.

D - A is false but R is true.

**7) Assertion** - In a function header, any parameter cannot have a default value unless all parameters appearing on its right have their default values.

**Reason** - Non- default arguments cannot follow default arguments.

A - Both A and R are true and R is the correct explanation of A.

B - Both A and R are true but R is not the correct explanation of A.

C - A is true but R is false.

D - A is false but R is true.

**8) Assertion** - Having keyword arguments after positional arguments will result into error.

**Reason** - an argument list must contain positional arguments followed by any keyword arguments.

A - Both A and R are true and R is the correct explanation of A.

B - Both A and R are true but R is not the correct explanation of A.

C - A is true but R is false.

D - A is false but R is true.

**9) Assertion** - A function ends the moment it reaches a return statement or all the statements in function body have been executed.

**Reason** - Return statement ends a function execution even if it is in the middle of the function.

A - Both A and R are true and R is the correct explanation of A.

B - Both A and R are true but R is not the correct explanation of A.

C - A is true but R is false.

D - A is false but R is true.

**10) Assertion**- One should avoid using global statements in python program.

**Reason** - Global statements cannot be reverted in a program run.

A - Both A and R are true and R is the correct explanation of A.

B - Both A and R are true but R is not the correct explanation of A.

C - A is true but R is false.

D - A is false but R is true.

**11) Assertion (A):-** If the arguments in a function call statement match the number and order of arguments as defined in the function definition, such arguments are called positional arguments.

**Reasoning (R):-** During a function call, the argument list first contains default argument(s) followed by positional argument(s). (2022-23)

(a) Both A and R are true and R is the correct explanation for A

(b) Both A and R are true and R is not the correct explanation for A

(c) A is True but R is False

(d) A is false but R is True

**Ans:** (c) A is True but R is False

**12) Assertion (A):** CSV (Comma Separated Values) is a file format for data storage which looks like a text file.

**Reason (R):** The information is organized with one record on each line and each field is separated by comma. (2022-23)

(a) Both A and R are true and R is the correct explanation for A

(b) Both A and R are true and R is not the correct explanation for A

(c) A is True but R is False

(d) A is false but R is True

**Ans:** (a) Both A and R are true and R is the correct explanation for A

**13) Assertion(A) :** The internet is a collection of interconnected computer networks, linked by transmission medium such as copper wires, fiber-optic cables, wireless connections etc.

**Reason (R) :** World Wide Web is a collection of interconnected documents.

1. Both (A) and (R) are true and (R) is the correct explanation of (A)

2. Both (A) and (R) are true and (R) is not the correct explanation of (A)

3. Both (A) is true. But (R) is false

3. Both (A) is False. But (R) is true

**Ans:** Both (A) and (R) are true and (R) is not the correct explanation of (A)

**14) Assertion(A):** Dictionaries are enclosed by curly braces { }

**Reason(R):** The key-value pairs are separated by commas ( , )

A. Both A and R are true and R is the correct explanation of A.

B. Both A and R are true but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

Ans B. Both A and R are true but R is not the correct explanation of A.

**15) Assertion(A):** The pop() method can be used to delete elements from a dictionary.

**Reason(R):** The pop() method deletes the key-value pair and returns the value of deleted element.

A. Both A and R are true and R is the correct explanation of A.

B. Both A and R are true but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

**Ans A.** Both A and R are true and R is the correct explanation of A.

**16) Assertion(A):** clear() method removes all elements from the dictionary

**Reason(R):** len() function cannot be used to find the length of a dictionary.

A. Both A and R are true and R is the correct explanation of A.

B. Both A and R are true but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

**Ans C.** A is true but R is false.

**17) Assertion(A):** keys() of the dictionaries must be unique.

**Reason(R):** The keys of a dictionary can be accessed using values.

A. Both A and R are true and R is the correct explanation of A.

B. Both A and R are true but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

**Ans C.** A is true but R is false.

**18) Assertion(A):** A dictionary consists of a collection of key-value pairs.

**Reason(R):** Each key-value pair maps the key to its associated value.

A. Both A and R are true and R is the correct explanation of A.

B. Both A and R are true but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

**Ans A.** Both A and R are true and R is the correct explanation of A.

**19) Assertion(A):** The random module is a built-in module to generate the pseudo-random variables.

**Reason(R):** The randrange() function is used to generate a random number between the specified range in its parameter.

A. Both A and R are true and R is the correct explanation of A.

B. Both A and R are true but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

**Ans B.** Both A and R are true but R is not the correct explanation of A.

**20) Assertion(A):** The randint() method returns an integer number selected element from the specified range.

**Reason(R):** The syntax for randint() is : random.randint(start: stop)

A. Both A and R are true and R is the correct explanation of A.

B. Both A and R are true but R is not the correct explanation of A.

C. A is true but R is false.

D. A is false but R is true.

**Ans C.** A is true but R is false.

~~~~~

XII COMPUTER SCIENCE

Towards Getting Centum in CS - See Yes!

MAGICAL MCQs

REVISION TOUR I

1. Which of the following is not a token?

- a) Keywords b) Operators
- c) Punctuators d) Comments

2. Which of the following is not a valid identifier?

- a) Myfile b) IF c) FILE_13 d) 29_FILE

3. is a special literal in Python

- a) Null b) Not
- c) None d) none of the above

4. A variable pointing to a value of a certain type, can be made to point to a value of different type. This is called

- a) Type Casting b) Static Typing
- c) Dynamic Typing d) none of the above

5. conversion is user defined conversion that forces an expression to be of specific type.

- a) Implicit b) Explicit
- c) both a and b d) none of the above

6. What will be the output of the given statement, if you will enter 9 a=b=int(input("Enter a Number"))

print(a,b)

- a) 9 9 b) 9 0
- c) Generate error d) none of the above

7. What will be the output of the given statement, if you will enter 9 7 a,b=int(input("Enter a Number")),int(input("Enter second number")) print(a,b)

- a) 9 7 b) 9 0
- c) Generate error d) none of the above

8. The statement is used to terminate the loop or statement in which it is present.

- a) pass b) break
- c) continue d) all of the above

9. The statement is opposite to the break statement, instead of terminating the loop, it forces to execute the next iteration of the loop.

- a) pass b) break
- c) continue d) all of the above

10. The statement in Python is used when a statement is required syntactically but you do not want any command or code to execute.

- a) pass b) break
- c) continue d) all of the above

11. In nested loop, a break statement terminates all the nested loop.

- a) True b) False

12. Which of the following is/are component(s) of variable?

- a) Id b) value
- c) type d) all of the above

13. By using function you can check the value of a variable

- a) id() b) type()
- c) print() d) all of the above

14. range() method is used to specify the length of a for-in loop.

- a) True b) False

15. What will be the output of the given statement, if value of n is 6

```
if n%2==0 and n%3==0:
```

```
    print("Buzz Fizz!", end=" ")
```

```
if n%2==0:
```

```
    print("Buzz", end=" ")
```

```
if n%3==0:
```

```
    print("Fizz!")
```

- a) Buzz Fizz! b) Buzz
c) Fizz d) Buzz Fizz! Buzz Fizz!

16. What will be the output of the given statement, if value of n is 6

```
if n%2==0 and n%3==0:
```

```
    print("Buzz Fizz!", end=" ")
```

```
elif n%2==0:
```

```
    print("Buzz", end=" ")
```

```
elif n%3==0:
```

```
    print("Fizz!")
```

- a) Buzz Fizz! b) Buzz
c) Fizz d) Buzz Fizz! Buzz Fizz!

17. What will be the output of the given statement, if value of n is 6

```
if n%2==0:
```

```
    print("Buzz ", end=" ")
```

```
elif n%2==0:
```

```
    print("Fizz ", end=" ")
```

```
elif n%2==0 and n%3==0:
```

```
    print("Buzz Fizz!")
```

- a) Buzz Fizz! b) Buzz
c) Fizz d) Buzz Fizz! Buzz Fizz!

18. What will be the output of the given statement

```
ans=0
```

```
for i in range(11,20,1):
```

```
    if i%2==0:
```

```
        ans+=5
```

```
    else:
```

```
        ans-=2
```

```
print(ans)
```

- a) 12 b) 10 c) 11 d) None of the above

19. What will be the output of the given statement
ans=20

```
while ans<50:
```

```
    if ans%10==0:
```

```
        ans+=5
```

```
    elif ans%5==0:
```

```
        ans+=10
```

```
    else:
```

```
        ans+=15 print(ans)
```

- a) 55 b) 45 c) 60 d) 65

20. Creating individual values from a tuple's elements is called

- a) Collection b) Unpacking
c) Packing d) None of the above

21. Read the following statements, and find the output :

```
T1=(((('x',5),'y','z'),'w',6),'u',7)
```

```
print(len(T1))
```

- a) 8 b) 3 c) 4 d) None of the above

22. What will be the output of the given statement, if the value of msg is "study4u@com"

```
txt=""
```

```

for i in range(0, len(msg)):
    if msg[i].isupper():
        txt+=msg[i].lower()
    elif text[i].islower():
        txt+=msg[i].upper()
    else:
        txt+='**' print(txt)

```

- a) STUDY**U**COM b) STUDYUCOM
c)Generate Error d)None of the above

23. Read the following statements, and find the output :

L=[7,12,16,5,4,28,25]

n1=n2=L[0]

for i in L:

 if n1<i:

 n1=i

 if n2>i:

 n2=i

print(n1,n2)

- a) ? 7 b) 28 4 c) 4 28 d) 25 25

24. Read the following statements, and find the output : L=[7,12,16,5,4,28,25]

n=L[0]

for i in range(len(L)-1):

 L[i]=L[i+ 1]

L[i+1]=n

print(L)

- a) [25,7,12,16,5,4,28] b) [12,16,5,4,28,25,7]
c) [28,25,7, 12,16,5,4] d) None of the above

25. Read the following statements, and find the output : L=[7,12,16,5,4,28,25,30]

for i in range(0,len(L)-1,2):

 L[i],L[i+1]=L[i+1],L[i]

print(L)

- a) [12, 7, 5, 16, 28, 4, 30, 25]
b) [12, 7, 5, 16, 28, 4, 30, 25]
c) Generate Error d) None of the above

Answers

1. (d) 2. (b) 3. (c) 4. (c) 5. (b) 6. (a) 7. (a)
8. (b) 9. (c) 10. (a) 11. (b) 12. (d) 13. (c) 14. (a)
15. (d) 16. (a) 17. (b) 18. (b) 19. (a) 20. (b) 21.(b)
22.(a) 23.(b) 24.(b) 25. (a)

REVISION TOUR II

1. A is a sequence of characters.

- a)Numeric b)Decimal
c)String d)None of the above

2.String index is Bi-Direction.

- a)True b) False

3.Which of the following is not a basic string operator

- a) + b) * c) in d) all of the above

4.print(("Study"+"Trigger"+"2021 ")*3)

print("Study"+"Trigger"+"2021 "*3)

Both the above statement will produce the same result.

- a)True b) False

5.What will be the output of the given statement.

print("Study"+5)

- a)StudyStudyStudyStudyStudy b) Study5
c) Generate Error d) None of the above

6. `print("Study"*3)`

`print(3*"Study")`

Both the above statement will produce the same result.

a) True b) False

7. Which of the following is not a membership operator

a) > b) not in c) in d) all of the above

8. "study" in "Study Trigger" will return.

a) True b) False

9. The opposite of `ord()` is

a) `chr()` b) `dro()` c) `char()` d) `str()`

10. The extraction of substring from string is called

a) Substring b) String c) Slicing d) None of the above

11. If "name" is a string and 'i' is index of that string, then `name[:i] + name[i:]` will return the complete string "name".

a) True b) False

12. Strings are immutable, i.e. You cannot change the individual letters of a string in place by assignment.

a) True b) False

13. What will be the output of the given statement.
`print("Study"[1])`

a) Study b) S c) t d) Generate Error

14. What will be the output of the given statement.
`print("Study Trigger 2021"[6:14])`

a) Study b) Trigger c) Generate Error d) None of the above

15. How many times the word "Study" printed in the following code :

`word="CBSE 2021"`

`for alpha in word[3:9]:`

`print('Study')`

a) 7 times b) 5 times c) 6 times d) Generate error

16. Which function is used to check whether the string contains a digit or not.

a) `digit()` b) `todigit()` c) `isdigit()` d) None of the above

17. function is used to check whether all letters of the string are in capital or not.

a) `capitalize()` b) `upper()`

c) `isupper()` d) `toupper()`

18. function is used to remove all the white spaces from the beginning of the string.

a) `remove()` b) `ltrim()` c) `rtrim()` d) `isspace()`

19. `lower()` and `islower()` are the same method in string.

a) True b) False

20. What will be the output of the given statement.

`print("Study Trigger 2021".lower().count("t"))`

a) 2 b) 1 c) Generate Error d) None of the above

21. What will be the output of the given statement.

`board="CBSE"`

`print(board[-1::-1])`

a) CBSE b) ESBC c) Generate Error

d) None of the above

22. What will be the output of the given statement.
`name="Study Trigger"`

`for i in range(len(name)):`

`if i%2==0:`


```

        print(name[i].upper(), end="")
    else:
        print(name[i].lower(), end="")

```

- a) StUdY TrlgGeR b) sTuDy tRiGgEr
c) Study Trigger d) Generate Error

23. What will be the output of the given statement.

```
name="StudyTrigger2021"
```

```
newName=""
```

```
for i in range(len(name)):
```

```
    if i%2==0:
```

```
        newName+=name[i+ 1]
```

```
    else:
```

```
        newName+=name[i-1]
```

```
print(newName)
```

- a) tSduTyirggre0212 b) 1utydrTgieg2r20
c) 2021TriggerStudy d) None of the above

24. What will be the output of the given statement.

```
name="Study Trigger"
```

```
newName=name[len(name)-1]
```

```
for i in range(len(name)-1 ):
```

```
    newName+=name[i]
```

```
print(newName)
```

- a) rStudy Trigge b) erStudy Trigge
c) Study Trigger d) tudyS riggerT

25. What will be the output of the given statement.

```
name="Study Trigger"
```

```
for i in range(0,len(name)-4,2):
```

```
    print(i,end="")
```

- a) Suy Tigr b) 02468 c) Suy d) 0246

Answers

1. (c) 2. (a) 3. (c) 4. (b) 5. (c) 6. (a) 7. (a)
8. (b) 9. (a) 10. (c) 11. (a) 12. (a) 13. (c) 14. (b)
15. (c) 16. (c) 17. (c) 8. (b) 19. (b) 20. (a) 21. (b)
22. (a) 23. (a) 24. (a) 25. (b)

REVISION TOUR III

1. List are i.e. you can change the elements of a list.

- a) mutable b) immutable c) both a and
d) None of the above

2. To create an empty list, L: L=[] //statement 1
L=list() //statement 2

- a) Both statements are true
b) only statement 1 is true
c) only statement 2 is true
d) both statements are false

3. List inside the list is called

- a) inner list b) nested list c) 2-D list d) none of the above

4. List index is bi-directional just like strings.

- a) True b) False

5. What will be the output of the given statement.
print([1,2,3,4,5]+5)

- a) [6,7,8,9, 10] b) [1,2,3,4,5,5]
c) Generate Error d) None of the above

6. What will be the output of the given statement.
print([1,2,3,4,5]+[5])

- a) [6,7,8,9, 10] b) [1,2,3,4,5,5]
c) Generate Error d) None of the above

7. What will be the output of the given statement.
print([1,2,3]*3)

a) [3,6,9] b) [1,2,3,3]

c) [1,2,3, 1,2,3,1,2,3] d) Generate Error

8.What will be the output of the given statement.
print([1,2,3]*[3])

a) [3,6,9] b) [1,2,3,3]

c) [1,2,3, 1,2,3,1,2,3] d) Generate Error

9.What will be the output of the given statement.
print([10,20,30,40,50].index(90))

a)No Output b) 0 c) Generate Error d) None of the above

10.append() and extend() are the same method.

a) True b) False

11.Which of the following method is not use for insert an/are element(s) into list:

a) insert() b) append()

c) extend() d) push()

12.How many parameter(s) are taken by insert():

a) 0 b) 1 c) 2 d) 3

13.What will be the output of the given statement.
print([1,2,3,4,5].reverse())

a) [5,4,3,2, 1] b) None

c) [1,2,3,4,5] d) Generate Error

14.What will be the output of the given statement.
L=["Study","Trigger",["Welcome","to","Study",
"Trigger"],"Always","With","You!"] print(L[2][0])

a) Study b) Trigger c) Welcome d) Generate Error

15.What will be the output of the given statement.
L=["Study","Trigger",["Welcome","to","Study",
"Trigger"],"Always","With","You!"]

print(L[2][0][2])

a) u b) l c) l d) Generate Error

16..... function is used to delete a given element from the list.

a) remove() b) clear() c) pop() d) none of the above

17..... function is used to add an element into beginning of the list.

a) push() b) insert() c) append()

d) extend()

18..... function is used to add an element into the end of the list.

a) push() b) insert() c) append() d) extend()

19.List are immutable and strings are mutable

a) True b) False

20.Strings store single type of elements and list can store different types of elements.

a) True b) False

21.To create an empty tuple, T: T=() //statement 1
T=tuple() //statement 2

a) Both statements are true b) only statement 1 is true

c) only statement2 is true d) both statements are false

22.User can also create nested tuple just like nested list

a) True b) False

23. Tuples are immutable and list are mutable.

a) True b) False

24.Creating tuple from a set of values is called

.....

- a) Collection b) Sequence c) Packing
d) None of the above

25. Read the following statements, and find the output :

T1=(2,3)

T2=('2','3')

print(T1==T2)

- a) True b) False c) Generate Error d) None of the above

Answers

1. (a) 2. (a) 3. (b) 4. (a) 5. (c) 6. (b) 7. (c)
8. (d) 9. (c) 10. (b) 11. (d) 12. (c) 13. (b) 14. (c)
15. (c) 16. (a) 17. (b) 18. (c) 19. (b) 20. (a) 21. (a)
22 (a) 23. (a) 24. (c) 25. (b)

REVISION TOUR IV

1. Dictionaries are mutable with elements in the form of a key:element pair.

- a) True b) False

2. To create an empty dictionary, D: D={}
//statement 1

D=dict() //statement 2

- a) Both statements are true b) only statement 1 is true
c) only statement 2 is true d) both statements are false

3. Dictionaries are also called associative arrays, mappings or hashes.

- a) True b) False

4. As dictionaries are mutable, so, a user can change the value of keys.

- a) True b) False

5. A dictionary operation that takes a key and finds the corresponding value, is called

- a) vlookup b) searching c) lookup
d) None of the above

6. To see all the values of a dictionaryfunction is used

- a) value() b) item() c) items() d) values()

7. To see all the keys of a dictionaryfunction is used

- a) key() b) item() c) items() d) keys()

8. If you want to add an element into a dictionary then key must not exist, but if you want to update an element then key must exist in a dictionary.

- a) True b) False

9. The membership operator (in, not in) work on dictionary with

- a) keys b) values
c) Both a and b d) None of the above

10. clear() and del are the same for deleting a dictionary

- a) True b) False

11. of a dictionary must be unique.

- a) keys b) values c) Both a and b d) None of the above

12. Choose the correct way to declare a dictionary :

- a) dic={'CBSE':1,'Exam':2,'2022':3}
b) dic={'CBSE':1 ;'Exam':2;'2022':3}
c) dic={'CBSE'=1,'Exam'=2,'2022'=3}
d) dic={'CBSE';1,'Exam';2,'2022'3}

13. What will be the output of the given statement.

dic={'Study': 1,'Trigger':2,'Student':3,'CBSE':4} t=""

for key in die:

if t<key:

t=key

print(t)

a) Trigger b) Study c) Student d) CBSE

14.What will be the output of the given statement.

a={{(1,2):"Study" ,(2,3):"Trigger"}}

print(a[1,2])

a) Study b) Trigger c) Generate Error d) None of the above

Answers

1. (b) 2. (a) 3. (a) 4. (b) 5. (c) 6. (d) 7. (d) 8. (a)
9. (a) 10. (b) 11. (a) 12. (a) 13. (a) 14. (a)

FUNCTIONS I

1.In Python, function can be return only one value.

a)True b) False

2.The variable declared outside a function is called a global variable.

a)True b) False

3.You can call a function only once after defining it.

a)True b) False

4.Value returning functions should be generally called from inside of an expression.

a)True b) False

5.Ais basically a block of code that will execute only when it is called.

a)function b) sub program c) block d) all

6.Functions provide amechanism.

The same function can be called any number of times.

a)call b) reuse c) define d) none of the above

7.Actual parameters are the parameters specified within a pair of parenthesis in the function definition

a)True b) False

8.A function in Python is waste without calling it.

a)True b) False

9.A local variable having the same name as global variable

a)True b) False

10.Python passes arguments to function by.....

a) reference b) value c) both d) none of the above

11.The variable declared inside the function is called a variable

a) global b) local
c) external d) none of the above

12.Python passes parameters by value.

a)True b) False

13.....is the blank space in the beginning of a statement within a block.

a) indentation b) space
c) body d) none of the above

14.Every Functions return a value if function do not explicitly return a value, then it will return.....

a)return b) None c) value d) all

15. Function header and function definition is the same thing

a) True b) False

16. These are predefined functions that are always available for use.

For using them we don't need to import any module

a) built function b) pre define function

c) user-defined d) none of the above

17. A function is said to be if it calls itself.

a) built function b) pre define function

c) recursive d) all

18. These are predefined functions that are available in a module.

For using them we need to import module

a) built function b) pre define function

c) user-defined d) function define in module

19. The of a variable is the area of the program where it may be referenced

a) external b) global c) scope d) local

20. If you want to communicate between functions i.e. calling and called statement, then you should use.....

a) values b) return c) arguments d) none of the above

21. In Python, parameters and arguments are the same thing.

a) True b) False

22. The syntax of function call and declaration is very similar, except that the keyword and are missing:

a) colon, def b) argument, def c) def, parameters d) none

23. Find the output of the following code :

```
x = 100
```

```
def study(x):
```

```
    x=50
```

```
    study(x)
```

```
    print("Value of x is ", x)
```

a) 100 b) 50 c) Error d) none of the above

24. Find the output of the following code :

```
x = 100
```

```
def study(x):
```

```
    x=50
```

```
print("Value of x is ", x)
```

a) 100 b) 50 c) Error d) none of the above

25. Find the output of the following code :

```
def study(x,y,z):
```

```
    print(x+y+z)
```

```
a=study(2,3,4)
```

```
print(a)
```

a) 9, None b) 9,9

b) c) None, None d) none of the above

Answers

1. (b) 2. (a) 3. (b) 4. (a) 5. (a) 6. (b) 7. (b) 8. (a)
9. (a) 10. (b) 11. (b) 12. (a) 13. (a) 14. (b) 15. (b)
18. (d) 16. (a) 17. (c) 19. (c) 20. (c) 21. (b) 22. (a)
23. (a) 24. (a) 25. (a)

FUNCTIONS II

1. User can change the functionality or name of Built-in functions.

a) False b) True

2. The first line of a function header begins with def keyword and ends with a semi-colon.

a) False b) True

3.Which of the following function headers is correct.

- a)def study(a=2,b=5,c):
study(a=2,b,c=5):
b) def
c) def study(a, b=2,c=5):
d) none of the above

4.Which one of the following is the correct way of calling a function:

- a)call function_name()
b) def function_name()
c)ret function_name()
d) function_name()

5.Positional arguments is also called Required Arguments.

- a)True b) False

6.A function in python begins with which keywords

- a)void b) return c) int d) def

7.Which statement is used to send back a value from a function

- a)print b) input c) return d) none of the above

8.A variable created or defined within a function body is

- a)local b) global c) built-in d) instance

9.If we initialize some value to parameter at function header, it's called of that parameter

- a)keyword b) required c) variable length d) default

10.In..... arguments, we can skip the default argument, but all the keyword arguments should match the parameters in the function definitions

- a)keyword b) required c)default d)none of the above

11.When large programs are broken down into smaller units known as.....

- a)sub program b) functions c) class d) none of the above

12.In function definition, the function name is called

- a)keyword b) identifier c) variable d) none of the above

13.By default if you return multiple value separated by comma, then it is returned as:

- a)list b) tuples c) string d) none of the above

14.A function cannot return a value in the form of

- a)Literals b) Expression c) Variable d) none of the above

15.A function that does not have any return value is known as.....

- a)library function b) void function c) fruitful function d) none of the above

16.means a function definition is starting.

- a) return b) function name c) def d) colon

17.There is aat the end of the def line, meaning it requires a block.

- a) colon b) return c) def d) none of the above

18. Write the output of the following code.

```
def Greet(name):
```

```
    print("Hello! ",name)
```

```
Greet('Mahesh')
```

- a)Hello! Mahesh b) Hello! c) Mahesh d) none of the above

19.How to call the given function:

```
def study(num=S):
```

```
    print(num+S)
```

- a)study() b)study() c)call study(9)
d) none of the above

20. What will be the output of the following code :

```
def calculate(a,b):  
    return a+b, a-b  
res=calculate(7, 7)  
print(res)  
a)(14,0) b) 14,0 C)14 d)0
```

21. What will be the output of the following code:

```
def study (n):  
    ans=0  
    for i in range(1,n):  
        ans+=i  
    return ans  
print(study(5))  
a) 10 b) 15 c) Error d) No Output
```

22. What will be the output of the following code:

```
x = 100  
def study():  
    global x  
    x=50  
print(x)  
a) 100 b) 50 c) Error d) none of the above
```

Answers

1. (a) 2. (a) 3. (c) 4. (d) 5. (a) 6. (d) 7. (c) 8. (a)
9. (d) 10. (a) 11. (b) 12. (b) 13. (b) 14. (d) 15. (b)
16. (c) 17. (a) 18. (a) 19. (a,b) 20. (a) 21. (a) 22. (b)

FUNCTIONS III

1. Find the flow of execution of the following code :

```
1. def calculate (a,b):  
2.     res=a**b
```

```
3.         return res  
4.  
5.     def study( a):  
6.         ans=calculate(a,b)  
7.         return ans  
8.  
9.     n=2  
10.    a=study(n)  
11.    print(a)
```

a) 1->5->9->10->6->2->3->7->11
b) 5->9-> 10->6->2->3->7->11
c) 9->10->5->1->6->2->3-> 7->11
d) None of the above

2. What will be the output of the following code :

```
def display(s):  
    l = len(s)  
    m=""  
    for i in range(0,l):  
        if s[i].isupper():  
            m=m+s[i].lower()  
        elif s[i].isalpha():  
            m=m+s[i].upper()  
        elif s[i].isdigit():  
            m=m+"$"  
        else:  
            m=m+"*"  
    print(m)
```

display("STU DY21@trigger.com")

a) study\$\$*TRIGGER*COM
c) study21*TRIGGER*COM
b) STUDY\$\$*trigger*COM
d) None of the above

3. Python resolves the scope of a name using the LEGB rule. a) True b) False

4. A python module can be imported in a program using the include statement.

a) True b) false

5. User can also change the functionality of built-in functions. a) True b) False

6. In which part of memory does the system store the parameter and local variables of function call.

a) Heap b) Stack
c) Both a and b d) None of the above

7. What will be the output of the following code:

```
def Display(str):  
    m=""  
    for i in range(0, len(str)):  
        if(str[i].isupper()):  
            m=m+str[i].lower()  
        elif str[i].islower():  
            m=m+str[i].upper()  
        else:  
            if i%2==0:  
                m=m+str[i-1]  
            else: m=m+"@"  
    print(m.swapcase())
```

Display('StudyTrigger\$2021 ')

a) StudyTriggerR@2@2
b) Studytrigger\$@2@2
c) StudyTrigger\$@2@2
d) None of the above

8. What is the output of the following statement
print(chr(ord(chr(97))))

a) a b) A c) 97 d) error

9. Function returning value is also known as non-fruitful functions.

a) True b) False

10. Find the flow of execution of the following code :

1. def calculate (a):

2. a+=1

3.

4. n=2

5. print(n)

6. calculate(n)

7. print(n)

a) 1->4->5->6->1->2->7
c) 1->4->5->6->1->2->6->7
b) 1->4->5->6->2->7
d) None of the above

11. You can define a function inside a function.

a) True b) False

12. A function can only return a single value.

a) True b) False

13. Positional argument is also called argument

a) default b) named c) keyword d) required

14. Named argument is also called argument

a) default b) positional c) keyword
d) required

15. When you use multiple type argument in function, then default argument take place

- a) at beginning b) at end c) anywhere
d) none of the above

16. A void function internally returns legal empty value

- a) None b) close() c) Return d) all

17. The values being passed through a function-call statement are called

- a) arguments b) parameters c) values d) none

18. The values received in the function definition / header are called

- a) arguments b) parameters c) values d) none

19. A can be skipped in the function call statements.

- a) named parameter b) default parameter
c) keyword parameters d) all of the above

20. What will be the output of the following code
total=0

```
def add(a,b):  
    global total  
    total=a+b  
    print(total)
```

add(6,6)

print(total)

- a) 12 12 b) 12 0 c) 0 12 d) None of the above

Answers:

1. (a) 2. (a) 3. (a) 4. (b) 5. (b) 6. (b) 7. (a) 8. (a)
9. (b) 10. (a) 11. (b) 12. (b) 13. (d) 14. (c) 15. (b)
16. (a) 17. (a) 18. (b) 19. (b) 20. (a)

LIBRARIES

4. behaves like a container which contains variable, class definitions, statements and function related to a particular task.

- a) Library b) Module c) Function d) Class

5. is useful for documentation purposes and should be written with triple quotes.

- a) classes b) Object c) statements
d) docstrings

6. function resides inside the random module which generate random number between 0.0 to 1.0 and return a floating point number.

- a) randint() b) rand() c) random() d) None of the above

7. You cannot create your own modules in Python.

- a) True b) False

8. function resides inside the random module which generate random number in a range and return a integer value.

- a) randint() b) rand() c) random() d) None of the above

9. If you want to import all names from a module into the current calling module, then you should use statement

- a) from b) from* c) import d) import *

10. returns the current system date from date class function.

- a) now() b) today() c) current() d) month()

11. Find the output(s) for the following codes:

```
import random
```

```
print(int(15+random.random()*5),end=' ')
```

```
print(int(15+random.random()*5),end=' ')
```

```
print(int(15+random.random()*5),end=' ')
print(int(15+random.random()*5))
```

- a) 19 15 16 19 b) 19 20 16 19
c) 20 14 16 19 d) 19 19 19 19

12.Find the minimum and maximum possible value from the code:

```
import random print(random.randint(3, 10)-3))
```

- a) min=0, max=? b) min=-3, max=10
c) min=-3, max=? d) none of the above

13.Find the output(s) for the following codes:

```
import random L=[2,3,4,5,6,7]
```

```
N1=random.randint(1,3)
```

```
N2=random.randint(2,4)
```

```
for i in range (N1, N2+1):
```

```
    print(L[i],end="#")
```

- a) 1#4#7# b) 3#4#5#
b) c) 5#6#7# d) 4#5#7#

14.Find the output(s) for the following codes:

```
import random
```

```
import math
```

```
print(int(math.pow(random.randint(2,4),2)),end='**')
```

```
print(int(math.pow(random.randint(2,4),2)),end='**')
```

```
print(int(math.pow(random.randint(2,4 ),2)))
```

- a)4*16*4 b) 4*4*32
c)16*16*32 d) 16*16*16

15.Find the output(s) for the following codes:

```
import random
```

```
Name="STUDYTRIGGER"
```

```
Num=random.randint(0,3)
```

```
N=9
```

```
while Name[N]!='R':
```

```
    print(Name[N]+Name[Num]+"#", end=' ')
```

```
    Num+=1
```

```
    N-=1
```

- a) GD# GY# IT# b) GS# GT# IU#
c) GU# GD# IY# d) All the above

Answers:

4. (b) 5. (d) 6. (c) 7. (a) 8. (a) 9. (d) 10. (b)
11. (a) and (d) 12. (a) 13. (b) 14. (a) and (d)
15. (d)

FILE HANDLING I

1.A relative path always begins with the root folder.

- a)True b) False

2.It is necessary to always create the file in the same default folder where Python has been installed.

- a) True b) False

3.Files in Python can be opened in one of the three modes:

- a) real, modify, write b) read, append, create
c) read, write, delete d) read, write, append

4.The method of a file object flushes any unwritten information and closes the file object, after which no more writing can be done.

- a)flush() b) close() c) write() d) read()

5.readlines() reads the entire file at a time.

- a)True b) False

6.A close() function breaks the link of the file object and the file on the disk.

a)True b) False

7.When you open a file in write mode, the given file must exist in the folder; otherwise Python will raise FileNotFoundError.

a)True b) False

8.The default file open mode is write mode.

a)True b) False

9.Opening a file in write mode erases the previous data.

a)True b) False

10..... by means the information in the file remains even if the computer is switched off.

a)Files b) Strings
c) Streams d) None of the above

11.Themethod writes a list of strings to a file.

a)write() b) writelines() c) append() d) dump()

12.readlines() function returns a list of strings, each separated by "\n".

a)True b) False

13.file() and open() functions have different usage.

a)True b) False

14.Data maintained inside the file is termed as "un-preserved" data.

a)True b) False

15.With statement ensures that all the resources allocated to the file objects get deallocated automatically.

a)True b) False

16.A file mode defines the type of operations that is to be performed on the file.

a)True b) False

17.....function creates a file object which serves as a link to a file residing on the computer.

a)close() b) open()
c) load() d) dump()

18.A stream is defined as a sequence of characters.

a)True b) False

19.A binary file stores information in ASCII or Unicode characters.

a)True b) False

20.In binary file, there is no delimiter for a line.

a)True b) False

Answer the following questions (21 to 24) on the basis of ABOUT.txt file having the below content:



21. What is the output of the statement :

file("about.txt", "r").readline().split()

a) ['STUDY', 'TRIGGER', '?']
b) ['STUDYTRIGGER', '?']
b) ['STUDYTRIGGER?']
d) None of the above

22. What is the output of the statement

```
file("about. txf","r").readline()
```

- a) STUDY TRIGGER ?\n
- b) ['STUDY TRIGGER ?\n']
- b) Whole content of the file.
- d) None of the above

23. What is the output of the statement

```
myfile=open("about1 .txf","r").readlines()
```

- a) Read whole data as a list c) Generates an error
- b) Read whole data as a string d) None of the above

24. What is the output of the statement:

```
myfile=openC'about. txt',"w")
```

```
myfile.write("Welcome to Study Trigger !")
```

- a) Erase existing data and write new line.
- b) Write new line with existing data.
- c) Generates an error
- d) None of the above

25. CSV stands for :

- a) Comma Split Value b) Common Split File
- c) Comma Separated Function
- d) Comma Separated File

26. The separator character of CSV files is called a

- a) end of record b) end of file
- c) end of row d) delimiter

27. To work with CSV file you need to import module in your program

- a) file b) pickle c) csv d) none of the above

28. Which of the following is/are function(s) not used for writing into csv files

- a) writer() b) writelines() c) writerow()
- d) writerows()

Answers

- 1. (b) 2. (b) 3. (d) 4. (b) 5. (a) 6. (a) 7. (b)
- 8. (b) 9. (a) 10. (a) 11. (d) 12. (a) 13. (b) 14. (b)
- 15. (a) 16. (a) 17. (b) 18. (b) 19. (a) 20. (a) 21. (a)
- 22. (a) 23. (c) 24. (a) 25. (d) 26. (d) 27. (c) 28. (b)

FILE HANDLING II

2. By using method, you can rename the file or folder.

- a) flush() b) filename()
- c) rename() d) fname()

3. To remove/delete a file, method can be used.

- a) filename() b) rename()
- c) remove() d) flush()

4. A file cannot be read by human directly, to read this type of file you have to create a reading program.

- a) text b) binary
- c) special d) None of the above

5. The statement automatically closes the file after the processing on the file gets over.

- a) try b) except
- c) catch d) with

6. The readline() returns a list of line from the file till.

- a) end of file b) end of word
- c) end of line d) end of characters

7.'n' characters from the file can be read by method.

- a) readline(n) b) reads(n)
- c) read(n) d) readlines(n)

8.If you want transfer data from buffer to file forcefully, then function is used.

- a) flush() b) with c) read() d) write()

9.To read 4 characters from a file object "myfile", which statement you should use :

- a) myfile.read(4)
- c) myfile.readline()
- b) myfile.read()
- d) myfile.readlines()

10.To read the complete content of a file as a string from a file object f, which statement you should use:

- a) f.read(S) b) f.read()
- c) f.readline() d) f.readlines()

11.A file object is also known as.....

- a) file recover b) file cursor
- c) file content d) file handle

12.When you open a file in this mode, the file pointer is placed at the beginning of the file and if the file is not existing, it will generate FileNotFoundError.

- a) r+ b) rb c) w d) r

13.It opens a binary file for reading only. File must exist.

- a) r+ b) rb c) w d) r

14.It opens a file for both reading and writing mode.

- a) r+ b) rb c) w d) r

15.The read() function reads data from the of a file.

- a) at the end b) from current position
- c) from cursor position d) beginning

16.The pickle module contains two main methods called..... and for writing and reading operation.

- a) flush() and get() b) dump() and load()
- c) read() and write() d) open() and close()

17.To open a file e:\study.txt for reading, which statement you should use :

- a) f1=open("e:\study.txt ", " r")
- b) f1=open("e:\\study.txt " , "r")
- c) f1=open(file=" e:\study.txt", " r")
- d) f1=open(file=" e:\\s study.txt ", "r")

18.To open a file d:\test.txt for writing, which statement you should use:

- a) fobj=open("d:\\test.txt", "w")
- b) fobj=open("d:\\test.txt", "r")
- c) fobj=open("d:\\test.txt", "w")
- d) fobj=open(file="d:\\test.txt", "w")

19.To open a file c:\test.txt for appending data, which statement you should use:

- a) fobj=open("c:\\test.txt", "a")
- b) fobj=open("c:\\test.txt", "rw")
- c) fobj=open(file="c:\\test.txt", "w")
- d) fobj=open(file="c:\\test.txt", "w")

20.How many parameters you can pass in open() method : a) 1 b) 2 c) 3 d) 4

21.How many parameters you can pass in read() method: a) 1 b) 2 c) 3 d) 4

22.Afunction is also built in function which breaks the link of file-object and the file on the disk

- a)flush() b) close()
c)load() d) none of the above

23.What is the output of the following code :

```
myfile=open("about. txf","w")
```

```
myfile.write("Hello!")
```

```
myfile.write("Study Trigger")
```

```
myfile.close()
```

- a)Hello!Study Trigger b) Hello!\nStudy Trigger
c) Generate Error d) None of the above

24.What is the output of the following code:

```
myfile=open("about. txf","w")
```

```
myfile.write("[Hello!","Hi"," How"])
```

```
myfile.close()
```

- a)Hello!HiHow b) [Hello!HiHow]
c) Generate Error d) None of the above

25. Serialisation is also called

- a) Unpickling b) Pickling
c) both a and b d) none of the above

26.is the process in which an object converts into a byte stream

- a) Unpickling b) Pickling
c) both a and b d) none of the above

27....is the process in which a byte stream converts into object.

- a) Unpickling b) Pickling
c) both a and b d) none of the above

Answers

2. (c) 3. (c) 4. (b) 5. (d) 6. (c) 7. (c) 8. (a)
9. (a) 10. (b) 11. (d) 12. (d) 13. (b) 14. (a) 15. (d)
16. (b) 17. (b) 18. (a) 19. (a) 20. (b) 21. (a) 22. (a)
23. (a) 24. (c) 25. (b) 26. (b) 27. (a)

FILE HANDLING III

1.A binary file can store data in the form of 0 and 1

- a)True b) False

2.File object and file handle is the same thing

- a)True b) False

3.r and r+ are the same mode

- a)True b) False

4.You may combine the file() or open() with the file object's function.

- a)True b) False

5.write() method can be used to write multiple lines simultaneously.

- a)True b) False

6.It opens a binary file for both reading and writing. The file pointer is placed at the beginning of the file.

- a)r+ b) rb c) rb+ d) r

7.It opens a binary file for writing only. Overwrite the file if the file exist. If file not exist then create new file.

- a)w b) w+ c) wb+ d) wb

8.It opens a file for both reading and writing. Overwrite the existing file if the file exists.

- a)w b) w+ c) wb+ d) wb

9.It opens a binary file for both reading and writing. The file pointer is placed at the beginning of the file.

- a)w b) w+ c) wb+ d) wb

10.It opens a binary file for writing only. The file pointer is placed at the beginning of the file.

- a)w b) w+ c) wb+ d) wb

11. It opens a file for appending. The file pointer is placed at the end of the file. If the file is not exist, it will create new file.

a) a b) ab c) a+ d) ab+

12. It opens a binary file for appending. The file pointer is placed at the end of the file.

a) a b) ab c) a+ d) ab+

13. It opens a file for both appending and reading.

a) a b) ab c) a+ d) ab+

14. It opens a binary file for both appending and reading. The file pointer is placed at the end of the file.

a) a b) ab c) a+ d) ab+

15. Which of the following is not a mode of a file

a) r b) r+ c) w+ d) wa+

16. Which of the following is not a function

a) open() b) close() c) splits() d) dump()

17. Which of the following is not used for writing data into file

a) write() b) writes() c) writeline() d) None of the above

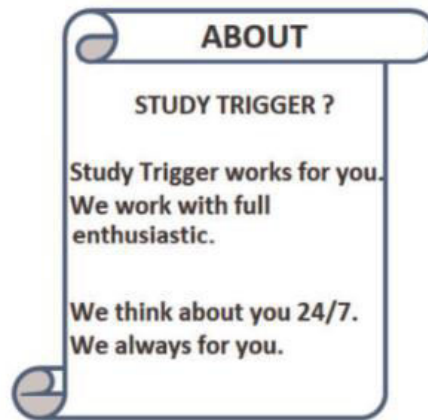
18. readline() method will return

a) string b) a list of integers
c) integer d) a list of strings

20. Directories and folders are the same thing

a) True b) False

Answer the following questions (21 to 24) on the basis of ABOUT.txt file having the below content:



21. What is the output of the statement :

```
data=open("about.txt").read().split()
print(data[2])
```

a) STUDY b) TRIGGER
c) ? d) None of the above

22. What is the output of the statement

```
data=open("about.txt").read().split()
print(data[0][3])
```

a) S b) T c) D d) ?

23. What is the output of the statement

```
data=open("about.txt").read().split()
print(data[1][3])
```

a) S b) T c) G d) ?

24. What is the output of the statement:

```
data=open("about.txt").read()
print(data.count('S'))
```

a) 2 b) 6
c) Generates an error d) None of the above

25. Reading data from Binary file is also called Pickling

a) True b) False

26..... .function places the file pointer at the specified position by in an open file.

- a) tell() b) seek()
c) both a and b d) none of the above

27.function return the current position of the file pointer.

- a) tell() b) seek()
c) both a and b d) none of the above

28. The csv files can only take comma as delimiter.

- a) True b) False

Answers

1. (b) 2. (a) 3. (b) 4. (a) 5. (b) 6. (c) 7. (c)
8. (b) 9. (c) 10. (d) 11. (a) 12. (b) 13. (c) 14. (d)
15. (d) 16. (c) 17. (b) 18. (a) 20. (a) 21. (b) 22. (c)
23. (c) 24. (a) 25. (b) 26. (a) 27. (a) 28. (b)

DATA STRUCTURE & STACK I

1.In Data Structure queue is a type of queue data structure.

- a) True b) False

2.Reversing a word/line is an application of queue.

- a) True b) False

3.The insertion and deletion from a stack takes place only from one ended called the 'TOP'.

- a) True b) False

4.A queue follows the LIFO principle.

- a) True b) False

5.When a queue is implemented with the help of a list ,it is termed as a linear queue

- a) True b) False

6..... is a data organization, management, and storage format that enables efficient access and modification.

- a) Data Manage b) Data Structure
c) Data Manipulation d) None of the above

7.A___ is a linear list, which follow LIFO principle.

- a) stack b) queue
c) both a & b d) None of the above

8.A___ is a linear list ,which follow FIFO principle.

- a) stack b) queue
c) both a & b d) None of the above

9.A___ is a mutable sequence of data elements indexed by their position .

- a) list b) tuple
c) both a & b d) None of the above

10.___ means accessing/processing each element of any data structure.

- a) searching b) traversing
c) insert d) None of the above

11.CPU uses queues to implement round-robin scheduling among waiting processes.

- a) True b) False

12.Pop operation is the Deletion of an element from the queue.

- a) True b) False

13.In Queue, an element is always added from the front end.

- a) True b) False

14.The most important application of queues are "Handling the processes of a network printer".

- a) True b) False

15. You can insert an element at the rear in queue using append () function.

a) True b) False

16. Insertion of elements in a stack is also termed as-----

a) pop b) push
c) remove d) None of the above

17. A queue contains elements "S1", "S2", "S3", and "S4". You want to delete elements one by one, then the order in which they will be removed is :

a) S4S3S2S1 b) S1S2S3S4
c) S2S3S1S4 d) None of the above

18. Insertion of elements in a queue is also termed as

a) Create b) Push c) Enqueue d) Pop

19. If a user tries to remove an element from the empty stack, then it is called -----

a) underflow of stack b) overflow of stack
c) both a & b d) None of the above

20. In search, each element of the list is compared with the given item to be searched for, one by one.

a) linear b) binary
c) both a & b d) None of the above

21. Consider a Stack with TOP=3

Diksha, Reena, Shawna, Teena,,

What will be the value of TOP when "Pihu" and "Khushi" are added to the stack.

a) 5 b) 6 c) 4 d) None of the above

22. Consider a Stack with TOP=3 Diksha, Reena, Shawna, Teena

What will be the value of TOP when "Teena" is deleted from the stack.

a) 5 b) 2 c) 4 d) None of the above

23. Consider a Stack with TOP=-1 and Size = 9

What will be the value of TOP when an element is deleted from the stack.

a) 0 b) 1 c) -1 d) underflow

24. Consider a Stack with TOP=8 and Size = 9

What will be the value of TOP when an element is insert from the stack.

a) 9 b) 8 c) overflow d) underflow

Answers

1. (b) 2. (b) 3. (a) 4. (b) 5. (a) 6. (b) 7. (a)
8. (b) 9. (a) 10. (b) 11. (a) 12. (b) 13. (b) 14. (a)
15. (a) 16. (b) 17. (b) 18. (c) 19. (a) 20. (a) 21. (a)
22. (b) 23. (c) 24. (c)

DATA STRUCTURE & STACK II

1. Removing an element from stack is called:

a) Create b) Push c) Dequeue d) Pop

2. An element can be deleted from a Stack using method

a) pop() b) del() c) remove() d) all of the above

3. Stack size is 5 and it already contains 5 elements, now if you push an element into this stack, then you will get :

a) User flow b) Crash c) Underflow
d) Overflow

4. If insertion can take place at one end (rear) and deletion can be done from another end (front) from a list, this list is called :

a) Queue b) Stack c) Tree d) Linked list

5. An element can be added to a Stack using method?

a) extend() b) insert() c) append() d) none of the above

6. is the best way to implement the type of sharing when a "resource is shared among multiple consumers".

a) stack b) queue
c) both a & b d) None of the above

7. When you are waiting for paying a bill in super market, a is formed to hold all the customers in a line until the billing executive is free.

a) stack b) queue
c) both a & b d) None of the above

8. Airport authorities make use of in situations of sharing a single runway of airport for both landing and take-off flights.

a) stack b) queue
c) both a & b d) None of the above

9. "Reversal of a String/Number or any sequence" is application of

a) stack b) queue
c) both a & b d) None of the above

10. A is a physical implementation that clearly defines a way of storing, accessing, manipulating data stored

a) Data Structure b) Data Type
c) both a & b d) None of the above

11. is/are example(s) of linear data structure

a) stack b) queue
c) both a & b d) None of the above

12. Which of the following operation are performed on data structure?

a) Traversal b) Sorting c) Merging d) All of the above

13. Insertion and deletions take place from different end in

a) Stack b) Queue c) both a & b
d) None of the above

14. data structures are single level data structures

a) Linear b) Non-Linear
c) both a & b d) None of the above

15. data structures are multi-level data structures.

a) Linear b) Non-Linear
c) both a & b d) None of the above

16. Which of the following is/are Python's built in linear data structure.

a) List b) Tuples
c) Dictionaries d) All of the above

17. A defines a set of values along with well-defined operations stating its input-output behaviours:

a) Data Structure b) Data Type
c) both a & b d) None of the above

18. Applications of Stack is/are:

a) A parentheses balancing program
b) Tracking of local variable at run time
c) Compiler syntax Analyzer
d) All of the above

Answers

1. (d) 2. (a) 3. (d) 4. (a) 5. (c) 6. (b) 7. (b)
8. (b) 9. (a) 10. (a) 11. (c) 12. (d) 13. (b) 14. (a)
15. (b) 16. (d) 17. (b) 18. (d)

CONNECTIVITY I

1.Fetching an information from the table is same as read operation on that table.

a)True b) False

2.fetch() and fetchall() methods gives the same result.

a)True b) False

3.Rowcount is a read-only attribute.

a)True b) False

4.We use connect() method to disconnect database connection

a)True b) False

5.To insert the data into a table "update" statement is used.

a)True b) False

6.The resultset return only one row as a tuple by using.....method.

a)fetchone() b) fetchmany()
c) fetchall() d) all of the above

7.How many parameters can be accepted by is_connected() method.

a)2 b) 3 c) 1 d) 0

8.A logical set of records that is fetched from the database refers to

a)resultset b) dataset c)both d)none of the above

9.....statement is used to open a connector to MySql database

a)connector b) connect c) password d) username

10. is called a session between the application program and the database.

a) bridge b) connection c) gap d) none of the above

11.A resultset is returned when a cursor object is used to query a table.

a)True b) False

12.execute() method will be use, after establishing a connection

a)True b) False

13.the next row of resultset is fetched via fetchone().

a)True b) False

14.'hostname' argument inside the connect() function define its name of database server or IP address.

a)True b) False

15.How many parameters can be accepted by connect() method.

a) 2 b) 3 c) 4 d) 5

16.The Is a property of cursor object that returns the number of rows allowed from the cursor so far.

a) fetchall() b) resultset
c) rowcount d) none of the above

17.package need to import before working with MySql.

a)mysql.connector b)mysql.connect
c)sql.connector d) none of the above

18. Which one is correct form to connect the database from Python code, if host is "localhost", user="Study", password="Trigger" and the database is "StudyTrigger".

a) connect(host="localhost", user="Study", password="Trigger", database="StudyTrigger")

b) connect(host="localhost", user="Study", passwd="Trigger", database="StudyTrigger")

c) connect(host="localhost", user="root", passwd="Trigger", database="StudyTrigger")

d) connector(host="localhost", user="Study", passwd="Trigger", database="StudyTrigger")

19. A is a special control structure that facilitates the row-by-row processing of records in the result set.

a) database cursor b) dataset cursor

c) database row d) none of the above

20. is_connected() method is used to check the connection is successfully connected or not.

a) True b) False

Answers

1. (a) 2. (a) 3. (a) 4. (b) 5. (b) 6. (a) 7. (d)

8. (a) 9. (b) 10. (a) 11. (a) 12. (a) 13. (a) 14. (a)

15. (c) 16. (c) 17. (a) 18. (a) 19. (b) 20. (a)

CONNECTIVITY II

1. To count the number of rows in a database, command is used

a) row b) rowcount c) count() d) row_count

2. To displaying only one row from resultset, we use method

a) fetchmany b) fetchno c) fetchall d) fetchone

3. method is used to execute all the rows from the result set.

a) fetchmany b) fetchone

c) fetchall d) none of the above

4. fetchmany(1) and fetchone() gives the same result

a) True b) False

5. Which of the following method require parameters to execute.

a) fetchall() b) fetchmany()

c) fetchone() d) none of the above

6. What will fetchone() method the return if there are no record in the database

a) error b) None

c) crash the program d) none of the above

7. Which command is used for cleaning up the environment.

a) my.close() b) is.close ()

c) con.close() d) mycon.close()

8. Read the given code, assume that all the necessary file already imported:

```
mycon=sql.connect(host="localhost", user="study",  
passwd="Trigger", database="Students")
```

```
cursor=mycon.cursor()
```

```
query="Select * from Students"
```

Which method will you use for executing the above query?

a) fetchall() b) cursor()

c) execute() d) None of the above

9. statement will be executed to ensure that changes you made in the code related to table are also being reflected in the database.

- a) fetchall() b) commit()
c) execute() d) None of the above

10. Whenever you run Insert, Update and Delete query using Python code, you must run method within the connection object.

- a) fetchall() b) commit()
c) execute() d) None of the above

Read the given code, and answers the following :

```
import mysql.connector as ..... //Line 1
mycon=mysql.connector.connect(host="local host",
user="study",
passwd="Trigger", database="Students")
cursor=mycon.cursor()
query=" ..... " //Line 2
cursor.execute( query) data=.....//Line 3

for row in data:
    print(row)
print("Total no of rows is ", ..... ) //Line 4
..... //Line 5
```

11. What is the correct statement for line 1 to import the mysql file

- a) sql b) mysql
c) connection d) None of the above

12. What is the correct statement for line 2 to select all student whose marks is greater than 70

- a) select * from Students where marks >=70
b) select Name from Students where marks >=70
c) select * from Students where marks >70
d) None of the above

13. What is the correct statement for line 3 to fetch all records

- a) fetchone() b) fetchall()
c) fetchmany() d) None of the above

14. What is the correct statement for line 4 to display total number of rows you will from the database

- a) rows b) rowcount
c) data.count d) data.length

15. What is the correct statement for line 5 to cleanup the environment as well as remove the connection

- a) clear() b) clean()
c) close() d) none of the above

Answers

1. (b) 2. (d) 3. (c) 4. (a) 5. (b) 6. (b) 7. (c)
8. (c) 9. (b) 10. (b) 11. (a) 12. (c) 13. (b) 14. (b)
15. (c)

SQL I

1. COUNT(*) functions ignore duplicates and null values while counting the records.

- a) True b) False

2. Max() function is return an integer field.

- a) True b) False

3. Count() and Count(*) are aggregate functions and can be used with any columns (any data types).

- a) True b) False

4. The SQL statement : SELECT Salary + Comm as Total From Emp; add two fields salary and comm from each row together and lists the results in a column named Total.

- a) True b) False

5.Total numbers of columns are called cardinality of a relation. a)True b) False

a)group b) single c) scalar d) all of the above

6.Count(*) function applicable on integer column only. a)True b) False

15. You can arrange the result set in specific order by using Function. However, it can not change the actual data (table).

7.Total numbers of attributes are called degree of a relation. a)True b) False

a)sort by b)order by
c) group by d) none of the above

8.IN and "Between_And" are used for pattern matching. a)True b) False

16.A..... is a column of a table which can be used as Primary key of a table.

9.....query is use to sort the name in reverse order.

a)Primary b) Foreign
c) Candidate d) Alternate

a)sort by b) sort by asc
c) sort by desc d) all of the above

17.Select the query used to open the STUDY database.

10.You can find the total number of rows, including duplicates and NULL in a table using function.

a)Use Study; b) Show Study;
c) Open Study; d) none of the above

a)count() b) count(*) c) countif()
d) all of the above

18.Select the query which is used to show all the tables present in the currently opened database.

11..... .function applied on integer (numeric) column which returns the lowest value from the column.

a)Desc Tables; b) Open Tables
c) Use Tables;d) Show Tables;

a)max() b) minimum() c) min() d) all of the above

19.In which of the following cases a DML statement is not executed.

12.Order by statement sort the result in order by default.

a)new row is added b)row in deleted
c)row is modified d)new column is added

a)ascending b) descending
c) none of the above d) both a & b

20.Select the query which is used to display the structure of the table STUDENT.

13. An AVG() function take parameter astype.
a) int b) string c) char d) depends on data

a)Desc Student; b) Show Student;
c)both a & b d) none of the above

14.You can work on multiple rows at a time and get (return) an aggregate value by using function.

Answers

1. (b) 2. (a) 3. (a) 4. (a) 5. (b) 6. (b) 7. (b)
8. (b) 9. (c) 10. (b) 11. (c) 12. (a) 13. (a) 14. (a)
15. (b) 16. (c) 17. (a) 18. (d) 19. (d) 20.(a)

SQL II

1.You can combine all the records that have identical values in a particular field or a group of fields by using ORDER BY statement.

a)True b) False

2.To filter the conditions for groups, WHERE clause is used. a)True b) False

3.Single row function and Function are the two type of functions supported by MySql.

a)scalar functions b) multi row functions
c) both a & b d) none of the above

4.....function is also termed as Group Functions.

a)aggregate b) scalar
c) both a & b d) none of the above

5..... statement helps you to consider only unique values from the given column.

a)all b) distinct
c) both a & b d) none of the above

6.To count the total number of records in a table, we use sum() function. a)True b) False

7.clause is used to specify filtering condition for groups.

a)where b)having
c)group by d)none of the above

8.Group functions can be applied on any data type i.e. numeric, date, string. a)True b) False

9.Which of the following is not an aggregate functions

a)Max() b) AVG() c)SUM() d) WITH()

10.....function applied on integer (numeric) column which returns the highest value from the column

a)MAX() b) MAXIMUM() c)LARGEST()
d) BIG()

11.A.....is a column of a relation which uniquely identifies each row of a table.

a)Primary Key b) Candidate Key
c) Alternate Key d) Foreign Key

12.....commands are used to change the structure of the table

a)TCL/DCL b)DDL
c)DML d)none of the above

13.To change data stored in the table, you will use..... commands

a)TCL/DCL b)DDL c)DML d)none of the above

14.DELETE and UPDATE commands comes under.....command

a)TCL/DCL b) DDL c) DML d) none of the above

15.Write a query to display the record of all students from "Study" table whose last name contains 6 letters ending with "A".

a)SELECT * FROM Study WHERE LNAME LIKE '%A';
b)SELECT * FROM Study WHERE LNAME LIKE '____ _A';
c)SELECT * FROM Study WHERE LNAME LIKE '____ _%A';
d)SELECT * FROM Study WHERE LNAME LIKE '%%%%%A';

16.Write a query to return the number of not null records in the Project field of " Study" table?

a)SELECT COUNT(PROJECT) FROM Study;
b)SELECT COUNT(*) FROM Study;

c)SELECT PROJECTS (NOT NULL) FROM Study;

d)SELECT COUNT(DISTINCT PROJECT) FROM Study;

17. Write a query to find the average salary of all employee from Salary field of "Employee" table

a)SELECT AVERAGE(SALARY) FROM EMP;

b)SELECT AVG(SALARY) FROM EMP;

c)SELECT SALARY(AVG) FROM EMPLOYEE;

d)None of the above.

18. Select correct SQL query from below to find the temperature in increasing order of all cities from weather table

a)SELECT city FROM weather ORDER BY temperature ;

b)SELECT city, temperature FROM weather ;

c)SELECT city, temperature FROM weather ORDER BY temperature;

d)SELECT city, temperature FROM weather ORDER BY city;

19.wild card (symbol) is used for matching a single character in a string.

a) @ b) # c) o/o d) _

20. The HAVING clause does which of the following

a)Acts EXACTLY like a WHERE clause.

b)Acts like a WHERE clause but is used for columns rather than groups.

c)Acts like a WHERE clause but is used for group rather than rows.

d)Acts like a WHERE clause but is used for rows rather than columns.

Answers

1. (b) 2. (b) 3. (b) 4. (a) 5. (b) 6. (b) 7. (b)

8. (a) 9. (d) 10. (a) 11. (a) 12. (b) 13. (c) 14. (c)

15. (b) 16. (a) 17. (d) 18. (d) 19. (d) 20. (c)

SQL III

1. When you execute SELECT statement, it will return sorted result but only by one column.

a)True b) False

2. When you don't want to consider duplicate rows, then you will use DISTINCT. a)True b) False

3. Behaviour of DISTINCT is different from ALL

a)True b) False

4.wild card (symbol) is used for matching multiple characters in a string.

a) @ b) # c) o/o d) _

5. Choose the option which is used to specify the list of values in SQL commands.

a)IN b) Like c) Or d) And

6. You can use ORDER BY with SELECT and as well as with GROUP BY Statements.

a)True b) False

7. Where is use with Select statement however Having is use with GROUP BY Statement.

a)True b) False

8. Which command is used to add new column in a table.

a)Add b) insert into
c) Alter table ...Add d) None of the above

9.clause is used to apply condition with Group By statement.

a)Group by b)With c)Where d)Having

10. Which command is used to update the size of column in a table.

a) Add b) update
c) Alter tableModify d) None of the above

11. clause of SELECT query allows us to select only those rows in the result that satisfy a specified condition.

- a) Where b) from c) having d) like

12. Drop and Delete are the same thing

- a) True b) False

13.command is used to delete all table with values from memory. After executing this command you will be able to create new table with the deleted table name.

- a) delete b) drop
c) Both a & b d) none of the following

14.command is used to delete values from the table, but it lies on memory. After executing this command you will not be able to create new table with the same table name.

- a) delete b) drop
c) Both a & b d) none of the following

15. Select the correct option, table name is "Study", Rno is field name, then:

- a) Select * from Study where Rno=10;
b) Show * from Study where Rno=10;
c) Select * from Study where Rno is 10;
d) Show * from Study where Rno in 10;

16. If table name is "Study", Rno is field name, then display all data except whose Rno is 10

- a) Select * from Study where Rno<>10;
b) Show * from Study where Rno Not in 10;
c) Select * from Study where Rno Not in 10;
d) Show * from Study where Rno != 10;

17. By mistake, a teacher wrote 75 as 57 in marks column of "Study" table where roll no (Rno) is 10. Now, choose the correct command to update the above marks

- a) Update Study Set Marks=75;
b) Update Table Study Set Marks=75;
c) Update Study Set Marks=75 where Rno=10;
d) Update From Study Set Marks=75;

18. In an institute some student didn't pay the fees and in fees column "Null" is automatically filled. Choose the correct statement to display all the students who haven't paid the fees. Table name is "Study", Field name is "Fees"

- a) Select * from Study where Fees=NULL;
b) Show * from Study where Fees IS 10;
c) Select * from Study where Fees==Null;
d) Select * from Study where Fees IS NULL;

19. In an institute some students left who didn't pay the fees, but instead of deleting the record of these students, teacher updated a field called "IsDeleted" to 1. Choose the correct statement to handle the above situation. Table name is "Study", Field name is "Fees" and "IsDeleted".

- a) Update Study Set IsDeleted =1 where Fees Is NULL;
b) Update Table Study Set IsDeleted =1 where Fees Is NULL;
c) Update Study Set IsDeleted =1;
d) Update From Study Set IsDeleted =1 where Fees Is NULL;;

20.is primary key of some other table and a non-key attribute whose values are derived from the existing table.

- a) Primary Key b) Candidate Key
c) Foreign Key d) None of the above

Answers

1. (b) 2. (a) 3. (a) 4. (c) 5. (a) 6. (a) 7. (a)
8. (c) 9. (d) 10. (c) 11. (a) 12. (b) 13. (b) 14. (a)
15. (a) 16. (a) 17. (c) 18. (d) 19. (a) 20. (c)

SQL IV

1. After applying Where clause in Select Statement you will get only one value.

a) True b) False

2. AS keyword is used to assign a meaningful column name to the result of queries.

a) True b) False

3. SQL is a programming language.

a) True b) False

4. If you want to see duplicate result in a query, then you should use DISTINCT

a) True b) False

5. Creating a database, including creating, altering, and dropping tables and establishing constraints are examples of Data Manipulation Language (DML).

a) True b) False

6. To work with range of values, LIKE keyword is used with WHERE statement.

a) True b) False

7. The keyword BETWEEN can be used while working with a range of values.

a) True b) False

8. keyword is used to specify the table that contains the data to be retrieved.

a) from b) where c) having d) like

9. What is the output of the following query : Select CHAR(70,65,67,69);

a) Face b) Fbce c) FACE d) '70656769'

10. To get all the columns, use instead of column names.

a) % b) \$ c) * d) #

11. clause contains the condition in SELECT statement.

a) having b) like c) group by d) where

12. If you want to execute a statement when both the conditions are true in Where clause, then you will use operator.

a) OR b) NOT IN c) NOT d) AND

13. If you want to execute a statement when one of the conditions is true in Where clause, then you will use operator.

a) OR b) NOT IN c) IN d) AND

14. Consider the query :

SELECT name FROM Study where Fees.....NULL;

Fill in the above blank with an appropriate operator

a) AND b) IS c) LIKE d) NOT LIKE

15. If you want to exclude one or more values (from list), then you will use

a) OR b) NOT IN c) IN d) AND

16. Pattern matching is possible by Keyword.

a) LIKE b) NOT LIKE
c) both a & b d) none of these

17. What is the output of the following query : Select SUBSTR('Study Trigger', 6, 7);

a) Study b) Trigger
c) Study Trigger d) none of these

18.What is the output of the following query : Select INSTR('Study Trigger', 'ri');

- a)8 b) 7 c) 6 d) none of these

19.What is the output of the following query : Select DATE('2021-08-03 01:02:03');

- a) 2021-08-03 b) 03
c) 08 d) 2021-08-03 01:02:03

20.What is the output of the following query : Select MOD(15,4);

- a) 0 b) 1 c) 2 d) 3

Answers

1. (b) 2. (a) 3. (b) 4. (b) 5. (b) 6. (b) 7. (a)
8. (a) 9. (c) 10. (c) 11. (d) 12. (d) 13. (a) 14. (b)
15. (b) 16. (a) 17. (b) 18. (a) 19. (a) 20.(d)

NETWORKING I

1.If data is transmitted at the same time from 2 devices on the same ethernet then it will result as a Collision a)True b) False

2.Different protocols are handled by repeater

- a)True b) False

3..... is known as an intelligent device on the network

- a)hub b)switch
c) both a and b d) none of the above

4..... is not an example of guided media

- a)Coaxial Cable b) Twisted Pair Cable
c) Microwave d) none of the above

5.HTTP and HTTPS are the same for every webpage.

- a)True b) False

6.To forwards data packets along networks device is preferable.

- a)Router b) Bridge
c) Repeater d) Gateway

7.A is a host computer which always depends on the server for request and services

- a)server b) client
c) both a & b d) none of the above

8.Thetype of transmission media works on waves.

- a)Guided b) Unguided
c) Both a & b d) None of the above

9.Thetype of transmission media works on physical path

- a)Guided b) Unguided
c) Both a & b d) None of the above

10..... device is an extension of hub and more efficient than hub.

- a)Hub b)Switch c)Modem
d)All of the above

11.Modem is used to segment networks into sub-networks or subnets.

- a)True b) False

12.SMTP is used by Email for sending and receiving messages. a)True b) False

13.The degeneration of a signal over distance on a network is called attenuation.

- a)True b) False

14.optical fiber is more expensive as well as more faster than coaxial cable. a)True b) False

15.....is a network of networks that spread all over the world.

- a)Internet b) Intranet
c) Networking d) None of the above

16.The standard measurement for data transfer speed is in terms of Megabits per second (Mbps).

- a)True b) False

17.....is a network device that connects dissimilar networks.

- a)bridge b)gateway
c) both a & b d) none of the above

18.WAN stands for

- a) Wide Area Network b) World Area Network
c) Working Area Network d) None of the above

19.Google Drive is an example of.....

- a)public cloud b) private cloud
c) both a & b d) none of the above

20.PAN stands for.

- a) Post Area Network b) Person Area Network
c) Personal Area Network d) None of the above

Answers

- 1.(a) 2. (b) 3. (b) 4. (c) 5. (b) 6. (a) 7. (b)
8. (b) 9. (a) 10. (b) 11. (b) 12. (a) 13. (a) 14. (a)
15. (a) 16. (a) 19. (b) 18. (a) 19. (a) 20. (c)

NETWORKING II

1.In secure communication no one can interfere when 2 devices are communicating with each other.

- a)True b) False

2.OSI Reference Model containslayers

- a) 4 b) 5 c) 6 d) 7

3.IP address and IPv4 address are the same.

- a)True b) False

4.Aset of rules that computers and other devices use when communicating with each other.

- a)Client b) Server c) Standards d) Protocol

5.Hub is a

- a)Broadcast device b) Unicast device
c)Multicast device d)None of the above

6.Any computer which has to be connected to a network must have a

- a)NIC b) NIA c) NID d) None of the above

7.The device that can operate in place of hub is a :

- a)Switch b) Bridge c) Router d) Gateway

8.A weak and corrupted signal is by repeater .

- a)Amplified b) Regenerated
c) Resembled d) Reroutes

9.A central controller is required bynetwork topology.

- a)Star b) Bus c) Mesh d) Tree

10.SMTP is the abbreviation of

- a)Simple Mail Transfer Protocol
b)Simplified Mail Transfer Protocol
c)Single Mail Transfer Protocol
d)Simple Mailed Transfer Protocol

11.VoIP is the abbreviation of

- a)Voice over Internet Protocol
- b)Voice over Intranet Protocol
- c)Video over Internet Protocol
- d)None of these

12DHCP is the abbreviation of

- a)Dynamic Host Control Protocol
- b)Dynamic Host Config Protocol
- c)Dynamic Hyper Controller Protocol
- d)Dynamic High Configuration Protocol

13.DNS is the abbreviation of

- a)Dynamic Name System
- b)Dynamic Network System
- c)Domain Name System
- d)Domain Network Service

14ADSL is the abbreviation of :

- a)Asymmetric Dual Subscriber line
- b)Asymmetric Dual System Line
- c)Asymmetric Duplex System Line
- d)Asymmetric Digital Subscriber Line

15.The transmission capacity of a communication channel is called

- a)Baud b) Speed c) Bandwidth d) None of the above

16. IMAP stand for

- a)Intranet Mail Access Protocol
- b)Internet Mailed Access Protocol
- c)Internet Mail Accessed Protocol
- d)Internet Mail Access Protocol

17.IoT is a phenomenon that connects the smart devices to the internet over wired or wireless connection.

- a)True b) False

18.Which of the following is not referred to as Mailer

- a) Mail Program b) Mailbox
- c) Mail Application d) Mail Client

19. is a network diagnostic tool used to track the pathway taken by a packet on an IP network from source to destination.

- a) traceroute b) tracing
- c) both a and b d) None of the above

20. MAC address and IP address are same in every system. a) True b) False

Answers

1. (a) 2. (d) 3. (b) 4. (d) 5. (a) 6. (a) 7. (a)
8. (b) 9. (a) 10. (a) 11. (a) 12. (a) 13. (c) 14. (d)
15. (c) 16. (d) 17. (a) 18. (b) 19. (a) 20. (b)

7 layers of OSI: (Application Layer, Presentation Layer, Session Layer, Transport Layer, Network Layer, Data Link Layer, Physical)

NETWORKING III

1.Multipoint connection on a single cable is required by topology.

- a)Star b)Bus c) Mesh d) Tree

2.Two friends create a network for sharing the practical files, this is a kind of network

- a)LAN b) WAN c) PAN d) MAN

3.A combination of any two or more than two topologies are called a Topology

- a)Combination b) Hybrid c) Mesh d) Tree

4. In topology all systems are placed in a circular way.

- a) Star b) Bus c) Mesh d) Ring

5. Data is converted in analog to digital and vice versa using device on network

- a) Modem b) Hub c) Switch
d) Router

6. Repeater is a device that regenerates and retransmits the weak signals.

- a) True b) False

7. If you want to connect networks of similar types (same protocols), then you will use device

- a) Hub b) Router c) Bridge d) Gateway

8. device connects dissimilar networks

- a) Hub b) Router c) Bridge d) Gateway

9. is a device which is used to decide the path (route) over the networks.

- a) Hub b) Router c) Bridge d) Gateway

10. In topology each node is directly connected to the server.

- a) Bus b) Star c) Ring d) Mesh

11. protocol is used for direct communication between two computers, typically a personal computer connected by phone line to a server.

- a) FTP b) PPP c) SMTP d) IP

12. is not an example of unguided media

- a) Coaxial Cable b) Radiowave
c) Microwave d) none of the above

13. is the costliest mode of transmission data.

- a) Satellite b) Radiowave
c) Microwave d) None of the above

14. In topology all the nodes are connected to a main cable.

- a) Hybrid b) Star c) Mesh d) Bus

15. By using, phone calls can be done using internet connection.

- a) GSM b) CDMA c) VoIP d) None of the above

16. Telnet is the internet protocol for creating a connection with a remote machine.

- a) True b) False

17. Which communication channel(s) is/are suitable in communication in hilly areas:

- a) Radiowave b) Bluetooth c) WiFi d) None of the above

18. Spam are the unwanted messages or mails.

- a) True b) False

19. ... receives and holds email for an individual until they pick it up.

- a) TCP/IP b) SMTP c) POP3 d) PPP

20. IP address of two computers on a network can be the same.

- a) True b) False

Answers

1. (b) 2. (c) 3. (b) 4. (d) 5. (a) 6. (a) 7. (c)
8. (d) 9. (b) 10. (c) 11. (b) 12. (a) 13. (a) 14. (d)
15. (c) 16. (a) 17. (a) 18. (a) 19. (c) 20. (b)

~~~~~

## XII COMPUTER SCIENCE

*Towards Getting Centum in CS – See Yes!*

### THE CS BIBLE

#### CBSE XII CS CURRICULUM 2022-23

##### REVISION OF PYTHON LEARNT IN XI

- Familiarization with the basics of Python programming: Introduction to Python, features of Python, executing a simple "hello world" program, execution modes: interactive mode and script mode, Python character set, Python tokens (keyword, identifier, literal, operator, punctuator), variables, concept of l-value and r-value, use of comments
- Knowledge of data types: number (integer, floating point, complex), boolean, sequence (string, list, tuple), none, mapping (dictionary), mutable and immutable data types
- Operators: arithmetic operators, relational operators, logical operators, assignment operator, augmented assignment operators, identity operators (is, is not), membership operators (in, not in)
- Expressions, statement, type conversion & input/output: precedence of operators, expression, evaluation of expression, python statement, type conversion (explicit & implicit conversion), accepting data as input from the console and displaying output
- Errors: syntax errors, logical errors, runtime errors
- Flow of control: introduction, use of indentation, sequential flow, conditional and iterative flow control
- Conditional statements: if, if-else, if-elif-else, flowcharts, simple programs: e.g.: absolute value, sort 3 numbers and divisibility of a number
- Iterative statements: for loop, range function, while loop, flowcharts, break and continue statements, nested loops, suggested programs: generating pattern, summation of series, finding the factorial of a positive number etc
- Strings: introduction, indexing, string operations (concatenation, repetition, membership & slicing),

traversing a string using loops, built-in functions: len(), capitalize(), title(), lower(), upper(), count(), find(), index(), endswith(), startswith(), isalnum(), isalpha(), isdigit(), islower(), isupper(), isspace(), lstrip(),rstrip(), strip(), replace(), join(), partition(), split()

- Lists: introduction, indexing, list operations (concatenation, repetition, membership & slicing), traversing a list using loops, built-in functions: len(), list(), append(), extend(), insert(), count(), index(), remove(), pop(), reverse(), sort(), sorted(), min(), max(), sum(); nested lists, suggested programs: finding the maximum, minimum, mean of numeric values stored in a list; linear search on list of numbers and counting the frequency of elements in a list
- Tuples: introduction, indexing, tuple operations (concatenation, repetition, membership & slicing), built-in functions: len(), tuple(), count(), index(), sorted(), min(), max(), sum(); tuple assignment, nested tuple, suggested programs: finding the minimum, maximum, mean of values stored in a tuple; linear search on a tuple of numbers, counting the frequency of elements in a tuple
- Dictionary: introduction, accessing items in a dictionary using keys, mutability of dictionary (adding a new item, modifying an existing item), traversing a dictionary, built-in functions: len(), dict(), keys(), values(), items(), get(), update(), del, clear(), fromkeys(), copy(), pop(), popitem(), setdefault(), max(), min(), count(), sorted(), copy(); suggested programs : count the number of times a character appears in a given string using a dictionary, create a dictionary with names of employees, their salary and access them
- Introduction to Python modules: Importing module using 'import <module>' and using from statement, Importing math module (pi, e, sqrt, ceil, floor, pow, fabs, sin, cos, tan); random module (random, randint, randrange), statistics module (mean, median, mode)

##### FUNCTIONS

Functions: types of function (built-in functions, functions defined in module, user defined functions), creating user defined function, arguments and parameters, default parameters, positional parameters, function returning value(s), flow of execution, scope of a variable (global scope, local scope)

## **FILE HANDLING**

Introduction to files, types of files (Text file, Binary file, CSV file), relative and absolute paths

Text file: opening a text file, text file open modes (r, r+, w, w+, a, a+), closing a text file, opening a file using with clause, writing/appending data to a text file using write() and writelines(), reading from a text file using read(), readline() and readlines(), seek and tell methods, manipulation of data in a text file

Binary file: basic operations on a binary file: open using file open modes (rb, rb+, wb, wb+, ab, ab+), close a binary file, import pickle module, dump() and load() method, read, write/create, search, append and update operations in a binary file

CSV file: import csv module, open / close csv file, write into a csv file using csv.writer() and read from a csv file using csv.reader()

## **DATA STRUCTURE & STACK**

☐ Data Structure: Stack, operations on stack (push & pop), implementation of stack using list.

## **DATA STRUCTURE & STACK**

Evolution of networking: introduction to computer networks, evolution of networking (ARPANET, NSFNET, INTERNET)

## **COMPUTER NETWORKS**

- Data communication terminologies: concept of communication, components of data communication (sender, receiver, message, communication media, protocols), measuring capacity of communication media (bandwidth, data transfer rate), IP address, switching techniques (Circuit switching, Packet switching)

- Transmission media: Wired communication media (Twisted pair cable, Co-axial cable, Fiber-optic cable), Wireless media (Radio waves, Micro waves, Infrared waves)

- Network devices (Modem, Ethernet card, RJ45, Repeater, Hub, Switch, Router, Gateway, WIFI card)

- Network topologies and Network types: types of networks (PAN, LAN, MAN, WAN), networking topologies (Bus, Star, Tree)

- Network protocol: HTTP, FTP, PPP, SMTP, TCP/IP, POP3, HTTPS, TELNET, VoIP

- Introduction to web services: WWW, Hyper Text Markup Language (HTML), Extensible Markup

Language (XML), domain names, URL, website, web browser, web servers, web hosting

## **DATABASE MANAGEMENT**

Database concepts: introduction to database concepts and its need

Relational data model: relation, attribute, tuple, domain, degree, cardinality, keys (candidate key, primary key, alternate key, foreign key)

Structured Query Language: introduction, Data Definition Language and Data Manipulation Language, data type (char(n), varchar(n), int, float, date), constraints (not null, unique, primary key),

create database, use database, show databases, drop database, show tables, create table, describe table, alter table (add and remove an attribute, add and remove primary key),

drop table, insert, delete, select, operators (mathematical, relational and logical), aliasing, distinct clause, where clause,

in, between, order by, meaning of null, is null, is not null, like, update command, delete command, aggregate functions (max, min, avg, sum, count), group by, having clause,

joins: cartesian product on two tables, equi-join and natural join

~~~~~